

WHO FOOD ADDITIVES SERIES: 91

Prepared by the one-hundredth meeting of the
Joint FAO/WHO Expert Committee
on Food Additives (JECFA)

Safety evaluation of certain food additives



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World Health Organization, Geneva, 2025



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PREFACE

The monographs contained in this volume were prepared at the one-hundredth meeting of the Joint Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO) Expert Committee on Food Additives (JECFA), which met at FAO headquarters in Rome, Italy on 10–19 June 2025. These monographs summarize the data on specific food additives reviewed by the Committee.

The ninety-ninth report of JECFA has been published by WHO as WHO Technical Report No. 1056. Reports and other documents resulting from previous meetings of JECFA are listed in [Annex 1](#), a list of acronyms and abbreviations used throughout is provided in [Annex 2](#), and the participants of the meeting are listed in [Annex 3](#). A summary of the conclusions of the Committee with respect to the food additives discussed at the meeting is provided in [Annex 4](#).

JECFA serves as a scientific advisory body to FAO, WHO, their Member States and the Codex Alimentarius Commission, primarily through the Codex Committee on Food Additives, the Codex Committee on Contaminants in Food and the Codex Committee on Residues of Veterinary Drugs in Foods, regarding the safety of food additives, residues of veterinary drugs, and naturally occurring toxicants and contaminants in food. Committees accomplish this task by preparing reports of their meetings and publishing specifications or residue monographs and dietary exposure and toxicological monographs, such as those contained in this volume, on substances that they have considered.

The monographs contained in this volume are based on working papers that were prepared by WHO and FAO experts. An acknowledgement is given at the beginning of each monograph to those who prepared the working papers. The monographs were edited by E. Rowan, Contin, United Kingdom of Great Britain and Northern Ireland.

The monographs are based on evaluations of original studies and the dossiers provided by the sponsor(s) of the compound, relevant published scientific literature and data submitted by Codex members. When consistent with the data from the original study, the monographs may contain parts of the text and tables of the dossier submitted by the sponsor(s), but not the conclusions of the sponsor(s). The monographs and their conclusions are based on independent reviews of the available data and do not constitute endorsement of the position of the sponsor(s). All experts participating in the one-hundredth meeting completed declaration of interest forms. No conflicts of interest were identified.

Any comments or new information on the biological or toxicological properties of or dietary exposure to the compounds evaluated in this publication should be addressed to: WHO Joint Secretary of the Joint FAO/WHO Expert Committee on Food Additives, Department of Food Safety and Zoonoses, World Health Organization, 20 Avenue Appia, 1211 Geneva 27, Switzerland (jecfa@who.int).

**SAFETY EVALUATION OF SPECIFIC FOOD ADDITIVES
(OTHER THAN FLAVOURING AGENTS)**

Adipates (addendum)

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1. Explanation

Adipic acid (International Numbering System for Food Additives [INS] No. 355) is a food additive permitted in the Codex General Standard for Food Additives (GSFA) as an acidity regulator. Adipic acid is also used as a flavouring agent.

At its Twenty-first meeting in 1977, the Committee reaffirmed an acceptable daily intake (ADI) for adipic acid of 0–5 mg/kg body weight (bw) (1), based on a reduced body weight gain in a long-term rat study that had been established at its Ninth meeting in 1965 (2). Details of the toxicological studies that supported the establishment of a group ADI of 0–5 mg/kg bw for adipic acid and its potassium (INS No. 357), sodium (INS No. 356) and ammonium (INS No. 359) salts were made available in the monograph of the Twenty-first meeting (3). Adipic acid was also evaluated by the Committee at its Fifty-third meeting in 1999 as a flavouring agent, when it was concluded that there were no safety concerns with current levels of exposure for this use (4).

At the Forty-eighth session of the Codex Committee on Food Additives (CCFA) (5), it was noted during the process of proposing provisions for adipic acid for the GSFA that the Committee had not undertaken a dietary exposure assessment (6). A circular letter was issued to Codex members and observers to request the submission of use level information for adipic acid to enable a dietary exposure assessment to be conducted (7). Replies to CL 2016/9-FA were

compiled in agenda items of the Forty-ninth session of the CCFA, namely CX/FA 17/49/8 (8), FA/49 CRD12 (9) and FA/49 CRD19 (10).

A literature search was undertaken to identify any publications including estimates of dietary exposure to, or use levels of, adipic acid. No publications were identified.

2. Dietary exposure

2.1 Introduction

The adipates considered in this assessment are those included in the group ADI, namely adipic acid (as the free acid form) and its sodium, potassium and ammonium salts.

The assessment by the current Committee included an evaluation of information submitted to the Committee that was available via Codex documents, and estimates of dietary exposure calculated by the Committee at the current meeting using national food consumption datasets.

As specific estimates of dietary exposure were able to be calculated using national food consumption datasets, the current Committee did not conduct or review estimates of dietary exposure based on screening methods such as the budget method. In addition, the Committee did not calculate international estimates of dietary exposure using the Global Environment Monitoring System (GEMS)/Food cluster diets, as permissions for use of adipic acid are for processed foods, and the consumption data in the cluster diets are primarily for raw commodities and semi-processed commodities.

A literature search was also undertaken to identify any publications that included estimates of dietary exposure to adipates. EBSCO Discovery Service was used, and databases including Medline, Food Science Source, Food Science and Technology Abstracts, and Science Direct were searched as well as a number of science, toxicological, food, nutrition and public health journals. Search terms included “adipate” or “adipic acid” and “dietary intake”, “intake”, “dietary exposure”, “consumption” or “risk assessment”. These terms were also used in a general internet search to capture other publications or reports not included in the scientific literature. No estimates of dietary exposure were found from these searches.

A literature search was also conducted to identify papers containing use levels or concentration data. Using the same databases as above, search terms included “adipate” or “adipic acid” and “concentration”, “content”, “occurrence”, “use level” or “quantification”. These search terms were also used in a general internet search to capture other publications or reports not included in the scientific literature. No use level data were found from these searches.

2.2 Use of adipates

2.2.1 Codex permissions

Adipates are permitted for use globally as a food additive. In the GSFA, adipic acid (INS No. 355) is currently permitted for use as an acidity regulator in three food categories all at 1500 mg/kg: 1.7 Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt),¹ 1.2.1.2 Fermented milks (plain), heat-treated after fermentation and 1.1.4 Flavoured fluid milk drinks¹ (see [Table 2.1](#)).

There are no specific uses in the GSFA indicated for potassium adipate (INS No. 357), sodium adipate (INS No. 356) or ammonium adipate (INS No. 359). The Forty-seventh meeting of the CCFA in 2015 agreed to remove the permissions for potassium adipate and sodium adipate from the GSFA (11). The Forty-eighth session of the CCFA noted that the three salts in the GSFA were previously removed from the group heading for adipates because of a lack of JECFA specifications (6).

There are permissions proposed for the use of adipic acid in a further 20 GSFA food categories through the Codex step process (12). These categories include processed cheeses, desserts, processed vegetables, jams and jellies, confectionery, decorations, cocoa-based spreads, pre-cooked pastas and noodles, seasonings, sauces, water-based flavoured drinks and snack foods. Proposed maximum levels range from 790 to 50 000 mg/kg. This includes a proposed increase to a maximum level for category 1.7 dairy-based desserts, which already has a permitted use. These proposed uses are all at Step 7 (13)² and are listed in [Table 2.1](#).

2.2.2 National/regional permissions and use levels

Permissions for the use of adipic acid (INS No. 355) and its salts also exist in specific countries, with some examples noted below.

In Australia and New Zealand, adipic acid is permitted for use at Good Manufacturing Practices (GMP) in over 50 food classes including dairy products, edible oils and oil products, processed fruit and vegetables and products, chocolate and cocoa products, confectionery, cereal products, processed meats and seafood, table-top sweeteners, special purpose foods, beverages and mixed foods (14). Ammonium adipate is permitted in food category 12.1.3 salt substitutes

¹ For use in products conforming to the Standard for Fermented Milks (standard CXS243-2003) only.

² Step 7 of the Codex Step Procedure for Elaborating Codex Standards is the second-last step for the proposed standard amendments. The draft standard at this stage has been through a round of comment by Codex members and observers, followed by review by the Codex Executive Committee, another round of comment and any required revisions. Step 8 is where the draft standard is submitted to the Executive Committee for review and the Commission for adoption.

Table 2.1
Permitted, proposed and use levels of adipic acid (INS No. 355)

Food category code	Food category name	GSFA current permitted maximum level (mg/kg)	Proposed GSFA maximum level (mg/kg) ^a	Range of maximum use levels (mg/kg) ^b
1.1.4	Flavoured milk drinks	1 500	NA	NA
1.2.1.2	Fermented milks (plain), heat treated after fermentation	1 500	NA	NA
1.6.4	Processed cheese	NA	5 000	200
1.6.5	Cheese analogues	NA	5 000	4 500
1.7	Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)	1 500	6 000	1 000–6 000 ^c
2.1	Fats and oils essentially free from water	NA	NA	3 000
2.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7	NA	30 000	1 000–6 000 ^c
4.1.2.5	Jams, jellies, marmalades	NA	2 000	100–5 500
4.1.2.9	Fruit-based desserts, including fruit-flavoured water-based desserts	NA	30 000	1 000–6 000 ^c
4.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses)	NA	NA	100
4.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweeds in vinegar, oil, brine or soybean sauce	NA	50 000	1 000
4.2.2.5	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g. peanut butter)	NA	NA	2 900
4.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweeds	NA	1 000	210–770
5.1	Cocoa products and chocolate products including imitations and chocolate substitutes	NA	NA	2 000 – 10 000 ^d
5.1.3	Cocoa-based spreads, including fillings	NA	2 000	NA
5.1.5	Imitation chocolate, chocolate substitute products	NA	NA	50
5.2	Confectionery including hard and soft candy, nougats, etc. other than food categories 05.1, 05.3 and 05.4 ^e	NA	30 000	200–20 000
5.2.1	Hard candy	NA	NA	6 500
5.3	Chewing gum ^f	NA	20 000	200–20 000
5.4	Decorations (e.g. for fine bakery wares), toppings (non-fruit) and sweet sauces	NA	2 000	2 000 ^g
6.4.3	Pre-cooked pastas and noodles and like products	NA	1 000	40
6.5	Cereal- and starch-based desserts (e.g. rice pudding, tapioca pudding)	NA	30 000	1 000–6 000 ^c
6.8.4.2	Deep-fried semi-dehydrated soybean curd	NA	NA	1 200

Table 2.1. (continued)

Food category code	Food category name	GSFA current permitted maximum level (mg/kg)	Proposed GSFA maximum level (mg/kg) ^a	Range of maximum use levels (mg/kg) ^b
7	Bakery wares	NA	NA	500
7.1.2	Crackers, excluding sweet crackers	NA	NA	2 000
7.2.1	Cakes, cookies and pies (e.g. fruit-filled or custard types)	NA	NA	2 000
7.2.3	Mixes for fine bakery wares (e.g. cakes, pancakes)	NA	NA	500
9.2.4.1	Cooked fish and fish products	NA	NA	100–1 660
9.2.4.2	Cooked molluscs, crustaceans and echinoderms	NA	NA	120–2 426
9.2.5	Smoked, dried, fermented and/or salted fish and fish products, including molluscs, crustaceans and echinoderms	NA	NA	570
9.3.2	Fish and fish products, including molluscs, crustaceans and echinoderms, pickled and/or in brine	NA	NA	100–840
9.3.3	Salmon substitutes, caviar and other fish roe products	NA	NA	360–900
12.2	Herbs, spices, seasonings and condiments (e.g. seasoning for instant noodles)	NA	NA	30
12.2.2	Seasonings and condiments	NA	50 000	282
12.6	Sauces and like products	NA	10 000	NA
12.6.2	Non-emulsified sauces (e.g. ketchup, cheese sauce, cream sauce, brown gravy)	NA	NA	18–50 000
14.1.4	Water-based flavoured drinks, including sport, energy or electrolyte drinks and particulated drinks	NA	NA	50
14.1.4.1	Carbonated water-based flavoured drinks	NA	790	NA
14.1.4.2	Non-carbonated water-based flavoured drinks, including punches and ades	NA	2 000	NA
14.1.4.3	Concentrates (liquid or solid) for water-based flavoured drinks	NA	10 000	10–10 000
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	NA	NA	10 000 ^c
15.1	Snacks – potato-, cereal-, flour- or starch-based (from roots and tubers, pulses and legumes)	NA	13 000	100–13 000

NA: not applicable for GSFA columns and no reported levels for use levels column.

^a All at Step 7.

^b Provided by countries as well as industry.

^c Includes 1000 mg/kg for dry powdered dessert mixes and 6000 mg/kg for gel-like desserts.

^d Lower end of the range is coatings; upper end of the range is fillings.

^e Range of minimum values reported 1000–2000 mg/kg; average use level reported 6534 mg/kg.

^f Range of minimum values reported 1000–2000 mg/kg; range of average use levels reported 6534–10 000 mg/kg.

^g Only fillings and toppings for fine bakery wares.

^h As a powder.

(at GMP) (15). Potassium adipate is also permitted in over 50 categories at GMP; however, its use is restricted to salt-reduced and low-sodium foods only (14).

In Singapore, adipic acid is permitted as a general-purpose food additive in the Eighth Schedule of the Food Regulations. The permission enables use of the food additive in foods at GMP (16).

In the United States of America (USA) adipic acid is considered Generally Regarded as Safe (GRAS) for use in foods at GMP (8). Specific GMP use levels are noted in regulation 21 CFR 184.1009(d) including for baked goods, non-alcoholic beverages, condiments and relishes, dairy product analogues, fats and oils, desserts, gravies, meat products and snack foods with maximum levels when used at GMP ranging from 4 to 50 000 mg/kg.

A search of a consumer packaged food label database showed that foods containing adipic acid in the USA (specific GMP use levels are noted in regulation 21 CFR 184.1009(d)) are cheeses (processed 200 mg/kg, analogues 4500 mg/kg), dairy-based desserts (5500 mg/kg), fruit-based desserts (5500 mg/kg), confectionery (200 mg/kg), chewing gum (200 mg/kg) and bakery wares (500 mg/kg) (8). There are some categories with GMP levels for which no use was reported, including non-alcoholic beverages, condiments and relishes, fats and oils, gravies, meat products and snack foods.

In the European Union (EU), adipic acid (E 355), sodium adipate (E 356) and potassium adipate (E 357) are permitted for use in 1.4 Flavoured fermented milk products including heat-treated products up to a maximum level of 1000 mg/L (only fruit-flavoured desserts), 5.4 Decorations, coatings and fillings up to 2000 mg/kg (only fillings and toppings for fine bakery wares), 14.1.4 Flavoured drinks up to 10 000 mg/L (only powders for home preparation of drinks), 14.1.5.2 Other up to 10 000 mg/L (only powders for home preparation of drinks) and 16 Desserts excluding products covered in categories 1, 3 and 4 up to 1000 mg/kg for dry powdered dessert mixes, 6000 mg/kg for gel-like desserts and 1000 mg/kg for fruit-flavoured desserts (17).

Use level data were provided in response to the CCFA circular letter (7) to enable JECFA to undertake a dietary exposure assessment.

Use levels were provided for a number of countries from three of the six WHO Regions, including the WHO Region of the Americas (Brazil, Chile, Costa Rica, Cuba and the USA), European Region (from EU countries) and Western Pacific Region (China, Japan, Malaysia and the Republic of Korea). No data were provided for countries of the WHO African Region, South-East Asia Region or Eastern Mediterranean Region.

Use levels were also provided by industry associations, including the International Confectionery Association (ICA), the International Chewing Gum Association (ICGA), the International Council of Grocery Manufacturers Associations (ICGMA) and the International Food Additives Council (IFAC).

The ICA (FA/49 CRD 19) noted that use in confectionery contributes to fruit and sour tastes and increases saliva flow while having low solubility. Maximum use levels provided by the ICA were 12 000 mg/kg for 5.2 Confectionery and 3900 mg/kg for chewing gum (8). This maximum use level for category 5.2 was also noted by the ICGMA (9). The ICGA noted use of adipic acid is not common in chewing gum; however, it is used in specific products for its contribution to fruit and sour tastes. It was noted that a level lower than maximum reported use levels is used, and it is not entirely chewed out of the gum. Maximum use levels were reported by ICGA of 3900–9820 mg/kg in chewing gum (8). A maximum level of 10 000 mg/kg was also proposed by ICGA for chewing gum. The IFAC provided suggested maximum use levels for a range of food categories including 1.6.5 Cheese analogues (4500 mg/kg), 1.7 Dairy-based desserts (5500 mg/kg), 2.1 Fats and oils essentially free of water (3000 mg/kg), 4.1.2.5 Jams, jellies, marmalades (5500 mg/kg), 7.2.3 Mixes for fine bakery wares (500 mg/kg), 12.6.2 Non-emulsified sauces (50 000 mg/kg), 14.1.4 Water-based flavoured drinks (50 mg/kg) and 15.1 Snacks (13 000 mg/kg) (8).

A summary of the provided use level data is listed in [Table 2.1](#). Data were reported in different ways, including minimum and maximum use levels or typical/average use levels. The majority of usage data provided were maximum use levels with average and minimum use levels provided for two categories. The only category with permitted uses in the GSFA for which usage data were provided was 1.7 Dairy-based desserts. Some data provided are higher than the GSFA maximum for category 1.7 of 1500 mg/kg. Use levels were reported for 35 food categories for which there are no permissions in the GSFA. Use levels were reported for 20 more food categories than those with proposed maximum permitted levels (MPLs) in the GSFA.

2.3 Estimates of dietary exposure to adipates

2.3.1 Estimates of dietary exposure

(a) Calculated by the Committee

(i) Methods

Estimates of dietary exposure to adipic acid (INS No. 355) were calculated by the current Committee using food consumption data from the FAO/WHO chronic individual food consumption database – summary statistics (CIFOcOss) database (18).

The CIFOcOss database contains summary food consumption data for a range of countries derived using 2 days or more of dietary survey data from individuals. Foods are coded according to FoodEx2 classifications (19). The Committee used the file mapping the GSFA classifications to FoodEx2 foods

using the exposure hierarchy that was generated by some members of CCFA (20) and submitted to the JECFA Secretariat in 2025, enabling concentration data to be assigned to foods consumed to estimate dietary exposure to adipic acid. Multiple coded foods were evaluated to determine whether they fit with the most appropriate category for this assessment, and a small number were recoded where needed. Where a major food category (e.g. 5.1) was assigned a use level, all subcategories (e.g. 5.1.1, 5.1.2 etc.) were also assigned the same use level, except where a subcategory had its own specific use level provided.

The Committee calculated estimates of dietary exposure for all countries represented in CIFOcOss where consumption data existed for the foods with permissions or use level data. Food consumption amounts for all respondents expressed on a body weight basis in the CIFOcOss database (accessed January 2025) were used; no division by a standard mean body weight was therefore required. Some countries had more than one survey for the same population group collected in different years. The oldest of these surveys were not included, allowing the most recent food consumption data to be used in the dietary exposure calculations. Surveys that included results referred to as “all population” that were in fact only relevant to adults, children or one sex were not included. Mean estimates of dietary exposure were calculated. Results are presented for males and females combined; however, in a few cases only females were surveyed and results are included for that group alone. High estimates of dietary exposure were calculated by the mean multiplied by 2, which approximates the 90th percentile (21).

Three scenarios were considered: (i) GSFA current maximum levels scenario, using the current permissions for use from the GSFA for three food categories at MPLs; (ii) GSFA current and proposed maximum levels scenario, using all three existing GSFA food categories (one with a higher proposed maximum level) and an additional 19 food categories with proposed maximum levels (at Step 7); and (iii) use levels scenario for 36 food categories for which use level data were submitted, using typical use levels where provided as a more realistic estimate of dietary exposure (for two food categories only), or the upper end of the range of maximum use levels where average values were not provided.

For the GSFA current maximum levels scenario and the GSFA current and proposed maximum levels scenario, the concentrations used in the dietary exposure assessment were as per [Table 2.1](#). For the use levels scenario, the concentrations used in the dietary exposure assessment are presented in [Table 2.2](#). These are derived from the concentrations shown in [Table 2.1](#). The source of the value used is described in the last column of [Table 2.2](#). For the majority of foods, the maximum use levels were used as there were only average use levels provided for two food categories. For many cases, the top end of the range of maximum use levels was used as the worst-case scenario. Some use levels were provided for dried or powdered versions of the foods. Where needed, these were converted to

Table 2.2

Concentrations of adipic acid (INS No. 355) used in the use level scenario dietary exposure assessment, and the source of the concentration

Food category code	Food category name	Concentration (mg/kg) as adipic acid	Source of concentration
1.1.4	Flavoured milk drinks	0	—
1.2.1.2	Fermented milks (plain), heat-treated after fermentation	0	—
1.6.4	Processed cheese	200	Maximum use level
1.6.5	Cheese analogues	4 500	Maximum use level
1.7	Dairy-based desserts	5 500 ^a	Top end of maximum range
2.1	Fats and oils essentially free from water	3 000	Maximum use level
2.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7	1 000 ^a	Maximum use level
4.1.2.5	Jams, jellies, marmalades	5 500	Top end of maximum range
4.1.2.9	Fruit-based desserts, including fruit-flavoured water-based desserts	6 000 ^a	Top end of maximum range
4.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses)	100	Maximum use level
4.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweeds in vinegar, oil, brine or soybean sauce	1 000	Maximum use level
4.2.2.5	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g. peanut butter)	2 900	Maximum use level
4.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweeds	770	Top end of maximum range
5.1	Cocoa products and chocolate products including imitations and chocolate substitutes	10 000	Top end of maximum range
5.1.3	Cocoa-based spreads, including fillings (fillings)	2 000	Top end of maximum range
5.1.5	Imitation chocolate, chocolate substitute products	50	Maximum use level
5.2	Confectionery including hard and soft candy, nougats, etc. other than food categories 05.1, 05.3 and 05.4	6 534	Average use level
5.2.1	Hard candy	6 500	Maximum use level
5.3	Chewing gum	10 000	Top end of average use levels
5.4	Decorations (e.g. for fine bakery wares), toppings (non-fruit) and sweet sauces	2 000	Maximum use level
6.4.3	Pre-cooked pastas and noodles and like products	40	Maximum use level
6.5	Cereal- and starch-based desserts (e.g. rice pudding, tapioca pudding)	1 000 ^a	Maximum use level
6.8.4.2	Deep-fried semi-dehydrated soybean curd	1 200	Maximum use level
7	Bakery wares	500	Maximum use level

Table 2.2. (continued)

Food category code	Food category name	Concentration (mg/kg) as adipic acid	Source of concentration
7.1.2	Crackers, excluding sweet crackers	2 000	Maximum use level
7.2.1	Cakes, cookies and pies (e.g. fruit-filled or custard types)	2 000	Maximum use level
7.2.3	Mixes for fine bakery wares (e.g. cakes, pancakes)	500	Maximum use level
9.2.4.1	Cooked fish and fish products	1 660	Top end of maximum range
9.2.4.2	Cooked molluscs, crustaceans and echinoderms	2 426	Top end of maximum range
9.2.5	Smoked, dried, fermented and/or salted fish and fish products, including molluscs, crustaceans and echinoderms	570	Maximum use level
9.3.2	Fish and fish products, including molluscs, crustaceans and echinoderms, pickled and/or in brine	840	Top end of maximum range
9.3.3	Salmon substitutes, caviar and other fish roe products	900	Top end of maximum range
12.2	Herbs, spices, seasonings and condiments (e.g. seasoning for instant noodles)	30	Maximum use level
12.2.2	Seasonings and condiments	282	Maximum use level
12.6.2	Non-emulsified sauces (e.g. ketchup, cheese sauce, cream sauce, brown gravy)	50 000	Top end of maximum range
14.1.4	Water-based flavoured drinks, including sport, energy or electrolyte drinks and particulated drinks	50	Maximum use level
14.1.4.3	Concentrates (liquid or solid) for water-based flavoured drinks	10 000	Top end of maximum range
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	10 000 (650)	Maximum use level, dried basis (made-up basis)
15.1	Snacks – potato-, cereal-, flour- or starch-based (from roots and tubers, pulses and legumes)	13 000	Top end of maximum range

^a Dry powdered dessert mixes assigned 1000 mg/kg and gel-based desserts assigned 6000 mg/kg.

the as-consumed version of the food to combine with the consumption data for the dietary exposure calculations. For category 14.1.5 including coffees, teas and infusions, a concentration of 10 000 mg/kg was provided for the dry powders, and a concentration of 650 mg/kg was assigned to made-up coffees and teas, assuming they were made from a powder as a worst-case scenario. It was assumed that leaves, barks, flowers and espresso did not contain the additive.

(ii) For each scenario

For all three scenarios assessed, the estimates of dietary exposure by country and population subgroup are shown in [Table 2.3](#).

Table 2.3

Estimates of dietary exposure to adipic acid (mg/kg bw per day) derived using the CIFOCCss food consumption database by WHO region, country and age group for the GSFA scenario, proposed GSFA scenario and use levels scenario

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
African Region								
Burkina Faso	2010	3–5 years	NA	NA	1.5	3.0	2.7	5.4
	2010	6–14 years	NA	NA	NA	NA	0.7	1.4
	2010	15–49 years	NA	NA	0.6	1.2	0.9	1.8
	2010	50–74 years	NA	NA	0.3	0.6	0.5	1.0
Democratic Republic of the Congo	2016	15–49 years	NA	NA	17	34	1.6	3.2
Ethiopia	2016	15–49 years	0.1	0.2	66	133	27	53
Mozambique	2010	6–14 years	NA	NA	11	23	1.3	2.6
	2010	15–49 years	0.01	0.02	11	21	1.3	2.5
	2010	All	0.01	0.02	117	21	1.3	2.5
Uganda	2010	15–49 years	NA	NA	NA	NA	0.5	1.0
	2010	50–74 years	NA	NA	NA	NA	0.7	1.4
Zambia	2009	0–35 months	NA	NA	12	25	2.0	4.0
	2009	3–5 years	0.1	0.2	305	59	5.7	11
	2009	15–49 years	0.1	0.2	7.6	15	0.9	1.8
	2009	50–74 years	NA	NA	12	24	0.9	1.8
	2009	All	0.1	0.2	18	35	3.1	6.2
Region of the Americas								
Bolivia (Plurinational State of)	2011	3–5 years	0.3	0.6	43	86	5.5	11
	2011	6–14 years	0.1	0.2	30	60	4.9	9.8
	2011	15–49 years	0.3	0.6	22	45	4.0	8.0
	2011	50–74 years	NA	NA	16	33	1.1	2.2
Brazil	2010	6–14 years	2.2	4.4	25	50	16	32
	2010	15–49 years	1.8	3.6	24	48	15	30
	2009	50–74 years	0.4	0.8	46	91	5.2	10
	2009	> 75 years	0.6	1.2	45	91	4.7	9.4
Guatemala	2016	15–49 years	0.1	0.2	47	94	6.4	13
Mexico	2012	0–35 months	2.3	4.6	13	26	18	36
	2012	3–5 years	2.8	5.6	18	36	22	44
	2012	6–14 years	1.0	2.0	14	27	16	33

Table 2.3. (continued)

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
Peru United States of America	2012	15–49 years	0.3	0.6	8.3	17	7.8	16
	2012	50–74 years	0.2	0.4	4.8	9.6	4.8	9.6
	2012	> 75 years	0.2	0.4	3.8	7.6	3.7	7.4
	2012	All	1.1	2.2	12	23	14	27
	2016	0–35 months	5.0	0.2	25	50	12	25
	2010	0–35 months	NA	NA	1.7	3.4	3.7	7.4
	2010	3–5 years	NA	NA	3.1	6.2	6.7	13
	2010	6–14 years	NA	NA	2.1	4.2	4.0	8.0
	2010	15–49 years	NA	NA	1.9	3.8	1.8	3.6
	2010	50–74 years	NA	NA	2.4	4.8	1.5	3.0
	2010	> 75 years	NA	NA	1.7	3.4	1.2	2.4
	2010	All	NA	NA	2.1	4.2	2.2	4.4
South-East Asia Region								
Bangladesh	2018	0–35 months	NA	NA	NA	NA	0.7	1.4
	2018	3–5 years	NA	NA	NA	NA	0.8	1.6
	2018	6–14 years	NA	NA	NA	NA	0.2	0.4
	2018	15–49 years		NA		NA	0.1	0.2
	2018	> 75 years	NA	NA	NA	NA	0.4	0.8
India	2015	15–49 years	0.0006	0.001	1.6	3.1	1.7	3.4
Indonesia	2016	15–49 years	0.1	0.2	66	133	27	53
European Region								
Austria	2010	3–9 years	3.8	7.6	52	104	33	27
	2018	10–17 years	0.7	1.4	36	71	17	33
	2014	65–74 years	0.7	1.4	90	179	19	38
	2010	65–74 years	0.4	0.8	64	129	13	27
	2010	≥ 75 years	0.8	1.6	95	189	20	39
Belgium	2002	12–35 months	16	33	44	89	62	125
	2014	3–9 years	4.8	9.6	54	108	46	91
	2014	10–17 years	1.6	3.2	31	61	26	52
	2014	65–74 years	0.8	1.6	61	122	19	38
	2004	65–74 years	0.4	0.8	84	168	17	35
Bulgaria	2004	> 75 years	0.4	0.8	81	162	14	29
	2007	3–11 months	0.7	1.4	50	96	11	22
	2007	12–35 months	2.0	4.0	76	152	30	61
	2007	3–9 years	1.7	3.4	66	132	29	57
	Croatia	2011	65–74 years	0.2	0.4	28	55	5.4

Table 2.3. (continued)

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
Cyprus	2014	3–11 months	0.2	0.4	20	39	2.8	5.6
	2014	12–35 months	2.0	4.0	19	39	14	29
	2014	3–9 years	2.4	4.8	20	40	20	39
	2014	10–17 years	1.3	2.6	15	29	10	21
	2014	65–74 years	0.3	0.6	28	57	5.8	12
	2014	65–74 years	0.1	0.2	25	50	3.8	7.6
Czechia	2003	3–9 years	1.3	2.6	30	60	27	53
	2003	10–17 years	0.6	1.2	23	45	16	32
	2003	65–74 years	0.2	0.4	15	30	7.5	15
Denmark	2006	3–11 months	3.1	6.2	17	33	15	30
	2006	12–35 months	6.5	13	33	67	38	76
	2005	3–9 years	4.3	8.6	35	69	39	79
	2005	10–17 years	2.5	5	27	54	20	40
	2005	18–64 years	0.9	1.8	115	229	17	33
	2005	65–74 years	0.8	1.6	126	251	15	30
	2005	≥ 75 years	0.6	1.2	152	304	17	33
	2005	≥ 75 years	0.6	1.2	152	304	17	33
Finland	2001	3–11 months	0.03	0.06	2.0	3.9	2.4	4.8
	2003	12–35 months	6.7	13	36	72	28	56
	2003	3–9 years	6.7	13	43	86	44	88
	2007	10–17 years	2.1	4.2	25	50	18	35
	2012	18–64 years	1.6	3.2	94	187	21	41
	2012	65–74 years	1.1	2.2	81	162	13	27
France	2014	3–11 months	1.1	2.2	82	164	31	62
	2014	12–35 months	6.4	13	178	357	77	155
	2014	3–9 years	4.5	9.0	89	177	54	109
	2014	10–17 years	1.8	3.6	30	60	27	54
	2014	18–64 years	1.1	2.2	83	165	19	38
	2014	65–74 years	0.6	1.2	87	175	15	30
	2014	≥ 75 years	0.5	1	71	142	15	30
Greece	2004	3–9 years	2.9	5.8	24	47	24	48
	2014	10–17 years	0.2	0.4	4.6	9.2	5.7	11
	2014	18–64 years	0.1	0.2	7.6	15	4.8	10
	2014	65–74 years	0.05	0.1	6.5	13	3.0	5.9
Hungary	2003	18–64 years	1.1	2.2	53	107	5.8	12
	2003	65–74 years	1.2	2.4	46	92	3.8	7.6
	2003	≥ 75 years	1.1	2.2	54	108	4.2	8.4

Table 2.3. (continued)

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
Ireland	2008	18–64 years	0.8	1.6	69	137	16	33
	2008	65–74 years	0.8	1.6	90	181	14	28
	2008	≥ 75 years	0.8	1.6	93	185	13	25
Italy	2006	0–35 months	3.4	6.8	25	51	20	40
	2006	3–5 years	2.4	4.8	26	52	20	39
	2006	6–14 years	1.1	2.2	22	43	13	25
	2006	15–49 years	0.5	1.0	24	4	5.0	10
	2006	50–74 years	0.4	0.8	23	47	3.7	7.4
	2006	≥ 75 years	0.3	0.6	21	43	3.7	7.4
	2006	All	0.6	1.2	23	47	5.6	11
	2006	≥ 75 years	0.3	0.6	21	43	3.7	7.4
Latvia	2012	3–11 months	3.3	6.6	87	174	29	58
	2012	12–35 months	5.3	11	159	319	63	127
	2012	3–9 years	3.0	6.0	205	410	75	150
	2012	10–17 years	1.2	2.4	98	197	39	77
	2012	18–64 years	0.4	0.8	94	188	21	41
	2012	65–74 years	0.4	0.8	79	158	15	29
	2012	> 75 years	0.8	1.6	74.3	148.6	18	37
	2012	> 75 years	0.8	1.6	74.3	148.6	18	37
Netherlands (Kingdom of the)	2012	12–35 months	11	21	162	323	72	143
	2012	3–9 years	10	21	124	248	65	129
	2012	10–17 years	3.9	7.8	77	153	36	73
	2012	18–64 years	2.0	4.0	127	254	25	50
	2012	65–74 years	1.6	3.2	138	275	25	50
	2012	≥ 75 years	1.5	3.0	124	248	22	45
	2012	≥ 75 years	1.5	3.0	124	248	22	45
	2012	≥ 75 years	1.5	3.0	124	248	22	45
Portugal	2015	3–11 months	2.5	5.0	6.2	12	4.5	9.0
	2015	12–35 months	3.1	6.2	23	45	11	22
	2015	3–9 years	8.0	16	35	70	20	39
	2015	10–17 years	2.6	5.2	17	33	10	20
	2015	18–64 years	1.1	2.2	33	65	6.3	13
	2015	65–74 years	0.6	1.2	31	62	4.6	9.0
	2015	≥ 75 years	0.4	0.8	30	61	4.9	10
	2015	≥ 75 years	0.4	0.8	30	61	4.9	10
Romania	2012	15–49 years	0.2	0.4	27	54	7.3	15
	2012	50–74 years	0.2	0.4	28	55	5.2	10
	2012	≥ 75 years	0.1	0.2	41	81	7.3	15
Slovenia	2017	3–11 months	1.8	3.6	68	135	30	60
	2017	12–35 months	4.9	9.8	147	293	45	90

Table 2.3. (continued)

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
Spain	2017	10–17 years	1.1	2.2	32	65	16	31
	2017	18–64 years	0.4	0.8	36	72	8.9	18
	2017	65–74 years	0.3	0.6	48	95	8.3	17
	2012	3–11 months	3.3	6.6	28	56	21	42
	2012	12–35 months	9.4	19	43	87	41	81
	2012	3–9 years	6.7	13	27	55	25	51
	2012	10–17 years	2.6	5.2	14	28	12	25
Sweden	2013	18–64 years	0.8	1.6	17	34	5.2	10
	2013	65–74 years	0.7	1.4	15	30	4.2	8.4
	2003	3–9 years	7.3	15	59	119	70	141
	2003	10–17 years	3.9	7.8	36	71	38	75
	2010	18–64 years	1.1	2.2	71	141	23	46
	2010	65–74 years	0.8	1.6	80	160	23	47
	2010	≥ 75 years	0.6	1.2	71	142	20	41
United Kingdom	2011	3–11 months	2.2	4.4	23	45	21	42
	2011	12–35 months	3.3	6.6	38	75	39	78
	2008	3–9 years	3.3	6.6	40	80	48	95
	2008	10–17 years	1.4	2.8	29	58	26	52
	2008	18–64 years	0.6	1.2	69	137	17	33
	2008	65–74 years	0.8	1.6	86	172	14	28
	2008	≥ 75 years	0.8	1.6	93	185	17	34
Eastern Mediterranean Region								
Pakistan	2016	15–49 years	NA	NA	1.0	2.0	1.2	2.4
Tunisia	1997	0–35 months	2.6	5.2	12	23	15	31
	1997	3–5 years	4.1	8.2	21	42	20	40
	2010	6–14 years	0.2	0.4	8.0	16	6.9	14
	2010	15–49 years	0.1	0.2	12	24	3.6	7.2
	2010	50–74 years	0.2	0.4	13	26	2.9	5.8
	1997	> 75 years	1.1	2.2	1.3	2.6	1.5	3.0
	2010	All	0.1	0.2	11	23	4.1	8.2
Western Pacific Region								
China	2002	0–35 months	NA	NA	6.4	13	9.9	20
	2002	3–5 years	NA	NA	1.8	3.6	7.0	14
	2002	6–14 years	NA	NA	2.2	4.4	4.1	8.2
	2002	15–49 years	NA	NA	3.6	7.2	2.2	4.4
	2002	50–74 years	NA	NA	1.8	3.6	2.1	4.2

Table 2.3. (continued)

Country	Year of survey	Age group	GSFA scenario		Proposed GSFA scenario		Use level scenario	
			Mean	High ^a	Mean	High ^a	Mean	High ^a
Lao People's Democratic Republic	2002	> 75 years	NA	NA	9.4	19	2.5	5.0
	2002	All	NA	NA	2.3	4.6	2.7	5.4
	2017	0–35 months	2.9	5.8	8.9	18	3.9	7.8
	2017	3–5 years	12	25	31	62	17	35
	2017	6–14 years	2.3	4.6	29	59	15	30
	2017	15–49 years	0.3	0.6	3.5	7.0	2.6	5.2
	2017	50–74 years	0.1	0.2	7.1	14	3.0	6.0
Malaysia	2017	All	3.0	6.0	14	27	7.2	14
	2013	15–49 years	0.3	0.6	33	65	17	33
	2013	50–74 years	0.1	0.2	28	56	14	28
Philippines	2000	6–14 years	0.1	0.2	4.0	8.0	2.9	5.8
	2016	15–49 years	0.02	0.04	4.2	8.5	1.2	2.4
	2016	50–74 years	0.1	0.2	3.7	7.4	1.4	2.8
Republic of Korea	2016	> 75 years	NA	NA	3.5	7.0	0.5	1.0
	2015	0–35 months	6.4	13	22	45	22	45
	2015	3–5 years	5.1	10	24	49	32	64
	2015	6–14 years	2.5	5.0	15	31	21	42
	2015	15–49 years	0.9	1.8	27	54	12	24
	2015	50–74 years	0.5	1.0	16	31	5.8	12
	2015	> 75 years	0.5	1.0	8.2	16	4.6	9.2
	2015	All	1.2	2.4	20	40	11	23

NA: no exposure calculated because there was no consumption of foods containing the additive for that population group (consumption of a food category may not have been reported, depending on how the food consumption survey data were grouped, summarized, coded and submitted to the CifOCoss database).

^a High exposure is mean $\times$ 2, approximating the 90th percentile (21). High may not be exactly a multiple of 2 because of rounding.

(iii) Summary

A summary of the national estimates of dietary exposure calculated by the Committee for the GSFA, proposed GSFA and use level scenarios is presented in [Table 2.4](#).

For the GSFA current maximum levels scenario, mean estimates for children (aged up to 14 or 17 years) ranged between 0.03 and 16 mg/kg bw per day, for adults (aged from 15 or 18 years) between 0.0006 and 2.0 mg/kg bw per day, and for the general population between 0.01 and 3.0 mg/kg bw per day. High estimates of dietary exposure for children (aged up to 14 or 17 years) ranged between 0.06 and 33 mg/kg bw per day, for adults (aged from 15 or 18 years) between 0.001 and 4.0 mg/kg bw per day, and for the general population between 0.02 and 6.0 mg/kg bw per day.

Table 2.4

Range of national estimates of dietary exposure to adipic acid (mg/kg bw per day) for the three concentration scenarios assessed

Scenario	Population group	Age range	Mean	High ^a
GSFA current maximum levels	Children	Included up to 14 or 17 years	0.03–16	0.06–33
	Adults	Included from 15 or 18 years	0.0006–2.0	0.001–4.0
	General population	All ages	0.01–3.0	0.02–6.0
GSFA current and proposed maximum levels	Children	Included up to 14 or 17 years	1.5–205	3.0–410
	Adults	Included from 15 or 18 years	0.3–152	0.6–304
	General population	All ages	2.1–66	4.2–133
Use levels	Children	Included up to 14 or 17 years	0.2–78	0.4–155
	Adults	Included from 15 or 18 years	0.1–27	0.2–53
	General population	All ages	1.3–27	2.5–53

GSFA: General Standard for Food Additives.

^a High exposure is mean $\times$ 2, approximating the 90th percentile.

For the GSFA current and proposed maximum levels scenario, mean estimates for children ranged between 1.5 and 205 mg/kg bw per day, for adults between 0.03 and 152 mg/kg bw per day, and for the general population between 2.1 and 66 mg/kg bw per day. High estimates of dietary exposure for children ranged between 3.0 and 410 mg/kg bw per day, for adults between 0.06 and 304 mg/kg bw per day, and for the general population between 4.2 and 133 mg/kg bw per day.

For the use levels scenario, mean estimates for children ranged between 0.2 and 78 mg/kg bw per day, for adults between 0.1 and 27 mg/kg bw per day, and for the general population between 1.3 and 27 mg/kg bw per day. High estimates of dietary exposure for children ranged between 0.4 and 155 mg/kg bw per day, for adults between 0.2 and 53 mg/kg bw per day, and for the general population between 2.5 and 53 mg/kg bw per day.

(b) Submitted to the Committee

ICA (10) and ICGMA (9) reported an estimate of dietary exposure for confectionery. At a use level of 12 000 mg/kg in food category 5.2 confectionery, the estimated dietary exposure for 5 g/day of candy consumption would be 1 mg/kg bw per day (assuming a body weight of 60 kg).

ICGA provided an estimate of dietary exposure for chewing gum: based on a daily serving of 3 g/day and a maximum level of 10 000 mg/kg, dietary exposure would be 0.5 mg/kg bw per day (assuming a body weight of 60 kg). It was noted this would be an overestimate based on lower levels typically being used, and because it is not all chewed out of the gum (8).

These estimates of dietary exposure based on single food categories fall within the range of estimates presented above calculated by the Committee for a range of populations that include a range of food categories.

2.3.2 Major food contributors to dietary exposure

For the GSFA current maximum levels scenario, the foods that contributed to high estimates of dietary exposure for children (> 5 mg/kg bw per day) were flavoured milks (including yoghurt drinks in most instances and flavoured milk in one instance) and yoghurt (in a small number of cases). There were no individual foods that contributed more than 5 mg/kg bw per day for adults.

For the GSFA current and proposed maximum levels scenario, there were many foods that individually resulted in estimates of dietary exposure that were more than 5 mg/kg bw per day for both children and adults. These included confectionery, fruit-based desserts, dairy desserts, cereal-based desserts, sauces, liquid drink bases, powdered drink bases and yoghurt.

For the use levels scenario, there were many foods that individually resulted in estimates of dietary exposure that were more than 5 mg/kg bw per day. These included sauces, coffee (for adults), desserts, liquid drink bases and flavoured yoghurts (for children). Of note is that the two foods with the highest contribution to dietary exposures under the GSFA scenario did not have any use levels submitted, and were therefore assumed to contain 0 mg/kg adipic acid for the use levels scenario and were not a contributor to dietary exposure.

3. Comments

3.1 Assessment of dietary exposure

The Committee evaluated information on use levels provided in response to the Codex circular letter and submitted to the Committee for consideration, and dietary exposure estimates submitted to and calculated by the Committee.

Adipic acid is permitted for use in the GSFA in three food categories (1.7 dairy-based desserts, 1.2.1.2 fermented milks (plain) heat-treated after fermentation and 1.1.4 flavoured fluid milk drinks) at a maximum level of 1500 mg/kg. There are also maximum levels for the GSFA being proposed by the CCFA in 20 food categories (one with a higher proposed maximum level and 19 new food categories) at levels ranging from 790 to 50 000 mg/kg (22). Adipic acid and its salts are also approved for use in many countries. Use levels were provided in response to the circular letter by a number of countries and industry associations for 36 food categories, of which one was a permitted GSFA food category and 20 were food categories for which there were no current or proposed GSFA maximum levels. Use levels provided ranged between 10 and 50 000 mg/kg. The Committee noted that some use levels reported were well

below the proposed maximum levels for some foods; however, some use levels were near to or exceeded the existing or proposed maximum levels for a number of other food categories.

The dietary exposure was assessed based on three scenarios of concentration levels: (i) GSFA current maximum levels (for three food categories); (ii) GSFA current and proposed maximum levels (three current [one with a higher proposed maximum level] and 19 additional food categories); and (iii) use levels (36 food categories). The FAO/WHO CIFOCC food consumption dataset was used to calculate estimates of dietary exposure. The FoodEx2 to GSFA food category mapping provided to the JECFA secretariat in 2025 was used to assign concentrations to food categories consumed. The range of estimated dietary exposures for each of the concentration scenarios for children, adults and the general population is provided in [Table 2.4](#). Estimated dietary exposures range between less than 0.001 and 410 mg/kg bw per day across all the mean and high estimates for all child and adult population subgroups and concentration scenarios assessed.

The main contributors to dietary exposure were assessed as individual foods that resulted in dietary exposures of more than 5 mg/kg bw per day. For the GSFA current maximum levels scenario, these were flavoured milks and yoghurts for children only. For the GSFA current and proposed maximum levels scenario, these were confectionery, desserts, sauces, beverages and yoghurts for both children and adults. For the use levels scenario, these were snack foods and sauces for children and adults, coffee for adults, and desserts, liquid drink concentrates and flavoured yoghurts for children.³

4. Evaluation

The high dietary exposure estimates calculated by the Committee for the current meeting for the GSFA current maximum levels scenario were up to 33 mg/kg bw per day for children and up to 4 mg/kg bw per day for adults. For the GSFA current and proposed maximum levels scenario, and the use levels scenario, high dietary exposures were up to 410 mg/kg bw per day for children and up to 304 mg/kg bw per day for adults.

The Committee noted that the estimates of dietary exposure calculated at the current meeting are overestimated for a number of reasons, including: the large number of food categories included; the magnitude of the concentrations; the assumption of MPLs, or the upper end of the range of reported use levels in the food categories; and the assumption that all foods that could contain adipic acid do contain this food additive.

³ Some minor errors were identified after the meeting; these were corrected during the editing process with the approval of the Committee.

The Committee concluded that most of these conservative dietary exposure estimates greatly exceed the current group ADI of 0–5 mg/kg bw. For the GSFA current maximum levels scenario, for children there was exceedance of the group ADI by up to 660% for high exposures. For the GSFA current and proposed maximum levels scenario, the group ADI was exceeded for children and adults by up to around 8000% for high exposures; for the use level scenarios, the group ADI was exceeded for children and adults by up to 3000% for high exposures.

Further, the Committee noted that there is the potential for exceedances of the current group ADI as a result of normal daily consumption amounts of an individual food. For example, using the current permitted maximum level for flavoured milks of 1500 mg/kg, children only need to consume 67 g/day and adults 200 g/day before the group ADI is exceeded. For food containing adipic acid at the proposed maximum level for the GSFA of 50 000 mg/kg, consumption of only a few grams would result in an exceedance of the current group ADI. Many of the foods with use levels that would result in exceedance of the group ADI are frequently or daily consumed foods.

The Committee noted that hazard assessment procedures have evolved and additional relevant data may have become available since the numerical ADI was established at its Ninth meeting in 1965, indicating that a re-evaluation of the toxicology of adipic acid is needed.

4.1 Recommendations

The Committee requests submission of any updated or additional data on use levels of adipic acid to refine the estimates of dietary exposure, including:

- minimum use levels that achieve the technological function of the additive in the range of relevant foods;
- any further or more specific information on typical use levels;
- updated information on proposed maximum levels;
- an updated list of food categories for which uses are actually required; and
- national estimates of dietary exposure based on individual dietary records and typical use levels.

The Committee recommends that an updated toxicological evaluation, including the reassessment of the current group ADI, be conducted.

Updated assessments of toxicological data, dietary exposure estimates and specifications would allow a full safety assessment for this food additive to be completed by the Committee.

Amyloglucosidase (JECFA95-4) from *Rasamsonia emersonii* expressed in *Aspergillus niger* (addendum)

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1. Explanation

At the request of the CCFA at its Forty-eighth session (5), the Committee evaluated the safety of amyloglucosidase (Enzyme Commission [EC] No. 3.2.1.3; Chemical Abstracts Service [CAS] No. 9032-08-0) from *Rasamsonia emersonii* (formerly *Talaromyces emersonii*) expressed in *Aspergillus niger* at its Ninety-fifth meeting (23). It was also at its Ninety-fifth meeting that the Committee allocated the unique JECFA enzyme identifier JECFA95-4⁴ to this enzyme preparation.

The safety evaluation could not be completed at the Ninety-fifth meeting because of the tentative nature of the specifications and the inability to complete

⁴ At its Ninety-fifth meeting the Committee decided that an identification system would be used for all enzyme preparations, consisting of the JECFA meeting number followed by the number reflecting the order of the enzyme in the report.

the allergenicity assessment. A temporary ADI “not specified”⁵ was established, pending the submission of data on digestibility and chemical and technical considerations for this amyloglucosidase.

The term “amyloglucosidase” refers to the amyloglucosidase enzyme and its amino acid sequence; the term “enzyme concentrate” refers to the fermentation product containing the enzyme of interest that is used in the toxicity studies; and the term “enzyme preparation” refers to the product formulated for commercial use.

The Committee has previously assessed the safety of several enzyme preparations derived from *A. niger* (e.g. asparaginase (24); phytase (25)) and established an ADI “not specified” for these enzyme preparations. The Committee also evaluated another amyloglucosidase from *A. niger* at its Thirty-fifth meeting (26), for which an ADI “not specified” was established. On this basis, at its Ninety-fifth meeting the Committee considered that this amyloglucosidase (JECFA95-4) from *R. emersonii* expressed in *A. niger* met the criteria of a Class I, Type iii enzyme, as described in Environmental Health Criteria (EHC) 240 (27). A Class I, Type iii enzyme preparation is produced by a Safe Food Enzyme Production Strain or a Presumed Safe Progeny Strain. Although toxicity data and a dietary exposure assessment are not required for Class I, Type iii enzymes, the previous Committee evaluated the submitted data.

Amyloglucosidase hydrolyses (1→4)- α as well as (1→6)- α glycosidic linkages in starch. This enzyme preparation is intended for use as a processing aid in starch processing.

The present Committee conducted a literature search and identified 370 references. None of the identified publications was considered relevant to the current evaluation.

1.1 Genetic background

No new information on the genetic background was submitted for the current evaluation. The production organism *A. niger* is a non-pathogenic filamentous fungus that is ubiquitous in the environment (28,29). It has a history of use in the production of enzymes intended for use in food processing and citric acid production. In their natural habitats, *A. niger* strains have the potential to produce mycotoxins and other secondary metabolites. However, the *A. niger* production strain expressing amyloglucosidase does not produce mycotoxins under the production conditions used.

The non-toxicogenic and non-pathogenic production strain, *A. niger* NZYM-BE, was constructed through a combination of chemical mutagenesis and genetic modifications of the parental strain. The expression plasmid contained

⁵ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

a promoter sequence obtained from *A. niger* BO-1, the amyloglucosidase gene obtained from *R. emersonii*, a terminator sequence obtained from *A. niger* BO-1 and an antibiotic marker. The stability of the introduced amyloglucosidase gene was confirmed by Southern blot hybridization. The production strain is deposited in Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH, Germany.

1.2 Chemical and technical considerations

Amyloglucosidase is produced by controlled fermentation of a pure culture of the *A. niger* production strain. The manufacture of the amyloglucosidase enzyme preparation includes fermentation, recovery and formulation. After fermentation, the broth containing the amyloglucosidase enzyme is separated from the biomass by sedimentation followed by several filtration steps. The resulting concentrate is formulated into a liquid preparation. The entire process is performed in accordance with current GMP and with food-grade raw materials. The enzyme concentrate is tested to be free from the production organism and any antibiotic activity.

The primary sequence of amyloglucosidase produced by *A. niger* consists of 591 amino acids; its molecular weight by calculation from the determined amino acid sequence is 62.8 kDa. The amyloglucosidase produced by *A. niger* is not expected to have any secondary or subsidiary activity.

The activity of amyloglucosidase is measured using the method for determining glucoamylase activity (amyloglucosidase activity) provided in the *Combined compendium of food additive specifications* (30). Amyloglucosidase activity is determined spectrophotometrically (400 nm) by measuring the hydrolysis of *p*-nitrophenyl- α -D-glucopyranoside (PNPG). The quantity of *p*-nitrophenol (PNP) liberated per unit of time against a standard PNP preparation is used to calculate the enzyme activity. Amyloglucosidase activity is measured using the unit Glucoamylase M. One Glucoamylase M is defined as the amount of amyloglucosidase required to liberate 0.1 mmol/min of PNP from the PNPG substrate solution under the assay conditions (pH 4.3; 50 °C). The mean total organic solids (TOS) of the five batches of the enzyme concentrate was 10.26% and the mean enzyme activity/TOS ratio was 4.9 Glucoamylase M/mg TOS.

Amyloglucosidase catalyses the hydrolysis of (1 $\rightarrow$ 4)- α and (1 $\rightarrow$ 6)- α glycosidic linkages in starch. The amyloglucosidase enzyme preparation is intended for use as a processing aid in starch/carbohydrate processing, alcoholic beverage processing, brewing and other cereal-based beverage processing, baking and other cereal-based processes, and fruit and vegetable processing. The amyloglucosidase enzyme preparation is used at a maximum level of 931 mg TOS/kg starch-based raw material for solid foods and 1090 mg TOS/kg starch-based raw material for non-milk beverages. The TOS includes the enzyme of interest

and residues of organic materials (e.g. proteins, peptides and carbohydrates) derived from the production organism during the manufacturing process. The amyloglucosidase enzyme is inactivated by heat or removed during processing and is not expected to have any technological function in the final food.

2. Biological data

2.1 Assessment of potential allergenicity

The Committee previously evaluated the potential for allergenicity of the enzyme preparation (23) by conducting a homology search of the amino acid sequence of amyloglucosidase with those of known allergens in the AllergenOnline (31) and Allergen (32) databases according to bioinformatics criteria recommended by EHC 240 (27). A search for matches with more than 35% identity in a sliding window of 80 amino acids, and a search for exact matches over contiguous stretches of eight amino acids and a full-length FASTA sequence with an E-value⁶ of 0.1 or less, were conducted to assess homology to any known allergens. Comparisons between highly homologous proteins yield E-values approaching zero, indicating very low probability that such matches would occur by chance (larger E-values indicate a lower degree of similarity). The Committee identified a match to a respiratory allergen, the amyloglucosidase Sch c 1 found in split-gill mushrooms (*Schizophyllum commune*), which is associated with occupational asthma in bakers (33–35). This allergen showed 60–84% identity with amyloglucosidase over a sliding window of 80 amino acids (with and without scaling); 47.9% identity over the full length of the protein; and 10 matches based on a 100% identity over eight contiguous amino acids. The Committee acknowledged that amyloglucosidase could pose a risk as a respiratory allergen that, when ingested, may also cause a food allergy (36,37). However, the Committee also noted that *S. commune* is consumed as a traditional food throughout Asia and Central America, and there are no published reports of *S. commune* being linked to a food allergen. The Committee did not have any data on the digestibility of the enzyme at the time, and was therefore unable to complete the assessment of the potential for allergenicity of dietary exposure to this enzyme.

In response to the previous Committee's recommendation to provide digestibility data (23), the sponsor submitted a report of an in vitro digestibility study (38) on this enzyme conducted using simulated gastric fluid (SGF) and simulated intestinal fluid (SIF) solutions. In this study, the enzyme was treated with or without pepsin from porcine gastric mucosa in SGF solution (pH 1.2–1.6) or pancreatin in SIF solution (pH 6.8), and analysed by using sodium dodecyl-sulfate polyacrylamide gel electrophoresis with Coomassie brilliant blue staining

⁶ The E-value selected for a search tends to be larger for searches of short sequences ($E < 0.1$) than of long sequences ($E < 1 \times 10^{-7}$), as the likelihood of random matches is greater in the search of shorter sequences.

and Western blotting. The enzyme was rapidly digested with pepsin such that the digestion was complete after 1 hour in SGF under the testing conditions. The enzyme was relatively slowly digested with pancreatin and was still detectable after 6 hours in SIF under the testing conditions. Even though the digestion of the enzyme in the SIF was slow and not complete at the end of the testing period, given that the enzyme was rapidly and fully digested in the SGF within 1 hour, the Committee concluded that the enzyme has the potential to be readily digested in the acidic conditions of the stomach.

The Committee noted that the enzyme preparation would be heat-denatured under conditions of food processing.

Although the Committee acknowledged that the enzyme is homologous to the known respiratory allergen Sch c 1, when taking into account the results of the digestibility study the Committee concluded that dietary exposure to the enzyme is not anticipated to pose a risk for allergenicity.

2.2 Toxicological studies

No new toxicological data on the enzyme were submitted for the current evaluation.

2.2.1 Acute toxicity

No data on the acute toxicity of the enzyme preparation were available.

2.2.2 Short-term studies of toxicity

No new short-term toxicity studies were submitted for the current evaluation. The previous Committee evaluated a report of a 13-week oral toxicity study (39) in male and female rats that received an amyloglucosidase enzyme concentrate at dose levels of up to 1470 mg TOS/kg bw per day. Based on the absence of any treatment-related adverse effects in this study, the previous Committee identified a no-observed-adverse-effect level (NOAEL) of 1500 mg/kg bw per day (rounded by the Committee from 1470 mg TOS/kg bw per day), the highest dose tested.

2.2.3 Genotoxicity

No new genotoxicity studies were submitted for the current evaluation. The previous Committee evaluated data from two genotoxicity studies conducted with the enzyme concentrate: an in vitro bacterial reverse mutation assay with some strains of *Salmonella typhimurium* and *Escherichia coli* (40), and an in vitro micronucleus induction assay with cultured human peripheral lymphocytes (41), both with or without metabolic activation. Based on negative results in both studies, the previous Committee had no concerns regarding the potential for genotoxicity of the enzyme concentrate (42).

2.2.4 Other studies

The previous Committee evaluated the enzyme for potential toxicity by conducting a homology search in which the amino acid sequence of the enzyme was compared with known toxins in the UniProt database (42). This search revealed no biologically relevant homology; the previous Committee therefore concluded that the enzyme was unlikely to be a toxin (23).

3. Dietary exposure

No new information for the dietary exposure assessment of the enzyme was submitted for the current evaluation. This enzyme is intended for use in starch processing, alcohol distilling processes, brewing processes, baking processes, and fruit and vegetable processing. The previous Committee considered these uses for the dietary exposure assessment and estimated a theoretical maximum daily intake (TMDI) of the TOS from this amyloglucosidase enzyme preparation as 5.8 mg TOS/kg bw per day for solid foods and 2.7 mg TOS/kg bw per day (the same as the sponsor) for non-milk beverages (23). This resulted in a total dietary exposure of 9 mg TOS/kg bw per day (rounded from 8.5 mg/kg bw per day). The previous Committee concluded that the dietary exposure estimate of 9 mg TOS/kg bw per day was appropriate for use in the evaluation.

For the dietary exposure assessment, it was assumed that 100% of the TOS from the enzyme preparation remains in the final food. The enzyme is heat-inactivated by high temperatures during processing of foods and food ingredients.

4. Comments

4.1 Assessment of potential allergenicity

The Committee previously evaluated the potential for allergenicity of the enzyme preparation (23) by conducting a homology search of the amino acid sequence of amyloglucosidase with those of known allergens in the AllergenOnline (31) and Allergen (32) databases, according to bioinformatics criteria recommended by EHC 240 (27). The previous Committee identified a match to a respiratory allergen, the amyloglucosidase Sch c 1 found in split-gill mushrooms (*S. commune*) that is associated with occupational asthma in bakers (33–35). This allergen showed 60–84% identity with amyloglucosidase over a sliding window of 80 amino acids (with and without scaling); 47.9% identity over the full length of the protein; and 10 matches based on a 100% identity over eight contiguous amino acids. The previous Committee acknowledged that amyloglucosidase could pose a risk as a respiratory allergen that, when ingested, may also cause a food allergy (34,35). However, the previous Committee also noted that *S. commune* is consumed as a traditional food throughout Asia and Central America, and there are no

published reports of *S. commune* being linked to a food allergen. In the absence of an in vitro digestibility study, the previous Committee could not complete the assessment of potential for allergenicity from dietary exposure to this enzyme.

The present Committee evaluated the submitted in vitro digestibility study (36), in which the enzyme was rapidly digested with pepsin such that no enzyme was detected after 1 hour in SGF, indicating its potential to be readily digested in the acidic conditions of the stomach.

The Committee noted that the enzyme preparation would be heat-denatured under conditions of food processing.

Based on the results of the digestibility study together with the previously evaluated information, the Committee concluded that dietary exposure to the enzyme is not anticipated to pose a risk for allergenicity.

4.2 Toxicological studies

No new toxicological data were submitted for the present evaluation.

4.3 Assessment of dietary exposure

No new information for the dietary exposure assessment of the enzyme was submitted. The previous Committee concluded that the dietary exposure estimate of 9 mg TOS/kg bw per day was appropriate for use in the evaluation.

5. Evaluation

The previous Committee acknowledged that this amyloglucosidase is homologous to the known respiratory allergen Sch c 1. However, when taking into account the results of the digestibility study that showed that this enzyme is readily digested in the acidic conditions of the stomach, the present Committee concluded that dietary exposure to this enzyme is not anticipated to pose a risk for allergenicity.

Based on the absence of any treatment-related adverse effects in the 13-week oral toxicity study, the previous Committee identified a NOAEL of 1500 mg TOS/kg bw per day, the highest dose tested. The previous Committee concluded that the dietary exposure estimate of 9 mg TOS/kg bw per day was appropriate for use in the evaluation. Comparison of the NOAEL of 1500 mg/kg bw per day with the dietary exposure estimate of 9 mg TOS/kg bw per day yields a margin of exposure (MOE) of more than 160.

Based on this MOE and a lack of concern for allergenicity and genotoxicity, the Committee removed the temporary designation of the previous ADI and established an ADI “not specified”⁷ for amyloglucosidase (JECFA95-4)

⁷ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

from *R. emersonii* expressed in *A. niger* when used at the levels specified and in accordance with current GMP.

The specifications monograph was revised and the tentative status was removed.

Ascorbyl palmitate (addendum)

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1. Explanation

Ascorbyl palmitate (INS No. 304) is a food additive used as an antioxidant in a wide range of foods and beverages, including infant formula, formula for special medical purposes (FSMP) for infants and food for young children.

At its Forty-third session, the Codex Committee on Nutrition and Foods for Special Dietary Uses (CCNFSDU) (43) concluded that the use of ascorbyl palmitate as an antioxidant at 1 mg/100 mL in all types of formula covered by food category (FC) 13.1.1 Infant formulae was technologically justified. However, the CCNFSDU noted that ascorbyl palmitate has a numerical ADI and has not been assessed by the Committee for infants younger than the age of 12 weeks. At its Fifty-fourth session (44), the CCFA requested a full safety evaluation of ascorbyl palmitate, in particular for infants younger than the age of 12 weeks.

Presently, in accordance with the GSFA provisions, ascorbyl palmitate may be used at levels of up to 10 mg/kg (as consumed) in FC 13.1.1 Infant formulae, 50 mg/kg (as consumed) in FC 13.1.2 Follow-up formulae, 10 mg/kg (as consumed) in FC 13.1.3 Formulae for special medical purposes for infants and 10 mg/kg (as consumed) in FC 13.3 Dietetic foods intended for special medical purposes (excluding products in FC 13.1).

In addition to the CCFA request, the sponsor requested an increase in the use levels of ascorbyl palmitate in each of the food categories, which include infant-specific uses, to 100 mg/kg (as consumed). For FC 13.3, the sponsor is also proposing to remove the restriction that ascorbyl palmitate may only be added in food products conforming to the *Guidelines for ready-to-use therapeutic foods* (products intended for infants aged > 6 months and children aged < 5 years) (45). This will expand the use of ascorbyl palmitate in FC 13.3 to include foods captured under this food category that can be consumed by population groups aged 5 years and older.

The safety of ascorbyl palmitate was considered previously by the Committee at its Sixth, Thirteenth and Seventeenth meetings (46–48). At its Thirteenth meeting, the Committee established an ADI of 0–1.25 mg/kg bw as ascorbyl palmitate or ascorbyl stearate, or the sum of both. The ADI was based on the level of ascorbyl palmitate that caused no significant toxicological effects in rats in a 2-year study at a dose of 125 mg/kg bw per day, the highest dose tested.

In response to the Committee's request for data, a dossier with chemical and technical, dietary exposure and toxicological data on ascorbyl palmitate was submitted by the sponsor. This dossier included references related to the safety of

ascorbyl palmitate previously reviewed by the Committee, studies published prior to 1973 but not previously reviewed by the Committee, and studies conducted since the previous evaluations.

A literature search relevant to the safety of ascorbyl palmitate was also conducted by the Committee in the database PubMed for the period 1973–2025 based on the search terms “(ascorbyl palmitate) AND (food* OR additive*)”, and in Google (no publication date criteria) based on the search terms “ascorbyl palmitate food additive”. The PubMed search retrieved 142 references. References from the first 10 pages of Google were reviewed. No additional publications identified in the literature search were considered relevant for inclusion in the evaluation.

The Committee also conducted a literature search to consider any studies on dietary exposure to ascorbyl palmitate. For this, the databases Embase, Scopus and PubMed were searched using the terms “ascorbyl palmitate”, “6-O-palmitoylascorbic acid”, “INS 304”, “E 304” OR “E304” AND “food”, “diet*”, “food additive” OR “nutrition” AND “intake”, “expos*”, “consum*”, “concentr*” OR “eat*” without a publication date limit. Only an EFSA opinion on dietary exposure to this food additive for Europe was identified for inclusion in the evaluation (49).

1.1 Chemical and technical considerations

Ascorbyl palmitate (CAS No. 137-66-6) is an ester formed from the reaction of the primary alcohol group of ascorbic acid (vitamin C) and the carboxylic group of palmitic acid. Its chemical formula is $C_{22}H_{38}O_7$ and its molecular weight is 414.5 g/mol. Ascorbyl palmitate can be produced enzymatically or by chemical synthesis. Chemical synthesis of ascorbyl palmitate involves esterification in the presence of sulfuric acid, followed by extraction with suitable solvents (petroleum ether, mesityl oxide and acetone) and further purification steps. The final product, a white or yellow-white powder with a citrus-like odour, is 98% pure on the dried basis.

Ascorbyl palmitate functions as an antioxidant, providing a fat-soluble alternative to ascorbic acid. It is used in various food categories, including infant formula, follow-up formula and FSMP for infants.

When stored in the dark under ambient conditions, ascorbyl palmitate is stable for at least 30 months. The oxidation mechanism of ascorbyl palmitate in food is expected to be that of ascorbic acid. The fate of ascorbyl palmitate in food is expected to be similar to that of ascorbic acid and palmitic acid.

2. Biological data

2.1 Biochemical aspects

2.1.1 Absorption, distribution and elimination

No studies on the absorption, distribution, biotransformation or elimination of ascorbyl palmitate were available to previous Committees at the Sixth, Thirteenth, Fifteenth or Seventeenth meetings. Studies not previously assessed by the Committee are summarized below.

2.1.2 Biotransformation

Ascorbyl palmitate, a fatty acid ester of vitamin C, is hydrolysed in the gastrointestinal tract (GIT), followed by absorption of the hydrolysis products (ascorbic acid and palmitic acid). Carboxylesterases are expected to play a role in the hydrolysis of ascorbyl palmitate. Carboxylesterases catalyse the hydrolysis of a wide variety of endogenous and exogenous chemicals, including esters, thioesters, carbamates and amides (50). Hydrolysis of an ester bond results in the formation of the corresponding carboxylic acid and alcohol (50–53).

The two most significant forms of carboxylesterases in humans are carboxylesterase-1 and carboxylesterase-2 (50–53). Carboxylesterases are principally located in tissues that maintain a barrier function including the small intestine and other parts of the GIT, liver, kidneys and lungs (50,52,53). In humans, carboxylesterase-1 is predominantly expressed in the liver with virtually none found in the small intestine. Carboxylesterase-2 is expressed mainly in the small intestine, but also in the liver, colon and kidney (50–53).

Expression of carboxylesterases in humans is age-dependent, with newborns and infants having lower levels of carboxylesterases than older children and adults (54–57). Carboxylesterase-1 and carboxylesterase-2 messenger ribonucleic acid (mRNA) expression in the liver increased rapidly following birth (54–58). Carboxylesterase-2 mRNA expression was detected in the infant duodenum within a few days after birth and increased 2.7-fold between the ages of 1 and 2 months. Duodenum carboxylesterase-2 protein levels also increased with age (56). Carboxylesterase hydrolytic activity has been detected in liver samples from infants, although activity is generally lower than that in older children and adults (54,55,58).

In a recent *in vitro* study, ¹⁴C-labelled ascorbyl palmitate (radiolabel in the ascorbic acid moiety) was incubated in SIF containing bile acid and pancreatic enzymes (pH 7.5) at 37 °C at concentrations of 0.1 and 0.2 mM. Incubation samples were collected after 0.05, 0.25, 0.5, 0.75, 1.0, 2.0 and 3.0 hours and analysed by liquid scintillation counting and high-performance liquid chromatography (HPLC) with radiodetection. In a second (repeat) study, samples were collected for up to 6.0 hours. Controls included incubation with ¹⁴C-labelled ascorbic acid and incubations in heat-deactivated intestinal fluid.

^{14}C -ascorbyl palmitate concentrations decreased with incubation time. Concomitantly, ^{14}C -ascorbic acid and two additional radiolabelled hydrolysis products, P1 and P2, were formed. Hydrolysis of ascorbyl palmitate was virtually complete after 5–6 hours. No ester cleavage was observed in the heat-deactivated control incubations, suggesting an enzymatic process. The radiolabelled products P1 and P2 were also observed in incubations with ascorbic acid. These products were tentatively identified by liquid chromatography (LC) and mass spectrometry (MS) analysis as dehydroascorbic acid and 2,3-diketogulonic acid, substances reported to be ascorbic acid degradation products (59,60). The half-lives of ascorbyl palmitate hydrolysis and the formation of ascorbic acid and its degradation products were estimated to be 15–45 minutes and 20–30 minutes, respectively. The authors concluded that ascorbyl palmitate is rapidly and completely hydrolysed to ascorbic acid and palmitate using SIF (61).

In a study in albino guinea pigs, groups of five males fed a diet containing no or very low levels of vitamin C received an equimolar dose of 0.015 mmoles/kg bw ascorbic acid (2.6 mg/kg bw per day) or ascorbyl palmitate (6.2 mg/kg bw per day) in 40% ethanol for 45 days. There were no significant differences in food intake, body weight or body weight gain between the two groups. Ascorbic acid levels in the liver, spleen and brain were similar between the two groups, while a slight but significantly higher amount of ascorbic acid was observed in the adrenal glands of animals given ascorbyl palmitate than those given ascorbic acid (16.5 ± 0.84 mg/100 g versus 15.1 ± 0.67 mg/100 g, respectively; *P*-value not reported) (62).

Hartley guinea pigs (10 males per group) were fed a diet containing no or very low vitamin C until signs of scurvy were observed. Animals then received the vehicle control (20% ethanol) or equimolar amounts of ascorbic acid (4.9 mg/kg bw per day) or ascorbyl palmitate (11.6 mg/kg bw per day) for 9 days. Mortality rates during the repletion period were 20, 10 and 50% for the ascorbic acid, ascorbyl palmitate and control groups, respectively. Mean weight loss was significantly greater in the vehicle group compared with either treatment group. Animals given ascorbyl palmitate had a greater weight loss than animals treated with ascorbic acid on days 5 and 6 of repletion, but the total body weight change over the repletion period was not significantly different between groups. Based on the areas under the body weight versus time curves during repletion, ascorbyl palmitate appeared to be about 50% as effective as ascorbic acid in reversing signs of scurvy. Mean ascorbic acid concentrations in plasma, liver, spleen and adrenal glands were similar in ascorbic-acid-treated and ascorbyl-palmitate-treated guinea pigs, and significantly greater than those in the vehicle control group (63).

In a human volunteer study, seven male participants were given an oral dose of L-ascorbic acid (300 mg). On a separate occasion the same volunteers were given ascorbyl palmitate (700 mg, equivalent to 285 mg L-ascorbic acid). Ascorbyl palmitate was given in a solid form as a suspension in water. Elimination

of ascorbic acid in the urine over a 24-hour period was similar following administration of L-ascorbic acid and ascorbyl palmitate (64).

In another study with human volunteers, six healthy non-smoking adults (one male and five female) consumed a self-selected diet excluding foods rich in vitamin C for 12 weeks. On day 27, a baseline 24-hour urine sample was collected. The following day, blood was collected and participants consumed either 2 g ascorbic acid or an equimolar amount of ascorbyl palmitate (4.7 g). A 24-hour urine sample was collected and blood was taken at intervals over the 24-hour period. Participants then consumed the same amount of ascorbic acid or ascorbyl palmitate daily for 2 weeks, followed by collection of a blood sample on day 42. This 6-week protocol was then repeated in a cross-over, double-blind manner.

Mean plasma vitamin C values following ascorbyl palmitate ingestion were 6–19% lower compared with ascorbic acid over the first 8 hours following the first dose. At 24 hours post-dose, plasma vitamin C values were similar for ascorbic acid and ascorbyl palmitate. Mean urinary vitamin C elimination was 33% lower following ascorbyl palmitate ingestion compared with ascorbic acid ingestion, but the difference was not statistically significant. After 2 weeks of supplementation, plasma levels of vitamin C were slightly higher following ascorbyl palmitate ingestion compared with ascorbic acid ingestion, although the difference was not statistically significant (63).

2.2 Toxicological studies

2.2.1 Acute toxicity

Ascorbyl palmitate was of low acute oral toxicity in mice and rats (65–68) (Tokita, unpublished report, University of Toho, 1968).

2.2.2 Short-term studies of toxicity

No new short-term toxicity studies were available.

In a study evaluated by the previous Committee, groups of 10 male Osborne-Mendel rats were fed control diets or diets containing 2 or 5% ascorbyl palmitate (equivalent to 2000 or 5000 mg/kg bw per day, respectively) for 9 months. The growth rate was significantly retarded in the high-dose group, and two of the 10 rats had numerous bladder stones and hyperplasia of the bladder epithelium. Another rat in the high-dose group showed an inflammatory condition in the kidney. There was a slight decrease in body weight gain in the low-dose group, but there were no significant differences between these rats and the controls with respect to mortality and histopathology (69).

2.2.3 Long-term studies of toxicity and carcinogenicity

No new long-term studies of toxicity and carcinogenicity were available.

A 2-year study in rats was evaluated at the Sixth meeting, and also considered at the Thirteenth meeting as part of the Committee's evaluation of ascorbyl stearate (46,47). Heat-treated lard containing 1 or 5% of ascorbyl palmitate (0.05 or 0.25% of the total diet; equivalent to 25 or 125 mg/kg bw per day, respectively) was fed to groups of 10 male Osborne-Mendel rats for 2 years. No adverse effects were observed in any of the experimental animals as determined by growth rate, mortality and pathological examination (69). The NOAEL was 125 mg/kg bw per day, the highest dose tested.

2.2.4 Genotoxicity

There is only one bacterial reverse mutation assay study on ascorbyl palmitate available, in which ascorbyl palmitate did not show any mutagenic activity at concentrations of 0.01–3.3 mg/plate (70).

Although genotoxicity studies examining the full range of end-points were not available, ascorbyl palmitate is metabolized pre-systemically to normal dietary components (ascorbic acid and palmitic acid), which does not raise any concerns for genotoxicity.

2.2.5 Reproductive and developmental toxicity

No reproductive or developmental toxicity studies were available for ascorbyl palmitate.

2.2.6 Special studies

(a) Tumour promotion

Female CF-1 mice (12 per group) were fed diets containing 0 or 2% ascorbyl palmitate. After 2 weeks of treatment, six mice per group were given subcutaneous injections of azoxymethanol once weekly for 6 weeks. Ascorbyl palmitate did not inhibit azoxymethanol-induced hyperproliferation of colonic epithelial cells. In the uninduced group that received 2% ascorbyl palmitate, body weights were comparable to controls and no overt signs of toxicity were observed (71).

In another study, male Fischer 344 (F344) rats (48 per group) were fed a control diet or diet containing ascorbyl palmitate at concentrations equivalent to 200 or 400 mg/kg bw per day for 52 weeks. After 2 weeks, 36 rats per group were given two subcutaneous injections of azoxymethane once per week for 2 weeks at 15 mg/kg bw, and 12 rats per group were given saline. All groups were continued on their respective dietary regimen until the termination of the experiment 52 weeks after treatment with azoxymethane. Body weights of animals given diets containing ascorbyl palmitate were similar to those of controls throughout

the study. Ascorbyl palmitate at 200 mg/kg bw per day inhibited multiplicities of non-invasive and total (invasive plus non-invasive) adenocarcinomas of the colon. Ascorbyl palmitate at 400 mg/kg bw per day significantly inhibited the incidence and multiplicities of invasive and total adenocarcinomas of the colon (72).

2.3 Observations in humans

2.3.1 Human trials

A specific carbohydrate dietary plan with supplementation of ascorbyl palmitate (dose not stated in the report) was concluded to be safe and effective for the reduction of autism spectrum disorder (ASD) symptoms in children in a non-randomized, 3-month dietary intervention study of children aged 2–18 years diagnosed with ASD (or with autistic symptoms waiting to be diagnosed) (73). In the absence of information on dosing, this study is not useful for the safety assessment of ascorbyl palmitate.

In a self-controlled 28-day study, 14 women aged 35–45 years received 100 g of wheat bread fortified with 30 mg of ferrous sulfate/kg of bread and incremental doses of ascorbyl palmitate at 0 (day 1; control), 222 (day 2), 445 (day 14) and 890 (day 15) mg/100 g of bread, equivalent to a 1:1, 2:1 and 4:1 ascorbyl palmitate to iron molar ratio, respectively (74). Ascorbyl palmitate at concentrations of 445 and 890 mg/100 g of bread increased iron absorption by 14.6 and 20.2%, respectively, compared with the control. No adverse events and symptoms related to the tolerance of ascorbyl palmitate were reported.

(a) Infants

No clinical studies designed to assess the safety or tolerability of ascorbyl palmitate in infants were available. The sponsor identified seven paediatric trials designed to evaluate the safety and tolerability of infant formula that were reported to include ascorbyl palmitate as an ingredient in the test and/or control formula (75–81). The concentrations of ascorbyl palmitate in the infant formula were not reported; these studies were therefore not useful for the evaluation of the safety of ascorbyl palmitate to infants. A summary of these paediatric trials is provided in [Table 2.1](#).

Table 2.1

A summary of the seven paediatric trials of infant formula containing ascorbyl palmitate (concentrations not reported)

Reference	Study design	Test groups	End-points	Results
Auestad et al. (75)	Randomized 12-month trial; 134 healthy, term (≥ 37 weeks) infants aged < 7 days at enrolment, and a non-randomized group of 63 breastfed infants	Formula containing linoleic acid and α -linolenic acid $\pm$ DHA and AA: (i) control: no added long-chain polyunsaturated fatty acids; (ii) DHA + AA; (iii) DHA Ascorbyl palmitate included in intervention and control formula (dose not reported)	Visual acuity, growth (bw, length, hc, including change over 12 months) and tolerance (formula intake; spit-up, vomiting and stool frequency)	No significant difference in visual function and growth between test groups, or in stool consistency between breastfed, control and DHA groups; firmer stools in DHA + AA group at 2 and 4 months
Borschel et al. (76)	Randomized, blinded parallel growth trial to age 112 days (4 months); 213 healthy, term infants aged ≤ 9 days at enrolment	(i) Control: hypoallergenic extensively hydrolysed casein-based formula; (ii) treatment: hypoallergenic amino acid-based formula Ascorbyl palmitate included in intervention (amino acid-based) formula (dose not reported)	Growth (bw, length, hc, including change at 14–112 days) and tolerance (formula intake; and stool frequency and consistency)	No significant difference in growth or in formula intake between test groups; higher number of stools/day in control group at 14 and 28 days
Borschel et al. (77)	Randomized, blinded, parallel growth trial to age 112 days (4 months); 195 healthy, term (37–42 weeks) infants aged ≤ 9 days at enrolment	(i) Control: extensively hydrolysed casein-based RTF formula; (ii) treatment: powdered form of control formula Ascorbyl palmitate included in intervention (powdered) formula (dose not reported)	Growth (bw, length, hc, including change at 14–112 days) and tolerance (formula intake; frequency of spit-up and/or vomit; stool frequency and consistency; and early exit because of intolerance)	No significant difference in growth between test groups; significantly higher formula intake in control group; no significant difference in stool consistency and spit-up and/or vomit between test groups; higher number of stools/day in control group; no significant difference in early exit because of intolerance between test groups
Borschel et al. (78)	Randomized, blinded, parallel growth trial to age 119 days (4 months); 209 healthy, term (37–42 weeks) infants aged ≤ 8 days at enrolment	(i) Control: partially hydrolysed whey-based formula containing palm olein oil; (ii) treatment: partially hydrolysed whey-based formula not containing palm olein oil Ascorbyl palmitate included in intervention (palm olein oil-free) formula (dose not reported)	Growth (bw, length, hc, including change at 14–119 days) and tolerance (formula intake; frequency of spit-up and/or vomit; and stool frequency and consistency)	No significant difference in growth at 14–119 days between test groups; significantly higher bw gains 14–28 days in control group; no significant difference in spit-up and/or vomit, and stools/day between test groups; significantly firmer stools in intervention group at five measured intervals, although no difference at 119 days; no significant difference in early exit from the study (reason unspecified) between test groups

Table 2.1. (continued)

Reference	Study design	Test groups	End-points	Results
Fleddermann et al. (79)	Randomized, blinded, parallel growth trial to age 120 days (4 months); 213 healthy, term (37–41 weeks) infants aged ≤ 28 days at enrolment, and a non-randomized group of 195 breastfed infants	(i) Control: whey-based formula, containing no long-chain polyunsaturated fatty acids; (ii) treatment: formula containing α-lactalbumin and long-chain polyunsaturated fatty acids Ascorbyl palmitate included in intervention and control formula (dose not reported) ^a	Growth (bw, length, hc, including change at 30–120 days) and tolerance (formula acceptance; frequency of colic, flatulence, regurgitation and vomiting; and stool frequency, consistency and colour)	No significant difference in bw and hc measures between test groups; significantly higher length gain (cm/day) at 30–120 days in intervention group; no significant difference in tolerance parameters between test groups; high loss to follow-up in breastfed group (93/185; 50.3%), primarily because of parental refusal (76/93; 81.7%)
Ahrens et al. (80)	Randomized, blinded, parallel growth trial to age 112 days (4 months), with additional follow-up at 12 months; 402 healthy, term (≥ 37 weeks) infants aged < 28 days at enrolment, and a non-randomized group of 101 breastfed infants	(i) Control: extensively hydrolysed, standard protein content, whey-based formula; (ii) partially hydrolysed, low protein content, whey-based formula containing no synbiotics; (iii) partially hydrolysed, low protein content, whey-based formula containing synbiotics; (iv) extensively hydrolysed, low protein content, whey-based formula containing synbiotics Ascorbyl palmitate included in intervention and control formula (dose not reported) ^a	Growth (bw, length, hc, including change at 28–112 days), tolerance (formula intake; frequency of reflux, flatulence, regurgitation and vomiting; and stool frequency, consistency, colour and odour) and biochemical markers	No significant difference in growth or tolerance parameters between test groups; differences in stool colour in intervention groups, although no differences attributed to synbiotics
Kantaras et al. (81)	Randomized, blinded, parallel growth trial to age 17 weeks (4 months); 345 healthy, full-term infants aged ≤ 28 days at enrolment	(i) Control: standard, intact protein-based formula; (ii) partially hydrolysed, whey-based infant formula Ascorbyl palmitate included in intervention and control formula (dose not reported)	Growth (bw, length, hc, body mass index, and anthropometric Z-scores, including change at 28 days–17 weeks) and tolerance (formula intake; frequency of vomits/regurgitation, colic, constipation, diarrhoea and crying episodes; and stool frequency, consistency, colour and odour)	No significant difference in growth between test groups; no significant difference in formula intake, vomits or regurgitation, colic, constipation, diarrhoea and crying episodes between test groups; significant differences in stool consistency and colour between test groups

AA: arachidonic acid; bw: body weight; DHA: docosahexaenoic acid; EFSA: European Food Safety Authority; hc: head circumference; RTF: ready to feed (liquid).

^a Ascorbyl palmitate reported as an ingredient in infant formula by EFSA (82).

2.3.2 Post-market surveillance

Ascorbyl palmitate is added to infant formula at the higher proposed use level of 100 mg/L (as consumed) in certain jurisdictions. The sponsor obtained post-market surveillance data for infant formula containing ascorbyl palmitate currently on the market and launched more than 10 years ago. In addition, the sponsor included information identified from a literature search conducted from 2022 through 2024 of adverse events related to one ascorbyl-palmitate-containing infant formula (Similac Sensitive Infant Formula). Overall, no adverse events attributable to the consumption of ascorbyl palmitate have been reported based on available post-market surveillance data representing over 175 million consumption days.

3. Dietary exposure

3.1 Introduction

The Committee has not previously evaluated the dietary exposure to ascorbyl palmitate as a food additive. Ascorbyl palmitate is endorsed for use as an antioxidant in 40 food categories as specified in the GSFA. In 36 food categories, use is endorsed together with ascorbyl stearate (INS No. 305) at maximum levels ranging from 80 to 5000 mg/kg, expressed as ascorbyl stearate. In five food categories, use is endorsed for only ascorbyl palmitate at maximum levels ranging from 10 to 200 mg/kg. These food categories include foods for infants of age up to 1 year (FC 13.1.1 Infant formulae, FC 13.1.2 Follow-up formulae, FC 13.1.3 Formulae for special medical purposes for infants and FC 13.2 Complementary foods for infants and young children) and foods for special dietary use that are specifically for patients and can only be used under medical supervision (FC 13.3 Dietetic foods intended for special medical purposes (excluding products of FC 13.1)). The maximum levels for FCs 13.1.1, 13.1.2 and 13.1.3 refer to the product as consumed.

For the current meeting, the Committee evaluated the request of one sponsor to increase the maximum levels for ascorbyl palmitate in FCs 13.1.1, 13.1.3 and 13.3 from 10 to 100 mg/kg (as consumed), and in FC 13.1.2 from 50 to 100 mg/kg (as consumed). This sponsor also proposed the removal of the restriction that ascorbyl palmitate may only be added in food products conforming to the *Guidelines for ready-to-use therapeutic foods* (products intended for infants aged > 6 months and children aged < 5 years) (45). This removal will expand the use of ascorbyl palmitate to also include dietetic foods that can be consumed by population groups aged 5 years and older.

The Committee also discussed the submission by two sponsors on the use of ascorbyl palmitate in food supplements and confectionery, as well as the dietary exposure to this additive in the general population for Europe as identified from the literature search (see [Section 3.4](#)).

The Committee did not assess dietary exposure with the use levels provided and food consumption data from the CIFOCC database (18). The main reason for this was that the use levels provided by the two sponsors did not cover all relevant sources of dietary exposure.

3.2 Exposure estimates in infants

One sponsor submitted different dietary exposure estimates of ascorbyl palmitate from its use in FC 13.1.1 Infant formulae, FC 13.1.2 Follow-up formulae and FC 13.1.3 Formulae for special medical purposes for infants based on: (i) enteral feeding volumes of pre-term infants; (ii) recommended consumption of infant formula by WHO and body weight-for-age standards; (iii) estimated energy requirements for fully formula-fed infants; (iv) reported consumed amounts of infant formula; and (v) consumption of infant formula proposed by the Scientific Committee of EFSA (83).

These assessments were based on an ascorbyl palmitate concentration in the consumed product that is equal to the current maximum level in the GSFA and to the requested maximum level by the sponsor (see Table 3.1). As these levels refer to the products as consumed, they are provided in units of milligrams per litre (mg/L). The consumption data were the same as those used by the Committee at its Eighty-seventh meeting when evaluating the dietary exposure to gellan gum (INS No. 418(i)) for its use in infant formula, except for the consumption data proposed by the Scientific Committee of EFSA.

The sponsor did not estimate the exposure based on the current maximum level of 50 mg/L for follow-up formula. Follow-up formula is relevant for infants from 6 months of age. Estimates of exposure to ascorbyl palmitate through the consumption of this type of formula were calculated by the Committee. The different assessments are summarized below.

Table 3.1
Maximum levels used to estimate the dietary exposure to ascorbyl palmitate in infants

GSFA food category		Maximum level (mg/L) ^a	
Number	Name	Current ^b	Requested ^c
13.1.1	Infant formulae	10	100
13.1.2	Follow-up formulae	50	100
13.1.3	Formulae for special medical purposes for infants	10	100

GSFA: General Standard for Food Additives.

^a Maximum levels refer to those in products as consumed.

^b As specified in the GSFA.

^c As requested by the sponsor.

3.2.1 Using enteral feeding volumes of pre-term infants

Ascorbyl palmitate is currently permitted for use in FSMP for infants, which includes formula for preterm infants (< 37 weeks gestation). Enteral feeding volumes for preterm infants are cited to be in the range of 150–180 mL/kg bw per day (84,85). At the current maximum level, these volumes would result in an exposure to ascorbyl palmitate of 1.5–1.8 mg/kg bw per day in preterm infants. At the requested maximum level, the exposure would increase to 15–18 mg/kg bw per day.

3.2.2 From WHO-recommended consumption of infant formula

Exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants was estimated using WHO-recommended average daily consumption amounts of infant formula (86). Based on the current maximum level for both types of formula, the estimated mean exposure to ascorbyl palmitate ranged from 0.6 mg/kg bw per day in the newborn infant to 1.5 mg/kg bw per day in infants aged 1–6 months (see Table 3.2). Based on the requested maximum level, the estimates would increase to 6.0 and 15 mg/kg bw per day, respectively.

Table 3.2
Estimated exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants at the current and requested maximum levels, based on WHO-recommended (86) consumption amounts of infant formula

Age (months)	WHO-recommended consumption of infant formula (mL/kg bw per day)	Estimated exposure to ascorbyl palmitate ^a (mg/kg bw per day)	
		Current maximum level	Requested maximum level
0 (newborn)	60	0.6	6.0
1	150	1.5	15
3	150	1.5	15
6	150	1.5	15

Source: sponsor.
bw: body weight; FSMP: formula for special medical purposes; WHO: World Health Organization.
^a See Table 3.1 for the current and requested maximum levels in infant formula and FSMP for infants.

3.2.3 Using estimated energy requirements for fully formula-fed infants

The sponsor also estimated the exposure to ascorbyl ascorbate based on consumption estimates of infant formula and FSMP for infants assuming that these are consumed in amounts that meet the estimated energy requirements (EERs) for fully formula-fed infants. Standard body weights and the median EERs for infants aged up to 1 year were obtained from the Joint FAO/WHO/United Nations University report on human energy requirements (87). A common

formula energy density of 67 kcal/100 mL (280 kJ/100 mL) was used to convert the EERs to the amount of formula consumed.

Using this approach, the estimated exposure to ascorbyl ascorbate through the consumption of infant formula and FSMP for infants at the current maximum level ranged from 1.2 mg/kg bw per day for boys aged 3–6 months and girls aged 4–6 months to 1.7 mg/kg bw per day in boys aged 0–1 month (see [Table 3.3](#)). For follow-up formula, relevant for infants from the age of 6 months, the estimated exposure at the current maximum level ranged from 5.8 mg/kg bw per day for girls aged 6–8 months to 6.0 mg/kg bw per day for boys aged 9–12 months (see [Table 3.3](#)). The Committee noted that infants from the age of 6 months are also likely to consume complementary foods and that the exposure to ascorbyl palmitate in these children may therefore be higher than reported here.

At the requested maximum level of 100 mg/L, the exposure in infants up to the age of 6 months and aged 6–12 months increased by a factor of 10 and 2, respectively, resulting in a maximum estimate of exposure of 16.9 mg/kg bw per day in boys aged 0–1 month (see [Table 3.3](#)).

The sponsor also considered formula-fed boys and girls aged 14–27 days with high (95th percentile) EERs of 149 and 146 kcal/kg bw per day, respectively (88). Using the same assumptions as for the calculations based on the median EERs, the high exposure to ascorbyl palmitate at the current and requested maximum level was estimated at 2.2 and 22 mg/kg bw per day for boys and girls, respectively. The sponsor noted that more recent EERs in infants were available (89). However, because these were lower, the sponsor did not consider them. The Committee agrees with this.

3.2.4 Using reported consumed amounts of infant formula

The sponsor also estimated the exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants based on the German Dortmund Nutritional and Anthropometrical Longitudinally Designed study (90). In this study, mean and high (95th percentile) consumptions of infant powder (dry) at the age of 3 and 6 months were reported. Using typical preparation instructions for infant formula (13 g of powdered infant formula to yield 100 mL of ready-to-eat formula), the mean exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants at 3 and 6 months was 1.3 and 0.7 mg/kg bw per day, respectively, based on the current maximum level. Corresponding high exposures were 1.9 and 1.2 mg/kg bw per day (see [Table 3.4](#)). At the requested maximum level in both types of formula, the high exposure increased to a maximum of 18.8 mg/kg bw per day for infants aged 3 months.

Table 3.3

Estimated exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants, and follow-up formula, at the current and requested maximum levels, based on standard median EERs for fully formula-fed infants

Age range (months)	Body weight (kg) ^a	Average EERs (kcal/ day) ^a	Volume of formula consumed (mL/day) ^b	Estimated exposure to ascorbyl palmitate ^c (mg/kg bw per day)			
				Infant formula and FSMP		Follow-up formula	
				Current ML	Requested ML	Current ML	Requested ML
Boys							
0–1	4.58	518	773	1.7	16.9	–	–
1–2	5.50	570	851	1.5	15.5	–	–
2–3	6.28	596	890	1.4	14.2	–	–
3–4	6.94	569	849	1.2	12.2	–	–
4–5	7.48	608	907	1.2	12.1	–	–
5–6	7.93	639	954	1.2	12.0	–	–
6–7	8.30	653	975	–	–	5.9	11.7
7–8	8.62	680	1015	–	–	5.9	11.8
8–9	8.89	702	1048	–	–	5.9	11.8
9–10	9.13	731	1091	–	–	6.0	12.0
10–11	9.37	752	1122	–	–	6.0	12.0
11–12	9.62	775	1157	–	–	6.0	12.0
Girls							
0–1	4.35	464	693	1.6	15.9	–	–
1–2	5.14	517	772	1.5	15.0	–	–
2–3	5.82	550	821	1.4	14.1	–	–
3–4	6.41	537	801	1.3	12.5	–	–
4–5	6.92	571	852	1.2	12.3	–	–
5–6	7.35	599	894	1.2	12.2	–	–
6–7	7.71	604	901	–	–	5.8	11.7
7–8	8.03	629	939	–	–	5.8	11.7
8–9	8.31	652	973	–	–	5.9	11.7
9–10	8.55	676	1009	–	–	5.9	11.8
10–11	8.78	694	1036	–	–	5.9	11.8
11–12	9.00	712	1063	–	–	5.9	11.8

Source: sponsor, Committee.

bw: body weight; EER: estimated energy requirement; FSMP: formula for special medical purposes; ML: maximum level.

^a Median body weights and average EERs reported by the Joint FAO/WHO/United Nations University expert report on human energy requirements (87).

^b Volume of consumed formula based on standard energy density of 67 kcal/100 mL (280 kJ/100 mL).

^c See Table 3.1 for the current and requested maximum levels in infant formula, follow-up formula and FSMP for infants.

Table 3.4

Estimated exposure to ascorbyl palmitate from its use in infant formula and FSMP for infants at the current and requested maximum levels based on actual consumption levels of infant formula by infants aged 3 and 6 months

Age (months)	Consumption of infant formula				Estimated exposure to ascorbyl palmitate ^a			
	Dry (g/kg bw per day) ^b		Ready-to-eat (mL/kg bw per day) ^c		Current ML (mg/kg bw per day)		Requested ML (mg/kg bw per day)	
	Mean	High ^d	Mean	High ^d	Mean	High	Mean	High ^d
3	17.1	24.4	132	188	1.3	1.9	13.1	18.8
6	8.6	15.8	66	122	0.7	1.2	6.6	12.2

Source: sponsor.

bw: body weight; FSMP: formula for special medical purposes; ML: maximum level.

^a See Table 3.1 for the current and requested maximum levels in infant formula and FSMP for infants.

^b Mean and high consumption levels of dry formula as reported by Kersting et al. (90).

^c Calculated consumption levels of ready-to-eat formula assuming a reconstitution factor of 13 g of powdered formula per 100 mL of ready-to-eat formula.

^d High: 95th percentile.

For follow-up formula, using the same preparation instructions as for infant formula, the mean exposure to ascorbyl palmitate at the age of 6, 9 and 12 months was 3.3, 2.0 and 1.8 mg/kg bw per day, respectively, based on the current maximum level. Corresponding high exposures were 6.1, 4.1 and 3.4 mg/kg bw per day (see Table 3.5). At the requested maximum level, the exposure would increase by a factor of 2 to a maximum high exposure of 12.1 mg/kg bw per day for infants aged 6 months.

Table 3.5

Estimated exposure to ascorbyl palmitate from its use in follow-up formula at the current and requested maximum levels, based on actual consumption levels of follow-up formula by infants aged 6, 9 and 12 months

Age (months)	Consumption of follow-up formula				Estimated exposure to ascorbyl palmitate ^a			
	Dry (g/kg bw per day) ^b		Ready-to-feed (mL/kg bw per day) ^c		Current ML (mg/kg bw per day)		Requested ML (mg/kg bw per day)	
	Mean	High	Mean	High ^d	Mean	High ^d	Mean	High ^d
6	8.6	15.8	66	122	3.3	6.1	6.6	12.1
9	5.3	10.7	41	82	2.0	4.1	4.1	8.2
12	4.6	8.9	35	68	1.8	3.4	3.5	6.8

Source: Committee.

bw: body weight; ML: maximum level.

^a See Table 3.1 for the current and requested maximum levels in follow-up formula.

^b Mean and high consumption levels of dry formula as reported by Kersting et al. (90).

^c Calculated consumption levels of ready-to-feed formula assuming a reconstitution factor of 13 g of powdered formula per 100 mL of ready-to-feed formula.

^d High: 95th percentile.

3.2.5 Using consumed infant formula from the Scientific Committee of EFSA

In 2017, the Scientific Committee of EFSA published guidance on the risk assessment of substances present in food intended for infants younger than 16 weeks (83). In this guidance, the Scientific Committee proposed the use of a mean and high daily consumption value of 200 and 260 mL infant formula per kg bw, respectively, when performing an exposure assessment for this age group. These values were based on all the available data at that time on the reported consumption of infant formula, and cover the potential high consumption of preterm infants on formula (83).

Based on these consumption values and the current maximum level in infant formula and FSMP for infants, the sponsor estimated the mean and high exposure to ascorbyl palmitate to be 2 and 2.6 mg/kg bw per day, respectively. This exposure would increase to 20 and 26 mg/kg bw per day, respectively, based on the requested maximum level.

In 2020, EFSA estimated the exposure to ascorbyl palmitate (E 304(i)) for its use in foods for infants younger than 16 weeks, including infant formula and “dietary foods for infants for special medical purposes and special formulae for infants” (82). This assessment was based on the same two maximum levels and resulted in identical estimates of mean and high exposure as reported here.

3.3 Exposure estimates through dietetic foods

The sponsor requested an increase of the maximum level from 10 to 100 mg/kg for FC 13.3 Dietetic foods intended for special medical purposes (excluding products of FC 13.1), referred to here as dietetic foods. These foods are intended for the exclusive or partial feeding of people whose nutritional requirements cannot be met by normal foods and are used under medical supervision. The sponsor provided estimates of exposure to ascorbyl palmitate through their consumption in the USA and Europe as part of a general population assessment. The Committee did not consider these estimates, because the number of consumers of these foods within the general population is too low to obtain an estimate of exposure through the consumption of these foods.

The Committee therefore estimated the exposure to ascorbyl palmitate through the consumption of only dietetic foods. For this, a typical consumption of one serving of 330 g/day and a high consumption of two servings per day as reported by the sponsor was used. Based on the current maximum level of 10 mg/kg, the typical and high exposure to ascorbyl palmitate equals 3.3 and 6.6 mg/day, respectively. Expressed per kg bw, this exposure would be 0.5 and 0.9 mg/kg bw per day for an adult of 70 kg. At the requested maximum level of 100 mg/kg, the corresponding estimates would be 4.7 and 9.4 mg/kg bw per day. These estimates include the expansion of the use of ascorbyl palmitate in FC 13.3 to also include dietetic foods captured under FC 13.3 that can be consumed by

population groups aged 5 years and older. The Committee noted that dietetic foods are intended for the exclusive or partial feeding of patients. In the case that patients also consume complementary foods, their exposure to ascorbyl palmitate may be higher than reported here.

3.4 Exposure estimates in the general population

The Committee evaluated the dietary exposure estimates of ascorbyl palmitate in the general population as provided by different sponsors, and for the European population (49).

3.4.1 Provided by sponsors

One sponsor provided use levels of ascorbyl palmitate in confectionery without corresponding exposure estimates (see Table 3.6). Another provided use levels in food supplements (see Table 3.7) and corresponding exposure estimates. A third sponsor provided exposure estimates for the general population of the USA and Europe based on maximum levels in the GSFA combined with individual food consumption data, as well as estimates for the general population of Europe based on use levels submitted to EFSA (49).

The Committee did not consider the exposure estimates provided by the sponsor for food supplements, because they did not cover all relevant sources of dietary exposure (see Section 3.4.2). The submitted estimates for the general population of the USA and Europe were also not considered because they were based on maximum levels, which produce highly conservative estimates of dietary exposure to a food additive for the general population. In addition, the Committee did not consider the European estimates that were based on use levels submitted to EFSA, because a more accurate assessment based on these levels was available from the 2015 EFSA opinion (see Section 3.4.2).

Table 3.6

Use levels of ascorbyl palmitate in confectionery provided by a sponsor

GSFA food category		Use level (mg/kg) ^a	
Number	Name	Minimum	Maximum
05.1	Cocoa products and chocolate products including imitation and chocolate substitutes	50	500
05.1.2	Cocoa mixes (syrops) ^b	10	500
05.1.5	Imitation chocolate, chocolate substitute products ^b	60	500
05.2	Confectionery including hard and soft candy, nougats, etc. other than food categories 05.1, 05.3 and 05.4	50	500

GSFA: General Standard for Food Additives.

^a Maximum level in GSFA is 500 mg/kg for all four food categories.

^b Used in chocolate syrups of particular flavours.

Table 3.7

Use levels of ascorbyl palmitate in food supplements provided by a sponsor

Form ^a	No. products	Use level (mg/kg) ^b
Hard capsules	5	50–54.5; 1 895–14 256 (NP)
Soft capsules	5	200–603; 0.00127 (NP)
Tablets	46	0.0099–12 938
Powders ^c	5	0.00002–7.3; 343 (NP)
Gummies	1	3.73

NP: niche product.

^a All food supplements are in solid form.^b Use levels of products as marketed.^c Powders are consumed after adding water or other suitable liquids for consumption.**3.4.2 Europe**

In 2015, EFSA performed a safety assessment of fatty acids of ascorbyl (E 340), including ascorbyl palmitate (E 304(i)) and ascorbyl stearate (E 304(ii)), in food in Europe (49). The Committee noted that the use of the additives is permitted in the EU and that the countries included in the safety assessment were members of the EU at that time.

For the safety assessment, EFSA received from the food industry use levels of ascorbyl palmitate in foods belonging to food categories for which the use of fatty acids of ascorbyl is permitted in the EU (91). For these food categories, the food industry submitted identical use levels for both fatty acids of ascorbyl, except for FC 6.3 Breakfast cereals, for which the industry provided only use levels for ascorbyl palmitate. For foods for infants aged up to 1 year (FCs 13.1.1–13.1.4, 13.5.1 and 13.5.2, as listed in annex II of Regulation (European Commission) No. 1333/2008 (91)), industry provided use levels for only ascorbyl palmitate, as the use of ascorbyl stearate in these food categories is not permitted in the EU (as in the GSFA). Because EFSA considered it unlikely that more than one antioxidant is used in a food item, they assumed the use levels to reflect the use of ascorbyl palmitate in the EU. Most use levels were reported per gram fat. The mean use level ranged from 50 to 300 mg/kg fat.

Use levels were provided for 71 food categories (including those related to foods for infants aged < 1 year) out of the 77 in which the use of ascorbyl palmitate is permitted in the EU. As ascorbyl palmitate is permitted at *quantum satis*, except in foods for infants aged up to 1 year, only these food categories could be considered in the assessment. Of these, 10 were not included because they are not referenced in the EFSA Comprehensive European Food Consumption Database (Comprehensive Database). In addition, 14 were not included because use levels for these categories were reported per gram fat, and foods belonging to these food categories do not contain fat (e.g. flavoured drinks, other sugars

and syrups, and vinegars). Overall, EFSA considered 47 food categories in the assessment.

EFSA calculated the dietary exposure to ascorbyl palmitate for a brand-loyal and a non-brand-loyal scenario. In the brand-loyal scenario, it was assumed that individuals are long-term exposed to ascorbyl palmitate at the maximum use level for all foods in the main contributing food category at the individual level, and at the mean use level for all foods in the remaining food categories. In the non-brand-loyal scenario, it was assumed that individuals are long-term exposed to this additive at the mean use level for all foods in all food categories. The use levels were combined with individual food consumption data from the Comprehensive Database (see also [Section 3.3](#)). At the time of the EFSA opinion, this database contained data from 33 dietary surveys in 19 European countries, covering infants (age < 1 year), toddlers (age 1–2 years), children (age 3–9 years), adolescents (age 10–17 years) and adults (age 18–64 and ≥ 65 years). Dietary exposure to ascorbyl palmitate was calculated by multiplying the relevant use level per food category by the summed consumption amounts of all foods within that category for each individual. Mean daily dietary exposure for each individual was determined by adding exposure estimates derived from each relevant food category for each survey day, dividing this by the individual's body weight and then averaging over the number of survey days. This resulted in a distribution of individual mean dietary exposure estimates per day. Subsequently, the mean and 95th percentile (P95) exposure estimates were calculated per survey and for each age group. Use levels expressed on fat basis were converted to whole weight based on fat content information obtained from the Comprehensive Database (49).

[Table 3.8](#) lists the estimated dietary exposure to ascorbyl palmitate. Food categories that contributed principally to the mean dietary exposure in both scenarios were processed fruits and vegetables in all age groups, and foods for infants and young children in infants. Additionally, fats and oils essentially free from water (excluding anhydrous milk fat) contributed largely in adults aged 65 years and older in both scenarios, and in the non-brand-loyal scenario in adults aged 18–64 years. Chewing gum could also contribute greatly to the exposure in children and adolescents in the non-brand-loyal scenario.

Despite not including 30 food categories in which ascorbyl palmitate is permitted in the EU, EFSA considered the dietary exposure estimates to overestimate the actual exposure to ascorbyl palmitate as a food additive in Europe as a result of the assumption that all foods within the considered food categories contain ascorbyl palmitate at the mean or maximum use level. In reality, not all foods are likely to contain ascorbyl palmitate as an antioxidant.

Table 3.8
Dietary exposure to ascorbyl palmitate (E 340(i)) from its use as a food additive in the European population for a brand-loyal scenario and a non-brand-loyal scenario^{a,b}

Exposure level ^c	Dietary exposure (mg/kg bw per day) per age group ^d					
	Infants (12 weeks– 11 months)	Toddlers (1–2 years)	Children (3–9 years)	Adolescents (10–17 years)	Adults (18–64 years)	Adults (≥ 65 years)
Brand-loyal scenario						
Mean	0.4–2.1	0.6–2.4	0.7–2.6	0.3–1.5	0.2–1.0	0.2–0.9
High	1.9–4.6	1.4–5.9	1.3–9.9	1.0–5.6	0.3–3.7	0.3–1.7
Non-brand-loyal scenario						
Mean	0.3–0.9	0.5–1.1	0.5–1.8	0.2–1.0	0.1–0.6	0.1–0.4
High	1.1–1.8	0.9–4.1	0.9–7.3	0.4–4.1	0.2–2.7	0.2–0.8

Source: EFSA (49).
 bw: body weight.
^a Exposure was calculated based on maximum and mean use levels for 47 food categories in which the use of the food additive is authorized, and for which use levels were provided by the food industry after a public call for data.
^b The European countries were Austria, Belgium, Bulgaria, Cyprus, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Romania, Spain, Sweden and the United Kingdom, all of which were members of the European Union at the time of the assessment.
^c High exposure: 95th percentile.
^d Range represents the lowest and highest estimate for each age group across the 33 dietary surveys included in the assessment.

3.5 Summary

Table 3.9 summarizes the estimates of dietary exposure to ascorbyl palmitate for infants aged up to 1 year from its use in infant formula, follow-up formula and FSMP for infants at the current maximum level of 10 or 50 mg/L and the requested maximum level of 100 mg/L in the product as consumed; for consumers of dietetic foods intended for special medical purposes at the current and requested maximum level of 10 and 100 mg/L, respectively; and for the European population based on use levels of ascorbyl palmitate in 47 food categories.

In infants, the dietary exposure to ascorbyl palmitate at current maximum levels of 10 or 50 mg/L ranged from 0.6 to 6.1 mg/kg bw per day and at the requested maximum level of 100 mg/L from 3.5 to 26 mg/kg bw per day. The highest high exposure of 26 mg/kg bw per day was based on a high daily consumption value of 260 mL infant formula per kg bw as proposed by the Scientific Committee of EFSA (83).

Based on the requested maximum level of 100 mg/kg in dietetic foods for special medical purposes, a high exposure to ascorbyl palmitate equalled 9.4 mg/kg bw per day based on a high consumption of two servings of 330 g/day and an adult body weight of 70 kg. This estimate of exposure includes the expansion of the use of ascorbyl palmitate in dietetic foods to also include those foods that can be consumed by population groups aged 5 years and older.

Table 3.9

Overview of the exposure estimates for ascorbyl palmitate from its use in infant formula, FSMP for infants and follow-up formula; dietetic foods at the current and requested maximum levels; and in the general European population

Population	Age	Source concentration ^a	Dietary exposure (mg/kg bw per day)	
			Mean	High ^b
Infants				
Enteral feeding ^c	< 0 ^d	Current ML	1.8	
		Requested ML	18	
WHO-recommended consumption	0–6 months	Current ML	0.6–1.5	–
		Requested ML	6.0–15	–
EERs of fully formula-fed infants	0–12 months	Current ML	1.2–6.0	–
		Requested ML	12–17	–
	< 1 month ^e	Current ML	–	2.2
		Requested ML	–	22
Reported consumption	3–12 months	Current ML	0.7–3.3	1.2–6.1
		Requested ML	3.5–13	6.8–19
Scientific Committee of EFSA	0–4 months	Current ML	2.0	2.6
		Requested ML	20	26
Consumers of dietetic foods ^f	19 to ≥ 65 years	Current ML	0.5	0.9
		Requested ML	4.7	9.4
General European population ^g	0 to ≥ 65 years	Use levels, mean ^h	0.2–1.8	0.2–7.3
		Use levels, maximum and mean ⁱ	0.2–2.6	0.3–9.9

Source: sponsor, Committee, EFSA (49).

bw: body weight; EER: estimated energy requirement; EFSA: European Food Safety Authority; FSMP: formula for special medical purposes; ML: maximum level.

^a See Table 3.1 for the current and requested maximum levels in infant formula, follow-up formula, FSMP for infants and dietetic foods.

^b High: 95th percentile, except for the estimates based on food consumption data of the Scientific Committee of EFSA. For these estimates the high consumption level is defined as "high".

^c Based on an enteral feeding volume of 180 mg/kg bw per day (see Section 3.2.1).

^d Age of < 0 months refers to preterm infants (< 37 weeks gestation).

^e Based on formula-fed boys and girls aged 14–27 days with high (95th percentile) EERs of 149 and 146 kcal/kg bw per day, respectively (88).

^f Estimated by the Committee based on a consumption of one serving of 330 g/day and a high consumption of 660 g/day by an adult of 70 kg.

^g The European countries were Austria, Belgium, Bulgaria, Cyprus, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Romania, Spain, Sweden and the United Kingdom, all of which were members of the European Union at the time of the assessment.

^h Estimates based on the non-brand-loyal exposure scenario and mean use levels in 47 food categories. For more details about this scenario, see Section 3.4.2.

ⁱ Estimates based on the brand-loyal exposure scenario and maximum and mean use levels in 47 food categories. For more details about this scenario, see Section 3.4.2.

The exposure for the general European population was calculated by EFSA for different age groups of 23 European countries according to brand-loyal and non-brand-loyal scenarios based on use levels in 47 food categories. The Committee determined that the most appropriate estimates of dietary exposure for Europe for the general population would be those from the non-brand-loyal scenario, because ascorbyl palmitate is added as an antioxidant and other additives are available for this purpose. The high (P95) dietary exposure estimate of ascorbyl palmitate could be up to 7.3 mg/kg bw per day for children aged 3–9 years in Europe, and up to 2.7 mg/kg bw per day in adults. The Committee recognized that for infants the estimates from the brand-loyal scenario were more appropriate as carers may be loyal to a certain brand of infant formula. The high (P95) dietary exposure estimate of ascorbyl palmitate in this age group could be up to 4.6 mg/kg bw per day. This estimate was based on reported use levels by food industry (ranging from 10 to 100 mg/L at the maximum level) and actual consumption levels of foods for infants of age up to 1 year, and of other foods that may contain the additive and are consumed by this age group.

The Committee noted that all estimates will overestimate the actual exposure to ascorbyl palmitate as a food additive in Europe because of the assumption that all foods within the food categories that could contain ascorbyl palmitate did contain the additive at the mean reported use level. Additionally, not all foods are likely to contain an antioxidant and, for those that do, other food additives with the same function may be used.

4. Comments

4.1 Biochemical aspects

Ascorbyl palmitate, an ester of ascorbic acid and palmitic acid, is extensively hydrolysed in the GIT to its constituent acids (61–64).

Hydrolysis is mediated by carboxylesterases, mainly carboxylesterase-2, present in the GIT (50–53). Although expression of carboxylesterases is lower in infants (age < 1 year) than in adults, the esterase is present in the neonatal liver and duodenum, and expression levels increase rapidly in infants during postnatal development (54–58).

Rapid and extensive hydrolysis of ascorbyl palmitate (0.1 and 0.2 mM, equivalent to gastrointestinal concentrations of approximately 40 and 80 mg/L, respectively) into ascorbic acid and palmitic acid was observed in a recent *in vitro* study with SIF containing bile acids and pancreatic enzymes (61).

Increased concentrations of ascorbic acid were observed in plasma, liver, spleen, adrenals and brain following oral administration of ascorbic acid or ascorbyl palmitate to guinea pigs (62,63). Increased urinary excretion of ascorbic acid was observed following intake of ascorbyl palmitate (700 mg or 4.7 g/day) by human volunteers (63,64).

The Committee concluded that any ascorbyl palmitate that was not hydrolysed in the GIT would be subject to hydrolysis to ascorbic acid and palmitic acid in the liver. Ascorbic acid and palmitic acid are normal components of the human diet and undergo typical metabolic pathways following absorption.

4.2 Toxicological studies

In a study previously evaluated by the Committee, no adverse effects were observed in rats fed ascorbyl palmitate for 2 years at 125 mg/kg bw per day, the highest dose tested (46,47).

Ascorbyl palmitate was not mutagenic in a bacterial reverse mutation assay at concentrations up to 3.3 mg per plate (70). Although genotoxicity studies examining the full range of end-points were not available, the Committee noted that ascorbyl palmitate is metabolized in the GIT to ascorbic and palmitic acid, which does not raise any concerns for genotoxicity.

Dietary administration of ascorbyl palmitate to mice and rats did not show a promoting effect on azoxymethanol- or azoxymethane-induced colon tumours (71,72).

4.3 Observations in humans

No adverse effects were associated with the consumption of bread fortified with ascorbyl palmitate at concentrations of up to 890 mg/100 g in a self-controlled, 28-day study in women aged 35–45 years (74).

In seven paediatric trials, no adverse effects attributable to feeding with infant formula containing ascorbyl palmitate were found (75–81). However, the concentrations of ascorbyl palmitate in the infant formula tested were not given; these studies were therefore not useful to the assessment of the safety of ascorbyl palmitate in infants.

No adverse events have been reported in post-market surveillance data representative of over 175 million consumption days of infant formula containing ascorbyl palmitate at levels of 100 mg/L (as consumed).

4.4 Assessment of dietary exposure

At its current meeting, the Committee evaluated the dietary exposure to ascorbyl palmitate based on the current and requested maximum levels in foods for infants aged up to 1 year (FC 13.1.1, 13.1.2 and 13.1.3) and dietetic foods for special medical purposes (FC 13.3). As the maximum levels refer to the products as consumed, they are provided in units of milligrams per litre (mg/L).

Dietary exposure was assessed using consumption data for infant and follow-up formula based on enteral feeding volumes for preterm infants (< 37 weeks gestation), WHO-recommended consumption levels, EERs, actual

reported levels and consumption levels of infant formula as proposed by the Scientific Committee of EFSA (83). Based on these consumption levels, dietary exposure to ascorbyl palmitate at current maximum levels of 10 or 50 mg/L ranged from 0.6 to 6.1 mg/kg bw per day, and at the requested maximum level of 100 mg/L from 3.5 to 26 mg/kg bw per day.

For dietetic foods at the current maximum level of 10 mg/L, a high exposure of 0.9 mg/kg bw per day was calculated based on a high consumption of two servings of 330 g/day and an adult body weight of 70 kg. At the requested maximum level of 100 mg/L, the dietary exposure would increase to 9.4 mg/kg bw per day. These dietary exposure estimates are based on the expansion of the use of ascorbyl palmitate in FC 13.3 to include dietetic foods captured under this food category that can be consumed by population groups aged 5 years and older.

Additionally, the Committee evaluated the dietary exposure to ascorbyl palmitate in the general European population (49). Most use levels used in this assessment were for both ascorbyl palmitate and ascorbyl stearate. Because it is unlikely that both antioxidants are used in a single food item, the use levels were assumed to reflect the use of ascorbyl palmitate. High (P95) dietary exposure to ascorbyl palmitate could be up to 7.3 mg/kg bw per day for children aged 3–9 years. For adults, the high exposure could be up to 2.7 mg/kg bw per day. The Committee noted that these estimates overestimate the actual exposure to ascorbyl palmitate as a food additive in Europe because of the assumption that all foods within the food categories that could contain ascorbyl palmitate did contain the additive at the mean reported use level. Additionally, not all foods are likely to contain an antioxidant and, for those that do, other food additives with the same function may be used.

5. Evaluation

The Committee noted new studies that showed that ascorbyl palmitate was rapidly and extensively hydrolysed pre-systemically to ascorbic acid and palmitic acid. Hydrolysis of ascorbyl palmitate is mediated by carboxylesterases, which are also expressed in neonates and infants, albeit at lower levels than in adults. On that basis, the Committee considered that systemic exposure to ascorbyl palmitate would be low for all population subgroups.

The Committee further noted that ascorbic acid and palmitic acid are normal components of the human diet. A group ADI “not specified”⁸ has been established by previous Committees for ascorbic acid and its sodium, potassium and calcium salts, and a group ADI “not specified” has been established for palmitic acid and its sodium, potassium and calcium salts. The Committee

⁸ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

therefore withdrew the previously established ADI of 0–1.25 mg/kg bw for ascorbyl palmitate or ascorbyl stearate, or the sum of both. The Committee noted that the previous ADI for ascorbyl palmitate or ascorbyl stearate, or the sum of both, was based on toxicological data for ascorbyl palmitate, and therefore established a group ADI “not specified” for ascorbyl palmitate or ascorbyl stearate, or the sum of both.

The requested increase of the maximum level from 10 to 100 mg/kg for ascorbyl palmitate in dietetic foods for special medical purposes (FC 13.3), and expansion of the use of ascorbyl palmitate in FC 13.3 to include all dietetic foods captured under this category, do not pose a concern for safety.

The Committee noted that ADIs do not apply to infants younger than 12 weeks because they might be at risk at lower levels of exposure compared with older age groups. However, based on the capacity for hydrolysis in young infants, the Committee did not identify any safety concerns for infants younger than 12 weeks at the currently permitted levels of ascorbyl palmitate or the requested increases in maximum levels for ascorbyl palmitate as an additive in foods for infants and young children (FCs 13.1.1, 13.1.2 and 13.1.3). The Committee noted that the amounts of ascorbic acid in infant formula must comply with established limits in FCs 13.1.1, 13.1.2 and 13.1.3.

At the present meeting the Committee only received information for chemically synthesized ascorbyl palmitate and prepared revised specifications accordingly. The Committee lowered the existing limit for lead, added specifications for arsenic and mercury, modified the method of assay to avoid the use of chloroform, increased the assay value from 95 to 98%, and added specifications for residual solvents (acetone, petroleum ether and mesityl oxide) and a method for their determination.

The Committee discussed lead specifications for ascorbyl palmitate and the other food additives considered for use in infant formula that were included in the Agenda. The current specification for ascorbyl palmitate restricts the level of lead to not more than 1 mg/kg. Considering the newly proposed maximum use level for ascorbyl palmitate in infant formula of 100 mg/L as consumed, and the maximum permissible level of lead in ascorbyl palmitate, the contribution of lead from ascorbyl palmitate in infant formula is expected to be less than 0.0001 mg/L.

Revised specifications and a chemical and technical assessment were prepared.

5.1 Recommendations

The Committee noted that ascorbyl stearate specifications need analogous revision and recommended that the additive be included in a future call for data.

Carob bean gum (addendum)

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1. Explanation

Carob bean gum (INS No. 410), also known as locust bean gum, carubin and algarroba, is used as an emulsifier, stabilizer, thickener and gelling agent. Carob bean gum as a food additive was evaluated by the Committee at its Thirteenth (47), Eighteenth (92), Nineteenth (93), Twenty-fourth (94) and Twenty-fifth meetings

(95). A temporary ADI “not specified”⁹ was established at the Nineteenth meeting (93), and the temporary status was extended at the Twenty-fourth meeting (94). The Committee established a full ADI “not specified” for carob bean gum at its Twenty-fifth meeting (95).

At its Fifty-fourth session the CCFA (44) requested the evaluation of the safety of carob bean gum for use as a thickener in infant formula and in FSMP intended for infants in the context of the (therapeutic) dietary management of gastro-oesophageal reflux. At its Eighty-second meeting in 2017 the Committee concluded that the studies available were not sufficient for the evaluation of carob bean gum for use in infant formula (96). The Committee requested toxicological data from studies in neonatal animals, adequate to evaluate the safety for use in infant formula, to complete the evaluation. At the Eighty-second meeting the Committee also conducted a dietary exposure assessment for the use of carob bean gum in infant formula for infants younger than 12 weeks (96). At a proposed maximum use level of 10 000 mg/L and a number of consumption scenarios, estimates of dietary exposure ranged from 600 to 1800 mg/kg bw per day and up to 2100 mg/kg bw per day for infants with high energy intakes.

At its Fifty-fourth session the CCFA included carob bean gum in the priority list of substances proposed for evaluation by JECFA. Data submitted for the present evaluation included information related to a neonatal toxicity study in Göttingen minipigs with formula adapted to include carob bean gum, also investigating the effect of carob bean gum thickener on the faecal microbiota composition in those minipigs. An overview of clinical studies conducted with carob bean gum and three publications were also provided. Using the search terms “locust bean gum” AND “toxicity”, “carob gum” AND “toxicity”, “carob” AND “gum” AND “toxicity”, “locust bean gum” AND “metabolism”, and “locust bean gum” AND “genotoxicity”, the databases PubMed, Scopus and Science Direct were searched by the Committee to retrieve information on carob bean gum toxicity published since 2016 (latest JECFA evaluation). This literature search identified four relevant publications not previously considered.

1.1 Chemical and technical considerations

No new information on chemical and technical considerations was submitted.

2. Biological data

2.1 Biochemical aspects

2.1.1 Absorption, distribution, metabolism and elimination

There was no new information on absorption, distribution, metabolism or elimination of carob bean gum.

⁹ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

2.2 Toxicological studies

2.2.1 Acute toxicity

No new information was identified on the acute toxicity of carob bean gum.

2.2.2 Short-term studies of toxicity

In a study not previously evaluated by the Committee, 16 male New Zealand white rabbits (age 6–8 months) received a carob bean water extract (prepared daily from 150 g carob beans) by daily gavage for 7 weeks in addition to their diet of standard commercial rabbit pellets (97). The composition of the extract was not defined. Rabbits were divided into control and treatment groups (eight per group) that received 10 mL of either tap water or carob water extract, respectively. Blood samples were collected at the end of the 7-week exposure period; animals were then killed and liver, kidney, lung, brain, heart and testes were collected for analysis. Feed intake, body weight gain, haematological and biochemical parameters, and histopathology were measured in collected samples. Clinical signs, body weight gain and feed intake were not affected by the treatment. No statistically significant effects were observed on haematological or biochemical parameters. Serum alkaline phosphatase (ALP), alanine aminotransferase (ALT) and aspartate aminotransferase (AST) concentrations were not affected by the treatment. Organ weights were not affected, and no microscopic changes were noted in observed organs.

2.2.3 Long-term studies of toxicity and carcinogenicity

No new information was identified on the long-term toxicity of carob bean gum.

2.2.4 Genotoxicity

Metabolically competent human cell lines LS-174T (colon) and HepG2 (liver), as well as human cell lines without metabolic capacity ACHN (kidney) and SH-SY5Y (neuron), were used to assess the genotoxicity of 32 food additives, including carob bean gum (98). Cell lines were exposed to five concentrations of carob bean gum (0.1, 0.3, 1.0, 3.0 and 10 mg/mL) individually or in mixtures for 24 hours, and cytotoxicity and genotoxicity were assessed using novel genotoxicity biomarkers γ H2AX/pH3 assay.¹⁰ Genotoxicity was determined by calculating the change in phosphorylation levels of the histones by dividing the γ H2AX and pH3 fluorescence intensities in treated cells by the intensities in untreated cells. Positive and negative controls were included. No statistically significant changes in γ H2AX or pH3 biomarkers were observed in any of the cell lines treated with carob bean gum.

¹⁰ Rabbit monoclonal anti- γ H2AX antibody is a biomarker of DNA double-strand breaks, and mouse monoclonal anti-pH3 antibody is a biomarker for late G2 and cell mitosis.

2.2.5 Reproductive and developmental toxicity

No new information was identified on the reproductive and developmental toxicity of carob bean gum.

2.2.6 Special studies

The current Committee received an unpublished report on a study of carob bean gum administered to juvenile minipigs in a piglet-adapted formula for 21 days from postnatal day (PND) 7 or 8 (99). This study aimed to investigate the effects of carob bean gum on the GIT, the gut microbiota, and the availability of vitamins A and D and minerals such as calcium, copper and zinc. The study was performed following the European Medicines Agency (EMA) guideline on the need for non-clinical testing in juvenile animals of pharmaceuticals for paediatric indications (100), the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) scientific guidelines on non-clinical safety studies for the conduct of human clinical trials on marketing authorization for pharmaceuticals (101), the EMA guideline on repeat-dose toxicity testing (102), and the EFSA guidance on the risk assessment of substances present in food intended for infants younger than 16 weeks (83). The toxicological part of the study was conducted in accordance with Good Laboratory Practice (GLP); the analysis of copper, zinc, and vitamins A and D in blood was not.

A total of 48 juvenile Göttingen SPF minipigs (24 males and 24 females) were delivered by nine minipig sows. Neonatal minipigs, weighing about 0.5–1.2 kg at PND 1, received piglet-adapted formula thickened with carob bean gum daily from PND 7. The formula was based on a commercially available sow's milk replacer fortified with whey protein concentrate, vitamin mixture and fat. Four groups (six males and six females per group) received either the control formula (0 g carob bean gum/L) or formula with 3.5, 5.0 or 8.0 g carob bean gum/L (an average of 0, 1.05, 1.50 or 2.40 g/kg bw per day). Evaluated parameters were mortality, body weight, organ weights and faecal consistency (classified as either hard, formed, loose or watery stools). Urinalysis at necropsy, gross pathology and microscopic examination, including the GIT, were also conducted. Clinical pathology, clinical chemistry and micronutrients vitamins A and D, zinc and copper were measured in blood.

Clinical signs, body weight gain and food consumption, as well as results from urinalyses and macroscopic and microscopic examinations, were not statistically significantly different between treated and control groups. For all tests, the level of significance was defined as $P < 0.05$. Mortality was unchanged, except for one high-dose male that was killed on PND 22 because of clinical signs of continuous watery faeces, subdued behaviour and unconsumed food remains. At necropsy, this animal showed red discolouration of the mucosa in the large intestine and in the colon. At the microscopic examination, marked, acute,

diffuse inflammation, including oedema, mixed inflammatory cell infiltration, haemorrhage and dilatation of lymphatics were recorded in the lamina propria (caecum) and submucosa/muscularis (colon) of the intestinal segments, corresponding to the red discolouration seen at necropsy. According to the report, faecal samples taken at necropsy demonstrated detectable concentrations of *Enterococci* and *Klebsiella pneumoniae*, not observed in the other animals tested, which could explain the poor clinical signs.

A positive effect was reported on the faeces consistency and frequency in the treated groups. The incidence of less hard stools or more formed faeces was increased in treated groups compared with controls, and the incidence of loose stools was lower in treated groups compared with controls. With the exception of the high-dose minipig that was killed, no incidences of watery faeces were reported in either control or treated groups.

Haematology parameters in treated groups did not differ from controls. Statistically significant decreases in triglycerides and cholesterol were reported in the clinical chemistry parameters of treated groups compared with those of controls. Slightly lower levels of triglycerides were reported for both sexes in treated groups (mean range males, 0.20–0.33 mmol/L; females, 0.24–0.38 mmol/L) compared with controls (males, 0.38 mmol/L; females, 0.42 mmol/L) and historical data (approximately 0.4 mmol/L). Slightly lower levels of cholesterol were reported for females in all treated groups (mean range, 2.15–2.38 mmol/L) compared with controls (2.50 mmol/L). As the levels were within the historical control data range (2.1–2.5 mmol/L) and no similar tendency was seen for males, this slight decrease was considered by the study authors to be incidental and of no toxicological relevance. Levels of ALP, ALT and AST were within concurrent control values. Increased levels of glucose were reported in both sexes, as shown in [Table 2.1](#).

The observed blood glucose increases in treated males compared with control males were not statistically significant, although a tendency to increase was observed. The increase in blood glucose in treated females relative to controls was statistically significant in the mid- and high-dose, but not low-dose, groups. The Committee noted that the blood glucose levels observed in high-dose non-fasted female piglets were about 1.5-fold higher (12.07 ± 1.60 mmol/L) than in controls (8.30 ± 0.76 mmol/L). The Committee also noted that the piglets were not fasted before blood was sampled, and that the blood glucose levels reported for both treated and control groups were, according to the study authors, higher than historical control data in fasted piglets (5.1–5.4 mmol/L).

Urinalysis did not show test-item-related glucose changes.

A statistically significant decrease in zinc blood levels was observed in high-dose males (mean, 20.00 μ mol/L) compared with controls (27.67 μ mol/L). Low- and mid-dose males did not show statistically significant decreases (mean, 24.83 and 23.33 μ mol/L, respectively). Zinc blood levels were statistically

Table 2.1

Blood glucose clinical chemistry results and group statistics from the juvenile minipigs fed a piglet-adapted formula containing carob bean gum

Dose groups (g/kg bw per day)	No. per group	Glucose (mmol/L $\pm$ SD)
Males		
0 (control)	6	9.02 $\pm$ 1.81
1.05 (low dose)	6	9.52 $\pm$ 1.69
1.50 (mid dose)	6	10.32 $\pm$ 1.32
2.40 (high dose)	5	11.56 $\pm$ 1.34
Females		
0 (control)	6	8.30 $\pm$ 0.76
1.05 (low dose)	6	8.78 $\pm$ 1.26
1.50 (mid dose)	6	10.42 $\pm$ 1.71*
2.40 (high dose)	6	12.07 $\pm$ 1.60***

bw: body weight; SD: standard deviation; * $P < 0.05$ (Dunnett two-sided test); *** $P < 0.001$ (Dunnett two-sided test).

significantly decreased in mid- and high-dose females (mean, 22.33 and 20.83 $\mu\text{mol/L}$, respectively) compared with controls (mean, 27.17 $\mu\text{mol/L}$), but the decrease in low-dose females was not statistically significant (mean, 25.50 $\mu\text{mol/L}$). The authors discarded this finding considering that “[d]ecreases in blood zinc concentration [were] considered of limited toxicological importance as it is not correlated with a reduced growth, reduced food intake, diarrhea, infections or a reverse effect on increased copper concentrations which are the most common expected signs of zinc deficiency”. The Committee considered that the zinc blood level decreases observed in the high-dose males and females were related to carob bean gum exposure, although no clear dose–response relationship could be identified in males.

Copper and calcium blood levels were not affected by the treatment. Copper levels in treated males (mean, 32.83, 33.50 and 27.00 $\mu\text{mol/L}$ in low-, mid- and high-dose groups, respectively) and females (mean, 32.67, 32.00 and 30.00 $\mu\text{mol/L}$ in low-, mid- and high-dose groups, respectively) were not statistically significantly different from control males (mean, 28.67 $\mu\text{mol/L}$) and females (mean, 28.67 $\mu\text{mol/L}$). Calcium levels in treated males (mean, 2.86, 2.79 and 2.82 mmol/L in low-, mid- and high-dose groups, respectively) and females (mean, 2.83, 2.74 and 2.76 mmol/L in low-, mid- and high-dose groups, respectively) were not statistically significantly different from control males (mean, 2.82 mmol/L) and females (mean, 2.82 mmol/L).

Levels of vitamins A and D in treated groups were not statistically significantly different from those in controls.

Increases in GIT organ weights were reported (jejunum, caecum and colon) in treated males and in the duodenum and jejunum of treated females. Statistically significant differences in organ weights and organ weights relative to body weight were mainly reported in the high-dose groups. Low- and mid-dose males did not show statistically significant differences from controls, but a dose-dependent increasing trend was observed in the jejunum, caecum and colon. In females, a dose-dependent increasing trend in absolute organ weight and organ weight relative to body weight was observed for the duodenum and jejunum. Statistical significance was reported in mid- and high-dose females in absolute and/or relative organ (duodenum, jejunum) weight increases relative to controls. However, no macroscopic or microscopic changes related to the treatment were reported in any of those organs. Other organs examined (gallbladder, kidney, pancreas, liver) did not show any statistically significant treatment-related histopathological differences from controls.

2.3 Observations in humans

Three new publications on paediatric trials were made available to the Committee by the sponsor (103–105).

A 4-week double-blind, randomized, controlled trial with a 4-week extension period reported tolerance to a novel formula containing 0.4 g/100 mL carob bean gum as thickener and partly fermented formula with postbiotics, short-chain galacto-oligosaccharides (scGOS) and long-chain fructo-oligosaccharides (lcFOS) (0.4 g/100 mL, ratio 9:1). Infants aged 21–91 days were eligible for the study ($n = 182$) if they had uncomplicated regurgitation, a normal birth weight and were fully formula fed. The control formula contained 0.4 g/100 mL carob bean gum and postbiotics, but no scGOS or lcFOS. Both types of formula were based on cow's milk. Outcomes were measured with the Infant Gastrointestinal Symptom Questionnaire (IGSQ) sum score, a validated 13-item questionnaire that allows parents to describe the frequency and intensity of their infant's gastrointestinal symptoms in the preceding 7 days. No effects were reported on stool characteristics or infant growth. Both control and treated groups showed improvement in regurgitation. Apart from a difference between groups in the occurrence of flatulence, there were no statistically significant differences between study groups in any other type of gastrointestinal event. IGSQ sum scores improved for the formula containing scFOS and lcFOS compared with the control formula. No adverse events were reported (103).

In a 3-month observational prospective study, infants younger than 4 months ($n = 183$) requiring prescription of anti-regurgitation formula were given the formula described in Bellaiche et al. (103). Evaluated outcomes included digestive tolerance (stool consistency and frequency), occurrence of colic symptoms and constipation, and severity of diarrhoea and regurgitation.

Statistically significant decreases in the number of infants with colic symptoms were reported after 1 and 3 months. Regurgitation severity was statistically significantly decreased in infants having up to two episodes per day. Those having more than two episodes per day did not show significant amelioration. Constipation episodes were reduced significantly. A total of 16 infants from the initial 183 experienced adverse events in the GIT (soft stools, constipation, infantile colic and flatulence). No other adverse events were reported after 3 months of feeding with the formula containing carob bean gum (104).

A multi-centre prospective randomized controlled trial in 13 paediatric centres in various countries tested the relative tolerance to and safety of the same formula (0.4 g/100 mL carob bean gum) as studied in Bellaiche et al. (103), compared with an unthickened control formula containing scGOS and lcFOS at a concentration of 0.8 g/100 mL without carob bean gum (105). Infants aged 21–63 days (average, 36.9 ± 12.9 days at baseline) were eligible ($n = 103$) if fully fed the control formula or the formula containing carob bean gum for at least 1 week prior to study participation. Infants were fed the formula for at least 8 weeks, with a cut-off of age 17 weeks. As with other studies conducted by the same research team, outcomes measured were digestive tolerance (stool consistency and frequency), constipation, diarrhoea, regurgitation frequency and volume, and infant growth and general well-being as reported by parents. The incidence of adverse events was comparable between treated and control infant groups. Some cases of mild diarrhoea (3.9% in the treated group versus 3.8% in the control group) and flatulence (2.0% in the test group versus 1.9% in the control group) were reported. Abdominal pain was reported in two subjects (3.8%) in the control group but not in the test group. No constipation or other adverse events were reported. No clinically relevant differences were observed between the study groups in the use of concomitant medications and nutritional supplements. Regurgitation frequency was reduced in infants fed the formula thickened with carob bean gum compared with controls (105).

A case report describes the influence of formula supplemented with carob bean gum on the absorption of L-thyroxine (L-T₄) in an infant with congenital hypothyroidism (106). The newborn girl showed blood levels of 496.0 µIU/mL thyroid-stimulating hormone (TSH) and of 0.13 ng/dL free thyroxine (FT₄) at the age of 8 days. To achieve normal levels of TSH and FT₄, 5 weeks of treatment with L-T₄ (32.2 µg/day) were given to the patient. Blood levels of TSH and FT₄ normalized to 2.72 µIU/mL and 2.08 ng/dL, respectively. However, 5 weeks after achieving normal levels, TSH and FT₄ worsened again despite increased dosage of L-T₄. A study of the personal medical history revealed that the infant had been given a formula thickened with carob bean gum to treat gastro-oesophageal reflux disease. After 14 days of cessation of feeding with the thickened formula, the continued treatment of the patient with L-T₄ resulted in normal levels of

TSH and LT₄ (0.38 μ IU/ml and 2.68 ng/dL, respectively). The authors suggested that formula thickened with carob bean gum could influence the absorption of L-T₄, but this has not been investigated. The authors recommended caution and frequent thyroid function evaluation when thickened formula is given to patients with congenital hypothyroidism (106).

2.4 Microbiological studies

A dose–response analysis of faecal microbiota composition was conducted using piglet stool samples obtained in the study with juvenile Göttingen minipigs fed a piglet-adapted formula containing carob bean gum or a non-thickened control formula for 21 days from PND 7 (see [Section 2.2.6](#)) (107). Faecal samples were collected from the piglets before the intervention (at PND 3–5) and from piglets fed 0 (control), 3.5, 5.0 and 8.0 g/L formula containing carob bean gum at the end of the intervention (PND 26–28). Seven faecal samples were available before the intervention and 11, 13, 11 and 10 faecal samples from the control and low-, mid- and high-dose groups, respectively, at the end of the intervention.

The bacterial composition of the faecal microbiota was determined by fluorescence in situ hybridization (FISH; used to identify and count populations of bacteria using specific 16S-rRNA targeted fluorescent DNA probes) to detect *Bifidobacterium* spp. (Bif164m), subset Enterobacteriaceae (Ec1531), *Eubacterium rectale* and *Blautia coccooides* group (Erec482), and the *Lactobacillus*–*Enterococcus* group (Lab158). The synthesized FISH probes covered several bacterial species and strains, but did not reveal statistically significant differences in relative abundances of the targeted microorganisms between control and treated groups. Ec1531 and Lab158 populations tended to show lower relative abundance values in treated groups compared with controls, but these differences were not statistically significant. The same pattern was observed in data on bacterial cell counts in the faecal samples: tendencies for lower total cell count values for Ec1531 and Lab158 probes, but no statistically significant differences. No dose–response relationship between carob bean gum and populations of targeted bacterial groups was observed under the conditions of the study.

3. Dietary exposure

3.1 Introduction

The dietary exposure to carob bean gum (INS No. 410) from infant formula was assessed by the previous Committee at its Eighty-second meeting in 2016 (108). The dietary exposure assessment for the Eighty-second meeting was conducted for infants aged 0–12 weeks based on the maximum proposed use level of 10 000 mg/L. The report from the Eighty-second Committee noted that the sponsor suggested a typical use level of 5000 mg/L.

A number of scenarios were assessed by the Eighty-second Committee, taking into account different consumption amounts of infant formula, body weights and energy requirements or intakes. The first scenario was based on WHO-recommended consumption amounts of infant formula and weight-for-age standards. Consumption amounts of formula ranged from 60 mL/kg bw per day in newborns to 150 mL/kg bw per day for infants aged 1–3 months. Another scenario was based on meeting FAO-estimated energy requirements for formula-fed infants with consumption amounts of around 180 mL/kg bw per day for infants aged 0–1 month, and around 150 mL/kg bw per day for infants aged 2–3 months. A final scenario was for extreme consumers based on high (90th percentile) reported energy intakes for formula-fed infants of around 210 mL/kg bw per day.

Based on a maximum proposed use level in infant formula of 10 000 mg/L, the Eighty-second Committee estimated dietary exposure to carob bean gum for infants aged 0–12 weeks of 600–1800 mg/kg bw per day, and dietary exposure for infants with high energy intakes of up to 2100 mg/kg bw per day (96).

3.1.1 Scope of the dietary exposure assessment

The purpose of the current assessment was to evaluate the safety of use of carob bean gum in infant formula (including for special medical purposes). The only population group assessed was infants up to 12 weeks because an ADI does not apply to this group of infants, infant formula is the sole source of nutrition at this age, and they are most at risk because they have the highest exposures on a body weight basis. Should older infants consume infant formula containing carob bean gum, their higher body weights would result in lower estimates of dietary exposure on a body weight basis assuming similar volumes of consumption.

The Committee acknowledges the submission of use level data by industry for foods other than infant formula that have permissions for use of carob bean gum in the GSFA (109). The Committee did not consider these data for the current dietary exposure assessment as the focus was dietary exposure from infant formula.

3.2 Use of carob bean gum in infant formula

Within the GSFA, carob bean gum (INS No. 410) is permitted for use as an emulsifier, gelling agent, stabilizer and thickener in a number of food categories (109). Specifically for infant formula, maximum levels are 1000 mg/L (as consumed) for 13.1.1 Infant formulae, 13.1.2 Follow-up formulae and 13.1.3 Formulae for special medical purposes for infants.

The Eighty-second Committee used a maximum use level of 10 000 mg/L of carob bean gum in infant formula in the dietary exposure assessment. Typical use levels of 5000 mg/L were noted.

Paediatric trial data summarized by the Eighty-second Committee (96) included carob bean gum use levels in infant formula of 3500–6000 mg/L, with levels commonly 4000–4500 mg/L (all expressed as reconstituted formula ready for consumption). Similar use levels were noted in more recent trials, with studies using 4000 mg/L (103–105).

EFSA (110) reported use levels of carob bean gum in infant formula provided from a call for data. The highest end of the range of use reported was equivalent to the EU MPL for category 13.1.5.1 Dietary foods for infants for special medical purposes and special formulae for infants of 10 000 mg/L. Typical use levels were reported as an average of 4347 mg/L. The lowest end of the range of uses at 650 mg/L was reported where the additive is used in combination with other additives.

3.3 Estimates of dietary exposure

The current Committee calculated additional dietary exposure estimates for other use level scenarios based on the consumption amounts used by the Eighty-second Committee.

At typical use levels of 5000 mg/L, estimated dietary exposures would range from 300 to 900 mg/kg bw per day for infants aged 0–12 weeks, to up to 1050 mg/kg bw per day for infants with high energy intakes.

Using concentration levels used for paediatric trials, if the commonly used use level of 4000 mg/L was considered for the dietary exposure assessment, estimated dietary exposures would range from 240 to 720 mg/kg bw per day for infants aged 0–12 weeks, to up to 840 mg/kg bw per day for infants with high energy intakes. If the upper end of the range of use levels from paediatric trials of 6000 mg/L was used for the dietary exposure assessment, estimated dietary exposures would range from 360 to 1080 mg/kg bw per day for infants aged 0–12 weeks, to up to 1260 mg/kg bw per day for infants with high energy intakes.

The MPL of carob bean gum in infant formula in the GSFA is 1000 mg/L, which is 10-fold lower than the level used by the Eighty-second Committee (10 000 mg/L). The current Committee therefore calculated estimated dietary exposures based on the GSFA maximum level. The estimated dietary exposures range from 60 to 180 mg/kg bw per day for infants aged 0–12 weeks, to up to 210 mg/kg bw per day for infants with high energy intakes.

EFSA (110) estimated dietary exposure to carob bean gum from infant formula for infants younger than 16 weeks. Consumption amounts of 200 and 260 mL of formula/kg bw per day were used for mean and high consumers. These conservative mean and high-level (95th percentile) consumption values for infants aged 2–4 weeks are recommended by the EFSA Scientific Committee to be used in risk assessments for substances in foods for infants (83). The high consumption level also covers the potential high consumption of preterm infants

on enteral formula feeding (83). For the regulatory maximum scenario of EFSA based on a concentration in infant formula of 10 000 mg/kg, estimated dietary exposures were 2000 and 2600 mg/kg bw per day at mean and high consumption amounts, respectively. The exposure estimates were the same for the maximum use level scenario as for the regulatory maximum level. For mean use levels reported by industry (4347 mg/L), estimated dietary exposures were 869 and 1130 mg/kg bw per day at mean and high consumption amounts, respectively.

The Committee noted the higher consumption amounts used by EFSA in their assessment, and therefore calculated dietary exposures based on these values as a worst-case scenario. For the MPL of carob bean gum for infant formula in the GSFA of 1000 mg/L, estimated dietary exposures would be 200 and 260 mg/kg bw per day for mean and high consumers, respectively. Based on the upper end of the range of concentrations used in paediatric trials of 6000 mg carob bean gum/L formula, estimated dietary exposure would be 1200 and 1560 mg/kg bw per day for mean and high consumers, respectively.

4. Comments

4.1 Biochemical aspects

No new information on absorption, distribution, metabolism or elimination of carob bean gum was available. During its previous evaluation, the Committee concluded that carob bean gum is largely non-digestible and not bioavailable. Carob bean gum is fermented by microbiota in the gut, producing oligosaccharides or monosaccharides that are converted to short-chain fatty acids. These short-chain fatty acids can then be absorbed and metabolized in normal biochemical pathways.

4.2 Toxicological studies

In its previous evaluation, the Committee considered carob bean gum used as a food additive to be a substance of low oral toxicity. Carob bean gum was not carcinogenic or mutagenic, and not associated with any reproductive or developmental toxicity (96). The NOAEL identified for carob bean gum in 13-week studies ranged from 2500 to 10 000 mg/kg bw per day in rats and from 7500 to 15 000 mg/kg bw per day in mice, the highest doses tested. At its Twenty-fourth meeting in 1980 the Committee established an ADI “not specified” for carob bean gum (94).

One new *in vitro* study with metabolically competent human cell lines LS-174T (colon) and HepG2 (liver), as well as human cell lines ACHN (kidney) and SH-SY5Y (neuron) without metabolic capacity, confirmed that carob bean gum lacks genotoxic activity under the conditions of the study (98).

In its previous evaluation of infant formula thickened with carob bean gum, the Committee evaluated a study in newly weaned cross-bred piglets (age 5 weeks) (96). No significant toxicological effects were observed in this study. However, the Committee concluded that results from the pig study were not sufficient for the evaluation of carob bean gum for use in infant formula since the animal model used did not mimic the infant gut over the 12-week developmental period.

In response to the reservations highlighted in the Committee's previous evaluation (96), a new unpublished study (99) with carob bean gum administered in a piglet-adapted formula to neonatal Göttingen minipigs fed for 21 days from PND 7 or 8, receiving 0, 1.05, 1.50 or 2.40 g carob bean gum/kg bw per day on average, was made available to the present Committee. The only effects observed related to carob bean gum were (i) a statistically significant decrease in serum zinc concentrations in high-dose males and in mid- and high-dose females; and (ii) dose-related increases in serum glucose levels in mid- and high-dose females.

During its previous evaluation, the Committee described results from studies that measured the potential of carob bean gum to decrease the bioavailability of minerals. Those studies suggested that infant formula with a carob bean gum concentration of more than 4 g/L reduced the levels of calcium, zinc and iron available for absorption; however, that effect was not considered to be unique to carob bean gum because it was also observed with other fibres such as oligofructose, maltodextrin and high-esterified pectin. Additionally, the previous Committee described an open randomized prospective study in healthy infants addressing the bioavailability of intestinal minerals and other nutrients with commercially available infant formula containing 4 g/L carob bean gum. Mean zinc levels were not statistically significantly different between treated and control groups.

Zinc deficiency has been described in animals and humans generally after a prolonged reduction in intake or excessive uncompensated loss (111). From the age of 6 months, infants should have started food diversification allowing consumption of other food sources of zinc, compensating for transient low serum zinc levels related to the consumption of thickened infant formula. During its previous evaluation the Committee confirmed that the highest exposure scenario to infant formula intake was for infants aged 14–27 days (96). At the present evaluation, the Committee considered that (i) potential serum zinc decreases in the infant population would be transient and would not represent a health concern over the long term, and (ii) no effect is observed on zinc levels in infants fed with infant formula containing 4 g/L carob bean gum in paediatric trials.

The Committee noted that the blood glucose levels observed at the highest dose in non-fasted female piglets were about 1.5 times higher (12.07 ± 1.60 mmol/L) than those in the respective controls (8.30 ± 0.76 mmol/L). Glucose levels in urine did not differ in this study. The Committee also noted that

the piglets were not fasted before sampling the blood, and that the blood glucose levels reported for both treated and control groups in this new study were, according to the authors, higher than historical control data in fasted piglets.

The Committee further noted that carob bean gum comprises galactose and mannose subunits (108) and therefore that the two sugar moieties will be released by fermentation of carob bean gum by microbiota (112). Galactose and mannose will interfere with the quantitative determination of glucose, leading to an overestimation of plasma glucose levels. The Committee therefore concluded that the dose-related increase in apparent glucose levels was likely to be related to the release of galactose and mannose from carob bean gum, and is not an adverse effect.

The Committee identified a NOAEL of 2.40 g/kg bw per day carob bean gum, the highest dose tested, as part of a thickened infant formula from the 21-day study in neonatal minipigs fed a piglet-adapted formula.

4.3 Observations in humans

In its previous evaluation the Committee concluded that the prevalence of oral hypersensitivity to carob bean gum in the overall population is expected to be very low and not of concern. During that evaluation, the Committee considered 13 paediatric tolerance trials conducted with infant formula containing carob bean gum as a thickening agent. No adverse events were reported in any of the studies, which included about 400 term infants fed formula from the age of 1 week to 3 months containing carob bean gum concentrations ranging from 3500 to 6000 mg/L. Those studies measured the effects of carob bean gum formula on regurgitation frequencies in infants with uncomplicated gastro-oesophageal reflux, growth, formula intake, and volume and characteristics of stool. Although those trials were of limited duration and were specifically designed to measure the effects of thickened formula on the treatment of gastro-oesophageal reflux, they demonstrate tolerance of infants exposed to carob bean gum.

Three new publications on paediatric trials were made available to the Committee (103–105). Those publications tested tolerance to an infant formula containing 4000 mg/L carob bean gum as a thickener in infants with uncomplicated regurgitation. Except for an increased incidence of sporadic flatulence, soft stools, constipation and abdominal pain in the test group, no other adverse events were reported in infants after 3 months of feeding for the longest trials. Regurgitation frequency was reduced in infants fed the formula thickened with carob bean gum compared with control infants.

4.4 Microbiological studies

The faecal microbiota composition was analysed from piglet stool samples obtained in the previously described study with neonatal Göttingen minipigs fed

a piglet-adapted formula containing carob bean gum for 21 days from PND 7 or 8 (107). Control piglets were fed a non-thickened formula. No response related to carob bean gum could be identified among the targeted bacterial groups under the conditions of the study.

4.5 Assessment of dietary exposure

The Committee assessed the dietary exposure to carob bean gum from infant formula.

For the purpose of the evaluation, the Committee used the highest consumption data available for infant formula of 200 and 260 g/kg bw per day for an average and high consumer, respectively, and dietary exposure estimates based on three concentration scenarios. (i) Carob bean gum is permitted for use in infant formula products in the GSFA at a maximum level of 1000 mg/L (as consumed) (109). The level of 1000 mg/L carob bean gum in infant formula leads to dietary exposures estimated by the Committee of 200 mg/kg bw per day (average consumer) and 260 mg/kg bw per day (high consumer) for infants. (ii) The upper end of the range of use levels of 6000 mg/L carob bean gum in infant formula from paediatric trials leads to dietary exposures estimated by the Committee of 1200 mg/kg bw per day (average consumer) and 1560 mg/kg bw per day (high consumer) for infants. (iii) The maximum use level of 10 000 mg/L provided by the sponsor that was used by the Eighty-second Committee resulted in estimates of dietary exposure to carob bean gum of 2000 mg/kg bw per day (average consumer) and 2600 mg/kg bw per day (high consumer) for infants.

5. Evaluation

The Committee previously assigned an ADI “not specified”¹¹ to carob bean gum. However, an ADI does not apply to infants younger than 12 weeks since they might be at risk at lower levels of exposure compared with older age groups. To complete the evaluation, the Committee therefore requested studies in neonatal animals adequate for the evaluation of the safety of carob bean gum in infant formula.

Newly submitted unpublished data obtained from a neonatal pig study evaluated at the present meeting allowed the Committee to identify a NOAEL of 2400 mg/kg bw per day, the highest dose tested.

Based on the estimated high dietary exposures for infants, the Committee noted that MOEs calculated for the use of carob bean gum at 1000 mg/L and 6000 mg/L in infant formula were 9 and 1.5, respectively. The MOE for the high exposure calculated for infants from the use of carob bean gum at 10 000 mg/L in infant formula was less than 1.

¹¹ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

To interpret an MOE related to exposure in infants, the Committee has previously established several considerations that need to be addressed (113). If these considerations are met, MOEs in the region of 1–10 may indicate low risk for the health of infants aged 0–12 weeks exposed to the food additive through infant formula (113). The considerations relevant for the current evaluation are: (i) no adverse effects were observed in any of the studies available, indicating that the toxicity of carob bean gum is low; (ii) the NOAEL was the highest dose tested in a study in neonatal pigs considered as a relevant animal model for human infants; (iii) a number of paediatric trials of infant formula containing carob bean gum show good tolerance of infants to carob bean gum at concentrations of up to 6000 mg/L (0.6%); (iv) the prevalence of hypersensitivity to carob bean gum orally is expected to be very low; (v) the highest exposure estimates consider high-intake scenarios; and (vi) the duration of the exposure will be short and followed by diversification in the diet of the infants.

Furthermore, the Committee observed that there is a variety of medical conditions among infants requiring FSMP, and that those infants would normally be under medical supervision.

The Committee concluded that the MOEs based on carob bean gum concentrations of 1000 and 6000 mg/L in infant formula indicate a low risk; however, the MOE based on a concentration of 10 000 mg/L indicates a potential risk.

Based on the available data and considering the short exposure period involved, the Committee concluded that there is no concern for the disruption of the intestinal microbiota by carob bean gum when used as a thickener in infant formula.

No new toxicological information was available for this evaluation that indicated a need for revision of the ADI “not specified” established previously by the Committee for carob bean gum as a food additive for the general population.

An errata item was prepared for the existing specifications monographs.

Diocetyl sodium sulfosuccinate

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1. Explanation

Diocetyl sodium sulfosuccinate (DSS) (INS No. 480) is permitted for use in FC 04.1.2.9 Fruit-based desserts, including fruit-flavoured water-based desserts, at a maximum level of 15 mg/kg, as specified in the GSFA. DSS was included in the agenda of the present meeting of the Committee as requested at the Fifty-fourth session of the CCFA (44). Only an evaluation of the dietary exposure was requested. Dietary exposure to DSS as a food additive has not previously been evaluated by the Committee.

A temporary ADI for DSS of 0–0.25 mg/kg bw per day was established by the Committee at its Thirty-seventh meeting in 1990, based on a NOAEL of 50 mg/kg bw per day from a reproductive toxicity study in rats and an uncertainty factor (114)¹² of 200 (115). The adverse effect was a reduction in parental body weight and weanling pup weight. The ADI was temporary, “pending the evaluation, in 1995, of the results of the long-term study in a rodent species that was requested earlier”. However, at its Forty-fourth meeting in 1995, no such data were available to the Committee (116). Because of this, the Forty-fourth Committee concluded that “diocetyl sodium sulfosuccinate could be evaluated on the basis of the limited data available because of a low intake anticipated”. The Committee withdrew its request for a long-term study and increased the uncertainty factor to 500 because of the limited toxicological database, resulting in an ADI of 0–0.1 mg/kg bw per day.

¹² According to EHC 240, chapter 5: ‘In the past, the term “safety factor” rather than “uncertainty factor” was often used, particularly in the establishment of ADIs. The term “uncertainty factor” is now preferred.’

The Committee also conducted a literature search to consider any studies on dietary exposure to DSS. For this, the databases Embase, Scopus and PubMed were searched using the terms “dioctyl sodium sulfosuccinate” OR “INS 480” AND “food”, “diet*”, “food additive” OR “nutrition” AND “intake”, “expos*”, “consum*” OR “eat*” without a publication date limit. No studies were identified for inclusion in the evaluation.

2. Dietary exposure

2.1 Use of DSS

DSS is approved for use in fruit and vegetable juice products, water-based flavoured drinks and formulated beverages (up to 10 mg/kg), and sauces and toppings (up to 50 mg/kg) in Australia and New Zealand, and in the finished beverage from fumaric-acid-acidulated dry beverage bases (up to 10 mg/kg) in Canada. In the USA, DSS is an approved food additive that is GRAS designated. Its use is approved as a wetting agent in fumaric-acid-acidulated foods (e.g. dry gelatine dessert at up to 15 mg/kg in the finished food, dry beverage base at up to 10 mg/kg in the finished beverage and fruit juice drink at up to 10 mg/kg), and as a diluent in colour additive mixtures (up to 9 mg/kg in the finished food). In carbonated and non-carbonated soft drinks, its use is approved as a wetting agent and as a solubilizer for flavour emulsion stabilizers (up to 10 mg/kg).

There is one permission proposed for the use of DDS in FC 14.1.4 Water-based flavoured drinks, including sport, energy or electrolyte drinks and particulated drinks through the Codex step process at a maximum level of 10 mg/kg (117). This permission is at Step 7 (see Adipates, [section 2.2.1](#)).

At the current meeting, the Committee evaluated the submission from one sponsor for the use of DSS as an emulsifier in carbonated soft drinks (FC 14.1.4.1 Carbonated water-based flavoured drinks) and non-carbonated soft drinks (FC 14.1.4.2 Non-carbonated water-based flavoured drinks, including punches and ades), including use levels and dietary exposure estimates. The sponsor indicated that DSS is not used in powders, syrups, mixes (on an as-consumed basis; FC 14.1.4.3 Concentrates [liquid or solid] for water-based flavoured drinks) or in hot beverages (coffees, teas and grain-based; FC 14.1.5 Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages). The maximum and average use levels in soft drinks provided by the sponsor are listed in [Table 2.1](#). The sponsor’s submission did not contain a discussion on the use of DSS in FC 04.1.2.9.

Table 2.1
Maximum and average use levels of diethyl sodium sulfosuccinate in carbonated soft drinks and non-carbonated soft drinks from the sponsor

GFSAs food category	Use level (mg/kg)	
	Maximum	Average
14.1.4.1 Carbonated soft drinks	7	4
14.1.4.2 Non-carbonated soft drinks	4.5	2

GFSAs: General Standard for Food Additives.

2.2 Estimates of dietary exposure

The Committee evaluated the submitted estimates of dietary exposure to DSS based on the use levels (see [Table 2.1](#)) and individual food consumption from the United Kingdom and the USA, two countries with high consumption levels of soft drinks. Additionally, the Committee calculated the dietary exposure to DSS by mapping the use levels to individual food consumption data of the European population, including only countries that are members of the EU. The use of DSS in soft drinks is not permitted in the United Kingdom and the EU. The estimates for the United Kingdom and Europe therefore represent potential high estimate exposure scenarios, considering the consumption of soft drinks in these countries.

The Committee did not consider submitted estimates of exposure based on market share volume-weighted maximum and on only average use levels. The market share volume-weighted use levels were based on market share information from North or Latin America on the percentage of soft drinks that are fruit flavoured, as industry indicated that DSS is only used in fruit-based flavoured soft drinks. Both weighted and average use levels were not considered appropriate as they do not account for the fact that people may be loyal to a certain food, such as soft drinks. Use of these levels would result in underestimates of exposure for such consumers, for whom it is important to determine the level of risk. This was also the reason why the Committee did not estimate dietary exposure to DSS using food consumption data from the CIFOCC database combined with these use levels. Further, the Committee did not estimate the dietary exposure based on CIFOCC data and the submitted maximum use levels, because maximum use levels produce highly conservative estimates of dietary exposure to DSS for the general population based on mean and high (e.g. P95) consumption levels per age group and country from the CIFOCC database.

2.2.1 Estimates for the United Kingdom and the USA provided by the sponsor

The sponsor estimated the dietary exposure to DSS for the United Kingdom and the USA by mapping individual food consumption data from both countries

to the maximum use levels, presented as the maximum exposure scenario, or to a combination of maximum and average use levels, presented as the refined exposure scenario. In the refined exposure scenario, the consumption of fruit-flavoured drinks within the two types of soft drinks was mapped to the maximum use levels, and the consumption of non-fruit-flavoured drinks was mapped to the average use levels. This scenario was based on information from the industry that DSS is only used in fruit-flavoured soft drinks.

Dietary exposure to DSS for the United Kingdom was based on food consumption data from the United Kingdom National Diet and Nutrition Survey Rolling Programme, covering the results from years 1 to 9 (2008/2009 to 2016/2017). The food consumption data for the USA were obtained from the What We Eat in America component of the National Health and Nutrition Examination Survey (NHANES) conducted in 2013–2014 and 2015–2016 (NHANES 2013–2016). The sponsor estimated the dietary exposure for individuals who reported the consumption of soft drinks, the so-called consumer-only, on at least one of the recording days in the dietary survey period (4 days for the United Kingdom and 2 days for the USA).

The sponsor calculated the exposure for different age groups, ranging from children aged 1 year up to adults, by mapping the use levels to the mean consumption amount of each type of soft drink per individual across the survey days. The resulting dietary exposure estimates for each type of soft drink were summed and adjusted for individual body weight to obtain the individual daily dietary exposure expressed per kilogram of body weight. The mean and 95th percentile (P95) exposure estimates were calculated from the resulting distribution of individual daily exposures, and are listed in [Table 2.2](#) for both exposure scenarios, both countries and all age groups.

The mean dietary exposure for the maximum exposure scenario ranged from 0.01 mg/kg bw per day for adults aged 65 years and older in the United Kingdom to 0.08 mg/kg bw per day for children aged 1–2 years in the USA. The corresponding high (P95) dietary exposure estimates ranged from 0.04 mg/kg bw per day for adults aged 65 years and older in the United Kingdom to 0.21 mg/kg bw per day for children aged 1–2 years in the USA. For the refined exposure scenario, the highest estimate of the high dietary exposure was 0.20 mg/kg bw per day for children aged 1–2 years in the USA.

2.2.2 Estimates for Europe as calculated by the Committee

The Committee used the DietEx Exposure tool (118) to estimate the dietary exposure to DSS for the European population based on the use levels submitted by the sponsor (see [Table 2.1](#)). At the time of the calculation, the DietEx Exposure tool contained information on food consumption from the EFSA Comprehensive European Food Consumption Database (Comprehensive Database) of 48 dietary

Table 2.2
Dietary exposure to DSS from carbonated soft drinks and non-carbonated soft drinks in the United Kingdom and the USA for consumers-only, according to two exposure scenarios from the sponsor^a

Country and age group (years)	Dietary exposure (mg/kg bw per day)			
	Maximum exposure scenario		Refined exposure scenario	
	Mean	High ^b	Mean	High ^b
United Kingdom ^c				
1–2 ^d	0.04	0.13	0.04	0.12
3–9	0.04	0.12	0.04	0.10
10–17	0.04	0.10	0.03	0.08
18–64	0.03	0.07	0.02	0.06
65–74	0.01	0.04	0.01	0.03
≥ 75	0.01	0.04	0.01	0.03
USA				
1–2 ^d	0.08	0.21	0.07	0.20
3–9	0.06	0.17	0.05	0.15
10–17	0.05	0.12	0.04	0.10
18–64	0.05	0.13	0.04	0.09
65–74	0.03	0.10	0.02	0.07
≥ 75	0.03	0.07	0.02	0.05

bw: body weight; DSS: diethyl sodium sulfosuccinate; USA: United States of America.
^a For the use levels, see Table 2.1.
^b High exposure: 95th percentile.
^c Estimates for the United Kingdom are potential estimates, because DSS is not permitted for use in soft drinks in this country.
^d This age group spans 18–35 months for the United Kingdom and 12–35 months for the USA.

surveys in 22 European countries (118). All countries are members of the EU. As stated above, the use of DSS in soft drinks is not permitted in the EU, and therefore the estimates for the European population represent the potential exposure to DSS through the consumption of these drinks.

The Committee calculated the dietary exposure by mapping the use levels in soft drinks with the corresponding mean consumption amounts of relevant drinks per individual across the survey days. The dietary exposure was calculated for the same age groups as those considered by the sponsor, and for the same two exposure scenarios (see Section 2.2.1). The resulting exposure estimates were subsequently summed and adjusted for individual body weight to obtain the mean individual daily exposure estimate expressed per kg bw. By doing this for all individuals per survey and age group, the distributions of individual exposure per survey and age group were obtained and the mean and the P95 of exposure

Table 2.3

Dietary exposure to dioctyl sodium sulfosuccinate from carbonated soft drinks and non-carbonated soft drinks in Europe,^a according to two exposure scenarios from the Committee using the DietEx Exposure tool^b

Age group (years)	Dietary exposure (mg/kg bw per day)			
	Maximum exposure scenario		Refined exposure scenario	
	Mean	High ^{c,d}	Mean	High ^{c,d}
1–2	< 0.01–0.08	0–0.30	< 0.01–0.08	0–0.30
3–9	< 0.01–0.07	0.02–0.19	< 0.01–0.06	< 0.01–0.19
10–17	< 0.01–0.05	0.03–0.15	< 0.01–0.04	0.01–0.13
18–64	< 0.01–0.03	0.02–0.10	< 0.01–0.02	0.01–0.07
65–74	< 0.01–0.01	< 0.01–0.04	< 0.01–0.01	< 0.01–0.03
≥ 75	< 0.01	0–0.03	< 0.01	0–0.03

bw: body weight.

^a Countries included are Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Portugal, Romania, Slovenia, Spain and Sweden, all of which are members of the European Union.

^b For the use levels, see Table 2.1.

^c High exposure: 95th percentile.

^d High exposure equal to 0 mg/kg bw per day occurs when consumption of carbonated and/or non-carbonated soft drinks is reported by less than 5% of the participants of a dietary survey.

were calculated. As the DietEx Exposure tool does not allow the estimation of consumer-only dietary exposure for foods belonging to different food categories, this assessment was performed for the general population, irrespective of whether individuals reported the consumption of soft drinks on at least one of the recording days in the dietary survey period (119). The DietEx Exposure tool also provided exposure estimates for infants aged less than 1 year. These estimates were lower than those reported for children aged 1–2 years in both scenarios and are not reported. Table A5.1 in Annex 5 lists the soft drinks included in the assessment. A summary of the exposure estimates is listed in Table 2.3.

The mean dietary exposure to DSS for the maximum exposure scenario ranged from 0.01 mg/kg bw per day or less for all age groups to 0.08 mg/kg bw per day for children aged 1–2 years. The corresponding high (P95) dietary exposure estimates were 0 mg/kg bw per day in children aged 1–2 years and adults aged 65 years and older, and 0.30 mg/kg bw per day in children aged 1–2 years. For the refined exposure scenario, the highest estimate of the high dietary exposure was 0.30 mg/kg bw per day for children aged 1–2 years. Table A5.2 in Annex 5 lists the dietary exposure estimates per country for the youngest age group with the highest exposure to DSS for the maximum exposure scenario.

The sponsor indicated that, based on information from the industry, DSS is only used in fruit-flavoured soft drinks. The Committee therefore also assessed the exposure to DSS for the European population on the assumption that non-fruit-flavoured soft drinks do not contain DSS instead of an average use level as assumed in the refined exposure scenario. Again, the high estimate of high dietary exposure was not affected and was 0.30 mg/kg bw per day for children aged 1–2 years.

2.3 Overview of estimated dietary exposure

Table 2.4 summarizes the estimates of dietary exposure to DSS from its use in soft drinks as considered at the present meeting. The exposure was estimated according to the two exposure scenarios, and for consumers-only from the United Kingdom and the USA and the general population from Europe. As the estimates for the two scenarios were similar, a summary of the estimates according to the refined exposure scenario is provided in Table 2.4. The estimates were similar because the exposure to DSS at the upper end of the exposure estimate ranges was largely determined by fruit-flavoured soft drinks. Furthermore, the Committee noted that the use of DSS in soft drinks is not permitted in the United Kingdom or the EU. The estimates for the United Kingdom and Europe therefore represent potential high estimate exposure scenarios, considering the consumption of soft drinks in these countries.

Table 2.4
Overview of estimates of dietary exposure to diethyl sodium sulfosuccinate from its use as a food additive in carbonated soft drinks and non-carbonated soft drinks according to the refined exposure scenario^a

Country	Age group (years)	Dietary exposure (mg/kg bw per day)	
		Mean	High ^b
United Kingdom	Children (1–17)	0.03–0.04	0.08–0.12
	Adults (≥ 18)	0.01–0.02	0.03–0.06
USA	Children (1–17)	0.04–0.07	0.10–0.20
	Adults (≥ 18)	0.03–0.05	0.07–0.13
Europe ^c	Children (1–17)	< 0.01–0.08	0–0.30
	Adults (≥ 18)	< 0.01–0.02	0–0.07

Source: sponsor (United Kingdom and the USA) and the Committee (Europe).
 bw: body weight; USA: United States of America.
^a Exposure was calculated by multiplying the consumption amounts of carbonated and non-carbonated soft drinks, averaged across the survey days, by maximum and average use levels as listed in Table 2.1.
^b High exposure: 95th percentile
^c The European countries included are Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Portugal, Romania, Slovenia, Spain and Sweden, all of which are members of the European Union.

The high (P95) dietary exposure according to the refined exposure scenario could be up to 0.30 mg/kg bw per day for children aged 1–17 years and 0.09 mg/kg bw per day in adults (see [Table 2.4](#)). The Committee noted that the range of exposure of 0–0.30 mg/kg bw per day for children aged 1–17 years from Europe, as calculated by the Committee, was based on 50 estimates across three age groups (1–2, 3–9 and 10–17 years) and 20 European countries. Of these estimates, 42 (84%) were at or below 0.10 mg/kg bw per day, 5 (10%) were between 0.11 and 0.13 mg/kg bw per day, and 3 (6%) were between 0.19 and 0.30 mg/kg bw per day. All estimates for the United Kingdom, the USA and Europe are overestimates of the actual mean or high dietary exposure to DSS via soft drinks, as it is assumed that all drinks that could contain DSS did contain DSS at the maximum or average use level. It is not likely that they all contain the additive as the emulsifier of choice and, if they do contain DSS, the actual levels may be lower than the use levels provided by the sponsor. In particular, using only maximum use levels in the exposure assessment (i.e. the maximum exposure scenario) resulted in comparable ranges of exposure estimates.

The Committee noted that DSS may also be present in food because of its use in FC 04.1.2.9 Fruit-based desserts, including fruit-flavoured water-based desserts at a maximum level of 15 mg/kg (GSFA). Including this use in the refined exposure scenario for Europe (Compote of fruit/vegetables (A01PD), Gelatine dessert (A036H) and Fruit dessert thickened (A16GL)) at this maximum level in the refined exposure scenario resulted in a negligible increase of the highest mean and high exposure to DSS compared with the estimates without considering its use in fruit-based desserts in this scenario.

To protect individuals who consume foods that may contain a food additive, exposure of the consumers-only population could be estimated by including only individuals who report consumption of these foods. The sponsor provided such estimates for the United Kingdom and the USA. The estimates for Europe also included individuals that did not report the consumption of soft drinks, because the DietEx Exposure tool does not permit consumers-only calculations for chemicals present in foods belonging to different food categories (118). Despite this, the highest high estimate of exposure was calculated for Europe (see [Table 2.4](#)).

Additionally, the Committee noted that exposure to additives that can be used in soft drinks should preferably be estimated using a brand-loyal scenario. However, an assessment based on only maximum use levels can be seen as a conservative brand-loyal assessment assuming that individuals are loyal to all soft drinks they consume. The assessment also included countries with high consumption levels of soft drinks. Because this scenario resulted in similar exposure results as the refined exposure scenario, these estimates were deemed sufficiently conservative to determine the level of risk, including those individuals that may be loyal to a certain soft drink.

3. Comments

3.1 Assessment of dietary exposure

At the current meeting, the Committee evaluated the submission from one sponsor for the use of DSS as an emulsifier in carbonated soft drinks (FC 14.1.4.1 Carbonated water-based flavoured drinks) and non-carbonated soft drinks (FC 14.1.4.2 Non-carbonated water-based flavoured drinks, including punches and ades), including use levels and dietary exposure estimates. The maximum and average use levels provided by the sponsor are listed in [Table 2.1](#). The sponsor's submission did not include a discussion on the use of DSS in FC 04.1.2.9.

The Committee evaluated the submitted estimates of dietary exposure to DSS based on the use levels and individual food consumption data from the United Kingdom and the USA. Additionally, the Committee calculated the dietary exposure to DSS by mapping the use levels to individual food consumption data from 47 dietary surveys in 22 European countries using the DietEx Exposure tool (118). The estimates were either calculated with the maximum use levels and presented as the maximum exposure scenario, or calculated with a combination of maximum and average use levels (presented as the refined exposure scenario). In the refined exposure scenario, the consumption of fruit-flavoured soft drinks within the two types of soft drinks was mapped to the maximum use levels, and the consumption of non-fruit-flavoured soft drinks was mapped to the average use levels. This scenario was based on information from the industry that DSS is only used in fruit-flavoured soft drinks. As the estimates of the two scenarios were similar, a summary of the estimates according to the refined exposure scenario is provided in [Table 2.4](#). The estimates were similar because the exposure to DSS at the upper end of the exposure estimate ranges was largely determined by fruit-flavoured soft drinks. Furthermore, the Committee noted that the use of DSS in soft drinks is not permitted in the United Kingdom or in the EU. The estimates for the United Kingdom and the EU therefore represent potential high estimate exposure scenarios, considering the consumption of soft drinks in these countries.

The high (P95) dietary exposure according to the refined exposure scenario could be up to 0.30 mg/kg bw per day for children aged 1–17 years and 0.09 mg/kg bw per day in adults (see [Table 2.4](#)). The Committee noted that the range of exposure of 0–0.30 mg/kg bw per day for children aged 1–17 years from Europe, as calculated by the Committee, was based on 50 estimates across three age groups (1–2, 3–9 and 10–17 years) and 20 European countries. Of these estimates, 42 (84%) were at or below 0.10 mg/kg bw per day, 5 (10%) were between 0.11 and 0.13 mg/kg bw per day, and 3 (6%) were between 0.19 and 0.30 mg/kg bw per day. All estimates for the United Kingdom, the USA and Europe are overestimates of the actual mean or high dietary exposure to DSS via soft drinks, as it is assumed that all drinks that could contain DSS did contain

DSS at the maximum or average use level. It is not likely that they all contain the additive as the emulsifier of choice and, if they do contain DSS, the actual levels may be lower than the use levels provided by the sponsor. In particular, using only maximum use levels in the exposure assessment (i.e. the maximum exposure scenario) resulted in comparable ranges of exposure estimates.

The Committee noted that DSS may also be present in food because of its use in FC 04.1.2.9 Fruit-based desserts, including fruit-flavoured water-based desserts at a maximum level of 15 mg/kg (GSFA). Including this use in the refined exposure scenario for Europe resulted in a negligible increase of the highest mean and high exposure to DSS compared with the estimates without considering its use in fruit-based desserts in this scenario.

4. Evaluation

At its Forty-fourth meeting in 1995, the Committee established an ADI of 0–0.1 mg/kg bw for DSS (116). The high (P95) dietary exposure estimates of DSS of up to 0.30 mg/kg bw per day for children aged 1–17 years could exceed this ADI. The estimates for adults were all below the ADI.

Considering that (i) the dietary exposure estimates are conservative because of the assumptions that all soft drinks that could contain DSS do contain this food additive (although other food additives that perform the same function in foods are available) and, when DSS is used, it will always be present at the maximum and/or average use levels; and (ii) the ranges of exposure estimates based on only maximum use levels were comparable to those based on a combination of maximum and average use levels, the Committee concluded that the use of DSS as a food additive in soft drinks and fruit-flavoured water-based desserts does not pose a safety concern.

4.1 Recommendations

The Committee noted that any further increases in exposure to DSS, for example as a result of higher use levels or an extension of its use in other food categories, would warrant an updated toxicological evaluation, including a reassessment of the current ADI (which was established 30 years ago).

Gardenia (Genipin) Blue

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1. Explanation

At the request of the CCFA at its Fifty-third session (120), the Committee evaluated the safety of Gardenia (Genipin) Blue (INS No. 165). Gardenia (Genipin) Blue is a water-soluble, dark purple to blue powder used as a colourant.

Gardenia (Genipin) Blue is intended to impart blue or green colour to different food types, including specific alcoholic beverages; sport and energy drinks; flavoured water; fruit drinks; carbonated soft drinks; powdered beverages; jams, jellies and marmalades; chewing gum; ready-to-drink teas; sugar coatings; icings; syrups for shaved ice; ice cream and frozen dairy desserts; hard and soft candy; flavoured milk; yogurt; processed cereals; sponge cakes; and food supplements (e.g. chewable tablets). Gardenia (Genipin) Blue is approved for use in foods in China; Taiwan, China; Japan; and the Republic of Korea.

The Committee has not previously evaluated the safety of Gardenia (Genipin) Blue. However, at its Eighty-fourth meeting, a previous Committee considered Gardenia Blue to be structurally related to Jagua (Genipin Glycine) Blue and evaluated data from a 104-week oral toxicity study on a Gardenia Blue polymer (121) to evaluate the safety of Jagua (Genipin Glycine) Blue (122). The Committee did not identify any treatment-related adverse effects in that study, but noted that it was a non-GLP study that did not describe the purity and content of the Gardenia Blue polymer in the test material.

The sponsor submitted a dossier containing chemical, technical, dietary exposure and toxicological data, including unpublished reports of several studies conducted with Gardenia (Genipin) Blue. These studies evaluated the biochemical aspects and different toxicological end-points, including short- and long-term oral toxicity, genotoxicity, and reproductive and developmental toxicity of Gardenia (Genipin) Blue.

The Committee evaluated the submitted data and conducted a literature search in the database PubMed (all fields) using search terms “Gardenia Blue” AND “safety” OR “toxicity”, which yielded 9 and 11 results, respectively. Of the retrieved references, eight publications were considered relevant to the present evaluation. These included seven publications (123–129) that discussed data from some unpublished study reports submitted by the sponsor, and one publication that discussed data from the 104-week oral toxicity study (121) conducted with a Gardenia Blue polymer that was reviewed by the previous Committee at its Eighty-fourth meeting (122).

The Committee conducted a literature search in PubMed (all fields) to locate relevant toxicological data on two impurities in Gardenia (Genipin) Blue: geniposide and genipin. The search terms used were “geniposide” AND “toxicity”, or “genipin” AND “toxicity”. Over 300 references were identified, 11 of which were considered relevant to the current evaluation. These studies evaluated the biochemical aspects (130) and potential for toxicity (123,131–139) of

geniposide and genipin; the hydrolysis of geniposide into genipin in the intestine (131,132,139); and the mechanisms of toxicity of geniposide (135,138,139). At the request of the Committee, the sponsor also submitted reports of six genotoxicity studies conducted with genipin. The Committee reviewed the submitted data and the identified publications on geniposide and genipin.

1.1 Chemical and technical considerations

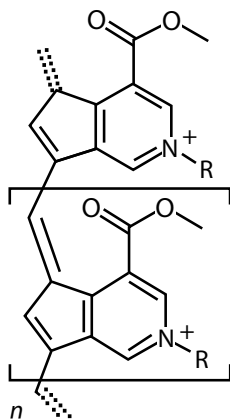
Gardenia jasminoides Ellis is an evergreen shrub belonging to the Rubiaceae family (140). It grows to a height of 2–3 m and produces dark green shiny leaves and white fragrant flowers. It is native to Asia and mainly grows in tropical and subtropical regions. The fruits are used in the production of yellow, blue, red and green pigments widely used as food colouring materials in Japan and East Asia (141). They have also been used in traditional Chinese medicines for centuries (142). The dried fruits are referred to as Zhizi Guo in China, Sanshishi in Japan and Cape Jasmine in the Republic of Korea.

Gardenia fruits are rich in iridoid glycosides, which are the primary bioactive constituents of the fruit (140). Iridoids are secondary plant metabolites and represent the largest group of monoterpenoids with a cyclopentane-[C]-pyran skeleton (143,144). Iridoids can be non-glycosylated or glycosylated (144). Approximately 48 iridoid glycosides from Gardenia fruit have been identified and characterized, with geniposide being the most abundant (141,145).

The starting material is ripe, dried and crushed Gardenia fruit, which is extracted with water or ethanol. The extract is filtered to remove the yellow pigments crocin and crocetin. The colourless geniposide in the filtrate is enzymatically hydrolysed by β -glucosidase to the genipin aglycone (123,141,142,144). This reaction is carried out under specific conditions of pH, temperature, time and enzyme-substrate ratio. Upon completion of the reaction, the solution containing genipin is heated to inactivate the β -glucosidase and cooled. Soy protein hydrolysate is added to this reaction mixture in the presence of oxygen to form the characteristic dark purple to blue polymer (141,146). The product is further refined through successive filtration steps to remove insoluble substances and low-molecular-weight precursors (e.g. genipin and geniposide). Food-grade dextrin or maltodextrin are added to the liquid concentrate and spray-dried to give a powdered product. Batches of Gardenia (Genipin) Blue are manufactured in accordance with GMP, resulting in an article of commerce with a colour value of 300 and consisting of the Gardenia Blue polymer (28.5–36.3%), maltodextrin and/or dextrin (55.6–61.1%), and water (4.1–6.6%).

The chemical formula for the Gardenia Blue polymer is $(C_{11}H_7NO_2R)_n$ where $n \geq 4$, and its structural formula is depicted in Fig. 1.1. The Gardenia Blue polymer is composed of genipin moieties containing peptide chains (146). The genipin moieties are linked via an alkene bridge. It has been quantified and

Fig. 1.1

Structural formula of the Gardenia Blue polymer

characterized by HPLC and nuclear magnetic resonance spectroscopy (¹³C) (146). The Gardenia Blue polymer has a molecular weight of 15 000–30 000 Da.

Geniposide and genipin, as potential impurities, are specified at no more than 80 and 5 mg/kg, respectively.

Gardenia (Genipin) Blue is stable for at least 63 months under storage conditions (1–10 °C) in a laminated bag.

2. Biological data

2.1 Biochemical aspects

2.1.1 Absorption, distribution and elimination

The Committee evaluated data from single- and repeat-dose oral absorption, distribution and elimination studies conducted with rats and mice. The Committee concluded that the test materials used in these studies were representative of the article of commerce.

(a) Animal studies

The sponsor submitted a report of absorption, distribution and elimination studies based on oral exposure to [¹⁴C]Gardenia Blue polymer in rodents (147), which were also discussed in a recent publication (129). The Committee evaluated the data on test material (lot no. 150501; 24.7% Gardenia Blue polymer, 69.4% dextrin and 4.6% water) and concluded that it is representative of the article of commerce. The studies (147) were conducted in compliance with GLP and designed in accordance with the Organisation for Economic Co-operation

and Development (OECD) Test Guideline No. 417 (148). In these studies, male and female Sprague Dawley rats and B6C3F1 mice were orally exposed to [^{14}C]Gardenia Blue polymer by administering Gardenia (Genipin) Blue at doses of 100 or 1000 mg/kg bw per day, and biological samples were collected at different timepoints. The volume of the administered treatment was 10 mL/kg bw.

The identity of Gardenia (Genipin) Blue was confirmed by analysis of its visible spectrum and by chromatographic comparison with unlabelled Gardenia Blue polymer using HPLC and radioactivity detection. The molecular weight of unlabelled Gardenia (Genipin) Blue was determined to be 17 851 Da. [^{14}C]Gardenia Blue polymer was prepared by the reaction of [2- ^{14}C]genipin with soy protein hydrolysate in an aqueous solution. The specific activity of [^{14}C]genipin was 27.46 mCi/mmol and the radiochemical purity was 98.8%. The specific activity of the prepared [^{14}C]Gardenia Blue polymer (lot no. 091720 of Gardenia (Genipin) Blue) was 17 $\mu\text{Ci}/\text{mg}$ with a total activity of 6650 μCi and a yield of 84%. The radiochemical purity of [^{14}C]Gardenia Blue polymer was greater than 99% and the molecular weight was 17 035 Da. The concentration of genipin (0.000165 $\mu\text{g}/\text{mL}$) and geniposide (0.00717 $\mu\text{g}/\text{mL}$) in 1 mg/mL aqueous solution of [^{14}C]Gardenia Blue polymer was found to be very low.

(i) Single-dose pilot study in rats

In a pilot study (129,147), four male rats received [^{14}C]Gardenia Blue polymer at a single oral dose of 100 mg/kg bw and urine, faeces, cage rinse, carbon dioxide (CO_2) and volatile organic chemicals (VOCs) were collected during the 168-hour study. The tissues (kidney, liver, stomach, caecum, large intestine and small intestine), blood, plasma, GIT contents, carcass digest and cage rinse were collected at 168 hours after dosing. The percentage of recovered dose of [^{14}C]Gardenia Blue polymer was determined from the collected samples. Overall, approximately 97% of radioactivity was recovered in faeces; over 90% was recovered in faeces within 72 hours of dosing and 0.34% of the radioactivity was recovered in the urine over the course of the study. Small amounts of radioactivity were recovered in exhaled breath over the course of the study as $^{14}\text{CO}_2$ (< 0.1%) or VOCs (< 0.05%), and 0.012% of radioactivity was recovered in the carcass, tissues and GIT. The average recovery of the administered dose at the end of the study was 97%. Based on the results of this pilot study, the length of follow-up absorption, distribution and elimination studies in rats and mice was reduced to 72 hours and exhaled $^{14}\text{CO}_2$ and VOCs were not collected.

(ii) Single-dose studies in rats and mice

In these studies (129,147), male and female rats and mice (four per sex per group) received [^{14}C]Gardenia Blue polymer via a single oral dose of 100 or 1000 mg/kg bw of Gardenia (Genipin) Blue. Urine and faeces were collected at different

timepoints until 72 hours and several tissues (kidney, liver, brain, heart, lung, adipose, bladder, muscle, skin, stomach, pancreas, small intestine, large intestine, spleen, thyroid, testes and uterus of rats; kidney, liver and spleen of mice), blood, plasma, GIT contents, carcass digest and cage rinse were collected and analysed at 72 hours after dosing. The percentage of recovered dose of [^{14}C]Gardenia Blue polymer was determined from the collected samples.

The majority of the dose in rats was recovered within 24 hours of dosing in faeces (79–92%), with an overall recovery of 89–97% in faeces at the end of the study. Only 0.25–0.36% of dose was recovered in the urine. Very little radioactivity was recovered in the tissues, with the highest individual values determined in the kidney (0.01–0.03%) and carcass digest (0.018–0.027%). The mean concentration values ($\pm$ standard deviation) of Gardenia Blue polymer radioactivity in the kidney of male and female rats treated with 100 mg/kg bw were 3570 ± 3337 and 1964 ± 475.2 ng/g, respectively; the mean concentration values of Gardenia Blue polymer radioactivity in the kidney of male and female rats treated with 1000 mg/kg bw were $15\,030 \pm 1947$ and $31\,280 \pm 25\,158$ mg/g, respectively.

Similarly to rats, in mice the majority of radioactivity was recovered in faeces within 24 hours with a cumulative recovery of 83–87% by 72 hours. Only 0.5–1.5% of the administered dose was recovered in the urine. Of the evaluated tissues, the kidney (0.013–0.027%) and the GIT (0.0024–0.04%) contained the highest percentages of recovered dose. The percentage of recovered dose in the GIT contents was 0.007–0.3%. The mean concentration values of Gardenia Blue polymer radioactivity in the kidney of male and female mice treated with 100 mg/kg bw were 1013 ± 185.9 and 1115 ± 69.26 ng/g, respectively; the mean concentration values of Gardenia Blue polymer radioactivity in the kidney of male and female mice treated with 1000 mg/kg bw were $10\,190 \pm 4281$ and $21\,790 \pm 17\,050$ mg/g, respectively.

(iii) Single-dose biliary elimination study in rats

In this study (129,147), four male rats with cannulated bile ducts received [^{14}C]Gardenia Blue polymer at a single oral dose of 100 mg/kg bw of Gardenia (Genipin) Blue, and bile was collected for up to 48 hours. The recovery of radioactivity in bile, urine, carcass and GIT contents was analysed. Approximately 0.5% of the administered dose was eliminated in the urine, and approximately 0.25% of the dose was recovered from the bile and carcass, both within 48 hours. The mean percentage absorption of [^{14}C]Gardenia Blue polymer was less than 2% of the administered dose.

(iv) Single- and repeat-dose toxicokinetic studies in rats

In these studies (129,147), male rats (eight per group) with jugular vein cannulas received a single or five daily oral doses of 1000 mg/kg bw of Gardenia (Genipin)

Blue with [^{14}C]Gardenia Blue polymer. Blood samples were collected at various timepoints until 48 hours after the single or the last dose. The kidney and carcass were also collected at 48 hours after dosing. The study authors noted that the radioactivity levels in blood and plasma were insufficient and variable, preventing a reliable toxicokinetic analysis. However, a noncompartmental toxicokinetic analysis was conducted to estimate toxicokinetic parameters to assess the absorption and elimination profiles of [^{14}C]Gardenia Blue polymer. In the single-dose part of the study, the time to reach maximum concentration ($T_{\max}$) was 0.5 or 1 hour, the peak concentration ($C_{\max}$) was 1174 or 1251 ng/mL, and the elimination half-life ($t_{1/2}$) was approximately 22 or 14 hours based on concentrations in plasma or blood. In contrast, in the repeat-dose part of the study, $T_{\max}$ was 121 or 132 hours, $C_{\max}$ was 1813 or 1534 ng/mL, and $t_{1/2}$ was 17 or 28 hours for plasma or blood.

The concentration of Gardenia Blue polymer radioactivity in the kidney (ng equivalent/g) decreased from approximately 19 400 to 14 900 ng/g from 6 to 48 hours in the single-dose part of the study and $t_{1/2}$ in the kidney was estimated as 112 hours. Similarly, in the repeat-dose part of the study, Gardenia Blue polymer radioactivity in the kidney decreased from approximately 91 400 to 57 900 ng/g during 6–48 hours and $t_{1/2}$ in the kidney was estimated as 64 hours. $C_{\max}$ values for the radioactivity in the kidney were 19 360 and 91 400 $\mu\text{g/mL}$ for the single- and repeat-dose studies, respectively. The Committee noted that the $C_{\max}$ values of radioactivity in the kidney after a single and repeated dose of Gardenia (Genipin) Blue were higher than those observed in the plasma or blood. However, the Committee also noted that the overall percentage of the administered dose detected in the kidney was small (approximately 0.012–0.015%) and $t_{1/2}$ in the repeat-dose part of the study was lower than in the single-dose part of the study. The Committee therefore concluded that Gardenia Blue polymer does not accumulate in the kidney.

Chromatographic profiling of samples from the rat and mouse absorption, distribution and elimination studies that contained more than 5% of the administered dose was conducted after pooling them by timepoint within a group. Urine and bile samples contained very low levels of radioactivity and were not selected for further analysis. The major peaks detected by analysing the pooled faeces samples from treated rats at the 24-hour and 48-hour timepoints had 94.9–96.5% and 93.6–95.6% of the radioactivity, respectively. The major peaks detected from the analysis of 24-hour and 48-hour mouse faeces samples contained 96.9–97.7% and 97.3% of radioactivity, respectively. No differences in the percentages of radioactivity were observed between sexes or doses.

Considering the results of all the absorption, distribution and elimination studies (129,147), the Committee concluded that Gardenia Blue polymer is poorly absorbed upon oral exposure. About 97% of the administered dose of Gardenia Blue polymer is eliminated unchanged in the faeces. A small fraction

(approximately 0.5%) of the administered dose is eliminated via urine. It becomes distributed in some tissues, including the kidneys. However, the fraction of administered dose in the kidney is only up to 0.03%. There was no indication of accumulation of Gardenia Blue polymer in all evaluated tissues, including kidney.

(b) Examination of rat tissues for particulate aggregates

To assess the distribution of Gardenia (Genipin) Blue in the GIT, mesenteric lymph nodes, kidneys and livers of Sprague Dawley rats upon oral exposure, tissue samples (two per tissue per sex) from selected animals from a 12-month oral toxicity study (127,149) were examined using enhanced darkfield microscopy and hyperspectral imaging (150). Enhanced darkfield microscopy analysis showed that Gardenia Blue polymer particles were present in lymph nodes of male and female rats exposed to the test material via diet at concentrations of 1.5, 3 and 5%. These particles were also observed in kidneys of female rats exposed to Gardenia (Genipin) Blue at 3 and 5%. Gardenia Blue polymer particles were not observed in the kidneys of male rats or in the livers of any treated rats.

The mean particle count was evaluated by hyperspectral imaging as a measure of abundance of Gardenia Blue polymer in the tissue samples. Sex-based differences were observed in Gardenia Blue particles in some tissues. The mean particle count of Gardenia Blue polymer was higher in kidneys of female (96.3 ± 110) versus male (3.3 ± 4.8) rats exposed to Gardenia (Genipin) Blue at 5%. The mean particle count of Gardenia Blue polymer was higher in lymph nodes of male (254 ± 73) versus female (94.4 ± 56.2) rats exposed to Gardenia (Genipin) Blue at 5%.

The images showed that Gardenia Blue polymer particles were not uniformly distributed throughout the evaluated tissues. Specific areas with the analysed tissues or cell types that had a high abundance of Gardenia Blue polymer particles were not identified in the study.

2.1.2 Biotransformation

No animal studies of metabolism of Gardenia (Genipin) Blue upon oral exposure were available and none of the evaluated absorption, distribution and elimination studies reported detection of any metabolites of Gardenia (Genipin) Blue. The Committee evaluated data from four in vitro studies that evaluated levels of geniposide in SGF upon incubation of Gardenia (Genipin) Blue (134,151) and the potential hydrolysis of geniposide into genipin at an intestinal pH (132,134,137), discussed in the following sections.

(a) Examination of digestibility under gastric conditions

The sponsor submitted a study report that examined the stability of Gardenia (Genipin) Blue in simulated gastric conditions with or without pepsin (151).

The tests were carried out in accordance with the protocol of dissolution tests described in the *Japanese Pharmacopoeia* (152). Gardenia (Genipin) Blue (lot no. 150501) was exposed to SGF at a pH of 1.2 (37 °C) for 5–240 minutes, with or without pepsin. At different timepoints, SGF was analysed for molecular weight and levels of total free amino acids, genipin and geniposide.

In experiments with or without pepsin, no significant changes in colour content and levels of free amino acids were observed between test and control samples over different timepoints. There were also no significant changes in average molecular weights between test and control samples in the absence of pepsin. However, in the presence of pepsin, the average molecular weights of test samples were approximately 21 kDa and those of control samples were approximately 26 kDa at all timepoints when assessed at 215 and 600 nm using gel permeation chromatography. At 215 nm, small-molecular-weight substances were observed that were not seen at 600 nm, which indicates that the release of small-molecular-weight substances could be the result of the degradation of peptide chains bound to the polymer backbone catalysed by pepsin. The levels of geniposide increased from 198 ng/g at 5 minutes to 463 ng/g at 240 minutes compared with controls (138 and 135 ng/g, respectively) or from 439 ng/g at 5 minutes to 912 ng/g at 240 minutes compared with controls (242 and 233 ng/g, respectively), in the absence or presence of pepsin, respectively. In contrast, the levels of genipin remained below the limit of quantification (LOQ; < 50 ng/g) in the experimental matrix throughout the study. Considering the stability of the polymer, the Committee concluded that geniposide detected in this study was a residual precursor and impurity of Gardenia (Genipin) Blue, and not a breakdown product.

(b) Studies on intestinal hydrolysis of geniposide into genipin

The Committee identified three publications (132,134,137) describing studies that reported the hydrolysis of geniposide into genipin in the intestine.

In one study (134), geniposide was incubated in SGF with a pH of 2.5 (37 °C) for an hour, after which the pH was adjusted to 7 (37 °C) to simulate intestinal digestion for 2 hours. Genipin was detected at the end of the study, indicating the hydrolysis of geniposide into genipin in the GIT.

In the second study (137), perfused livers from male Wistar rats were used to prepare a homogenate and subcellular fractions. Three β -D-glucosidases from a human intestinal anaerobe (*Eubacterium sp.* A-44) and two β -D-glucosidases from a human gut anaerobe (*Ruminococcus sp.* PO1-3), as well as carboxyesterases from *Eubacterium sp.* A-44 and pig liver, were used for the enzyme assay. β -D-glucosidase and carboxyesterase activities for geniposide and genipin as substrates were measured. No hydrolysis of geniposide by carboxyesterases was observed; however, one of the β -D-glucosidases from *Eubacterium sp.* A-44 hydrolysed geniposide into genipin.

In the third study, geniposide was incubated in SGF at a pH of 1.8 (37 °C) or with rat intestinal contents (132). Geniposide did not hydrolyse in the gastric juice but genipin was detected in the intestinal contents, which indicated that geniposide hydrolyses to form genipin in the intestine.

The Committee concluded from the biotransformation studies that geniposide may undergo enzymatic hydrolysis catalysed by β -D-glucosidase in the intestine to form genipin. The Committee therefore considered genipin to be a potential breakdown product of geniposide.

(c) Toxicokinetics of genipin

The Committee identified a publication (130) that evaluated the toxicokinetic profile of genipin and its metabolites upon oral exposure of a decoction of Gardenia fruit in rats. Male Sprague Dawley rats (six per group) received an oral decoction of Gardenia fruit at a concentration of 10 or 20 g/g, which corresponded to doses of 490 or 980 mg/kg bw, respectively, of geniposide. Blood samples were collected at several timepoints from 5 to 10 080 minutes post-dosing. No genipin or geniposide was detected throughout the study. Genipin sulfate was detected in the plasma at both doses within 5 minutes of dosing, and its levels were reduced to the baseline before the end of the study. Noncompartmental toxicokinetic modelling of the plasma concentrations of genipin sulfate over time indicated that the $C_{\max}$ values were 4.2 ± 0.6 nmol/mL and 8.8 ± 0.6 nmol/mL for the low- and high-dose groups, respectively. The areas under the plasma concentration versus time curve (AUC_{0-t} values) were $10\,087.7 \pm 1887.6$ and $26\,729.9 \pm 4526.3$ nmol minutes/mL, and the mean residence times (MRT_{0-t} values) were 2773.8 ± 225.4 and 3020.2 ± 322.6 minutes for the low- and high-dose groups, respectively. This study also evaluated levels of genipin and genipin sulfate in the plasma at several timepoints up to 2880 minutes after oral administration of genipin at doses of 100 and 200 mg/kg bw to male Sprague Dawley rats (four per group). No genipin or geniposide was detected in the plasma throughout the duration of the study. Genipin sulfate was detected in the plasma at 5 min and the concentrations rapidly dropped close to baseline at both doses.

The study authors discussed results from previously published studies (153,154) that demonstrated the elimination of genipin conjugates in the urine upon oral administration of genipin. By combining the results of this study (130) with previously published information, the authors concluded that genipin sulfate formed in the body from genipin enters the systemic circulation and is eliminated via urine.

The Committee also identified an additional publication that reported the elimination of genipin conjugates including genipin sulfate (155) in the urine upon exposure to genipin. On reviewing this study, the Committee concluded that genipin, being highly reactive, rapidly conjugates with sulfate upon oral

administration and there is no systemic exposure to genipin. Genipin sulfate reaches the systemic circulation but is subsequently rapidly eliminated via urine.

2.2 Toxicological studies

The Committee evaluated the toxicological studies on Gardenia (Genipin) Blue submitted by the sponsor. The test materials used in the submitted studies either contained 24.7% Gardenia Blue polymer, 69.4% dextrin and 4.6% water (lot no. 150501), or 32.3% Gardenia Blue polymer, 62.7% dextrin and 3.2% water (lot no. 180910). The Committee noted that these test materials were representative of the article of commerce. In addition, the Committee identified and evaluated relevant publications discussing toxicological data on geniposide and genipin.

2.2.1 Acute toxicity

(a) Gardenia (Genipin) Blue

No studies on the acute toxicity of Gardenia (Genipin) Blue were submitted. The median lethal dose (LD_{50}) of Gardenia Blue polymer has been reported to be more than 16.7 g/kg bw in ddY mice and 10 g/kg bw in Sprague Dawley rats (156).

(b) Geniposide

The Committee identified a publication that reported an oral LD_{50} of 1431 mg/kg bw of geniposide in Sprague Dawley rats (134).

In a follow-up oral acute toxicity study (134), male and female Sprague Dawley rats were divided into seven groups (six per sex per group) and exposed to either vehicle (water), carbon tetrachloride (CCl_4) administered at 2 mL/kg, or geniposide at doses of 143.5, 287, 574, 861 or 1148 mg/kg bw. The study reported a significant increase in ALT levels at geniposide doses of 574 mg/kg bw and higher at 24 hours. A significant increase in AST and ALP levels was also reported at doses of 861 mg/kg bw and higher, and total bilirubin (TBIL) levels were significantly increased at the highest dose tested at 24 hours. The relative liver weights of rats exposed to 861 and 1148 mg/kg bw of geniposide were significantly increased by about 29.8 and 32.4%, respectively. Histopathological examination revealed significant changes, including necrosis and infiltration of inflammatory cells, in the livers of rats exposed to CCl_4 and 1148 mg/kg bw of geniposide.

As a part of the same study, additional male and female rats (six per sex per group) were orally exposed to vehicle or 861 mg/kg bw of geniposide, and blood and liver samples were collected at 12, 24, 48 and 72 hours. Compared with controls, significant elevation in ALP, ALT and AST levels was observed at 24 and 48 hours in rats exposed to geniposide. Histopathological changes

were also observed in the liver of rats exposed to geniposide from the 24-hour timepoint. An additional oral acute toxicity study was conducted in which male and female rats (six per sex per group) were exposed to vehicle, CCl₄ or 861 mg/kg bw of geniposide. The levels of hepatic malondialdehyde (MDA) were significantly increased by 51.9% in males and 59.6% in females, and hepatic superoxide dismutase (SOD) activity was significantly decreased by 16.8% in males and 20.3% in females exposed to geniposide at 24 hours. These changes were comparable to the CCl₄ groups.

The Committee concluded that a single oral exposure to 574 mg/kg bw or more of geniposide in these studies caused hepatotoxic effects within 24 hours of exposure. The Committee also noted that the altered MDA and SOD levels in these studies suggested a role of oxidative stress in geniposide-induced hepatotoxicity.

(c) Genipin

The Committee identified a publication that reported an LD₅₀ of 510 mg/kg bw of genipin based on an oral acute toxicity study in Kunming mice (137).

A follow-up oral acute hepatotoxicity study (137) of genipin was conducted with male and female Kunming mice (five per sex per group) at doses of 125, 250 or 500 mg/kg bw via intragastric gavage. An additional four groups of mice were exposed to 70 mg/kg *a*-naphthalene isothiocyanate (ANIT) via intragastric gavage to induce hepatotoxicity prior to oral exposure to genipin at the same dose levels. The Committee noted that this study followed an unconventional protocol.

Mice that were not exposed to ANIT were examined to assess the potential for hepatotoxicity of genipin. No mortality was reported in any groups. Compared with controls, the liver index (%) was significantly higher ($P < 0.01$ or 0.05) in all groups exposed to genipin. Significant increases in levels of ALT ($P < 0.01$) and AST ($P < 0.05$) were observed at the highest dose tested compared with controls. Histopathological examination of livers of mice exposed to genipin showed dose-dependent oedematous degeneration of the hepatocytes that resulted in significantly higher ($P < 0.01$ or 0.05) histopathological scores for the treated groups compared with controls.

Mice exposed to ANIT and genipin were examined to assess the effects of genipin on ANIT-induced liver injury. Mortality was reported in two mice in the ANIT control group and in two mice in each of the low- and mid-dose groups. Mortality was reported in all five mice in the high-dose group. Compared with controls, the liver index was significantly higher in the high-dose group ($P < 0.01$) and ANIT control group ($P < 0.05$). The levels of ALP, ALT and AST were significantly higher in the ANIT control group compared with controls ($P < 0.01$). However, no significant increases in levels of ALP, ALT or AST were observed in groups exposed to genipin compared with controls. Histopathological examination of the livers revealed that mice in the ANIT

control group had abnormally loose sinusoidal spaces with inflammatory cell infiltration and a significantly higher ($P < 0.01$) histological score compared with controls. Compared with controls, mid- and high-dose groups showed a dose-dependent increase in inflammatory cell infiltration and signs of liver injury, including necrosis, as well as a significantly higher ($P < 0.05$) histological score. No significant differences in any of the parameters were observed between mice exposed to ANIT and genipin compared with ANIT control.

The Committee concluded that oral exposure to genipin at all doses in this study caused hepatotoxic effects in a dose-dependent manner in healthy mice as well as in mice with ANIT-induced liver injury.

2.2.2 Short-term toxicity studies

(a) Gardenia (Genipin) Blue

The sponsor submitted reports of two 90-day oral toxicity studies (149,157) conducted in rats, discussed below. One of the 90-day studies (149) was conducted as part of a 12-month oral toxicity study and is discussed in [Section 2.2.3\(a\)\(i\)](#). The Committee also identified a publication describing a 90-day oral toxicity study of Gardenia (Genipin) Blue in rats (158), discussed below.

(i) 90-day oral toxicity study in Sprague Dawley rats

A 90-day oral toxicity study was conducted in which male and female Sprague Dawley rats (eight per sex per group) (offspring generation, i.e. F1, Cohort A) were exposed to 0.5% (equal to 349.3 and 359.7 mg/kg bw per day, respectively), 1.5% (equal to 1033 and 1109.3 mg/kg bw per day, respectively), 3% (equal to 2035.9 and 2336.7 mg/kg bw per day, respectively) or 5% (equal to 3690.0 and 4065.1 mg/kg bw per day, respectively) of Gardenia (Genipin) Blue (lot no. 150501; 24.7% Gardenia Blue polymer) via the carrier diet, or 5% methyl cellulose or the carrier diet (control) (157). The selection of the route of administration and battery of tests conducted to assess the potential of subchronic toxicity followed United States Food and Drug Administration (FDA) Redbook test guidelines (159) and OECD Test Guideline No. 408 (160). The Committee noted that this was not a conventional 90-day oral toxicity study, but a combined reproductive and developmental and immunotoxicological study, with aspects of a subchronic toxicity study design.

Male and female rats were exposed to the diets for 10 and 4 weeks, respectively, prior to mating. On PND 4, the litters were randomly selected and male and female F1 (offspring; 10 per sex per group) animals, described as Cohort A in the study, were exposed to the assigned diets of the P0 generation for 90–93 days after weaning to assess the potential for subchronic toxicity. To examine the effect of oral exposure to Gardenia (Genipin) Blue on the immune system, additional randomly selected male and female rat pups (10 per sex per

group), described as Cohort B in the study, were exposed to the assigned diets of the parents for 42 days and blood samples were collected to evaluate T-cell-dependent antibody response. Blood samples from F1 offspring not subjected to either study, described as Cohort C, were collected within 3 days of weaning for haematological evaluation.

No statistically significant changes in anogenital distance, pinna unfolding, eye opening or incisor eruption in male or female F1 pups pre-weaning were observed after exposure to Gardenia (Genipin) Blue.

Cohort A. All rats, except one male rat exposed to 3% Gardenia (Genipin) Blue, survived until being killed at the end of the dosing period. The cause of mortality could not be identified, but the Committee considered this an incidental finding and not related to the treatment. One female rat exposed to 0.5% Gardenia (Genipin) Blue developed a mass in the neck area on day 77, which was identified as mammary gland adenocarcinoma upon further analysis and was not considered to be treatment related. Faeces discoloration and stained bedding were observed in all treated groups as a result of the nature of the test material. The final body weights, ophthalmological and neurological observations, plasma clotting times, most tissue weights, haematological parameters, and clinical chemistry and urinalysis end-points were not affected by the treatment.

Compared with controls: the relative kidney weights of male rats exposed to 1.5% Gardenia (Genipin) Blue were significantly lower (93% of control; $P < 0.05$); the relative spleen weights of male rats exposed to 1.5 and 3% Gardenia (Genipin) Blue were significantly higher (110 and 111% of controls, respectively; $P < 0.05$); the absolute spleen weights of male rats exposed to 3% Gardenia (Genipin) Blue were significantly higher (112% of control; $P < 0.05$); and the relative and absolute thyroid weights in male rats exposed to 3% Gardenia (Genipin) Blue were significantly lower (both 79% of control; $P < 0.05$). None of these changes was observed at the highest dose tested.

The absolute thyroid weights of female rats exposed to 1.5, 3 and 5% Gardenia (Genipin) Blue were significantly lower (82, 70 and 84% of controls, respectively; $P < 0.05$) compared with controls; however, these changes were small and within the historical control ranges. The relative thyroid weights of female rats exposed to 1.5 and 3% Gardenia (Genipin) Blue were significantly lower (83 and 72% of control, respectively; $P < 0.05$) compared with controls. Because these changes in males and females were not observed at the highest dose tested, the Committee did not consider these findings to be toxicologically relevant. Levels of thyroid hormones, triiodothyronine (T_3), thyroxine (T_4) and TSH, were also evaluated, and no significant differences were noted between treated and control groups.

Blue discoloration of various tissues was observed in some rats of each group exposed to Gardenia (Genipin) Blue. This discoloration of tissues was not apparent on microscopic evaluation and not associated with any histopathological

changes. Minimal to mild microscopic changes were observed in various tissues including kidney, liver, lung, heart and epididymis of some rats exposed to Gardenia (Genipin) Blue, but these changes were also observed in some control animals. Similarly, a small number of cysts were observed in the thymus gland of female control group and male and female high-dose groups. The Committee concluded that these findings were incidental and not treatment related, as they also occurred in control animals.

No significant differences in parameters of haematological evaluation or clinical chemistry, except serum cholesterol levels in females exposed to 1.5% Gardenia (Genipin) Blue (129% of control), were noted between treated and control male and female rats. The changes in serum cholesterol levels were not observed at higher doses; the Committee therefore did not consider this finding to be toxicologically relevant or treatment related.

Cohort B. No significant differences in mean immunoglobulin (Ig) M concentrations were observed between treated and control groups, which indicated that Gardenia (Genipin) Blue did not induce immunosuppression in the treated animals.

Cohort C. No dose-dependent or treatment-related differences in the evaluated haematological parameters were identified between rats exposed to Gardenia (Genipin) Blue and controls.

Considering the results from Cohorts A–C, the Committee did not identify any treatment-related or toxicologically relevant adverse effects of oral exposure to Gardenia (Genipin) Blue at the tested dose levels. The Committee therefore identified a NOAEL of 5% Gardenia (Genipin) Blue (equal to 3690 mg/kg bw per day [corresponding to a blue polymer dose of 911 mg/kg bw per day] and 4065 mg/kg bw per day [corresponding to a blue polymer dose of 1004 mg/kg bw per day] in males and females, respectively), the highest dose tested in this study.

(ii) 90-day oral toxicity study in F344 rats

A 90-day oral toxicity study was conducted in male and female F344 rats (10 per sex per group) exposed to doses of 0, 0.6, 1.25, 2.5 and 5% of Gardenia (Genipin) Blue in the diet (158) to determine maximum tolerable dose for a subsequent 104-week oral toxicity study (121). The 104-week oral toxicity study was previously reviewed by the Committee at its Eighty-fourth meeting during the review of Jagua Blue (122).

No mortality, changes in body weight or food consumption, or abnormal clinical observations were reported in rats exposed to Gardenia (Genipin) Blue compared with controls. No dose-dependent changes in absolute and relative organ weights were reported in rats exposed to Gardenia (Genipin) Blue compared with controls.

Examination of haematological and biochemical parameters in females revealed significant reduction ($P < 0.05$ or 0.01) in platelet counts at all tested dose levels (0.6%: $83.27 \pm 3.29 \times 10^4/\mu\text{L}$; 1.25%: $81.84 \pm 4.93 \times 10^4/\mu\text{L}$; 2.5%: $80.48 \pm 4.27 \times 10^4/\mu\text{L}$; and 5%: $77.01 \pm 6.0 \times 10^4/\mu\text{L}$) compared with controls ($98.4 \pm 15.87 \times 10^4/\mu\text{L}$). The levels of glutamic oxaloacetic transaminase and glutamic pyruvic transaminase were significantly increased ($P < 0.001$, 0.1 or 0.05) in females exposed to 2.5% (94.4 ± 4.39 and 45 ± 6.04 IU/L, respectively) and 5% (104.4 ± 6.11 and 46 ± 4.18 IU/L, respectively) Gardenia (Genipin) Blue compared with controls (83.6 ± 3.05 and 37.6 ± 3.36 IU/L, respectively). These values were within the normal biological range and were therefore not considered toxicologically relevant. Some sporadic changes in haematological and biochemical parameters were noted in males but were not dose-dependent; the Committee therefore did not consider these findings to be toxicologically relevant.

No treatment-related abnormal observations or accumulation of Gardenia (Genipin) Blue in tissues was noted upon histopathological examination.

Based on a lack of treatment-related adverse effects, the study authors identified a maximum tolerable dose of Gardenia (Genipin) Blue for both sexes of 5% in the diet (equal to 2173 mg/kg bw per day in males and 2533 mg/kg bw per day in females). The Committee identified a NOAEL of 5% (equal to 2173 mg/kg bw per day in males and 2533 mg/kg bw per day in females), the highest dose tested in this study.

(b) Geniposide

The Committee identified two publications (134,139) that discussed short-term toxicity studies of geniposide.

(i) 4-week oral toxicity study in C57BL/6J mice

The Committee evaluated data from a 4-week oral toxicity study (139) conducted with male and female C57BL/6J mice (five per sex per group) exposed to geniposide at dose levels of 60, 200 or 600 mg/kg bw per day, or vehicle (distilled water) via gavage. The ratio of liver to body weight was significantly higher ($P < 0.05$) in high-dose groups compared with controls. The levels of ALT, AST and TBIL were all significantly elevated ($P < 0.05$), but albumin and total protein concentrations were significantly lower ($P < 0.01$), in high-dose groups compared with controls. Histological examination revealed erythrocytes in the portal lumen, hepatic cords with structural ambiguity, and hepatocytes with vacuolization and oedema in liver sections of mid-dose groups compared with controls. Alterations in liver histopathology, including oedema, nuclei with pustular appearance and vacuolated cytoplasm in the hepatocytes, as well as increased infiltration of inflammatory cells around the bile ducts, were observed in high-dose groups compared with controls.

To investigate the mechanisms of geniposide-induced hepatotoxicity, liver tissues of mid- and high-dose groups were treated with protease, which revealed the presence of genipin and covalent conjugates or adducts with lysine (lys) in the post-digestion mixture. The intensities of genipin-lys adducts detected by mass spectrometry were increased and significantly higher ($P < 0.01$) in the groups exposed to genipin compared with controls. To gain further insight into underlying mechanisms, the effects of genipin and genipin-lys adducts on markers of oxidative stress in HepG2 cells were evaluated. Compared with controls, both genipin and genipin-lys adducts significantly increased ($P < 0.001$, 0.01 or 0.05) levels of reactive oxygen species (intensity) and MDA, and significantly reduced ($P < 0.001$, 0.01 or 0.05) levels of SOD, glutathione and mitochondrial membrane potential (fluorescent aggregates/monomers).

The Committee concluded that geniposide-induced hepatotoxic effects in this study were likely mediated by the formation of genipin and genipin-lys adducts that stimulated markers of oxidative stress.

(ii) 90-day oral toxicity study in Sprague Dawley rats

The Committee evaluated data from a 90-day oral toxicity study (134) of male and female Sprague Dawley rats (five per sex per group) exposed to 24.3 or 74.9 mg/kg bw per day of geniposide, or vehicle (water). No significant changes in ALP, ALT, AST, TBIL or relative liver weights were observed in geniposide-exposed groups compared with controls. The Committee did not identify any toxic effects of oral exposure to geniposide in this study at the dose levels tested.

(c) Genipin

The Committee evaluated data from an oral toxicity study of genipin at doses of 50 and 100 mg/kg bw per day conducted with male and female Kunming mice (30 per sex per group) for a duration of 7, 14 and 21 days (137). Additional groups of mice were followed until 28 days, but the intragastric administration of genipin was stopped at 21 days and the last 7 days served as a recovery period. Similar to the oral acute toxicity study of genipin discussed in [Section 2.2.1\(c\)](#), intragastric administration of ANIT at 70 mg/kg bw was performed every 3 days to evaluate the effects of genipin exposure on mice with ANIT-induced liver injury. For each segment of this study, animals were randomly divided into six groups (five per sex per group) as follows: controls; 50 mg/kg bw per day genipin; 100 mg/kg bw per day genipin; ANIT controls; ANIT and 50 mg/kg bw per day genipin; and ANIT and 100 mg/kg bw per day genipin. The Committee noted that this study followed an unconventional protocol.

In groups not exposed to ANIT, mortality was reported in four animals exposed to 50 mg/kg bw per day of genipin and six animals exposed to 100 mg/kg bw per day of genipin over a duration of 28 days. No significant differences

in body weights between groups were observed. Significant increases ($P < 0.05$) in liver index at 7, 14 and 21 days were observed in mice exposed to 100 mg/kg bw per day of genipin compared with controls. However, no significant differences in liver index were observed between the 28-day groups. The levels of ALT were significantly elevated ($P < 0.05$) in mice exposed to 100 mg/kg bw per day of genipin at 21 days compared with controls. No significant differences in ALT levels were observed between the 28-day groups. No dose-dependent differences in ALP, AST and TBIL were noted between groups at all timepoints. Histopathological examination revealed localized inflammatory cell aggregation in animals exposed to 100 mg/kg bw per day of genipin, such that the liver histological score was significantly higher ($P < 0.05$) in these groups compared with controls at all timepoints.

Mice exposed to ANIT showed mortality in three animals over a duration of 28 days. There were no significant differences in the body weights of animals in the ANIT groups compared with controls at all timepoints. ANIT groups showed significant elevation ($P < 0.05$) in the liver index at 14 and 21 days compared with controls. There were significant increases ($P < 0.05$) in ALT and TBIL levels in the ANIT groups at 7, 14 and 21 days compared with controls. Additionally, ALP levels were significantly elevated ($P < 0.05$) in the ANIT group compared with controls at 14 days. AST levels were also significantly elevated in ANIT groups at 7, 21 and 28 days compared with controls. Severe inflammatory cell infiltration was observed in the ANIT groups upon histopathological examination, such that the liver histological scores were significantly higher ($P < 0.01$) in the ANIT groups compared with controls at all timepoints.

Mice exposed to ANIT and genipin showed mortality in 11 animals for each dose over a duration of 28 days. There were no significant differences in body weights between groups exposed to both ANIT and genipin groups and control groups. Significant increases in the liver index were noted in ANIT and 100 mg/kg bw per day genipin groups at 7 days ($P < 0.05$), 14 days ($P < 0.01$) and 21 days ($P < 0.05$) compared with controls. Significant increases ($P < 0.05$) in ALT and TBIL levels were observed in ANIT and 100 mg/kg bw per day genipin groups at 7, 14 and 21 days compared with controls. Significant increases ($P < 0.05$) in AST levels were observed in both ANIT and genipin groups at 7 and 21 days compared with controls. In contrast, significant increases ($P < 0.05$) in AST levels were noted only in ANIT and 100 mg/kg bw per day genipin group at 14 days compared with controls. ALP levels were significantly elevated ($P < 0.05$) in both ANIT and genipin groups compared with controls at 14 days. No significant differences between ANIT and genipin groups and control groups were observed at 28 days. The histopathological examination revealed signs of liver injury, including inflammatory cell infiltration and necrosis, in the ANIT and genipin groups compared with controls. The liver histological scores were significantly higher in ANIT and 100 mg/kg bw per day genipin group at 21 days

compared with controls. No significant differences in liver histological scores between ANIT and genipin groups and control groups were observed at 28 days.

Proteomic analysis of liver samples from the 21-day groups was conducted to examine underlying mechanisms of genipin-induced hepatotoxicity. This involved protein quantification followed by gene ontology and pathway analyses of selected differential proteins and network analysis to assess protein–protein interactions. Molecular function and biological processes associated with the differentially expressed proteins included monocarboxylic acid metabolism, small molecule catabolism, organic hydroxy metabolism, oxidoreductase activities, terpenoid metabolism, glucuronidation, generation of cellular precursor metabolites and energy-related processes. Furthermore, predominant pathways associated with differentially expressed proteins included ascorbate and aldarate metabolism, ketone metabolism, steroid hormone biosynthesis, glyoxylate and dicarboxylate metabolism, pentose and glucuronate interconversions, butanoate degradation, retinol metabolism, chemical carcinogenesis, fatty acid metabolism and metabolism by cytochrome P450 (CYP) enzymes.

The Committee noted that repeated oral exposure to genipin at dose levels of up to 100 mg/kg bw per day induced hepatotoxic effects in healthy mice as well as mice with ANIT-induced liver injury, which was reversible after cessation of genipin exposure.

2.2.3 Long-term studies of toxicity and carcinogenicity

(a) Gardenia (Genipin) Blue

The Committee previously reviewed a 104-week oral toxicity study of Gardenia (Genipin) Blue (121) via diet in male and female F344 rats (50 per sex per group) at dose levels of 0, 2.5 and 5% in the diet (equal to 0, 1077 and 2173 mg/kg bw per day in males and 0, 1267 and 2533 mg/kg bw per day in females) to evaluate the safety of Jagua Blue. No treatment-related adverse effects were observed in this study. The previous Committee identified a NOAEL of 5% of Gardenia (Genipin) Blue (equal to 2173 mg/kg bw per day in males and 2533 mg/kg bw per day in females), the highest dose tested in this study (122). However, the previous Committee noted that this was a non-GLP study, and the purity and content of the Gardenia Blue polymer was not described in the study report.

The sponsor submitted reports of an oral chronic (12-month) toxicity study (149), a combined oral chronic toxicity (12-month) and carcinogenicity (24-month) study (161), and an oral carcinogenicity (26-week) study in transgenic rasH2 mice (162), which were also described in several other publications (124,126–128).

(i) 12-month oral toxicity study in Sprague Dawley rats

A 12-month oral toxicity study with components of short-term exposure (126,149) was conducted in which male and female Sprague Dawley rats (50 per sex per group) (P0) were exposed to carrier diet or Gardenia (Genipin) Blue (lot no. 150501; 24.7% Gardenia Blue polymer) at doses of 0.1% (equal to 57.9 and 69.3 mg/kg bw per day, respectively), 0.5% (equal to 286.3 and 353.5 mg/kg bw per day, respectively), 1.5% (equal to 861.3 and 1030.5 mg/kg bw per day, respectively), 3% (equal to 1727.3 and 2101.9 mg/kg bw per day, respectively) or 5% (equal to 2854.5 and 3465.4 mg/kg bw per day, respectively), for 10 weeks and 4 weeks, respectively, prior to mating. The in utero exposure phase of P0 and F1 to the test substance in this study was planned in accordance with FDA Redbook test guidelines (159). The Committee noted that this was not a conventional oral chronic toxicity study, but a combined reproductive and developmental and immunotoxicological study with aspects of chronic toxicity study design.

P0 and F1 offspring (pre-weaning). No treatment-related mortality or signs of moribundity were observed in P0 and F1 offspring pre-weaning. No significant differences in body weights or food consumption in P0 animals exposed to Gardenia (Genipin) Blue compared with controls were observed. Furthermore, no treatment-related effects in parameters of reproductive performance of P0 animals were noted. P0 animals as well as F1 offspring exposed to Gardenia (Genipin) Blue had blue or black faeces throughout the treatment period, which was related to the test substance but not considered adverse. F1 male rats showed statistically significant increases ($P < 0.05$) in mean anogenital distance at all dose levels except 1.5% Gardenia (Genipin) Blue compared with controls. Both F1 male (14.6 ± 1.1 days versus 15.3 ± 1.1 days) and female (14.5 ± 1.1 days versus 15.2 ± 1 days) rats exposed to 5% Gardenia (Genipin) Blue had significantly shorter ($P < 0.05$) mean onset of eye opening than the controls. However, these changes were small and within the historical control ranges, and therefore not considered toxicologically relevant.

Dividing F1 offspring into different cohorts. On PND 4, the litters were randomly selected and male and female F1 offspring were split into different cohorts exposed to the assigned diets of the P0 generation for 42 days (Cohort A: 18 per sex per group), 90 days (Cohort B: 10 per sex per group), 182 days (Cohort C: 10 per sex per group) or 12 months (Cohort D: 10 per sex per group) to evaluate its potential for immunotoxicity, subchronic toxicity or long-term toxicity. An additional group of male and female F1 offspring (Cohort E: 10 per sex per dose) was also exposed to the assigned diets for 12 months, and subjected to ophthalmologic examination within the first 2 weeks and 1 week before termination of the study. The doses of Gardenia (Genipin) Blue in terms of mg/kg bw per day that the different cohorts were exposed to are provided in [Table 2.1](#). Because of an analytical error, 20% more Gardenia (Genipin) Blue was added

Table 2.1
Doses of Gardenia (Genipin) Blue in mg/kg bw per day

Cohort	Male					Female				
	0.1% (0.12%)	0.5% (0.6%)	1.5% (1.8%)	3% (3.6%)	5% (6%)	0.1% (0.12%)	0.5% (0.6%)	1.5% (1.8%)	3% (3.6%)	5% (6%)
A	96.5 (107.8)	502.4 (568.0)	1399.4 (1638.9)	2874.1 (3285.0)	4823.7 (5609.6)	108.2 (110.2)	537.3 (565.5)	1607.5 (1640.5)	3017.2 (3284.6)	5322.9 (5653.5)
B	77.9 (91.5)	372.8 (468.2)	1022.5 (1357.6)	2325.7 (2956.3)	3706.4 (4767.1)	81.5 (96.5)	427.3 (494.3)	1184.8 (1481.0)	2536.8 (3103.1)	4181.0 (5136.1)
C	65.6 (86.4)	314.6 (424.4)	938.9 (1259.0)	1851.9 (2594.9)	3321.4 (4452.3)	84.8 (93.1)	389.4 (472.4)	1171.5 (1391.9)	2175.0 (2797.5)	4132.1 (4901.0)
D	73.3 (101.9)	338.2 (534.7)	869.1 (1436.3)	1818.6 (3105.3)	3166.7 (5218.5)	79.0 (105.5)	441.7 (530.9)	1208.5 (1668.3)	2154.6 (3123.2)	3823.2 (5446.2)
E	64.8 (101.5)	308.2 (512.0)	924.6 (1520.9)	1812.4 (3065.4)	3113.5 (5130.0)	84.6 (104.6)	380.7 (537.0)	1178.4 (1601.8)	2526.1 (3216.3)	4049.6 (5432.2)

Source: Maronpot et al. (127), unpublished report (149).

to the test diets, resulting in higher concentrations of Gardenia (Genipin) Blue (i.e. 0.12, 0.6, 1.8, 3.6 and 6%) for half of the study. Two values for doses (in mg/kg bw per day) of Gardenia (Genipin) Blue are therefore provided in Table 2.1 for each percentage test concentration.

The toxicological parameters assessed in each cohort are discussed with the results below.

Cohort A. No treatment-related or dose-dependent effects on body weights, food consumption, tissue weights and histopathological lesions were reported. Blue discoloration of mesenteric lymph nodes in a few rats treated with 0.5% and higher doses of Gardenia (Genipin) Blue, and blue discoloration in the small intestine of most rats, was observed. Assessment of immunotoxic potential by T-cell-dependent antibody response analysis did not show any significant differences in serum IgM concentrations between treated and control groups.

Cohort B. No mortality or signs of morbidity were observed in any animals. No treatment-related effects on body weights, food consumption or behavioural parameters from neurotoxicity screening, including motor activity assessment, were observed. Blue discoloration of some tissues, including mesenteric lymph node, duodenum, jejunum, ileum, caecum, colon, stomach and kidney, was observed upon macroscopic examination in a few animals exposed to 0.5% or higher doses of Gardenia (Genipin) Blue. This effect was considered to be related to the test article, but was not observed upon microscopic examination or considered adverse.

Statistically significant changes ($P < 0.05$) in absolute organ weight of prostrate of male rats (1.1 ± 0.23 g versus 0.82 ± 0.11 g) and relative weight of the thymus of female rats (0.09 ± 0.01 versus 0.11 ± 0.01) were reported upon exposure to 5% Gardenia (Genipin) Blue compared with controls. The relative weights of prostrate in male rats exposed to 5% (0.25 ± 0.05 versus 0.19 ± 0.03) and liver in female rats exposed to 3% (2.73 ± 0.32 versus 2.48 ± 0.19) Gardenia (Genipin) Blue were also statistically significantly different ($P < 0.05$) from the control rats. Haematologic examination showed statistically significant increases ($P < 0.05$) in cholesterol levels of males (153 ± 31.2 mg/dL versus 122 ± 22.30 mg/dL), and in mean corpuscular haemoglobin concentrations (32.1 ± 0.96 g/dL versus 30.9 ± 0.78 g/dL) and eosinophil counts (100 ± 50 cells/ μ L versus 60 ± 30 cells/ μ L) in females, both exposed to 5% Gardenia (Genipin) Blue, compared with controls. All these changes were small and within the historical control ranges, and therefore not considered toxicologically relevant by the Committee.

Histopathologic examination revealed some inflammatory and necrotic microscopic changes in the nose in several males and females of control and high-dose groups. Microscopic changes in some tissues, including chronic progressive nephropathy, mixed cell infiltrate in the liver, cardiomyopathy and pulmonary alveolar histiocytosis were noted in control and high-dose groups. All these observations were considered incidental and not treatment related.

Cohort C. All rats, except three male rats exposed to 5% Gardenia (Genipin) Blue, survived the treatment period. No treatment-related effects on body weights and food consumption were observed. Some statistically significant changes in organ weights between treated and control groups were noted in males and females, but these changes occurred sporadically and were not dose dependent. Haematologic examination showed statistically significant changes ($P < 0.05$) in prothrombin time (11.3 ± 0.74 sec versus 10.1 ± 0.58 sec) and g-glutamyl transpeptidase levels (0 ± 0 g/dL versus 0.3 ± 0.48 g/dL) in females exposed to 5% Gardenia (Genipin) Blue compared with controls. These changes were small and not considered toxicologically relevant. Histopathological examination revealed findings such as inflammatory and necrotic microscopic changes in the nose and microscopic changes in some tissues in control and high-dose groups, similar to those observed in Cohort B of this study. These findings were considered incidental and not treatment related.

Pig-a mutation analysis (163) was conducted using blood samples from control and high-dose groups (at least six per sex per group) and the results of this study are discussed in [Section 2.2.4\(a\)](#).

The serum and kidney samples from treated and control groups were analysed for geniposide and genipin levels, which were below the limit of detection (LOD) or LOQ.

Cohort D. All rats, except four male rats treated with the lower dose of Gardenia (Genipin) Blue, survived the treatment period. No dose-dependent or

treatment-related effects on body weights, food consumption and organ weights were observed in treated males and females compared with controls. Statistically significant changes in some haematologic parameters were observed compared with controls, but most of these changes were sporadic and not dose dependent. The only statistically significant ($P < 0.05$) and dose-dependent change was in the percentage of large unstained white blood cells (WBC) in week 10 of all groups of exposed male rats (0.1%: 0.9 ± 0.26 ; 0.5%: 0.8 ± 0.14 ; 1.5%: 0.7 ± 0.19 ; 3%: 0.7 ± 0.22 ; 5%: 0.8 ± 0.19) compared with controls (1.2 ± 0.32). However, these changes were small and not observed at later timepoints of analysis; they were therefore not considered toxicologically relevant.

Similarly to Cohorts B and C, some microscopic changes were observed in various tissues of control as well as high-dose groups upon histopathological examination. The Committee considered these findings to be incidental and not treatment related.

Cohort E. Mortality was reported in four control animals and four to seven animals in treated groups, which was associated with chronic progressive nephropathy, tumours in mammary gland, thymic lymphoma or an unknown cause. Because these incidences were observed sporadically in all groups, these findings were considered spontaneous and not related to the treatment. No treatment-related effects on body weights and food consumption were reported. Sporadic ophthalmologic changes, including mild corneal crystals, mild keratitis, mild cataracts and mild blepharitis, were observed in some animals in control and high-dose groups 1 week prior to termination of the study, and not considered treatment related.

Compared with controls, statistically significant differences ($P < 0.05$) in absolute and relative thyroid with parathyroid weights were reported in groups treated with 1.5 and 3% Gardenia (Genipin) Blue for both males (0.031 ± 0.005 and 0.039 ± 0.007 g, versus 0.026 ± 0.005 g) and females (0.027 ± 0.005 and 0.029 ± 0.006 g, versus 0.022 ± 0.004 g), respectively. These changes were not observed at the highest dose tested or in other cohorts of this study. The Committee therefore did not consider these findings to be toxicologically relevant. No treatment-related effects on haematologic and urinalysis parameters were reported.

A variety of neoplastic lesions commonly seen in aged rats, such as mammary fibroadenomas, pituitary pars distalis adenomas, adrenal cortical adenomas, hepatocellular adenomas and cholangiomas were observed in treated as well as control rats. Non-neoplastic lesions such as spontaneous hyperplastic, hypertrophic, degenerative, infiltrative and inflammatory lesions occurred with similar incidences in control and treated animals. These observations were not considered to be treatment related.

Summary. Considering the P0 and Cohort A–E results together, no treatment-related or toxicologically relevant effects of oral exposure to Gardenia

(Genipin) Blue were observed at the tested dose levels. The Committee therefore identified a NOAEL of 5% Gardenia (Genipin) Blue. This is equal to 2854 mg/kg bw per day (corresponding to a blue polymer dose of 705 mg/kg bw per day) and 3465.4 mg/kg bw per day (corresponding to a blue polymer dose of 856 mg/kg bw per day) in P0 males and females, respectively, prior to mating, and 3114 mg/kg bw per day (corresponding to a blue polymer dose of 769 mg/kg bw per day) and 4050 mg/kg bw per day (corresponding to a blue polymer dose of 1000 mg/kg bw per day) in F1 Cohort E males and females, respectively, the highest dose tested.

(ii) Combined oral chronic toxicity (12-month) and carcinogenicity (24-month) study in Sprague Dawley rats

A combined oral chronic toxicity and carcinogenicity study (127,161) of Gardenia (Genipin) Blue (lot no. 180910; 32.3% Gardenia Blue polymer) was conducted in male and female rats in accordance with OECD Test Guideline no. 453 (164).

Oral chronic toxicity study. Male and female Sprague Dawley rats (20 per sex per group) were exposed to 0, 0.5 (equal to 251.7 and 336.5 mg/kg bw per day, respectively), 2.5 (equal to 1257.5 and 1541.3 mg/kg bw per day, respectively) or 5% (equal to 2478.7 and 3282.7 mg/kg bw per day, respectively) Gardenia (Genipin) Blue via a carrier diet for at least 12 months. Most animals survived the treatment period; the few animals that died or showed signs of morbidity were from all groups, including controls. The causes of death were determined to be chronic progressive nephropathy in one control female and one high-dose male, and lymphoma in one high-dose female. One animal from each of the mid- and high-dose groups showed signs of moribundity as a result of sustained eye damage likely caused by the retroorbital blood collection procedure, and were therefore killed. The animals exposed to Gardenia (Genipin) Blue showed blue or black faeces, related to the test article but not considered adverse. No treatment-related or dose-dependent effects on clinical observations, body weights, feed consumption, ophthalmologic parameters or neurotoxicity scores were reported.

No dose-dependent significant differences in organ weights of males were noted between treated and control groups. Brain weights were significantly lower ($P < 0.05$) in high-dose females (1.832 ± 0.106 g) compared with controls (1.915 ± 0.072 g). Thyroid with parathyroid weights of females were significantly ($P < 0.05$) different between 0.5 (0.008 ± 0.001 g), 2.5 (0.006 ± 0.002 g) and 5% (0.010 ± 0.002 g) Gardenia (Genipin) Blue groups and controls (0.009 ± 0.001 g). These changes in organ weights were small and not accompanied by any dose-dependent histopathological changes; the Committee therefore did not consider these to be toxicologically relevant.

Haematological examination at 3 months revealed a statistically significant reduction ($P < 0.05$) in creatinine levels in mid- (0.41 ± 0.04 mg/dL) and high-dose (0.42 ± 0.05 mg/dL) females compared with controls

(0.46 ± 0.06 mg/dL). The activated partial thromboplastin time (APTT) of mid- (24.74 ± 2.85 sec) and high-dose (27.15 ± 6.53 sec) males was significantly higher ($P < 0.05$) than for controls (20.92 ± 2.07 sec) at 6 months. In addition, there were significant increases ($P < 0.05$) in levels of sodium (150.79 ± 6.91 mmol/L versus 146.3 ± 1.89 mmol/L), potassium (5.36 ± 0.37 mmol/L versus 5 ± 0.25 mmol/L), chloride (106.05 ± 5.07 mmol/L versus 102.55 ± 1.96 mmol/L), total protein (6.94 ± 0.50 g/dL versus 6.57 ± 0.44 g/dL), globulin (3.58 ± 0.34 g/dL versus 3.3 ± 0.29 g/dL) and calcium (10.29 ± 0.52 mg/dL versus 9.83 ± 0.37 mg/dL) in high-dose males compared with controls at 6 months. The sodium (152.05 ± 4.72 mmol/L versus 147.84 ± 2.59 mmol/L), chloride (109.58 ± 4.34 mmol/L versus 105.84 ± 2.54 mmol/L) and calcium (10.71 ± 0.4 mmol/L versus 10.33 ± 0.25 mmol/L) concentrations in high-dose females were also significantly higher ($P < 0.05$) than in controls at 6 months. Haematologic examination at 12 months revealed significant differences ($P < 0.05$) in certain parameters, including mean corpuscular volume (57.99 ± 1.41 fL versus 59.53 ± 2.15 fL), APTT (41.18 ± 5.79 sec versus 50.01 ± 6.17 sec), calcium levels (11.61 ± 0.67 mg/dL versus 10.96 ± 0.34 mg/dL), total bile acids (37.09 ± 29.19 μ mol/L versus 14.75 ± 16 μ mol/L) and lymphocytes ($4.71 \pm 2.07 \times 1000/\mu$ L versus $3.01 \pm 2.03 \times 1000/\mu$ L) between high-dose males and controls. In females, significant increases ($P < 0.05$) in glucose levels were reported in low- (241.89 ± 67.35 mg/dL), mid- (270.78 ± 62.79 mg/dL) and high-dose (240.29 ± 45.10 mg/dL) groups compared with controls (176.71 ± 39.88 mg/dL) at 12 months. Significant differences ($P < 0.05$) in blood urea nitrogen, creatinine and calcium levels in both mid- (18.22 ± 2.67 , 0.51 ± 0.06 and 11.56 ± 0.36 mg/dL, respectively) and high-dose (17.94 ± 1.86 , 0.51 ± 0.02 and 11.53 ± 0.48 mg/dL, respectively) females relative to controls (20.12 ± 1.8 , 0.58 ± 0.04 and 11.24 ± 0.68 mg/dL, respectively) were also observed at 12 months. However, all of these changes in haematological parameters were small and within normal physiological ranges; the Committee therefore did not consider them to be toxicologically relevant.

No treatment-related effects on the urinalysis parameters were observed at any timepoint.

Histopathological macroscopic examination showed blue discoloration of some tissues, including mesenteric lymph node, stomach, duodenum, jejunum, ileum, caecum and colon, and dark discoloration of the kidneys of some rats exposed to Gardenia (Genipin) Blue. However, these changes were not observed upon microscopic examination. Several incidental microscopic changes, including chronic progressive nephropathy, mixed cell infiltrates in the liver, cardiomyopathy, focal hypertrophy and/or hyperplasia in the cortex of the adrenal gland and inflammation in various tissues were observed in control as well as high-dose groups, and were not considered treatment related.

Based on a lack of treatment-related or toxicologically relevant effects of oral exposure to Gardenia (Genipin) Blue at the tested dose levels, the Committee identified a NOAEL for chronic toxicity of 5%, equal to 2478.7 mg/kg bw per day (corresponding to a blue polymer dose of 800.6 mg/kg bw per day) in males and 3282.7 mg/kg bw per day (corresponding to a blue polymer dose of 1060.3 mg/kg bw per day) in females, the highest dose tested.

Oral carcinogenicity study. Male and female Sprague Dawley rats (50 per sex per group) were exposed to 0, 0.5 (equal to 262.7 and 357.6 mg/kg bw per day, respectively), 2.5 (equal to 1185.2 and 1564.1 mg/kg bw per day, respectively) or 5% (equal to 2175.3 and 3075.4 mg/kg bw per day, respectively) of Gardenia (Genipin) Blue via a carrier diet for 24 months. Most animals survived during the treatment period and the survival rates were similar between treated and control groups, indicating that any mortality or moribundity observed was not treatment related. The animals exposed to Gardenia (Genipin) Blue showed blue or black faeces related to the test article but not considered adverse. No treatment-related effects on clinical observations, body weights or feed consumption were reported.

Assessment of organ weights showed a significant increase ($P < 0.05$) in paired kidney weights in high-dose females compared with controls (2.405 ± 0.194 g versus 2.141 ± 0.219 g), which was small and not considered to be toxicologically relevant. No other significant changes in organ weights were noted between treated and control groups.

Haematological examination showed a significant reduction ($P < 0.05$) in the percentage of large unstained WBC in females in 0.5% ($1.21 \pm 0.7\%$) and 5% ($1.15 \pm 0.43\%$) groups compared with controls ($1.78 \pm 0.89\%$). These changes were not observed at the middle dose and were within the historical control range; the Committee therefore did not consider these changes to be toxicologically relevant.

No treatment-related effects on the urinalysis parameters were observed at any timepoint.

The histopathological examination revealed several incidental neoplastic or non-neoplastic findings, such as chronic progressive nephropathy, in control and high-dose groups, which were not considered to be treatment related. Blue or black discoloration observed in several tissues macroscopically was not observed microscopically.

Based on a lack of treatment-related or toxicologically relevant effects, the Committee did not identify a concern for carcinogenicity of Gardenia (Genipin) Blue. The Committee identified a NOAEL of 5%, equal to 2175 mg/kg bw per day (corresponding to a blue polymer dose of 703 mg/kg bw per day) in males and 3075 mg/kg bw per day (corresponding to a blue polymer dose of 993 mg/kg bw per day) in females, the highest dose tested.

(iii) Carcinogenicity (26-week) study in transgenic rasH2 mice

An oral carcinogenicity study (128,162) was conducted in male and female transgenic CByB6F1-Tg (HRAS)2Jic (rasH2) mice (25 per sex per group) that were exposed to 0, 0.5 (664.8 and 1182.7 mg/kg bw per day, respectively), 2.5 (3341 and 5561.1 mg/kg bw per day, respectively) or 5% (6623.2 and 10 440.03 mg/kg bw per day, respectively) of Gardenia (Genipin) Blue (lot no. 180910; 32.3% Gardenia Blue polymer) in the diet for at least 26 weeks. The rasH2 mice used in the study belong to a transgenic mouse strain carrying the c-Ha-ras oncogene, and the promoter/enhancer of that gene is within the genetic background of a BALB/cBy x C57BL/6J F1 mouse (165). This transgenic strain has an overexpressed c-Ha-ras gene, making it highly susceptible to accelerated tumour development.

This study was designed as an alternative to the OECD Test Guideline no. 451 (166). The dose levels for the study were selected based on the results of a 28-day dose-range-finding study conducted with rasH2 mice exposed to Gardenia (Genipin) Blue in the diet up to doses of 5%, which did not show any abnormal clinical observations or significant changes in body weight. An additional group of male and female rasH2 mice (10 per sex) received 75 mg/kg N-methyl-N-nitrosourea via a single intraperitoneal administration on day 1 as a positive control.

Most animals survived during the treatment period and the survival rates were similar between Gardenia (Genipin) Blue treated and control groups, indicating that any mortality or moribundity observed was not treatment related. No treatment-related effects on clinical observations, body weights or feed consumption were reported.

Assessment of organ weights showed a significant increase ($P < 0.05$) in absolute (0.253 ± 0.044 g versus 0.225 ± 0.021 g) and relative (0.696 ± 0.115 versus 0.613 ± 0.075) heart weights in male rats exposed to 2.5% Gardenia (Genipin) Blue compared with controls. Female rats exposed to 0.5% Gardenia (Genipin) Blue showed a significant increase ($P < 0.05$) in absolute (0.00374 ± 0.00116 g versus 0.00301 ± 0.00075 g) and relative (0.0145 ± 0.0048 versus 0.0114 ± 0.0036) thyroid with parathyroid weights compared with controls. These changes were not dose dependent and were therefore not considered to be toxicologically relevant. The relative liver weights in males significantly increased ($P < 0.05$) after exposure to 0.5% (4.339 ± 0.586), 2.5% (4.178 ± 0.281) and 5% (4.116 ± 0.384) compared with controls (3.941 ± 0.304). Even though this effect was observed at all treated dose levels, the differences between treated and control groups were small and within the historical control range; these changes were therefore not considered to be toxicologically relevant. Some additional sporadic statistical differences in relative organ weights of male rats at some dose levels of Gardenia (Genipin) Blue compared with controls were observed, but these effects did not exhibit dose-dependence and were therefore not considered toxicologically relevant.

Evaluation of clinical chemistry indicated significant increases ($P < 0.01$) in percentage and number of basophils of high-dose males ($0.65 \pm 0.27\%$ and $0.04 \pm 0.02 \times 1000/\mu\text{L}$, respectively) compared with controls ($0.36 \pm 0.18\%$ and $0.02 \pm 0.01 \times 1000/\mu\text{L}$, respectively). Furthermore, urinalysis revealed that urobilinogen level was elevated in high-dose males ($0.4 \pm 0.4 \text{ mg/dL}$). These changes were not associated with any corresponding histopathological findings and were within the historical control ranges; they were therefore not considered to be toxicologically relevant.

Histopathological macroscopic examination showed dose-dependent blue discoloration of the GIT and mesenteric lymph nodes and darkening of the kidneys that was considered to be treatment related, but not observed upon microscopic examination and reversible after discontinuation of the treatment. Sporadic incidences of mass lesions, organ enlargement and spontaneous neoplasms, including haemangioma, haemangiosarcoma, bronchoalveolar adenoma, lymphoma, hepatocellular adenoma and squamous papilloma, were observed in both control and high-dose groups and therefore not considered treatment related. Neoplastic lesions were observed in the stomach and thymus of animals exposed to N-methyl-N-nitrosourea.

Based on a lack of toxicologically relevant effects of Gardenia (Genipin) Blue treatment, the Committee identified a NOAEL of 5%, equal to 6623 mg/kg bw per day (corresponding to a blue polymer dose of 2139 mg/kg bw per day) in males and 10 440 mg/kg bw per day (corresponding to a blue polymer dose of 3372 mg/kg bw per day) in females, the highest dose tested in this study.

(b) Geniposide

The Committee evaluated data from a 26-week oral toxicity study (136) of geniposide in male and female Sprague Dawley rats (15 per sex per group) at dose levels of 0, 50 and 100 mg/kg bw per day via intragastric gavage. No mortality or signs of morbidity were observed in any group. There were no significant differences in body weights and food consumption between geniposide-exposed and vehicle (distilled water) groups.

The study reported that the colour of the urine of geniposide-exposed rats changed to green or dark green in a dose-dependent manner throughout the duration of the study, and the urine output significantly increased in high-dose rats at 26 weeks. Furthermore, the authors stated that the levels of urinary bilirubin and urobilinogen were significantly elevated in high-dose males and mid- and high-dose females compared with controls at 26 weeks. The urinalysis data were not reported in the publication.

Evaluation of biochemical parameters in high-dose males at 26 weeks showed significant increases ($P < 0.05$) in AST ($114.8 \pm 24.5 \text{ U/L}$ versus $80.8 \pm 19.1 \text{ U/L}$) and potassium levels ($4.364 \pm 0.316 \text{ mmol/L}$ versus

3.79 ± 0.326 mmol/L) compared with controls. In males, dose-dependent and significant decreases ($P < 0.01$ or 0.05) in TBIL levels (0.28 ± 0.63 , 0.14 ± 0.31 and 0 $\mu\text{mol/L}$ in low-, mid- and high-dose groups, respectively) were observed relative to controls (1.0 ± 0.56 $\mu\text{mol/L}$). Significant decreases ($P < 0.05$) in total bile acids were also observed in mid- (8.32 ± 3.39 $\mu\text{mol/L}$) and high-dose (8.08 ± 3.47 $\mu\text{mol/L}$) males compared with controls (16.94 ± 9.95 $\mu\text{mol/L}$). In females, significant decreases in TBIL ($P < 0.001$ or 0.01) and creatinine levels ($P < 0.01$ or 0.05) were observed in mid- (0.04 ± 0.09 and 24.6 ± 0.9 $\mu\text{mol/L}$, respectively) and high-dose (0 and 26.6 ± 4.2 $\mu\text{mol/L}$, respectively) groups compared with controls (0.94 ± 0.54 and 36.8 ± 5.7 $\mu\text{mol/L}$, respectively).

Assessment of haematological parameters at 26 weeks showed significant decreases ($P < 0.05$) in red blood cell (RBC) counts in low- ($8.10 \pm 0.53 \times 10^{12}/\text{L}$), mid- ($8.10 \pm 0.46 \times 10^{12}/\text{L}$) and high-dose ($7.96 \pm 0.33 \times 10^{12}/\text{L}$) males compared with controls ($8.66 \pm 0.12 \times 10^{12}/\text{L}$). Significant decreases ($P < 0.05$) in haemoglobin levels (14.02 ± 0.67 g/dL versus 14.84 ± 0.44 g/dL) and increases ($P < 0.01$) in WBC counts ($9.39 \pm 4.78 \times 10^9/\text{L}$ versus $4.31 \pm 1.20 \times 10^9/\text{L}$) were observed in high-dose males compared with controls. Significant increases ($P < 0.01$ or 0.05) in RBC, haemoglobin, plateletcrit, percentage platelets and reticulocytes were observed in high-dose females ($6.67 \pm 0.14 \times 10^{12}/\text{L}$, 12.96 ± 0.27 g/dL, $968.20 \pm 221.13 \times 10^9/\text{L}$, $0.57 \pm 0.13\%$ and $4.48 \pm 0.74\%$, respectively) compared with controls ($7.66 \pm 0.47 \times 10^{12}/\text{L}$, 13.95 ± 0.62 g/dL, $724.75 \pm 124.70 \times 10^9/\text{L}$, $0.44 \pm 0.03\%$ and $2.49 \pm 0.29\%$, respectively). No significant differences in the evaluated haematological parameters were observed in low- and mid-dose females compared with controls.

Analysis of organ weights at 26 weeks showed significant increases ($P < 0.05$) in relative weights of liver ($2.913 \pm 0.358\%$ versus $2.509 \pm 0.372\%$) and spleen ($0.177 \pm 0.031\%$ versus $0.144 \pm 0.019\%$) in high-dose males compared with controls. The relative organ weights were also significantly increased in kidneys ($0.823 \pm 0.076\%$ versus $0.646 \pm 0.055\%$) and brain ($0.687 \pm 0.041\%$ versus $0.609 \pm 0.055\%$) in high-dose females compared with controls. In contrast, the relative weight of thymus gland was significantly decreased in high-dose females ($0.070 \pm 0.022\%$) compared with controls ($0.104 \pm 0.019\%$). No significant differences in the evaluated organ weights were observed in low- and mid-dose groups compared with controls.

Histopathological examination revealed changes in livers and kidneys of four out of five high-dose rats when compared with controls. These changes included intrahepatic bile duct proliferation, slight hepatic necrocytosis and the presence of hyaline droplets in the liver, and renal tubular injury with tubular epithelial desquamation and protein exudation in the renal tubular lumen.

The Committee noted significant changes in a few biochemical and haematological parameters observed in low- and mid-dose groups, but no

significant effects on organ weights or adverse histopathological observations at these doses compared with controls. Rats exposed to the high dose showed significant changes in several biochemical and haematological parameters, with effects on weights of some organs and signs of liver and renal injury upon histopathological examination. The Committee identified the middle dose, 50 mg/kg bw per day, as the NOAEL for this study.

(c) Genipin

The Committee did not identify any oral long-term toxicity studies on genipin.

2.2.4 Genotoxicity

(a) Gardenia (Genipin) Blue

The sponsor submitted study reports of a bacterial reverse mutation assay (167), in vitro chromosome aberration assay (168), in vitro micronucleus assay (169), in vivo erythrocyte micronucleus or Comet assay (170), in vivo reverse Comet assay (170) and in vivo *Pig-a* mutation assay (163) conducted with Gardenia (Genipin) Blue (lot no. 150501; 24.7–24.8% Gardenia Blue polymer). These studies were summarized in a publication on the evaluation of genotoxic potential of Gardenia (Genipin) Blue (123).

Most studies were conducted in accordance with the OECD test guidelines indicated in [Table 2.2](#) below, and all studies were negative. The study design and results of the studies are also provided in [Table 2.2](#). Based on the negative results of the reviewed genotoxicity studies, the Committee did not identify a concern for genotoxicity of Gardenia (Genipin) Blue.

(b) Geniposide

The Committee identified a publication (133) that discussed data from bacterial reverse mutation, Rec and sister chromatid exchange (SCE) assays conducted with geniposide. Adequate information was not provided in the publication to determine the regulatory compliance of the design of each study. The details and results of the studies are shown in [Table 2.3](#). Based on the negative results of the reviewed genotoxicity studies, the Committee did not identify a concern for genotoxicity of geniposide.

Table 2.2
Genotoxicity studies of Gardenia (Genipin) Blue

Study	Test guideline	Test system	Concentrations	Results	Reference
Bacterial reverse mutation assay	OECD Test Guideline No. 471 (171)	<i>Salmonella typhimurium</i> strains TA97a, TA98, TA100 and TA1535, and <i>Escherichia coli</i> WP2 uvrA PKM 101	0, 178.57, 357.14, 712.29, 1785.7, 3571 or 7142.89 µg/mL (± S9)	Negative ^a	Hobbs et al. (123), unpublished report (167)
In vitro chromosome aberration assay	OECD Test Guideline No. 473 (172)	Chinese hamster ovary (CHO)-WBL cells	0, 250, 500, 1250, 2500 or 5000 µg/mL (± S9), ^b and 0, 1250, 2500 or 5000 µg/mL (± S9) ^c	Negative ^d	Hobbs et al. (123), unpublished report (168)
In vitro micronucleus assay	OECD Test Guideline No. 487 (173)	Human TK6 lymphoblastoid cells (TK6)	0, 250, 500, 1250, 2500 or 5000 µg/mL (± S9) ^e	Negative ^f	Hobbs et al. (123), unpublished report (169)
In vivo erythrocyte micronucleus assay; in vivo micronucleus Comet assay	OECD Test Guideline Nos 474 (174) and 489 (175)	Male and female B6C3F1 mice (five per sex per group): peripheral blood and bone marrow; liver, duodenum and stomach	1000, 1500 or 2000 mg/kg bw per day orally	Negative ^{g,h}	Hobbs et al. (123), unpublished report (170)
In vivo reverse Comet assay ⁱ	None	Livers of five male B6C3F1 mice ^j	2000 mg/kg bw per day orally for 3 days; ± S0	Negative ^k	Hobbs et al. (123), unpublished report (170)
<i>Pig-a</i> mutation assay	None	Peripheral blood collected from Cohort C of 12-month oral toxicity study (127)	0 and 5% in the diet	Negative ^l	Unpublished report (163)

MN-RET: micronucleated reticulocytes; S0: styrene oxide.

^a No increase in number of mean revertants per plate and no cytotoxicity observed at all concentrations tested.

^b Actual concentrations: 0, 248.5, 496, 1278, 2518 or 5075 µg/mL (4 ± 0.5 hours ± S9 and 20 ± 1 hours – S9).

^c Actual concentrations: 1278, 2518 or 5075 µg/mL (4 ± 0.5 hours ± S9 and 20 ± 1 hours – S9).

^d Dose-dependent and statistically significant decreases ($P < 0.05$) in mitotic index under conditions of exposure for 4 hours (+ S9) and 20 hours (– S9) were observed at 1250 (1.1%), 2500 (0.6%) and 5000 µg/mL (0.9%) compared with controls (3.4%), and at 2500 (1.6%) and 5000 µg/mL (0.9%) compared with controls (3.7%), respectively; however, the percentage of damaged cells was not significantly higher than the control at any of the test concentrations and all observations fell within the historical control range.

^e Actual concentrations: 0, 248, 495, 1288, 2450 or 4933 µg/mL (4 hours ± S9 and 20 hours – S9).

^f No observation of cytotoxicity and negative for induction of micronuclei compared with controls.

^g The frequencies of MN-RET were significantly higher ($P < 0.05$) in female mice treated with 2000 mg/kg bw per day (3.27 ± 0.3 counts/1000) compared with controls (2.12 ± 0.23 counts/1000). However, no significant changes in MN-RET frequencies were observed in the bone marrow of these mice. The Committee therefore concluded that the overall result of the erythrocyte micronucleus assay was negative.

^h Some significant trends in increase or decrease in %tail DNA were observed in the liver of male or female mice, respectively, exposed to 2000 mg/kg bw per day compared with controls. These changes were not statistically significant and the Committee therefore concluded that the overall result of the Comet assay was negative.

ⁱ In vivo alkaline Comet assay.

^j In situ exposure to S0 (3600 µM) for 1 hour at 4 °C.

^k There was no indication of DNA cross-linking upon exposure to Gardenia (Genipin) Blue, based on no significant differences in %tail DNA between Gardenia (Genipin) Blue group and control (± S0).

^l No increases in *Pig-a* mutant-phenotype polychromatic erythrocytes or red blood cells of male or female rats exposed to Gardenia (Genipin) Blue compared with controls.

Table 2.3
Genotoxicity studies of geniposide

Study	Test system	Concentrations	Results
Bacterial reverse mutation assay	<i>Salmonella typhimurium</i> TA98 and TA100	0, 6250, 12 500 or 25 000 µg/plate (± S9)	Negative ^a
Rec assay	M45 Rec ⁻ and H17 Rec ⁺ strains of <i>Bacillus subtilis</i>	3750 or 7500 µg	Negative ^b
SCE assay	V79 Chinese hamster cells	0, 1250, 2500, 5000 or 10 000 µg/mL (± S9)	Negative ^c

SCE: sister chromatid exchange.

^a No increase in number of mean revertants per plate and no cytotoxicity observed at all concentrations tested.

^b No DNA-damaging activity at the concentrations tested.

^c No significant increases in SCE frequencies, tetraploid index and average generation time.

(c) Genipin

The sponsor submitted study reports of a bacterial reverse mutation assay (176), in vitro chromosome aberration assay (177), in vitro micronucleus assay (178), in vitro conventional and reverse Comet assay (179), in vivo erythrocyte micronucleus assay (180) and in vivo reverse Comet assay (181) conducted with genipin. The Committee noted that these data are also discussed in a separate publication (123). The Committee identified another publication that discussed data from bacterial reverse mutation, Rec and SCE assays (136) conducted with genipin. The details and results of the studies are provided in Table 2.4.

As described in Table 2.4, genipin was negative for mutagenicity in bacterial reverse mutation assays conducted with different strains of *Salmonella typhimurium* and *Escherichia coli* WP2 uvrA (123,133,176), and induced cytotoxicity at higher test concentrations.

In four in vitro genotoxicity studies (i.e. Rec, SCE, chromosome aberration and micronucleus assays) genipin tested positive, suggesting its potential to cause DNA damage in vitro (133,177,178). Cytotoxicity was reported at higher test concentrations, with the cell viability reduced to less than 50% in all these assays.

The Committee reviewed a report of an in vitro standard Comet assay (179), in which no significant decreases were observed in %tail DNA after 4 and 24 hours of exposure at 2000 and 400 µM of genipin, respectively. The Committee noted that the relative viability of cells (%) decreased at the highest two doses of genipin such that it was down to 13 and 5% after 4 and 24 hours, respectively, of exposure at the highest concentration tested. In a reverse Comet assay (179), after 4 hours of genipin exposure at concentrations of 667 and 2000 µM, significant decreases ($P < 0.05$) in %tail DNA versus styrene oxide control were observed. The relative viability of cells was over 60% at 667 µM and down to 10% at the highest concentration of genipin. The authors stated that, as per the scoring criteria (i.e. exclusion of hedgehogs and cells with $\geq 90\%$ tail DNA), the %tail DNA values underrepresented the extent of DNA damage induced at these concentrations.

Table 2.4
Genotoxicity studies of genipin

Study	Test guideline	Test system	Concentrations	Results	Reference
Bacterial reverse mutation assay	OECD Test Guideline No. 471 (171)	<i>Salmonella typhimurium</i> strains TA97a, TA98, TA100 and TA1535, and <i>Escherichia coli</i> WP2 <i>uvrA</i> PKM 101	0, 9.14, 28.57, 91.43, 285.71, 914.29, 2857.14 or 7142.86 µg/mL ^a (± S9)	Negative ^b	Hobbs et al. (123), unpublished report (176)
In vitro chromosome aberration assay	OECD Test Guideline No. 473 (172), FDA Redbook (182)	Chinese hamster ovary (CHO)-WBL cells	0, 2.5, 5, 10, 25, 50 or 75 µg/mL (± S9, 4 hours); 0, 2.5, 5, 10, 25 or 50 µg/mL (– S9, 20 hours)	Positive ^c	Hobbs et al. (123), unpublished report (177)
In vitro micronucleus assay	OECD Test Guideline No. 487 (173)	Human TK6 lymphoblastoid cells (TK6)	0, 2.5, 5, 10, 25, 50 or 75 µg/mL (+ S9); 0, 4, 6, 8 or 10 µg/mL (– S9)	Positive ^d	Hobbs et al. (123), unpublished report (178)
In vitro standard and reverse Comet assays	None	TK6 cells	0, 2.7, 8.2, 25, 74 222, 667 or 2000 µM (4 hours); 6.25, 12.5, 25, 50, 100, 200 or 400 µM (24 hours); versus S0 or dimethylsulfoxide	Equivocal ^e	Hobbs et al. (123), unpublished report (179)
In vivo micronucleus assay; in vivo micronucleus Comet assay erythrocyte	OECD Test Guideline No. 474 (174), FDA Redbook (182), and Testing Guidelines of the 4th International Workgroup of Genotoxicity (183)	Male (six per group) and female (five per group) B6C3F1 mice: peripheral blood; liver (males and females); duodenum and stomach (females only)	0, 8, 25, 74, 222, 667 or 2000 mg/kg bw per day orally to males and 0, 3, 8, 25, 74 or 222 mg/kg bw per day orally to females for 3 days	Negative ^f	Hobbs et al. (123), unpublished report (180)
In vivo reverse Comet assay ^g	None	Liver from five female B6C3F1 mice	222 mg/kg bw per day orally for 3 days	Equivocal ^h	Unpublished report (181)
Bacterial reverse mutation assay	Not reported	<i>S. typhimurium</i> TA98 and TA100	0, 250, 500 or 1000 µg/plate	Negative ⁱ	Tian et al. (136)
Rec assay	Not reported	M45 Rec [–] and H17 Rec ⁺ strains of <i>Bacillus subtilis</i>	2250 or 4500 µg	Positive ^j	Tian et al. (136)
SCE assay	Not reported	V79 Chinese hamster cells	0, 2, 46, 8 or 10 µg/mL	Positive ^k	Tian et al. (136)

FDA: United States Food and Drug Administration; MN-RET: micronucleated reticulocytes; SCE: sister chromatid exchange; S0: styrene oxide; T1: tetraploid index.

^a Two additional concentrations (2 and 6000 µg/plate) were also tested.

^b No increase in number of mean revertants per plate was observed in all *S. typhimurium* strains and *E. coli* WP2 *uvrA* strain ± S9, except *S. typhimurium* TA97a. A dose-dependent increase in number of mean revertants per plate was observed in *S. typhimurium* TA97a strain ± S9 (from 102 ± 3.5 at 0 µg/mL to 159 ± 10.1 at 914.29 µg/mL – S9, and from 146 ± 17.5 at 0 µg/plate to 220 ± 12.1 at 914.29 µg/mL + S9). However, since the number of mean revertants per plate for the TA97a strain ± S9 were not doubled at concentrations of 0–914.29 µg/mL and no increases were observed at any of the other strains, the Committee concluded that, overall, the results of this assay were negative. A decrease in number of mean revertants per plate was observed at concentrations of ≥ 7142.6 µg/plate in all *S. typhimurium* strains ± S9 and at concentrations of ≥ 5000 µg/plate in *E. coli* WP2 *uvrA* strain ± S9, indicating that genipin caused cytotoxicity at high test concentrations.

^c A statistically significant increase ($P < 0.05$) in the polyploidy index (%) was observed in a dose-dependent manner under all testing conditions from 3.5–4% at 0 µg/mL to 8.5–10% at 50 µg/mL. The percentage of metaphase cells with at least one structural chromosome aberration excluding gaps increased from 0 to 12.5% (+ S9) and 22% (– S9) after 4 hours of exposure, and 19% genipin after 20 hours of exposure at 50 µg/mL. Genipin was cytotoxic at concentrations of ≥ 50 µg/mL with the cell viability decreasing to < 50% of control. There were no statistically significant differences in the mitotic index (%) at non-cytotoxic concentrations.

Table 2.4. (continued)

- ^d A statistically significant increase ($P < 0.05$) in mean percentage induction of micronuclei was observed at 5 and 10 µg/mL compared with controls + S9 (1.49 and 2.30%, respectively, versus 1.03%) after 4 hours of exposure + S9, and at 6, 8 and 10 µg/mL compared with controls (1.07, 1.71 and 2.81%, respectively, versus 0.49%) after 24 hours of exposure – S9. The relative survival of cells decreased from 100% at 0 µg/mL ($\pm$ S9) to 33.4% at 10 µg/mL after 4 hours of exposure (+ S9) and to 44.5% at 10 µg/mL after 24 hours of exposure (– S9), which indicated that genipin induced cytotoxicity at concentrations ≥ 10 µg/mL.
- ^e In a standard Comet assay, no decrease in %tail DNA was observed at up to 2000 µM of genipin after 4 hours and up to 400 µM of genipin after 24 hours, which indicated that genipin may not have the potential to induce DNA interstrand cross-linking in the test conditions. A decrease in relative viability of cells was observed at the highest two doses of genipin after 4 hours (down to 13%) and 24 hours (down to 5%) of exposure. In a reverse Comet assay, significant decreases ($P < 0.05$) in %tail DNA were noted at 667 and 2000 µM after 4 hours exposure to genipin versus S0 control. The relative viability of cells was about 10% after 4 hours exposure to genipin at 2000 µM. The authors stated that, as per the scoring criteria (i.e. exclusion of hedgehogs and cells with $\geq 90\%$ tail DNA), the %tail DNA values underrepresented the extent of DNA damage induced at these concentrations. The relative viability of cells after 4 hours of exposure to 667 µM of genipin was over 60%. No significant decreases in %tail DNA was observed after 24 hours exposure to genipin at all concentrations.
- ^f Mortality or moribundity was observed in 2, 4 or 6 male mice exposed to 74, 222 or 667 mg/kg bw per day of genipin. All female mice survived the duration of the study. No statistically significant differences in mean MN-RET frequencies were observed in the peripheral blood of male and female mice exposed to genipin. A significantly positive trend ($P < 0.05$) was observed in female mice exposed to 222 mg/kg bw per day such that the mean MN-RET frequency was 3.58 ± 0.23 (versus 2.49 ± 0.13 in the control). However, these mice showed overt adverse clinical symptoms. Furthermore, the authors stated that MN-RET frequencies in female mice exposed to 74 mg/kg bw per day and 222 mg/kg bw per day fell outside the confidence interval of historical control data. No decrease in percentage of RET was observed in groups exposed to genipin compared with controls, which indicated a lack of suppression of the bone marrow in mice exposed to genipin. Overall, the Committee concluded that the increase in the mean MN-RET frequency in female mice exposed to 222 mg/kg bw per day was biologically insignificant. There were no significant or dose-dependent increases in %tail DNA in evaluated tissues of male and female mice, and all values were within the historical control data. The Committee concluded that genipin did not induce DNA damage in the Comet assay.
- ^g In vivo alkaline Comet assay.
- ^h There were statistically significant ($P = 0.05$) differences in the %tail DNA between genipin-exposed liver samples and vehicle ($\pm$ S0; 2400 µM); however, this study had some limitations and the Committee considered the response equivocal.
- ⁱ A decrease in number of mean revertants per plate was observed at 500 and 1000 µg/plate in TA98 and TA100 strains – S9 and at 1000 µg/plate in TA98 and TA100 + S9 because of cytotoxicity induced by genipin at higher test concentrations.
- ^j Genipin produced a killing zone of diameter 17 and 22 mm in M45 Rec⁺ and 13 and 19 mm in H17 Rec⁺ strains at low and high concentrations, respectively, indicating that genipin caused DNA damage in *Bacillus subtilis* under test conditions.
- ^k A significant increase ($P < 0.001$) in SCE frequency was observed at all test concentrations of genipin compared with controls (2.6 times increase at 10 µg/mL). The TI (%) was significantly increased ($P < 0.01$) at all test concentrations of genipin compared with controls with the highest TI of 95% at 8 µg/mL. The average generation time also increased from 14 to 18.9 hours at 0 and 10 µg/mL, respectively, of genipin.

The significant decrease in %tail DNA at 667 µM of genipin with over 60% of viable cells in the test system indicated that genipin may induce DNA cross-linking at this concentration. However, no significant decrease in %tail DNA was observed after 24 hours of exposure to genipin at all test concentrations. Because of significant changes noted at high test concentrations after 4 hours of genipin exposure in the reverse Comet assay, the Committee concluded that the result of this assay was equivocal.

The Committee evaluated a report of in vivo micronucleus and Comet assays (180) on male and female B6C3F1 mice orally exposed to genipin at different dose levels up to 2000 and 222 mg/kg bw per day, respectively, for three consecutive days. Because of mortality, moribundity or other abnormal clinical observations, blood and liver from male mice exposed to 222 mg/kg bw per day and more of genipin were not assessed. The Committee noted that there were no statistically significant differences in mean micronucleated reticulocyte (MN-RET) frequencies in the peripheral blood of male mice exposed to genipin at all doses. A significantly positive trend ($P < 0.05$) was observed in female mice exposed to 222 mg/kg bw per day such that the mean MN-RET frequency was increased (3.58 ± 0.23) compared with controls (2.49 ± 0.13). However, these mice showed overt adverse clinical symptoms suggesting genipin-induced

toxicity at the highest dose. The authors stated that MN-RET frequencies in female mice exposed to 74 and 222 mg/kg bw per day fell outside the confidence interval of historical control data. Furthermore, no decrease in percentage of RET was observed at any dose, which indicated a lack of suppression of the bone marrow in mice exposed to genipin. The Committee concluded that the increase in the mean MN-RET frequency in female mice exposed to the highest dose was not toxicologically relevant. There were also no significant or dose-dependent increases in %tail DNA in the livers of male mice and stomach and duodenum of female mice, and all values were within the historical control data. The Committee therefore concluded that genipin showed a negative response in the in vivo micronucleus and Comet assays.

The Committee evaluated data from an in vivo reverse Comet assay (181) using liver samples of female B6C3F1 mice exposed to 222 mg/kg bw per day of genipin for 3 days. The treatment of vehicle slides with styrene oxide increased the %tail DNA to 2.3-fold over the baseline. Similar treatment with styrene oxide of slides prepared using livers of animals exposed to genipin showed a 1.6-fold increase in %tail DNA over the baseline. The study authors reported a *P*-value of 0.05, which indicated that the difference between genipin-exposed and vehicle groups was statistically significant. However, the Committee noted that this study was not conducted in compliance with guidelines, and the frozen liver samples used in this study had been collected 2 years before. The Committee therefore considered this study to be unreliable for the evaluation of the potential for genotoxicity of genipin in vivo.

From the results of the genotoxicity studies, the Committee concluded that genipin showed positive results in all in vitro genotoxicity studies. However, it induced negative results in the in vivo micronucleus and Comet assays, which indicated that it did not induce significant DNA damage or exhibit DNA cross-linking in vivo.

2.2.5 Reproductive and developmental toxicity

The sponsor submitted three reports of reproductive and developmental toxicity studies conducted with Gardenia (Genipin) Blue (lot no. 180910; 32.3% Gardenia Blue polymer): an extended one-generation reproductive toxicity (EOGRT) study in rats (184), prenatal developmental toxicity study in rats (185) and prenatal developmental toxicity study in rabbits (186). The Committee noted that these data had also been published in research articles (124,125).

(a) EOGRT study in rats

Prior to conducting an EOGRT study, a dose-range-finding study (187) was conducted in which male and female (P0) Sprague Dawley rats (five per sex per group) were exposed to Gardenia (Genipin) Blue at dose levels of 0, 1.5, 3 or

Table 2.5

Doses of Gardenia (Genipin) Blue (mg/kg bw per day) from the EOGRT study

Cohort	Male			Female		
	0.5%	2.5%	5%	0.5%	2.5%	5%
P0	226.6	1150.2	2349.7	354.6	1730	3420.7
P0 (gestation)	—	—	—	335.7	1683	3461.2
P0 (lactation)	—	—	—	989.1	4981.2	9855.9
1A	402.7	2026.3	3914.6	449.2	2311.3	4349.3
1B	372.7	1877.4	3735.8	416.1	2104.9	4017.4
2A	428	2052.8	4095.1	449.8	2388	4490.2
2B	Not assessed	Not assessed	Not assessed	Not assessed	Not assessed	Not assessed
3	475.3	2301.3	4674.7	489.5	2509	4959.5

Source: Hobbs et al. (123), unpublished report (184)

5% in their feed for 14 days before and during mating, gestation and lactation. F1 offspring were exposed to Gardenia (Genipin) Blue in their feed at the same dose levels throughout lactation and until the end of the study (PND 21). No treatment-related effects on body weights, food consumption or reproductive parameters of P0 animals and organ weights of the F1 animals were observed in treated compared with controls groups. Blue or black discoloration of the faeces was observed in all treated animals. Dose-dependent blue discoloration of caecum, duodenum, jejunum, ileum, kidneys and stomach were observed in treated groups; however, this finding was not considered toxicologically relevant. Based on a lack of treatment-related adverse effects, doses of up to 5% (3063.5 and 6081.67 mg/kg bw per day in males and females, respectively) were considered appropriate for conducting a GLP-compliant EOGRT study in rats.

An EOGRT study was conducted in which male and female (P0) Sprague Dawley rats (30 per sex per group) were exposed to Gardenia (Genipin) Blue at dose levels of 0, 0.5, 2.5 and 5% in their feed for 14 days before mating, and for the entire duration of the study for P0 and F1 offspring (123,184). This study was conducted in accordance with OECD Test Guideline No. 443 (188) and GLP. At weaning, the F1 offspring were allocated to different cohorts as described below and in Table 2.5.

Cohorts 1A and 1B. One male and one female per litter assigned on PND 21 to yield at least 20 males and 20 females per dose per cohort to assess potential effects of Gardenia (Genipin) Blue on reproductive and developmental parameters. Cohort 1A and 1B animals were killed on PND 91–93 and PND 105 or 112, respectively.

Cohorts 2A and 2B. At least 10 males and 10 females per group assigned on PND 21 to assess the potential effects of Gardenia (Genipin) Blue on the nervous system. Cohort 2A and 2B animals were killed on PND 77 and PND 22, respectively.

Cohort C. At least 10 males and 10 females per group were assigned to Cohort 3 on PND 21 to assess the potential effects of Gardenia (Genipin) Blue on the developing immune system, and were killed on PND 62.

P0 rats. There were no treatment-related adverse effects on mortality, morbidity, clinical observations or reproductive parameters. There were no significant differences in food consumption or food efficiency of treated P0 males compared with controls. Mean food consumption in P0 females was significantly increased ($P < 0.05$) in the high-dose group (69.2 ± 7.4 g/kg bw per day) compared with controls (66.2 ± 3.1 g/kg bw per day), but there were no differences in feed efficiency between groups during gestation.

No treatment-related effects on mean oestrous cycle length or regularity in cycles for P0 females were observed. No dose-dependent or statistically significant changes in sperm motility, concentration or morphology in P0 males were observed. No treatment-related effects on the reproductive performance of P0 rats were observed.

An 18-hour urine sample was collected from 10 male and 10 female P0 animals and analysed to assess certain urinary parameters. Significant increases ($P < 0.05$) in urobilinogen levels were observed in P0 females exposed to 2.5% (0.86 ± 0.55 mg/dL) and 5% Gardenia (Genipin) Blue (1.47 ± 1.1 mg/dL) compared with controls (0.2 ± 0 mg/dL). This change was small and within the normal biological range; it was therefore not considered toxicologically relevant.

No dose-dependent changes in haematological parameters were observed in treated P0 rats compared with controls. The examination of clinical chemistry revealed statistically significant increases ($P < 0.001$) in levels of TSH in P0 males exposed to 5% Gardenia (Genipin) Blue (16.97 ± 8.13 ng/mL) compared with controls (8.89 ± 1.62 ng/mL).

Assessment of organ weights showed statistically significant increases ($P < 0.001$) in absolute and relative thyroid with parathyroid weights in P0 females exposed to 5% Gardenia (Genipin) Blue (0.029 ± 0.005 and 0.012 ± 0.002 g, respectively) compared with controls (0.022 ± 0.004 and 0.009 ± 0.001 g, respectively). In addition, the relative spleen weight was statistically decreased ($P < 0.05$) in P0 males exposed to 5% Gardenia (Genipin) Blue (0.157 ± 0.015) compared with controls (0.171 ± 0.026). All these changes were small and within the range of historical controls; they were therefore not considered toxicologically relevant.

Histopathological examination of tissues in P0 animals exposed to Gardenia (Genipin) Blue revealed blue or black discoloration of some organs,

including mesenteric lymph nodes, duodenum, jejunum, ileum, caecum, colon, stomach and kidney, which was not considered an adverse effect. Several microscopic findings, such as hepatocellular adenoma, chronic progressive nephropathy, cardiomyopathy, and cysts in the thymus and thyroid glands, were observed sporadically in treated as well as control groups, and therefore not considered toxicologically relevant.

F1 offspring (pre-weaning, PND 0–21). No dose-related effects on mortality, morbidity, clinical observations or body weights were observed. There were also no significant differences in food consumption or feed efficiency of F1 treated animals compared with controls. No significant differences in anogenital distance on PND 1 and 4 for male and female offspring, or mean number of male offspring with retained nipples on PND 12, were observed between exposed and control groups. There were also no significant differences in T_4 levels on PND 4 or T_4 and TSH levels on PND 21 in male and female offspring exposed to Gardenia (Genipin) Blue compared with controls. The only statistically significant change observed after examining organ weights was a decrease ($P < 0.01$) in absolute heart weight in F1 offspring exposed to 2.5% (0.3013 ± 0.0339 g) and 5% (0.2981 ± 0.0328 g) of Gardenia (Genipin) Blue compared with controls (0.3316 ± 0.0306 g).

F1 offspring (all cohorts). No treatment-related effects on mortality, morbidity, clinical signs, body weights, food consumption or feed efficiency were observed.

No statistically significant differences in F1 vaginal opening or preputial separation and body weight on that day were observed between treated and control groups. No dose-related effects on the mean oestrous cycle length or regularity in cycles were observed. The mean total sperm concentration was significantly increased ($P < 0.05$) in F1 males exposed to 5% (437.8 ± 109.4 million/g) Gardenia (Genipin) Blue compared with controls (353.9 ± 97.3 million/g); however, there were no significant changes reported in sperm mortality or morphology between treated and control groups.

F1 offspring (Cohort 1A). No statistically significant differences in urinalysis parameters, splenic lymphocyte or haematological counts were reported between treated and control groups.

Examination of clinical chemistry showed a statistically significant increase ($P < 0.01$) in TSH levels in females exposed to 5% (13.25 ± 3.2 ng/mL) Gardenia (Genipin) Blue compared with controls (8.59 ± 1.72 ng/mL), with a non-significant increasing trend at lower doses. All TSH values were within the historical control range; the Committee therefore did not consider these changes in TSH levels at the highest dose tested to be toxicologically relevant.

The assessment of organ weights showed significant increases ($P < 0.001$) in absolute thyroid with parathyroid weights in females exposed to 0.5% (0.025 ± 0.003 g), 2.5% (0.022 ± 0.002 g) and 5% (0.021 ± 0.003 g) Gardenia

(Genipin) Blue compared with controls (0.018 ± 0.002 g). These changes were small and within the historical control range; the Committee therefore did not consider these findings toxicologically relevant.

Histopathological examination showed a significant increase ($P < 0.05$) in the number of medium-sized ovarian follicles in females exposed to 5% (18.1 ± 6.7) Gardenia (Genipin) Blue compared with controls (13.9 ± 6.3). There were no changes in numbers of primordial or large follicles or changes in ovarian morphology or weights; the increase in the number of medium-sized ovarian follicles was therefore not considered toxicologically relevant. Blue discoloration of some organs, including intestine kidneys, was reported but not considered adverse. Several sporadic microscopic changes were observed in control as well as high-dose groups that were considered incidental.

F1 offspring (Cohort 1B). There were no statistically significant differences in mean or absolute organ weights of treated compared with control groups. No differences or abnormalities were observed in the kidneys, liver and mesenteric lymph nodes in animals exposed to 5% Gardenia (Genipin) Blue compared with controls.

F1 offspring (Cohort 2A). No treatment-related effects on the parameters assessed using the auditory startle test or motor activity were noted. The only significant effect in the functional observational battery test was an increase ($P < 0.05$) in forelimb grip strength in males exposed to 5% (978.7 ± 252.3 g) Gardenia (Genipin) Blue compared with controls (719.4 ± 202.7 g). These changes were not considered toxicologically relevant. No significant differences in brain weights of treated compared with controls groups were noted. The only significant change in morphometric brain measurements was a significant increase ($P < 0.05$) in the thickness of outer layer of cerebellar folia in males exposed to 5% (185.42 ± 2.95 μ m) Gardenia (Genipin) Blue compared with controls (182.33 ± 3.78 μ m). No significant histopathological changes were observed in animals exposed to Gardenia (Genipin) Blue compared with controls.

F1 offspring (Cohort 2B). There were significant decreases ($P < 0.05$) in the total body weight and significant increases ($P < 0.05$) in relative brain-to-body ratio in post-fixed samples from females exposed to 2.5% (48.7 ± 5.1 g and $2.94 \pm 0.24\%$, respectively) and 5% (50 ± 2.7 g and $2.82 \pm 0.09\%$, respectively) Gardenia (Genipin) Blue compared with controls (55.2 ± 5 g and $2.61 \pm 0.19\%$, respectively). However, no significant changes in post-fixed brain weights were noted between groups; the significant changes in total body weights and relative brain-to-body ratios were therefore not considered toxicologically relevant. There was a significant increase in cerebellar height in females exposed to 5% (5.14 ± 0.21 μ m) Gardenia (Genipin) Blue compared with controls (4.56 ± 0.38 μ m). However, because there were no significant changes in morphometric brain measurements between groups, the increase in cerebellar height in the high-dose group was not considered toxicologically relevant.

Sporadic histopathological observations were noted in control and high-dose animals; however, none of these changes was considered treatment related.

F1 offspring (Cohort 3). There were no significant differences in T-cell-dependent antibody responses of the immunotoxicity assay between treated and control groups.

Summary. Considering the combined results of the EOGRT study, no treatment-related or toxicologically relevant adverse effects of oral exposure to Gardenia (Genipin) Blue were observed in the P0 and F1 generations. The Committee identified a NOAEL of 5%, equal to 2350 mg/kg bw per day in males (corresponding to a polymer dose of 759 mg/kg bw per day) and 3421, 3461 and 9856 mg/kg bw per day in females (corresponding to a polymer dose of 1105, 1118 and 3183 mg/kg bw per day during premating, gestation and lactation, respectively), the highest dose tested, for the P0 generation. The Committee identified a NOAEL of 5%, equal to 39 156 mg/kg bw per day (corresponding to a polymer dose of 1264 mg/kg bw per day) in Cohort 1A males and 4349 mg/kg bw per day (corresponding to a polymer dose of 1405 mg/kg bw per day) in Cohort 1A females, the highest dose tested, for the F1 generation.

(b) Prenatal developmental toxicity study in rats

Prior to conducting an oral prenatal developmental toxicity study, a dose-range-finding study was conducted in which female Sprague Dawley rats (eight per group) received Gardenia (Genipin) Blue at doses of 250, 500, 1000 or 2000 mg/kg bw per day via oral gavage during gestational days (GD) 6–21 (189). Apart from blue discoloration of the faeces and staining of body surfaces, such as cranium and tail, no treatment-related effects were observed in this study. Notably, no significant changes in any of the evaluated maternal or developmental parameters in the fetus were observed between treated and control groups. Based on a lack of treatment-related adverse effects, oral doses of up to 2000 mg/kg bw per day were considered appropriate for conducting a prenatal developmental toxicity study in rats.

A prenatal developmental toxicity study was conducted in which female Sprague Dawley rats (25 per group) received Gardenia (Genipin) Blue at doses of 500, 1000 or 2000 mg/kg bw per day via oral gavage during GD 6–21 (124,185). This study was conducted in accordance with United States Environmental Protection Agency Health Effects Test Guidelines (190) and OECD Test Guideline no. 414 (191).

No treatment-related effects on mortality, morbidity or clinical observations, except for black faeces and blue discoloration of the fur and body surfaces, were observed. The discoloration of the faeces and body surfaces was treatment related but not considered adverse. No dose-related effects on maternal body weights, body weight gains and gravid uterine weights were observed. Over the entire dosing period, the mean food consumption was lower in all treated

groups compared with controls. However, given that there were no significant changes in mean body weights or body weight gains of treated groups compared with controls, the effect on mean food consumption was not considered adverse. There were no significant differences in maternal T_3 , T_4 and TSH concentrations between treated and control groups. No treatment-related effects on organ weights were observed. Dark discoloration of the kidney and dark blue discoloration of the tail was observed in all treated groups. Given that there were no treatment-related effects on survival or clinical observations, this change was not considered adverse. Some microscopic observations were noted in control as well as treated groups, but none was considered treatment related.

There were no treatment-related effects on any ovarian or uterine parameters, including litter sizes, weights, sex ratios, implantation sites and litter proportions of pre-implantation loss. Morphological evaluation of fetuses did not reveal any treatment-related external, visceral or skeletal malformations. A significantly higher mean litter proportion of supernumerary lobes of the liver (2.38% of the litter) was observed in the high-dose group compared with controls (0%). The mean litter proportion of moderate dilation of ureter in groups exposed to 500 mg/kg bw per day (0.67% litter) and 2000 mg/kg bw per day (1.24% litter) were significantly lower ($P < 0.05$) than for controls (8.12%). However, because all these values were within the historical control ranges, these findings were not considered toxicologically relevant.

Based on a lack of treatment-related adverse effects, the Committee identified a NOAEL of 2000 mg/kg bw per day, the highest dose tested, for maternal and prenatal developmental toxicity.

(c) Prenatal developmental toxicity study in rabbits

Prior to conducting a prenatal developmental toxicity study, a dose-range-finding study was conducted in which female New Zealand white rabbits (six per group) received Gardenia (Genipin) Blue at dose levels 250, 500, 1000 or 2000 mg/kg bw per day via oral gavage during GD 7–28 (192). Except for blue discoloration of the faeces and body surfaces, as well as dark blue discoloration of some organs such as intestine and stomach, no treatment-related effects were observed in this study. Notably, no significant differences in any of the evaluated maternal or developmental parameters in the fetus were observed between treated and control groups. Based on a lack of treatment-related adverse effects, oral doses of up to 2000 mg/kg bw per day were considered appropriate for conducting a prenatal developmental toxicity study in rabbits.

A prenatal developmental toxicity study was conducted in which female New Zealand white rabbits (24 per group) received Gardenia (Genipin) Blue at doses of 500, 1000 or 2000 mg/kg bw per day via oral gavage during GD 7–28 (124,186). The study was performed in accordance with GLP, the United States

Environmental Protection Agency Health Effects Test Guidelines (190) and OECD Test Guideline no. 414 (191).

No treatment-related effects on mortality, morbidity or clinical observations, except for black faeces and blue discoloration of body surfaces such as head, forepaws, hindpaws, tail and urogenital area, were observed. The discoloration of the faeces and body surfaces was treatment related but not considered adverse. The maternal body weight gain was significantly higher ($P < 0.05$) in rabbits exposed to 2000 mg/kg bw per day of Gardenia (Genipin) Blue on GD 4–7 and 7–8 compared with controls. However, no dose-related effects on maternal body weights, body weight gains and gravid uterine weights were observed over the entire dosing period; the transient increases in the maternal weight gain at 2000 mg/kg bw per day of Gardenia (Genipin) Blue were therefore not considered toxicologically relevant. The mean food consumption was significantly lower ($P < 0.05$) in the high-dose group (100 ± 37 g) compared with controls (129 ± 34 g) at GD 23–24. However, no significant differences in the mean food consumption between treated and control groups were observed over the entire dosing period; this effect on GD 23–24 was therefore not considered to be toxicologically relevant.

Dark discoloration of the kidney and dark blue discoloration of the tail was observed in all treated groups and of the mesenteric lymph nodes in some animals exposed to 1000 or 2000 mg/kg bw per day of Gardenia (Genipin) Blue. Given that there were no treatment-related effects on survival or clinical observations, this change was not considered adverse. Some microscopic observations were noted in control as well as in treated groups, but none was considered treatment related.

No dose-related effects on any ovarian or uterine parameters, including mean number of live fetuses, fetal body weights, sex ratios and mean litter proportions of post-implantation loss were observed. Morphological evaluation of fetuses did not reveal any treatment-related or dose-dependent external, visceral or skeletal malformations.

Based on a lack of treatment-related adverse effects, the Committee identified a NOAEL of 2000 mg/kg bw per day, the highest dose tested, for maternal and prenatal developmental toxicity.

The Committee did not identify any reproductive or development toxicity studies on geniposide or genipin that were relevant to the current evaluation.

2.2.6 Special studies

(a) Gardenia (Genipin) Blue

(i) Reversibility of discoloration of kidneys in rats

The sponsor submitted a study that evaluated the potential for reversibility of the dark discoloration in the kidneys after oral exposure to Gardenia (Genipin)

Blue, observed in all studies discussed above. Male Sprague Dawley rats (16 per group) were exposed to Gardenia (Genipin) Blue (lot no. 180910) in the diet at concentrations of 0, 0.5 (equal to 380.3 mg/kg bw per day) or 5% (equal to 3927.5 mg/kg bw per day) (193). Exposure to Gardenia (Genipin) Blue was discontinued after day 45 (5% group) or day 55 (0.5% group). A subset of the animals was observed for kidney discoloration on days 46, 103, 165 and 166.

No significant differences in body weights, food consumption or kidney weights were observed between treated and control groups. The control group did not show any kidney discoloration throughout the duration of the study. Kidney discoloration was observed in the 5% group on day 46 and in the 0.5% group on day 55. The kidney discoloration in the 5% group reduced in intensity on day 103 and was considered to be minimal on days 165 and 166. No kidney discoloration was observed in the 0.5% group on day 103. There were no significant differences in kidney weights between treated and control groups.

The Committee concluded that kidney discoloration observed in rats following 55 or 45 days of exposure to 0.5 or 5%, respectively, of Gardenia (Genipin) Blue was reversible following discontinuation of exposure.

(ii) Analysis of residual allergens

The sponsor submitted a study report that evaluated the potential for allergenicity of Gardenia (Genipin) Blue as a result of the use of soy protein hydrolysates as a raw material in its formation (194). The samples (lot no. 150510 with 24.7% polymer content, 180910 with 32.3% polymer content, 201109 with 30.8% polymer content, 201112 with 30.6% polymer content and 201113 with 30% polymer content) were analysed for the presence of residual allergens using a kit based on double antibody (sandwich) enzyme-linked immunosorbent assay (ELISA) and an LOD for soy protein residue of 1.25 mg/kg. The results from five representative lots of Gardenia (Genipin) Blue showed non-detectable levels of soy protein residue (i.e. < 1.25 mg/kg). The sponsor stated that allergens of soy protein are not expected to be detected in the article of commerce because the unreacted substances are removed as amino acids in the enzymatic process.

(b) Geniposide

The Committee identified additional studies that evaluated mechanisms of geniposide-induced hepatotoxicity.

(i) 3-day oral toxicity study in Sprague Dawley rats

The Committee evaluated data from a 3-day oral toxicity study in which male and female Sprague Dawley rats (six per sex per group) were exposed to geniposide at dose levels of 0, 100 and 300 mg/kg bw per day via intragastric gavage (135). Some clinical signs, including weight loss, diarrhoea and weakness followed

by mortality, were observed in one rat exposed to 300 mg/kg bw per day of geniposide.

Compared with controls, levels of AST (108.75 ± 18.39 U/L versus 131.50 ± 16.62 U/L), γ -glutamyltransferase (8.44 ± 7.25 U/L versus 1.17 ± 1.68 U/L), cholesterol (2.33 ± 0.5 mmol/L versus 1.75 ± 0.53 mmol/L) and TBIL (0.01 ± 0.04 μ mol/L versus 0.83 ± 0.25 μ mol/L) were significantly elevated ($P < 0.001$ or 0.05) in low-dose groups. Compared with controls, significant increases ($P < 0.01$ or 0.05) in levels of ALT (291 ± 294.36 U/L versus 25 ± 4.87 U/L), AST (474.86 ± 283.96 U/L versus 131.50 ± 16.62 U/L), ALP (455.71 ± 169.89 U/L versus 278.63 ± 42.08 U/L), γ -glutamyltransferase (15.76 ± 15.91 U/L versus 1.17 ± 1.68 U/L) and cholesterol (2.60 ± 0.37 mmol/L versus 1.75 ± 0.53 mmol/L) were also observed in high-dose groups.

Compared with controls, levels of taurochenodeoxycholic acid (TCDCA; 0.0669 ± 0.0152 μ g/mL versus 0.0448 ± 0.0259 μ g/mL) and hyodeoxycholic acid (HDCA; 0.619 ± 0.182 μ g/mL versus 1.492 ± 1.066 μ g/mL) were significantly altered in low-dose groups. Significant increases ($P < 0.001$ or 0.05) in levels of TCDCA (0.24 ± 0.026 μ g/mL versus 0.0448 ± 0.0259 μ g/mL), β -murocholic acid (1.128 ± 0.33 μ g/mL versus 0.46 ± 0.434 μ g/mL) and taurocholic acid (TCA; 0.712 ± 0.571 μ g/mL versus 0.138 ± 0.133 μ g/mL), and significant decreases ($P < 0.05$) in levels of HDCA (0.427 ± 0.123 μ g/mL versus 1.492 ± 1.066 μ g/mL), were observed in the serum of high-dose compared with control groups.

Levels of taurohyodeoxycholic acid (THDCA; 11.927 ± 4.966 μ g/g versus 20.798 ± 10.187 μ g/g) and HDCA (0.0817 ± 0.0324 μ g/g versus 0.470 ± 0.420 μ g/g) were significantly altered ($P < 0.05$) in low-dose groups.

Significant changes ($P < 0.001$, 0.01 or 0.05) in levels of tauro- α -muricholic acid (61.974 ± 14.241 μ g/g versus 36.112 ± 15.073 μ g/g), THDCA (7.73 ± 2.746 μ g/g versus 20.798 ± 10.187 μ g/g), TCA (126.849 ± 32.844 μ g/g versus 47.005 ± 21.61 μ g/g), TCDCA (11.199 ± 6.825 μ g/g versus 4.269 ± 2.075 μ g/g) and HDCA (0.0804 ± 0.0346 μ g/g versus 0.470 ± 0.420 μ g/g) were also observed in the livers of high-dose rats compared with controls.

The evaluation of the expression of genes involved in hepatic bile acid synthesis, metabolism and transport in the liver showed significant suppression ($P < 0.01$) of farnesoid X receptor mRNA expression in low-dose but significant upregulation ($P = 0.01$) in high-dose groups compared with controls. Both doses of geniposide significantly suppressed ($P < 0.001$) small heterodimer partner mRNA expression compared with controls. Expression of bile salt export pump mRNA was significantly inhibited ($P < 0.01$) in high-dose compared with control groups. Increased expression of additional genes involved in bile acid transport, including sodium taurocholate co-transporting polypeptide ($P < 0.01$), multidrug resistance protein (MRP) 2 ($P < 0.01$) and MRP3 ($P < 0.001$), were observed in low-dose groups (which decreased at the high dose) compared with controls.

Histopathological examination showed signs of liver injury, including degeneration, necrosis, lymphocyte infiltration and fat droplets in hepatocytes, in rats exposed to the high dose of geniposide that were not observed at the low dose.

The Committee concluded that oral exposure to geniposide at the tested doses induces hepatotoxic effects as well as corresponding changes in the expression of bile-acid-regulating genes and histopathological changes.

(ii) Systematic review and meta-analysis on potential for hepatotoxicity

The Committee identified a meta-analysis (138) based on a systematic review of published literature on reported liver effects of geniposide to assess the relationship between duration of exposure and dose, and the type of hepatic (i.e. protective or toxic) effects of geniposide. The literature searches for the period 2005–2023 were conducted in five databases (Web of Science, PubMed, China National Knowledge Infrastructure Database, Wanfang Database and Chinese Science and Technology Journal Database). The search terms included “geniposide” and “liver injury” or “liver protection” or “hepatotoxicity” or “hepatoprotection”. Overall, 103 studies were identified that included 61 studies on hepatotoxicity and 42 on hepatoprotection of geniposide. After further screening by applying defined inclusion and exclusion criteria to identify articles with relevant animal models, dose levels, exposure route and duration, and outcomes, 25 studies were selected. These included 13 studies on hepatotoxicity and 12 studies on hepatoprotection of geniposide. Three-dimensional (3D) dose/time-effect and mechanistic analyses were performed to evaluate therapeutic and toxic effects of geniposide.

The 3D dose/time-effect analysis reported that geniposide at oral doses of 28–150 mg/kg bw per day for 5–28 days effectively protected the liver without inducing toxicity. However, when administered at oral doses of 200 mg/kg bw per day or more for 7–30 days, geniposide induced ($P < 0.001$) hepatotoxicity in the evaluated studies. When the exposure duration was shorter (i.e. ≤ 5 days), geniposide significantly induced hepatotoxic effects at oral doses of 400 mg/kg bw per day or more compared with lower doses of less than 200 mg/kg bw per day. When the exposure duration was more than 30 days, geniposide caused liver toxicity at a low dose of less than 100 mg/kg bw per day. The mechanistic analysis indicated that inflammatory processes, including tumour necrosis factor- α and nuclear factor- κ B pathway, as well as changes in markers of oxidative stress such as MDA and SOD, are responsible for the hepatic effects.

2.3 Observations in humans

No data on observations of exposure to Gardenia (Genipin) Blue, genipin or geniposide in humans were available.

3. Dietary exposure

3.1 Introduction

Gardenia (Genipin) Blue is currently approved for use in food in China; Taiwan, China; Japan (where it has been used for > 30 years); and the Republic of Korea (195–198). Gardenia (Genipin) Blue is not listed in Codex 192-1995, and its dietary exposure was evaluated based on a request to assess this food additive at the Fifty-fourth session of the CCFA (44). The Committee reviewed the dietary exposure estimates submitted by the sponsor based on the proposed maximum GMP use level, as no actual concentrations of Gardenia (Genipin) Blue in the final products were provided or identified in the literature. The Committee did not consider the dietary exposure estimate using the budget method, as more refined dietary exposure estimates were already available.

3.2 Levels of use of Gardenia (Genipin) Blue

The effect of a colour additive varies depending on the colour tone of the food type to which it is added. In China, Gardenia (Genipin) Blue is approved for various food categories, including drinks, jams, jellies, pickled vegetables, fried nuts and seeds, bakery wares, candies and condiments, with maximum levels ranging over 0.2–1.0 g/kg (196). In Japan, the use of Gardenia (Genipin) Blue adheres to current GMP, with use levels depending on the food type and ranging over 0.001–1.0% (197).

The sponsor proposed the maximum GMP use levels for Gardenia (Genipin) Blue of 0.025–1.0% in 20 food categories and subcategories of the GSFA, corresponding to the intended colours of the final foods as presented in [Table 3.1](#). Gardenia (Genipin) Blue is proposed for use in various foods including alcoholic beverages; carbonated soft drinks; chewing gum; flavoured milks; flavoured enhanced water; fruit drinks; hard and soft candy; ice cream and frozen desserts; icing; jams, jellies and marmalades; powdered beverages; processed cereals; ready-to-drink teas; sugar coatings; sponge cakes; sport and energy drinks; teas; syrups for shaved ice; yogurt; and food supplements (chewable tablets).

Label searches on the Mintel database on 26 August 2021 for food and drink products on the Japanese market indicated that Gardenia (Genipin) Blue was used in 0.87% of processed food (380 coloured out of 43 468 food products) and 0.35% of drinks and beverages (34 coloured out of 9760 drinks and beverages). However, this information is not consistent with Japan's National Health and Nutrition Survey (NHNS) data (199). The Mintel survey is limited to processed food products, and it is assumed that Gardenia (Genipin) Blue is not used in any foods listed under the “small classification” in the NHNS. To further assess the actual use of Gardenia (Genipin) Blue in foods in Japan, another label survey on the Mintel database was conducted on 22 September 2022 to analyse

Table 3.1

Proposed maximum GMP use levels of Gardenia (Genipin) Blue for GSFA food categories, and the adjustment of those levels based on Mintel data for the USA and EU

Code	GSFA food category name	Proposed food use	Proposed maximum GMP use level in finished food product (%)	USA		EU	
				Percentage of products with blue colour (%) ^a	Adjusted maximum use level at GMP (%) ^b	Percentage of products with blue colour (%) ^a	Adjusted maximum use level at GMP (%) ^b
1.1.4	Flavoured fluid milk drink	Flavoured milk, milk shakes, dairy drinks	0.05	11	0.0055	2	0.001
1.7	Dairy-based desert (e.g. pudding, fruit or flavoured yogurt)	Yogurt, regular and Greek	1	2	0.02	2	0.02
		Ice cream and frozen dairy	1	11	0.11	9	0.09
2.3	Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions	Flavoured milk, milk shakes, dairy drinks	0.05	—	—	2	0.001
2.4	Fat-based desserts excluding dairy-based dessert products of food category	Non-dairy ice cream substitutes	1	—	—	9	0.09
		Non-dairy yogurt substitutes	1	—	—	2	0.02
3	Edible ices	Frozen desserts	0.1	8	0.008	9	0.009
4.1.2.5	Jams, jellies, marmalades	Jams, jellies, marmalades, confiture and spreads	0.05	1	0.005	1	0.0005
5.2.1	Hard candy	Boiled sweets, lollipops, medicated confectionery	0.05	59	0.0295	37	0.0185
5.2.2	Soft candy/nougats and marzipans	Liquorice, marshmallows, other sugar confectionery	1	34	0.34	41	0.41
5.2.3							
5.3	Chewing gum	Chewing gum	1	53	0.53	45	0.45
5.4	Decorations, toppings and sweet sauces	Sugar coating	1	34	0.34	69	0.69
		Syrup for saved ice	0.1	8	0.008	76	0.076
		Icing	0.5	76	0.38	69	0.345

Table 3.1. (continued)

Code	GSFA food category name	Proposed food use	Proposed maximum GMP use level in finished food product (%)	USA		EU	
				Percentage of products with blue colour (%) ^a	Adjusted maximum use level at GMP (%) ^b	Percentage of products with blue colour (%) ^a	Adjusted maximum use level at GMP (%) ^b
6.3	Breakfast cereals, including rolled oats	Processed breakfast cereals	0.5	14	0.07	1	0.005
7.2.1	Cakes, cookies and pies (e.g. fruit-filled or custard types)	Sponge cake	0.5	13	0.065	8	0.04
13.6	Food supplements	Chewable tablet	1	36	0.36	1 ^c	1
14.1.3	Fruit and vegetable nectar ^d	Fruit and vegetable nectar	0.025	—	—	—	—
14.1.4.1	Carbonated water-based flavoured drinks	Carbonated water-based flavoured drinks	0.05	—	—	3	0.0015
		Carbonated drinks	0.05	6	0.003	4	0.002
14.1.4.2	Non-carbonated water-based flavoured drinks	Fruit drink	0.05	6	0.003	6	0.003
		Flavoured enhanced water	0.05	3	0.0015	3	0.0015
		Sport drinks	0.05	13	0.0065	10	0.005
		Ready-to-drink-tea	0.05	3	0.0015	2	0.001
14.1.4.3	Concentrates (liquid or solid) for water-based flavoured drinks	Powdered beverage	0.05	29	0.0145	5	0.0025
14.1.5 ^c	Coffee, coffee substitutes, tea, herbal infusions and other hot cereal and grain beverages excluding cocoa	Coffee, coffee substitute	0.05	—	—	—	—
14.2.7	Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler-type beverages, low alcoholic refreshers)	Other alcoholic drink (e.g. cocktails, alcopops, mixed drinks)	0.05	1	0.0005	3	0.0015

EU: European Union; GMP: Good Manufacturing Practices; GSFA: General Standard for Food Additives; USA: United States of America.

^a Rounded up to nearest whole number.

^b Product of the maximum GMP use level of Gardena (Genipin) Blue and percentage of products with blue and green colour.

^c Not included in Mintel data search (100% assumed).

^d No information.

food and drink products available on the Japanese market. The actual use of colouring was 42.6% in processed cloudy food (18 437 coloured out of 43 299 food products) and 16.4% in clear drinks (1578 coloured out of 9618 drinks). Among the colorants used, Gardenia (Genipin) Blue was estimated to account for 16%. Specifically, the usage of Gardenia (Genipin) Blue was noted to be 0.07% in processed cloudy foods and 0.03% in clear drinks.

The sponsor also provided information on proportions of finished products within proposed food use in the USA and European (EU and United Kingdom) markets that could contain Gardenia (Genipin) Blue. The percentages of foods within each food type likely to contain Gardenia (Genipin) Blue were assumed to represent the proportions of blue or green coloured foods consumed in each category. These data were obtained from Mintel label searches in August 2024 (USA) and June 2022 (Europe). The search terms included various blue and green colorants such as Gardenia (Genipin) Blue, Huito, Spirulina Blue, Butterfly Pea, Brilliant Blue FCF, Indigotine, Fast Green FCF, chlorophylls, chlorophyllins and their complexes.

The proposed GMP maximum levels for each food category were then adjusted based on these label searches to reflect the typical usage of blue or green colouring in the food type or category in the market. These adjustments accounted for two main aspects: (i) the proportion of specific food components likely to contain Gardenia (Genipin) Blue (e.g. the coating of a candy); and (ii) the proportion of foods within a category that are commonly coloured blue or green, based on market data. Depending on the food category, adjustment factors of 1–76% were applied to the proposed GMP maximum use level of Gardenia (Genipin) Blue (Table 3.1).

3.3 Estimates of dietary exposure to Gardenia (Genipin) Blue

3.3.1 Assessments based on production volume data in Japan

An estimate of dietary exposure to Gardenia (Genipin) Blue (with colour value $E_{1\text{cm}}^{10\%}$ [i.e. absorbance of a 10% (w/v) solution in a 1-cm cell] 50), based on a survey of production volumes conducted by the Japan Food Additive Association, was reported to be 3 mg/person per day (200). The Sponsor provided further refined dietary exposure estimates to Gardenia (Genipin) Blue (colour value 300) based on production volumes, which included both manufactured and imported amounts. In 2014, the estimated exposure was 0.496 mg/person per day (equivalent to 0.009 mg/kg bw per day, based on an average body weight of 55.1 kg; 2.98 mg/person with colour value 50), with production volumes of 28 817 kg (converted from Gardenia (Genipin) Blue with a colour value 50, totalling 172 902 kg). In 2017, the estimated exposure was 0.386 mg/person per day (equivalent to 0.007 mg/kg bw per day; 2.3 mg/person with colour value 50), with production volumes of 22 735 kg (converted from Gardenia (Genipin) Blue

with a colour value 50, totalling 136 415 kg). It was assumed that 100% of the amount of Gardenia (Genipin) Blue used remains in finished food products, and that 80% of the food products were consumed.

3.3.2 Assessments based on country-specific dietary records

The sponsor submitted data on dietary exposure to Gardenia (Genipin) Blue based on country-specific dietary records from the general population of Brazil, Japan, the United Kingdom and the USA. These dietary exposures to Gardenia (Genipin) Blue were derived by multiplying national consumption data for each food type that could be coloured by the corresponding adjusted maximum use levels at GMP for that food.

(a) Brazil

The Brazilian Inquérito Nacional de Alimentação (INA) (2017–2018) dataset (201) was reviewed to identify food categories suitable for the proposed uses of Gardenia (Genipin) Blue. However, this dataset did not include food types such as icing and chewable tablets, which could be categorized under the proposed uses for Gardenia (Genipin) Blue. As a result, these food types were excluded from the assessment for the Brazilian population. For foods with a proposed use component (e.g. sugar coating on a bakery product), only the portion that could be coloured was considered in the assessment, rather than the entire food item. This approach aligns with recipe databases from the United Kingdom (202), as no relevant recipes were available in the Brazilian INA (2017–2018).

Mean and percentile exposure estimates using adjusted maximum use levels at GMP for the USA were calculated to provide representative exposure for the Brazilian population. The mean and high 95th percentile dietary exposure estimates for children aged 10–12 years were 0.9 and 3.4 mg/kg bw per day, respectively. For adults aged 19 years and older, the mean and 95th percentile estimates were 0.4 and 1.7 mg/kg bw per day, respectively, as listed in [Table 3.2](#).

(b) Japan

The sponsor provided NHNS dietary exposure data and maximum use levels for Gardenia (Genipin) Blue (199). Foods assumed to contain Gardenia (Genipin) Blue were investigated according to NHNS food categories. For food items such as Japanese buns, precooked noodles, buckwheat products, corn products, potato products, starch products, fish paste products, fermented milk and lactic acid bacteria beverages, other dairy products, pickles, fruit juices, fruit juice beverages, jams and confectionaries, a usage ratio of 0.07% was applied to its consumption. For drinks such as vegetable juice, wine, spirits and other beverages, the ratio was set at 0.03% for its consumption. Use levels based on the required colour, ranging from transparent foods to the surface of opaque foods, varies. For clear beverages

Table 3.2

Dietary exposure estimates for Gardenia (Genipin) Blue for the general population in Brazil, Japan, the United Kingdom and the USA (mg/kg bw per day)

Population group (years)	Brazil			Japan		United Kingdom			USA		
	Mean	90th	95th	Mean	90th ^a	Mean	90th	95th	Mean	90th	95th
Children (> 1) ^b	0.5	1.1	2.0	6.9	13.8	0.6	1.5	2.5	1.2	2.8	4.2
Children (2–5)	—	—	—	—	—	2.5	5.3	6.7	3.5	7.0	10.1
Children (6–12) ^c	0.9	1.8	3.4	3.2	6.4	1.7	3.5	5.1	2.5	5.8	8.1
Adolescents (13–18) ^d	0.7	1.6	2.6	2.3	4.6	0.7	1.6	2.4	1.2	3.1	4.0
Adults (> 19) ^e	0.4	0.9	1.7	1.9	3.8	0.3	0.8	1.2	0.8	1.9	2.6

^a High exposure is mean $\times$ 2, approximating the 90th percentile (27).

^b For Japan, 1–6 years; for Brazil, 2–9 years.

^c For Japan, 7–14 years; for Brazil, 10–12 years.

^d For Japan, 15–19 years.

^e For Japan, $\geq$ 20 years.

0.1% was applied to the maximum use level, and 1.0% was applied for cloudy or opaque foods.

Table 3.2 lists the dietary exposure estimates for Gardenia (Genipin) Blue for different age groups. The estimates per age group were divided by a mean body weight per age group from the NHNS to derive the estimates expressed in mg/kg bw per day. The highest mean estimate was up to 6.9 mg/kg bw per day from children aged 1–6 years. The Committee calculated the high dietary exposure using these mean estimates, with the 90th percentile estimates reaching 13.8 mg/kg bw per day for same age group. The food categories contributing the most to dietary exposure to Gardenia (Genipin) Blue was discretionary drinks (e.g. fermented milk and lactic acid bacteria beverages; 25.5%).

(c) United Kingdom

The United Kingdom National Diet and Nutrition Survey (2008/2009–2018/2019) dataset (203), containing 4-day estimated diet diaries from approximately 1000 individuals (500 adults and 500 children), was reviewed to identify food categories that may contain Gardenia (Genipin) Blue. Conservatively, foods that could potentially be coloured were included, except for those meeting United Kingdom national exclusion criteria such as baby foods, chocolate-flavoured products, chocolate candy and portions of candy containing chocolate. Foods typically prepared at home were also excluded. For foods with a proposed use component (e.g. icing on a bakery product), only the component that could be coloured was included in the assessment rather than the entire food item, reflecting publicly available data such as supermarket sources and recipe databases (204).

Mean and high percentile exposure estimates using adjusted maximum use levels at GMP for the EU were calculated to provide representative exposures for the United Kingdom population (Table 3.2). The mean and high 95th percentile dietary exposure estimates for children aged 2–5 years were 2.5 and 6.7 mg/kg bw per day, respectively. For adults aged 19 years and older, the mean and 95th percentile estimates were 0.3 and 1.2 mg/kg bw per day, respectively. The food category that contributed the most to the dietary exposure to Gardenia (Genipin) Blue was soft candy, accounting for up to 44.2%.

(d) USA

Consumption data from the United States National Health and Nutrition Examination Survey (2017–March 2020) (205), which included two non-consecutive 24-hour recalls, were reviewed to identify food categories that might contain Gardenia (Genipin) Blue. Conservatively, foods that could be coloured were included, except those meeting United States national exclusion criteria such as baby foods, chocolate-flavoured products, chocolate candy and portions of candy containing chocolate. Additionally, foods typically prepared at home were excluded. For foods where only a specific part was intended to be coloured (e.g. icing on a bakery item), only the fraction of the food to be coloured was considered according to data from the Food and Nutrient Database for Dietary Studies of the United States Department of Agriculture (206,207) and the Food Patterns Equivalents Database (208), rather than the entire food item. For chewable tablet consumption, the Dietary Supplement Database (205) was used, assuming each chewable supplement/tablet had a default mass of 1 g. This value was multiplied by the serving size and reported consumption of the supplement to calculate the 2-day average intake of chewable tablets.

Dietary exposure estimates for Gardenia (Genipin) Blue are summarized in Table 3.2. The mean and high 95th percentile dietary exposure estimates for children aged 2–5 years were 3.5 and 10.1 mg/kg bw per day, respectively. For adults aged 19 years and older, the mean and 95th percentile estimates were 0.8 and 2.6 mg/kg bw per day, respectively. The food category that contributed the most to dietary exposure to Gardenia (Genipin) Blue was ice cream, accounting for up to 36.0%.

3.3.3 Assessments based on EFSA and FAO/WHO databases

The sponsor submitted dietary exposure to Gardenia (Genipin) Blue based on summary food consumption data from two major databases: EFSA Comprehensive Food Consumption Database (51 surveys from 23 European countries) (209) and FAO/WHO CIFOSS (18). Both food consumption datasets were based on FoodEx2 categories for each country and age class, and for the general population or for consumers only.

The dietary exposure estimates were calculated by multiplying the proposed use levels of Gardenia (Genipin) Blue by their respective consumption amount in the Comprehensive Database. All identified FoodEx2 categories from the EFSA Comprehensive Database and CIFOCCoss food consumption data were assumed to be coloured with Gardenia (Genipin) Blue. As for the country-specific assessment, the same conservative assumptions were applied including cases where a food contained a proposed use component (e.g. icing on a bakery product).

The dietary exposure assessments for the general population and consumer-only groups were carried out by the sponsor. The mean and the 95th percentile exposure, the approach used in version 1 of the EFSA Food Additive Intake Model that consists of adding the highest high level of exposure from one food category to the mean exposure values for the remaining categories, are calculated per survey and population group. For the general population, the maximum use levels were adjusted based on the portion of food categories that would be expected to be coloured with Gardenia (Genipin) Blue available on the markets of the USA or the EU (Table 3.3). For the consumer-only scenario, the proposed maximum use levels were applied (Table 3.4). However, the Committee noted that the consumer-only scenario represents an extreme and unrealistic worst-case scenario.

Table 3.3

Dietary exposure estimates for Gardenia (Genipin) Blue for the general population using adjusted maximum GMP use levels (mg/kg bw per day)

Population group	Age	EFSA Comprehensive Food Consumption Database		FAO/WHO CIFOCCoss	
		Mean	95th ^a	Mean	95th ^b
Infants	≤ 11 months	0.2–3.1	0.7–12.9	0.3–6.7	0.5–13.0
Toddlers	1–3 years	0.8–3.7	2.7–9.5		
Other children	3–9 years	0.7–3.3	2.6–8.3	< 0.1–9.0	< 0.1–12.1
Adolescents	10–17 years	0.1–1.4	0.3–4.7		
Adults	18–64 years	0.1–0.8	0.6–2.7	< 0.1–3.7	< 0.1–10.4
Young older adults	65–74 years	0.1–0.6	0.4–2.4		
Old older adults	≥ 75 years	< 0.1–0.6	0.2–1.8	No data	No data

CIFOCCoss: FAO/WHO chronic individual food consumption database – summary statistics; EFSA: European Food Safety Authority; GMP: Good Manufacturing Practices.

^a The 95th percentile of exposure is only calculated by the tool for sufficiently large population groups (i.e. > 50 subjects).

^b Some estimates are based on a small number of consumers.

Table 3.4

Dietary exposure estimates for Gardenia (Genipin) Blue for the consumer-only population using proposed maximum GMP use levels in GSFA (mg/kg bw per day)

Population group	Age	EFSA Comprehensive Food Consumption Database		FAO/WHO CIFOCCoss	
		Mean	95th ^a	Mean	95th ^b
Infants	≤ 11 months	4.1–128.9	22.0–643.8	3.9–178.3	5.5–240.8
Toddlers	1–3 years	30.5–145.0	104.0–409.7		
Other children	3–9 years	20.3–95.6	62.4–233.7	0.3–166.6	0.3–181.7
Adolescents	10–17 years	4.5–24.3	22.6–80.7		
Adults	18–64 years	4.0–15.6	21.1–53.3	0.1–59.3	0.2–70.5
Young older adults	65–74 years	3.3–13.6	17.5–43.5		
Old older adults	≥ 75 years	2.0–12.3	10.9–38.3	No data	No data

CIFOCCoss: FAO/WHO chronic individual food consumption database – summary statistics; EFSA: European Food Safety Authority; GMP: Good Manufacturing Practices; GSFA: General Standard for Food Additives.

^a The 95th percentile of exposure is only calculated by the tool for sufficiently large population groups (i.e. > 50 subjects).

^b Some estimates are based on a small number of consumers.

For the general population of EU children and adolescents, the mean and high 95th percentile dietary exposure estimates were 0.1–3.7 and 0.3–12.9 mg/kg bw per day, respectively. For the general population of EU adults aged 18 years and older, the mean and 95th percentile estimates were < 0.1–0.8 and 0.2–2.7 mg/kg bw per day, respectively (Table 3.3). For consumer-only EU children and adolescents, the mean and high 95th percentile dietary exposure estimates were 4.1–145 and 22.0–643.8 mg/kg bw per day, respectively. For consumer-only EU adults aged 18 years and older, the mean and 95th percentile estimates were 4.0–15.6 and 10.9–53.3 mg/kg bw per day, respectively (Table 3.4). It should be noted that these assessments represent the worst-case scenario.

From the CIFOCCoss food consumption data, overlapping data for Brazil, the United Kingdom, the USA and the EU were excluded and data from 19 countries (Bangladesh, Benin, Burkina Faso, China, Democratic Republic of the Congo, Ethiopia, India, Indonesia, Kenya, Lao People's Democratic Republic, Malaysia, Mozambique, Nigeria, Pakistan, the Philippines, the Republic of Korea, Tunisia, Uganda and Zambia) were assessed. For the general population of children and adolescents, the mean and high 95th percentile dietary exposure estimates were < 0.1–9.0 and < 0.1–13.0 mg/kg bw per day, respectively. For the general population of adults aged 18 years and older, the mean and 95th percentile estimates were < 0.1–3.7 and < 0.1–10.4 mg/kg bw per day, respectively (Table 3.3). For consumer-only children and adolescents, the mean and 95th percentile estimates were 0.1–178.3 and 0.2–240.8 mg/kg bw per day, respectively. For consumer-only adults aged 18 years and older, the mean and 95th percentile estimates were 0.1–59.3 and 0.2–70.5 mg/kg bw per day, respectively (Table 3.4).

3.4 Estimates of dietary exposure to geniposide

According to the specification of Gardenia (Genipin) Blue, geniposide levels should be not more than 80 mg/kg in the article of commerce. The Committee concluded that the theoretical high dietary exposure to geniposide as an impurity is 1 µg/kg bw per day.

4. Comments

4.1 Biochemical aspects

The Committee evaluated data from single- and repeat-dose studies on absorption, distribution and elimination conducted with Gardenia (Genipin) Blue in rats and mice (147). The Committee noted that the test material used in these studies had a Gardenia Blue polymer content of 24.7%, which was representative of the article of commerce. The absorption, distribution and elimination studies indicated that the Gardenia Blue polymer was poorly absorbed (< 2%) upon oral exposure, and about 97% of the administered dose was eliminated unchanged in the faeces. Up to 0.5% of the administered dose was eliminated in the urine. The Committee also noted that Gardenia Blue polymer did not accumulate in any of the evaluated tissues, including kidneys.

No animal studies on the metabolism of Gardenia (Genipin) Blue were available. The sponsor submitted an in vitro digestibility study conducted with Gardenia (Genipin) Blue in SGF at a pH of 1.2 that evaluated levels of geniposide and genipin in the presence or absence of pepsin (151). Geniposide was detected in the SGF in the absence or presence of pepsin, and the level of genipin was below the LOQ (< 50 ng/g) in this experimental matrix. Considering the stability of the Gardenia Blue polymer, the Committee concluded that there was no possibility of degradation of the Gardenia Blue polymer into geniposide under physiological conditions when consumed in food. The Committee therefore considered any geniposide detected as an impurity of Gardenia (Genipin) Blue.

The Committee also reviewed data from publications that reported the potential hydrolysis of geniposide into genipin at intestinal pH in the presence of β-D-glucosidase (131,132,139). The Committee therefore considered genipin also as a potential hydrolysis product of geniposide in the body. The Committee evaluated a study in rats (130) that assessed the levels of genipin and genipin sulfate in the plasma after single oral administration of 480 or 980 mg/kg of geniposide present in Gardenia fruit decoction. The Committee noted that no geniposide or genipin were detected in the plasma throughout the study; however, genipin sulfate was detected, which was rapidly eliminated via urine. The Committee considered that, being highly reactive, genipin from the Gardenia fruit decoction rapidly conjugates with sulfate. There is therefore no systemic exposure to free genipin upon oral administration. The Committee concluded

that if any geniposide from Gardenia (Genipin) Blue hydrolyses to genipin in the intestine, genipin would rapidly undergo sulfate conjugation pre-systemically and then be eliminated completely as genipin sulfate.

4.2 Toxicological studies

4.2.1 Gardenia (Genipin) Blue

The Committee noted that the acute toxicity of Gardenia Blue polymer was low. The oral LD₅₀ was more than 16 g/kg in mice and more than 10 g/kg in rats (156).

The sponsor submitted several unpublished oral toxicity studies conducted with test materials with a Gardenia Blue polymer content of 24.7 or 32.3%. The Committee concluded that the test materials used in the studies were representative of the article of commerce. These studies evaluated various toxicological end-points, including short- and long-term oral toxicity, genotoxicity, and reproductive and developmental toxicity. In most studies, blue discoloration of faeces and some tissues, including kidneys, was observed in exposed animals. The Committee considered these findings to be treatment related but not adverse, as this observation was not associated with any histopathological changes. This conclusion was further supported by additional toxicological studies submitted by the sponsor that showed that the discoloration of kidneys observed in rats upon prolonged oral exposure was reversible upon discontinuation of exposure (193).

The Committee did not identify any treatment-related adverse effects in the evaluated short- and long-term oral toxicity studies conducted with Gardenia (Genipin) Blue, and the NOAEL was the highest dose tested in each study. The NOAELs of Gardenia (Genipin) Blue from these studies expressed in terms of the Gardenia Blue polymer content are provided in [Table 4.1](#) (149,157,158,161).

The Committee also evaluated a 26-week oral toxicity study in transgenic ras-H2 mice (162) conducted with Gardenia (Genipin) Blue at doses of up to 10 440 mg/kg bw per day in the diet. No increased tumour incidences were found in these mice that were highly susceptible to accelerated tumour development. The Committee considered this study as supporting evidence for the conclusion that Gardenia (Genipin) Blue does not exhibit a potential for carcinogenicity.

The Committee evaluated study reports of a bacterial reverse mutation assay (167), an in vitro chromosome aberration assay (168), an in vitro micronucleus assay (169), an in vivo erythrocyte micronucleus or Comet assay (170), an in vivo reverse Comet assay (170) and an in vivo *Pig-a* mutation assay (163) conducted with Gardenia (Genipin) Blue, all of which were negative. The Committee therefore did not identify a concern for genotoxicity of Gardenia (Genipin) Blue.

Table 4.1

NOAELs from the evaluated short- and long-term toxicity studies on Gardenia Blue

Study (reference)	NOAEL (mg/kg bw per day)	
	Gardenia (Genipin) Blue	Gardenia Blue polymer
90-day oral toxicity study in rats (157) ^a	3690 ^b	911 ^b
90-day oral toxicity study in rats (158)	2173	Not specified
52-week oral toxicity study in rats (149) ^c	2854 ^d , 3113 ^b	705 ^d , 769 ^b
104-week oral toxicity study in rats (161)	2175	703

NOAEL: no-observed-adverse-effect level.

^a The Committee noted that the design of this study did not follow the conventional protocol for a 90-day oral toxicity study, but represented a combined reproductive and developmental, and immunotoxicological study with an in utero exposure phase and aspects of subchronic toxicity study design.

^b Offspring generation.

^c The Committee noted that this study was not a conventional oral chronic toxicity study, but a combined reproductive and developmental, and immunotoxicological study with an in utero exposure phase followed by some aspects of chronic toxicity study design. The Committee also noted that, because of an analytical error, 20% more Gardenia (Genipin) Blue was added to the test diets, resulting in higher concentrations of Gardenia (Genipin) Blue for half of the study.

^d Parental generation.

The Committee evaluated reports of three reproductive and developmental toxicity studies conducted with Gardenia (Genipin) Blue: an EOGRT study in rats (184), a prenatal developmental toxicity study in rats (185) and a prenatal developmental toxicity study in rabbits (186). No treatment-related adverse effects were observed in these studies. The Committee identified NOAELs of 3915 mg/kg bw per day, corresponding to the Gardenia Blue polymer content of 1264 mg/kg bw per day (184) and 2000 mg/kg bw per day (185,186), the highest doses tested.

The sponsor submitted a study that analysed the presence of residual allergens in Gardenia (Genipin) Blue from the use of soy protein hydrolysates as a raw material in its formation (176). The Committee noted that there was no soy protein detected in the test material.

4.2.2 Geniposide and genipin

The Committee evaluated toxicological data on geniposide and genipin and noted that the oral LD₅₀ values for geniposide and genipin were 1431 mg/kg bw in rats and 510 mg/kg bw in mice, respectively (176).

A 90-day oral toxicity study of geniposide in rats did not report any treatment-related adverse effects at doses of up to 73 mg/kg bw per day (134). In a 26-week oral toxicity study of geniposide in rats at doses of up to 100 mg/kg bw per day (136), there were significant increases ($P < 0.05$) in some biochemical and haematological parameters, as well as liver and spleen weights, accompanied by histopathological changes in the liver and kidneys, in the high-dose group compared with controls. The Committee therefore identified the middle dose, 50 mg/kg bw per day, as the NOAEL for this study.

Data from a 21-day oral toxicity study of genipin in mice showed hepatotoxic effects at a dose of 100 mg/kg bw per day (137). The Committee identified the low dose, 50 mg/kg bw per day, as the NOAEL for this study. No longer-term oral toxicity studies on genipin were available.

Based on negative results in bacterial reverse mutation, Rec and SCE assays conducted with geniposide, the Committee did not identify a concern for genotoxicity of geniposide (133).

Genipin tested negative in bacterial reverse mutation assays (133,176), but showed a positive or equivocal response in several in vitro genotoxicity studies, including in vitro chromosome aberration (177), in vitro micronucleus (178), in vitro Comet (179), Rec (133) and SCE assays (133). The Committee noted that genipin tested negative in in vivo erythrocyte micronucleus and Comet assays, conducted using blood and tissues (stomach, duodenum and livers), respectively, from mice that were collected following oral exposure to genipin at doses of up to 222 mg/kg bw per day for 3 days (180). The Committee noted that this study was conducted according to OECD Test Guideline No. 474 (174) and FDA Redbook (182) test guidelines, and showed that there were no genotoxic effects of genipin in vivo. As stated above, the Committee noted that any genipin potentially formed in the body following oral exposure to geniposide would be rapidly conjugated before it reaches the systemic circulation. The Committee therefore concluded that no free genipin would be available to interact with DNA. The Committee did not identify a concern for genotoxicity of genipin in vivo.

4.3 Observations in humans

No data on observations of exposure to Gardenia (Genipin) Blue in humans were available.

4.4 Assessment of dietary exposure

Gardenia (Genipin) Blue is proposed for use in 20 food categories of GSFA with maximum levels at GMP, including alcoholic beverages; sport and energy drinks; flavoured water; fruit drinks; carbonated soft drinks; powdered beverages; jams, jellies and marmalades; chewing gum; ready-to-drink teas; sugar coatings; icing; syrups for shaved ice; ice cream and frozen desserts; hard and soft candy; flavoured milks; yogurt; processed cereals; sponge cakes; and food supplements (chewable tablets).

The Committee reviewed the dietary exposure estimates of Gardenia (Genipin) Blue submitted by the sponsor that were based on adjusted maximum use levels at GMP combined with food consumption data from Brazil, Japan, the United Kingdom and the USA, as well as estimates based on the EFSA Comprehensive Database and CIFOSS food consumption data. The Committee also calculated the high dietary exposure estimates using the mean

dietary exposure for Japan. The mean and high dietary exposure estimates were up to 9.0 and 13.8 mg/kg bw per day in children, and up to 3.7 and 10.4 mg/kg bw per day for adults, respectively.

Based on the highest dietary exposure for Gardenia (Genipin) Blue of 13.8 mg/kg bw per day, and considering the maximum Gardenia Blue polymer content of up to 36% from the specifications, the Committee calculated a dietary exposure expressed on the blue polymer basis of up to 5.0 mg/kg bw per day for children.

Based on the specifications of geniposide in the article of commerce of 80 mg/kg, the Committee calculated a theoretical high dietary exposure to geniposide of 1 µg/kg bw per day.

The Committee concluded that the conservative estimate of high exposure of 5.0 mg/kg bw per day for children should be considered in the safety assessment for Gardenia Blue polymer.

5. Evaluation

The Committee evaluated several reports of toxicological studies submitted by the sponsor to evaluate the safety of Gardenia (Genipin) Blue for the first time. The Committee noted that the test materials used in all the evaluated studies were representative of the article of commerce.

The Committee concluded that Gardenia Blue polymer is poorly absorbed upon oral exposure and extensively eliminated unchanged in the faeces.

The Committee did not identify any treatment-related adverse effects of oral exposure to Gardenia (Genipin) Blue at all tested dose levels in the short-term toxicity, long-term toxicity, genotoxicity, carcinogenicity, and reproductive and developmental toxicity studies, and the NOAEL was the highest dose tested in each study.

The Committee also evaluated data on biochemical aspects and toxicological studies of geniposide and genipin.

The Committee noted that no geniposide or genipin was detected in the plasma of rats upon oral exposure to a decoction of Gardenia fruit containing geniposide. The Committee also noted that genipin rapidly conjugated with sulfate in this study prior to reaching the systemic circulation and was then eliminated via urine as genipin sulfate.

The Committee identified a NOAEL of 50 mg/kg bw per day for geniposide based on hepatotoxic effects observed at the next higher dose in a 26-week oral toxicity study in rats. The Committee also identified a NOAEL of 50 mg/kg bw per day for genipin based on hepatotoxic effects observed at the highest dose in a 21-day oral toxicity study in mice. No longer-term oral toxicity studies on genipin were available.

The Committee did not identify a concern for genotoxicity of geniposide, based on negative results in the evaluated genotoxicity studies.

The Committee noted that genipin showed a positive or an equivocal response in several *in vitro* genotoxicity assays, but tested negative in *in vivo* erythrocyte micronucleus and Comet assays. The Committee noted that genipin is a highly reactive cross-linking agent, and would be rapidly conjugated with sulfate and eliminated in the urine. The Committee concluded that the reported *in vitro* genotoxicity of genipin is not observed *in vivo*, including in site-of-contact tissues, and therefore had no concerns for genotoxicity of genipin *in vivo*.

The Committee selected the NOAEL of 700 mg/kg bw per day for the Gardenia Blue polymer from the carcinogenicity study as a point of departure for the safety evaluation of Gardenia (Genipin) Blue. The Committee established an ADI of 0–7 mg/kg bw per day for Gardenia (Genipin) Blue, expressed on the basis of the Gardenia Blue polymer, by applying an uncertainty factor of 100 to the NOAEL to account for inter- and intraspecies variabilities.

The Committee considered that the highest estimated dietary exposure to the Gardenia Blue polymer of 5.0 mg/kg bw per day for young children was appropriate for its safety evaluation. The Committee also concluded that the theoretical high dietary exposure to geniposide as an impurity is 1 µg/kg bw per day.

Comparison of the NOAEL of 50 mg/kg bw per day with the theoretical high dietary exposure to geniposide of 1 µg/kg bw per day yields an MOE of 50 000. Based on this MOE and no concern for genotoxicity, the Committee concluded that dietary exposure to geniposide as an impurity in Gardenia (Genipin) Blue does not pose a safety concern.

The Committee concluded that the estimated dietary exposure to Gardenia (Genipin) Blue does not pose a safety concern.

A new specifications monograph and a chemical and technical assessment were prepared.

Glycolipids

Prepared by

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1. Explanation

Glycolipids are found in the membranes of various organisms. The article of commerce, also referred to as “IMD AM-1” or “AM-1”, referred to throughout as IMD AM-1, consists of structurally related glycolipids obtained from fermentation of the jelly mushroom *Dacryopinax spathularia*.



At its Fifty-third session, the CCFA requested an evaluation of the safety of glycolipids (120). Glycolipids have not previously been evaluated by the Committee.

Glycolipids are approved as a preservative in non-alcoholic beverages in Australia and New Zealand, Canada and the EU. In the USA, glycolipids have GRAS status when used as an antimicrobial agent in non-alcoholic beverages.

The Committee evaluated the submitted data, which included studies on pharmacokinetics, acute and short-term toxicity, long-term toxicity and carcinogenicity, genotoxicity, reproductive and developmental toxicity, dietary exposure, and chemical and technical considerations. The Committee conducted literature searches in the databases PubMed and Web of Science, using the search terms “glycolipids” and various toxicological end-points (e.g. “toxicity”, “adverse effects”, “carcinogenicity”, “mutagenicity”, “allergenicity” and “hypersensitivity”). These search terms identified 213 and 127 references in PubMed and Web of Science, respectively, but only five were considered relevant to the current evaluation.

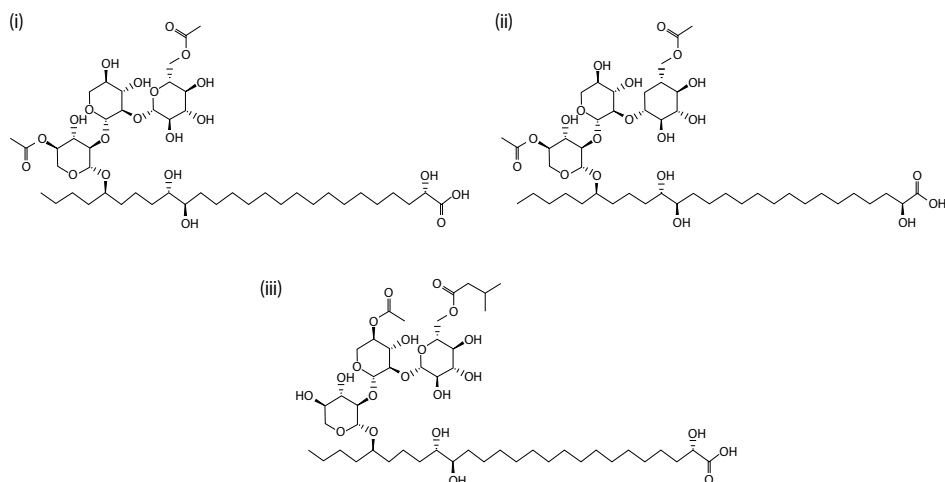
The Committee also performed a literature search to identify any studies on dietary exposure to glycolipids that were not included in the submission by the sponsor, that is, published since 2019. The databases Embase, Scopus and PubMed were searched using the terms “glycolipids”, “E246” OR “E246” AND “food”, “diet*”, “food additive” OR “nutrition” AND “intake”, “expos*”, “consum*”, “concentr*” OR “eat*”. No additional studies were identified for inclusion in the evaluation.

1.1 Chemical and technical considerations

Glycolipids from *Dacryopinax spathularia* (glycolipids; CAS No. 2205009-17-0) are a mixture of congeners of long-chain glycolipids. They are obtained by fermentation of the non-toxicogenic *D. spathularia* strain MUCL 53181. *D. spathularia* belongs to the phylum Basidiomycota, recognizable by its edible, orange-coloured, spatula-shaped fruiting bodies. The three primary glycolipid congeners (IMD-028835, IMD-026045 and IMD-027455) found in glycolipids from *D. spathularia*, shown in Fig. 1.1, are composed of a saturated C₂₆ fatty acid backbone and a glucopyranosyl-(1→2)-xylopyranosyl-(1→2)-xylopyranosyl trisaccharide moiety. IMD-028835 and IMD-026045 have the same molecular weight and are typically present together at 40%; IMD-027455 is typically present at 29%. The minor glycolipid congeners differ primarily from the three major congeners by the presence and location of acyl units (typically acetate, isovalerate and, in some congeners, 3-hydroxy-3-methylglutarate) on the trisaccharide.

Fig. 1.1

Molecular structure of the three primary glycolipid congeners: (i) IMD-028835: $C_{46}H_{82}O_{21}$ (molecular weight 971), (ii) IMD-026045: $C_{46}H_{82}O_{21}$ (molecular weight 971) and (iii) IMD-027455: $C_{49}H_{88}O_{21}$ (molecular weight 1031)



Glycolipids are produced by aerobic submerged fermentation using glucose and autolysed yeast extract as feedstock. The downstream process includes filtration and microfiltration to remove mycelial cells, precipitation and washing with acidified water. The final product is a pasteurized, spray-dried beige to light brown powder. The entire process is performed in accordance with current GMP. The total glycolipid content is specified as not less than 93% on a dry weight basis. The remainder consists of protein ($\leq 3\%$), fat ($\leq 2\%$) and sodium (1.7–3.3%). Glycolipids are surfactants intended for use as a preservative in select non-alcoholic beverages. They exhibit antifungal activity against yeast and mould in food.

The stability of glycolipids in dry powder, aqueous solutions and non-alcoholic beverage formulations under typical processing and storage conditions has been demonstrated.

2. Biological data

2.1 Biochemical aspects

An in vitro study was conducted to assess the stability of the test item (IMD AM-1) in SGF with pepsin at different timepoints and pH values (pH 2, 3.2 and 4.5) (210). The differing acidity levels were selected because the test article has poor solubility at the optimal pepsin pH of 2. Samples taken after incubating in SGF for 1 or 8 hours were analysed for glycolipid levels and composition. The levels

and composition were unchanged relative to controls without enzyme at pH 2, 3.2 and 4.5 after 1 hour, and also after 8 hours at pH 4.5. Samples incubated at pH 2 or 3.2 for 8 hours had lower glycolipid concentrations that were most likely the result of partial deacylation of the test article under acidic conditions, but no metabolites. At pH 4.5, where pepsin still has about 35% of its maximal activity, there were also no metabolites present. This outcome suggests that the test article is not a substrate for pepsin. The lower glycolipid concentrations observed at low pH values (pH 2 and 3.2) may also be because of a degree of precipitation that could affect analytical recovery.

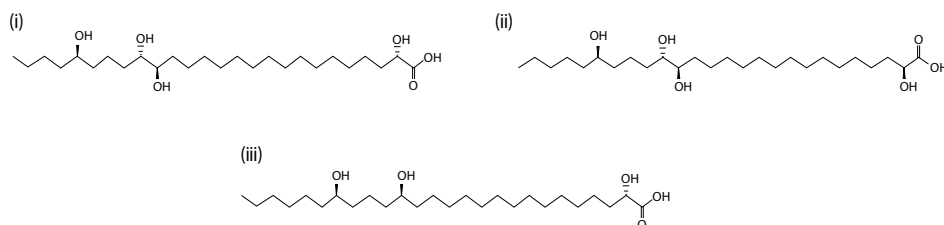
2.1.1 Absorption, distribution and elimination

A pharmacokinetic study not compliant with GLP was performed to evaluate the disposition of IMD AM-1 following intravenous and oral administration in rats, and to identify the main metabolites in plasma (211). IMD AM-1, suspended in a suitable vehicle (dimethyl sulfoxide [DMSO] and polyethylene glycol [PEG] 300 at a ratio of 10:90, or 1% aqueous carboxymethyl cellulose), was administered to male Sprague-Dawley rats (three per group) intravenously (10 mg/kg bw) or by gavage (100 mg/kg bw). Plasma samples were collected at seven timepoints during the 24 hours after administration. A total of 15 putative metabolites were identified from the three major glycolipid components (IMD-028835 + IMD-026045 combined, and IMD-027455) in IMD AM-1 and deacylated glycolipids (STMA-00274-06-01; glycolipid mixture derived by hydrolysis of IMD AM-1 with main components of molecular weight 886 Da and 870 Da, respectively) by LC-MS. The free fatty acids that are part of the chemical structures of the IMD AM-1 glycolipids (Fig. 2.1) were not detected in any of the samples. At 30 minutes after oral administration, the maximal plasma concentration of IMD-028835 + IMD-026045 combined and IMD-027455 was 1.1 ng/mL and 0.2 ng/mL, respectively. The absolute oral bioavailability of the main three glycolipids, calculated from the plasma levels following oral and intravenous dosing, was 0.01% for IMD-028835 + IMD-026045 and 0.001% for IMD-027455.

A second study in male Sprague-Dawley rats investigated absorption (by mass balance), elimination and metabolism of the four main components (IMD-028835, IMD-026045, IMD-027455 and IMD-026049) in a ^{13}C -labelled glycolipid mixture (IMD AM-1) following oral administration (212). According to the sponsor, the use of ^{13}C -glucose as the feed source in the fermentation of IMD AM-1 was the likely cause of an unintentional and unexpected change in glycolipid profile that resulted in an approximately sixfold enrichment (i.e. 4.4–24.7%) of a normally minor component (IMD-026049) together with a concomitant twofold reduction in IMD-027455 relative to the normal IMD-glycolipid distribution observed in other AM-1 preparations that had been used in all other studies. This ^{13}C -labelled test article (IMD AM-1; 10 mg/kg bw), suspended in 0.5% carboxymethyl cellulose, was administered by gavage

Fig. 2.1

Chemical structure of the main components of the fatty acids mixture derived from IMD AM-1: (i) and (ii) $C_{26}H_{52}O_6$, exact mass 460.38 and molecular weight 460.70; and (iii) $C_{26}H_{52}O_5$, exact mass 444.38 and molecular weight 444.70



to 15 rats; three animals per group were assigned to each of five sampling points (30 minutes and 1, 3, 8 and 24 hours). The ^{13}C -IMD AM-1 was quantified by LC-MS analysis using the four main test item components as markers. The plasma concentrations of ^{13}C -IMD AM-1 ranged around the lower LOQ throughout the 24-hour time profile. With the exception of low concentrations in plasma and trace amounts in one 0–8-hour urine sample, quantifiable amounts of the test item were exclusively detected in faeces and the GIT 24 hours post-dose. Most of the glycolipids (> 90%) identified in faeces were not metabolized. Compared with the aqueous solution of ^{13}C -IMD AM-1 with four main components and six metabolites, five main components and a total of 14 metabolites were identified in pooled 24-hour faeces, GIT and/or carcass samples of animals. In conclusion, ^{13}C -IMD AM-1 is not absorbed after oral administration in rats. It appeared to be completely eliminated by faecal elimination and metabolism within 24 hours.

AGLP-compliant study was conducted to investigate the pharmacokinetics, elimination and tissue distribution of [^{14}C]-labelled glycolipids (AM-1) following single or repeated administration to Sprague-Dawley rats, in comparison with [^{14}C]-long-chain fatty acids (LCFA) (213,214). Rats were randomly assigned to one of 12 treatment groups. The pharmacokinetics part of the study included four dose groups: groups 1 and 2 (nine per sex per group) received a single oral dose of [^{14}C]-AM-1 at 100 mg/kg bw or [^{14}C]-LCFA at 46 mg/kg bw, as well as a target radioactivity of 100 mCi/kg bw; group 3 received a single intravenous dose of [^{14}C]-AM-1 at 10 mg/kg bw and a target radioactivity of 60 mCi/kg bw; and group 4 received a single intravenous dose of [^{14}C]-LCFA at 4.6 mg/kg bw and a target radioactivity of 24 mCi/kg bw. Blood was collected at specified intervals for groups 1–4. For the single-dose elimination balance part of the study, groups 5 and 6 (three per sex per group) received single oral doses of [^{14}C]-AM-1 at 100 mg/kg bw and [^{14}C]-LCFA at 46 mg/kg bw, respectively. Urine, faeces and expired air were collected for these groups. For the single-dose quantitative whole-body autoradiography (QWBA) part of the study, groups 7 and 8 received the same doses as groups 5 and 6, respectively, but seven rats per sex per group

were treated and their carcasses were used for QWBA. For the repeat-dose elimination balance and QWBA parts of the study: groups 9 and 10 (three per sex per group) received oral doses of 100 mg/kg bw per day AM-1 for 14 days followed by a single oral dose of [^{14}C]-AM-1 at 100 mg/kg bw; and groups 11 and 12 (seven per sex per group) received oral doses of 46 mg/kg bw per day LCFA for 14 days followed by a single oral dose of [^{14}C]-LCFA at 46 mg/kg bw. Urine, faeces and expired air samples were collected from groups 9 and 10, and blood and carcass were collected from groups 11 and 12.

The plasma concentration time courses in male and female rats after oral administration of the test materials (AM-1 at 100 mg/kg bw and LCFA at 46 mg/kg bw) indicate partial absorption, with [^{14}C] peak levels occurring earlier for LCFA (2 and 4 hours) than for the AM-1-related radioactivity (8 and 24 hours). The calculated oral bioavailability for AM-1 metabolites was 10.7% for males and 12.7% for females.

For both AM-1 and LCFA, the exposure of tissues to test-item-related equivalents following treatment was generally within twofold of the plasma exposure as determined by tissue:plasma ratios. Higher ratios were generally found for those tissues associated with the GIT and with elimination, and lower for poorly perfused tissues. Tissue concentrations and the magnitude of the exposure were generally similar for AM-1 and LCFA equivalents and were low relative to the administered dose. QWBA was performed post-dosing and concentrations of radioactivity were calculated for adrenal gland, bone (femur), bone marrow (femur), brain, pituitary gland, caecum, eye, fat, skin, heart, stomach, small intestine, large intestine, liver, lung, muscle (femoral), pancreas, spleen, thymus, thyroid gland, kidney, kidney (cortex), kidney (medulla), urinary bladder, ovaries and uterus (females), and testes and prostate (males). Results from QWBA of rats killed at specified intervals post-administration showed that [^{14}C]-AM-1 equivalents were predominantly located in the GIT. No target tissues were identified.

Approximately 98% (overall average, males and females combined) of the administered dose was recovered in expired air, urine, faeces and carcass following a single oral administration of [^{14}C]-AM-1 at 100 mg/kg bw. Approximately 92% of the dose was recovered following repeated administration of AM-1. Approximately 16–20% of the recovered radioactivity was expired, approximately 1% was in the urine and 74–83% of the dose was eliminated in faeces. A maximum of 2% of the recovered radioactivity was found in the individual carcasses of dosed animals.

Overall, the data show that AM-1 glycolipids and its GIT-formed metabolites are poorly absorbed by the oral route and are mostly eliminated in the faeces. The pharmacokinetic, tissue distribution and excretion balance data are consistent with an interpretation that, following ingestion, AM-1 glycolipids

are partially hydrolysed to their components, namely glucose, xylose, acetate, isovalerate and LCFA.

2.2 Toxicological studies

2.2.1 Acute toxicity

A GLP-compliant acute oral toxicity study was performed in female Crl:WI(Han) rats treated with the test item (IM11, purity $\geq 90\%$) in accordance with OECD Test Guideline No. 423 (215,216). The glycolipid preparation (IM11) was suspended in water and administered by oral gavage to two groups of three rats at a dose of 2000 mg/kg bw. There were no clinical signs of toxicity in either group during the 14-day observation period. None of the animals showed any weight loss during the 15-day observation period. At necropsy, no treatment-related macroscopic findings were observed in any of the animals.

A study was conducted in 10 male and 10 female Sprague-Dawley rats according to the Chinese guideline GB15193.3-2014 (217). The glycolipid preparation (IMD AM-1) was suspended in water and administered by oral gavage at a dose of 10 000 mg/kg bw. There were no clinical signs observed during the 14-day observation period.

2.2.2 Short-term studies of toxicity

(a) Dogs

A 90-day GLP-compliant study was performed in young Beagle dogs (age 4–5 months) in accordance with OECD Test Guideline No. 409 (218). Three groups of dogs (four males and four females per group) received daily oral doses of glycolipids in gelatin capsules containing the glycolipids (IMD AM-1; 95% purity) at 150, 500 or 1000 mg/kg bw per day (219,220). A control group received empty capsules. The dose selection for this 90-day study was based on the results of a range-finding study in which no adverse effects were observed at the highest dose tested of 1000 mg/kg bw per day. All animals were killed after 91 or 92 days. Clinical examinations and individual food weights were performed and recorded daily, and individual body weights recorded weekly. A modified functional observational battery was performed before dosing and during week 12. Clinical pathology parameters (haematology, coagulation, serum chemistry and urinalysis) were analysed during weeks 2 and 6 and at necropsy. Ophthalmic examinations were performed during weeks 1 and 12. Complete necropsies were performed on all dogs and selected organs were weighed. Selected tissues were only examined microscopically from animals in the control and high-dose groups. There were no test-item-related effects on survival, clinical observations, ophthalmoscopy, haematology, coagulation, serum chemistry, urinalysis, neurobehavioral parameters, organ weight, or macroscopic or microscopic tissue

examination. The only test-item-related effect observed was an approximately 30% reduction in feed intake with a corresponding reduction in body weight gain in high-dose females relative to control females. The study authors speculated that the known surfactant properties of glycolipids on the mucosa of the GIT may have caused inappetence among the female dogs. Given the absence of any systemic exposure to glycolipids, the Committee considered that this observation was unlikely to be adverse and concluded that the NOAEL for this study was 1000 mg/kg bw per day, the highest dose tested.

(b) Rats

The sub-chronic toxicity of glycolipids (AM-1) was studied in Crl:CD(SD) rats in accordance with OECD Test Guideline No. 408 (160,221,222): 20 rats per group per sex received 0, 0.15, 0.5 or 1.5% of the test item in drinking water, for which there was a mean intake of 0, 133, 413 or 1201 mg/kg bw per day and 0, 166, 502 or 1423 mg/kg bw per day in male and females, respectively, for 90 days. There was also a 4-week recovery period for two of the groups (10 rats per group per sex) that received 0 or 1.5% test item. No toxicologically relevant test-item-related findings were observed in clinical signs, ophthalmological examination, organ weights, macroscopic or histological findings, and no test-item-related deaths were reported. No test-item-related adverse effects were noted for haematology or clinical chemistry, or urinalysis parameters. Statistically significant changes in parameters measured in the neurological screening were not dose related, lacked correlation with histopathological findings and were within historical control data of the laboratory.

Body weights of male rats treated with the highest dose were slightly decreased (5–7% lower than controls) in weeks 1–6, and the body weight gain during day 1–90 was 5.6% lower than in controls (not statistically significant). The consumption of drinking water of both sexes decreased in the mid-dose (by up to 11%) and the high-dose (by up to 22%) groups in the first 2 weeks compared with the controls. The authors suggested that this effect may be the result of the high viscosity and/or reduced palatability of the drinking water containing higher concentrations of the test item.

Overall minor treatment-related effects were observed that were mainly within the ranges of the historical control data of the laboratory. The minor reductions in body weight and body weight gain were most pronounced at the start of the study and are not considered adverse. The Committee identified a NOAEL of 1201 and 1423 mg AM-1/kg bw per day for male and female rats, respectively.

Table 2.1
Genotoxicity studies on glycolipids

End-point	Test system	Concentration	Results	Reference
In vitro bacterial reverse mutation	<i>Salmonella typhimurium</i> TA98, TA100, TA1535 and TA1537, and <i>Escherichia coli</i> WP2uvrA	Experiment 1: 62–5000 µg/plate ± S9 Experiment 2: 19–300 µg/plate ± S9 (TA1537 only), 62–5000 µg/mL – S9 (TA1535 only) Experiment 3: 62–5000 µg/mL ± S9 (TA1535 only)	Negative ^a	van den Wijngaard (223); Bitzer et al. (224)
In vitro micronucleus induction	Cultured binucleated human lymphocytes	25–700 µg/mL – S9 and 25–1000 µg/mL + S9	Negative ^b	Usta (225); Bitzer et al. (224)
In vitro gene mutation	L5178Y mouse lymphoma cells	Experiment 1: 50–450 µg/mL – S9, 100–800 µg/mL + S9 Experiment 2: 150–500 µg/mL – S9	Negative ^c	Verspeek-Rip (226); Bitzer et al. (224)
In vivo micronucleus induction	Male and female ICR mice	2500, 5000 or 10 000 mg/kg bw (by gavage)	Negative	Lijing et al. (217)

^a The first experiment was performed using the plate incorporation method and the second and third experiments were performed using the preincubation method. In the first experiment, the test substance was toxic to strain TA1537, in both the absence and presence of S9 at and above 185 µg/plate. The test substance was not toxic to strains TA1535, TA98 and WP2uvrA in both the absence and presence of S9 up to the highest concentration tested (5000 µg/plate). In the second experiment, the test substance was toxic to strain TA1537 in both the absence and presence of S9 at 300 µg/plate. In the second and third test with strain TA1535, no toxicity was observed up to the highest concentration tested (5000 µg/plate). Toxicity was evidenced by a decrease in the mean number of revertants and/or a clearing of the background lawn of bacterial growth compared with the negative controls.

^b A dose-range-finding test was performed to select the treatment concentrations. In the presence of metabolic activation (S9) the treatment and recovery times were 4 and 20 hours, respectively (pulse treatment). In the absence of S9, the treatment and recovery times were 4 and 20 hours (pulse treatment) and 20 and 28 hours (continuous treatment), respectively.

^c A dose-range-finding test was performed to select the treatment concentration. In the first experiment, the test item was tested up to concentrations of 450 and 800 µg/mL in the absence and presence of S9, respectively. The incubation time was 3 hours. Relative total growth was 15 and 18% in the absence and presence of S9, respectively. In the second experiment, the test item was tested up to concentrations of 500 µg/mL in the absence of S9. The incubation time was 24 hours. The relative total growth was 9%.

2.2.3 Genotoxicity

The test item was negative in a bacterial reverse mutation test, an in vitro micronucleus assay in human lymphocytes, a gene mutation study in L5178Y mouse lymphoma cells and a mammalian in vivo micronucleus assay (Table 2.1). Consequently, the Committee did not identify a concern for the genotoxicity of glycolipids.

2.2.4 Reproductive and developmental toxicity

(a) Multigeneration reproductive toxicity

The reproductive toxicity of glycolipids was investigated in a GLP-compliant two-generation study with Crl:CD (SD) rats in accordance with OECD Test Guideline No. 416 (227–229). The test item IMD AM-1 (purity 95%) was suspended in water and administered daily by gavage to groups of rats (25 per sex per group) at doses of 0, 150, 500 or 1000 mg/kg bw per day for the F0 and F1 generations. The control group received the vehicle (reverse-osmosis-treated

water) on a comparable regime. The dose volume for all groups was 10 mL/kg bw. F1 parental animals were dosed for a minimum of 70 days prior to mating and then throughout mating, gestation and lactation until termination after the weaning of litters on PND 21.

There were no test-item-related effects on body weight, body weight gains, feed consumption or feed efficiency in the F0 generation (including during gestation and lactation) in any of the groups. Occasionally at daily cage-side examinations, and more frequently 1–2 hours following gavage administration, a dose-related increase in the incidence of respiratory noises (rales) and red and/or clear material around the nose and/or mouth were observed for F0 and F1 males and females among all treatment groups. The study authors attributed these observed effects to the surfactant properties of glycolipids and the mode of administration, and were therefore not considered to be toxicologically relevant; the Committee concurred.

A lower mean body weight gain was observed in males in the treated groups during the first week of the F1 generation. No differences in body weight gain in these groups were observed throughout the remainder of the study. As a result of the initial lower mean body weight gains, mean body weights for F1 males were 4.0–7.1% lower than the control group throughout the study. These differences were not considered adverse because the mean body weights in these groups at termination (PND 161) were 5.4–6.1% lower than the control group, indicating that the initial effects were ameliorated over the course of the study. Furthermore, there were no test-item-related effects on feed consumption or feed efficiency in F1 males and females. There were no effects on body weight or body weight gain in the treated F1 females.

No effects related to the test item were seen in F0 and F1 male and female mating and fertility, male copulation and female conception indices, oestrous cycle length, pre-coital intervals, gestation length, the process of parturition or spermatogenesis parameters (motility, progressive motility, testicular and epididymal sperm concentration, sperm production rate and the percentage of morphologically normal sperm). In the animals of both generations, no effects on organ weight, macroscopic or microscopic findings, or on the F1 female primordial follicle counts were observed. In the F1 pups there were no effects on body weight or on attainment of balanopreputial separation and vaginal patency. For both generations, the number of pups born, sex ratio, postnatal survival and pup weight were comparable between all groups. On PND 21 no macroscopic findings or differences in organ weight were seen in the F1 and F2 pups of any of the groups. The Committee considered that the NOAEL for maternal and reproductive toxicity was 1000 mg/kg bw per day, the highest dose tested.

(b) Developmental toxicity

The prenatal developmental toxicity of glycolipids (IMD AM-1) was investigated in a GLP-compliant study with Crl:CD (SD) rats in accordance with OECD Test Guideline No. 414 (191,228,230). The test item IMD AM-1 (purity 95%) was suspended in water and administered daily by gavage to four groups of 24 female rats at doses of 0, 150, 500 or 1000 mg/kg bw per day in a dose volume of 10 mL/kg during GD 6–19. The dosing levels were selected based on the results of a previous range-finding study in which pregnant female rats were exposed to 150, 500 or 1000 mg/kg bw per day.

There were no clinical signs, and all animals survived until the scheduled necropsy on GD 20. There were no significant reductions in body weight, feed consumption, reproductive parameters or fetal weight in any of the treated groups. Dose-dependent rales were noted 1–2 hours after administration in the treated groups sporadically throughout the treatment period (GD 6–19). Although treatment related, these respiratory noises were not considered to be adverse because they did not usually persist to the next daily cage-side observation. Similarly, clear and/or red material around the nose and/or mouth observed among rats in the mid- and high-dose groups 1–2 hours after gavage during GD 7–16 was attributed to the surfactant properties of the dose formulation and not considered to be adverse.

Lower mean feed consumption was noted in the high-dose group during GD 6–9 and 15–20 and in the mid-dose group during GD 15–20. These changes were considered to be related to the test item but not adverse, because the effects were transient and the changes had no effect on mean group body weight throughout the treatment period. At Caesarean section on GD 20, no differences between the control and the treated groups were observed in the number of implantations, resorptions, or live and dead fetuses; in fetal weight; or in the macroscopic, visceral or skeletal abnormalities of the fetuses. The Committee considered that the NOAEL for developmental toxicity was 1000 mg/kg bw per day, the highest dose tested.

2.2.5 Special studies

(a) Allergenicity

A literature search did not identify any immune mediated adverse reactions to “glycolipids like jelly mushroom glycolipids” or to “*Dacryopinax spathularia*”, which could indicate a concern for allergenicity.

2.3 Observations in humans

No data for oral exposure to glycolipids in humans were available. The sponsor submitted a skin sensitization study of the test item in which no adverse effects were observed (231).

3. Dietary exposure

3.1 Introduction

The Committee considered studies of dietary exposure to long-chain glycolipids from *Dacryopinax spathularia*, hereafter referred to as glycolipids. Glycolipids are currently not included for use as a food additive in the GSFA.

As of March 2021, Australia and New Zealand permit glycolipids in non-alcoholic beverages at MPLs ranging from 10 to 100 mg/kg, depending on the beverage. Glycolipids are included in the list of permitted preservatives in Canada as of 10 June 2022, allowing their use as preservatives in different non-alcoholic beverages at maximum levels of use ranging from 20 to 100 mg/kg. In the USA, glycolipids are included in the list of GRAS notices as of May 2018; their use as antimicrobial preservatives in non-alcoholic beverages at use levels ranging from 2 to 100 mg/kg is considered to be safe. Glycolipids are as of 20 July 2022 endorsed for use in the EU in three food categories: FC 14.1.4 Flavoured drinks (excluding dairy based drinks), FC 14.1.5.2 Other (only liquid tea concentrates and liquid fruit and herbal infusion concentrates) and FC 14.2.1 Beer and malt beverages (only alcohol-free beer and malt beverages) as antimicrobial preservatives. Maximum levels in these food categories are 50, 20 and 50 mg/kg, respectively.

For the current meeting, the sponsor requested the use of glycolipids as an antimicrobial preservative in non-alcoholic beverages at maximum use levels ranging from 25 to 100 mg/kg. For this, the sponsor provided estimates of dietary exposure to glycolipids from non-alcoholic beverages for Brazil, the United Kingdom, the USA and Europe, and estimates based on consumption levels from the CIFOCCOss. The Committee did not consider the dietary exposure estimates provided for the European population, as a more accurate estimate of exposure was available from EFSA based on use levels applicable to the market of the EU. The Committee also did not consider the submitted estimates for Brazil and those based on the CIFOCCOss, as these estimates were based on mean and high (90th or 95th percentile) consumption statistics of non-alcoholic beverages. Dietary estimates based on individual food consumption data are considered to provide more accurate estimates of exposure, and were available for the United Kingdom, the USA and Europe. Furthermore, the Committee considered dietary estimates of glycolipids for Australia from Food Standards Australia and New Zealand (FSANZ) in 2020 (232). These estimates were also based on individual food consumption data.

3.2 Dietary exposure estimates

3.2.1 Australia (provided by FSANZ)

In 2020, FSANZ assessed an application to permit the use of glycolipids (so-called “jelly mushroom glycolipids”) as a preservative in non-alcoholic beverages at MPLs ranging from 2 to 100 mg/kg (see [Table 3.1](#)). FSANZ assessed the dietary exposure for three scenarios: general MPL, specific MPL and usual use levels. The exposure was assessed for both the Australian and New Zealand populations. However, the Committee did not consider the results for New Zealand because these were based on only 1 day of consumption and therefore do not adequately reflect repeated exposure.

Table 3.1

MPLs and usual use levels of glycolipids in non-alcoholic beverages as submitted to FSANZ

Food category	Subcategory		General MPL (mg/kg)	Specific MPL (mg/kg)	Usual use level (mg/kg)
	Number	Name			
14.1.2 Fruit and vegetable juices and fruit and vegetable juice products	14.1.2.1	Fruit and vegetable juices, clear	100	20	8
	14.1.2.1	Fruit and vegetable juices, turbid	100	100	75
	14.1.2.1	Fruit and vegetable nectars, clear	100	20	4
	14.1.2.1	Fruit and vegetable nectars, turbid	100	100	55
	14.1.2.2	Fruit and vegetable juices products (carbonated and non-carbonated drinks with >7% juice)	100	50	15
14.1.3 Water-based flavoured drinks	14.1.3	Carbonated soft drinks, sport drinks and electrolyte drinks, formulated caffeinated beverages	50	20	5
	14.1.3	Non-carbonated soft drinks, sport drinks and electrolyte drinks, formulated caffeinated beverages	50	20	5
	14.1.3	Carbonated and non-carbonated fruit juice drinks	50	50	15
	14.1.3	Cordials and concentrates (liquid and solid) for water-based flavoured drinks	50	20	5
	14.1.3.1	Brewed soft drinks	50	30	7
14.1.4 Formulated beverages			20	20	5
14.1.5 Coffee, coffee substitutes, tea, herbal infusions and similar products	14.1.5	Non-carbonated ready-to-drink teas	10	20	5
14.2.1 Beer and related products	14.2.1	Non-alcoholic beer (< 0.5% alcohol by volume)	100	100	—

Source: FSANZ (232).

FSANZ: Food Standards Australia New Zealand; MPL: maximum permitted level.

Table 3.2

Consumer-only dietary exposure to glycolipids from non-alcoholic beverages in Australia for three exposure scenarios^{a,b}

Age (years)	Percentage of consumers (%)	Dietary exposure (mg/kg bw per day) ^c					
		General MPL (mg/kg)		Specific MPL (mg/kg)		Usual use level (mg/kg) ^d	
		Mean	High	Mean	High	Mean	High
≥ 2	74.4	0.46	1.0	0.30	0.67	0.20	0.45

Source: FSANZ (232).

bw: body weight; FSANZ: Food Standards Australia New Zealand; MPL: maximum permitted level.

^a For the food categories and use levels used in this assessment, see Table 3.1.^b The consumer-only population includes only those who reported consumption of non-alcoholic beverages that could contain glycolipids on at least 1 of 2 recording days in the dietary survey period.^c High exposure: 90th percentile.^d Usual use level refers to the common use level at which the additive is expected to be used in foods within a food category.

The estimates for the population from Australia were based on food consumption data from the 2011–2012 Australian National Nutrition and Physical Activity Survey. The consumption data were averaged across the 2 days of recording and multiplied by the levels for the three scenarios (see Table 3.1). FSANZ calculated the mean and high (P90) exposure for the whole survey population aged 2 years and older that reported the consumption of non-alcoholic beverages, the so-called “consumer-only”, on at least 1 of the 2 recording days. The resulting exposure is listed in Table 3.2.

The mean estimates of dietary exposure ranged from 0.20 to 0.46 mg/kg bw per day over the three exposure scenarios. The high (P90) estimates ranged from 0.45 to 1.0 mg/kg bw per day. The two major contributing food categories to the dietary exposure across the three exposure scenarios were water-based flavoured drinks, excluding powder, iced teas and brewed soft drinks (31–45%), and fruit and vegetable juices (26–46%).

3.2.2 United Kingdom and the USA (provided by the sponsor)

The sponsor assessed the dietary exposure to glycolipids for the United Kingdom and the USA using maximum use levels for non-alcoholic beverages belonging to six food categories in the GSFA. The levels are listed in Table 3.3. The use of glycolipids in fruit and vegetable juices and nectars is not permitted in the United Kingdom (or in the EU; see Section 3.2.3). The estimates for the United Kingdom therefore represent a potential high-estimate exposure scenario, also considering the consumption of fruit and vegetable juices and nectars in this country.

Table 3.3

Maximum use levels of glycolipids in non-alcoholic beverages, as provided by the sponsor

GSFA Food category		Food types	Maximum use level (mg/kg)
Number	Name		
14.1.2	Fruit and vegetable juices	Fruit juices (including concentrates)	100
14.1.3	Fruit and vegetable nectars	Fruit nectars	80
14.1.4.1	Carbonated water-based flavoured drinks	Carbonated soft drinks	25
		Enhanced water	25
		Fruit drinks	80
		Sports and energy drinks	50
14.1.4.2	Non-carbonated water-based flavoured drinks, including punches and ades	Enhanced water	25
		Fruit drinks	80
		RTD tea	50
		Sport and energy drinks	50
14.1.4.3	Concentrates (liquid or solid) for water-based flavoured drinks	Concentrates/cordials	50 (RTD)
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	RTD coffee, without milk	50

GSFA: General Standard for Food Additives; RTD: ready to drink.

The food consumption data for the United Kingdom were obtained from the National Diet and Nutrition Survey rolling programme, covering the results from years 1 to 11 of the programme (2008/2009–2018/2019). The food consumption data for the USA were collected in the What We Eat in America component of the NHANES conducted during 2017–March 2020 (i.e. before the coronavirus disease pandemic). The sponsor estimated the dietary exposure for individuals who reported the consumption of non-alcoholic beverages, the so-called consumer-only, on at least one of the recording days in the dietary survey period (4 days for the United Kingdom and 2 days for the USA). The overall percentage of consumers was 81 and 74%, respectively.

Dietary exposure to glycolipids for both countries was calculated for children of age 2–5 years up to adults (≥ 19 years) by multiplying the mean consumption amount of the different types of non-alcoholic beverages per individual across the survey days by the maximum use level. The resulting dietary exposure estimates were summed and adjusted for individual body weight to obtain the individual daily dietary exposure expressed per kilogram of body weight. The mean and 95th percentile (P95) exposure estimates were calculated from the resulting distribution of individual daily exposures and are listed in [Table 3.4](#) for both countries and the different age groups.

Table 3.4

Consumer-only dietary exposure to glycolipids from non-alcoholic beverages in the United Kingdom and the USA based on maximum use levels, as provided by the sponsor^{a,b}

Country and age group (years)	No. participants in dietary survey	Consumer-only population (%)	Dietary exposure (mg/kg bw per day)	
			Mean	High ^c
United Kingdom ^d				
2–5	2 031	92	1.1	3.0
6–12	2 855	94	0.7	1.8
13–18	2 500	93	0.4	1.0
≥ 19	7 999	71	0.2	0.7
Total	15 385	81	0.4	1.2
USA				
2–5	668	81	1.2	3.3
6–12	1 194	82	0.6	1.6
13–18	929	83	0.4	1.0
≥ 19	5 139	72	0.3	0.9
Total	7 930	74	0.4	1.2

bw: body weight; USA: United States of America.

^a For the non-alcoholic beverages and maximum use levels included in the assessment, see Table 3.3.

^b The consumer-only population includes only those who reported consumption of non-alcoholic beverages that could contain glycolipids on at least one of the recording days in the dietary survey period (4 days for the United Kingdom and 2 days for the USA).

^c High exposure: 95th percentile.

^d Estimates for the United Kingdom are potential estimates because they include the exposure to glycolipids via fruit and vegetable juices and nectars, in which the food additive is not permitted for use in the United Kingdom.

The mean dietary exposure ranged from 0.2 mg/kg bw per day for adults in the United Kingdom to 1.2 mg/kg bw per day for children aged 2–5 years in the USA. The corresponding high (P95) dietary exposure estimates were 0.7 mg/kg bw per day for adults in the United Kingdom and 3.3 mg/kg bw per day for children aged 2–5 years in the USA.

3.2.3 Europe (provided by EFSA)

In 2021, EFSA assessed the safety of the use of glycolipids as a preservative in non-alcoholic beverages (233). As a result of this safety assessment, glycolipids have been endorsed for use in the EU in beverages according to annex II to Regulation (European Commission) No. 1333/2008 (91) and were assigned the E-number E 246.

As part of the assessment, EFSA estimated the dietary exposure to glycolipids based on maximum and typical use levels for three food categories of non-alcoholic beverages in annex II to Regulation (European Commission)

Table 3.5

Maximum and typical use levels of glycolipids in non-alcoholic beverages, as provided by industry to EFSA (233)

Food category		Restrictions/exceptions	Use level (mg/L)	
Number	Name		Maximum	Typical ^a
14.1.4	Flavoured drinks	—	50	10
14.1.5.2	Other ^b	—	20	10
14.2.1	Beer and malt beverages	Only alcohol-free	50	10

EFSA: European Food Safety Authority.

^a Typical use level refers to the common use level at which the additive is expected to be used in foods within a food category.

^b This food category covers tea, herbal and fruit infusions, flavoured tea infusions, flavoured instant coffee, flavoured roast coffee, coffee substitutes (2-in-1 or 3-in-1), coffee mixes and mixes for hot beverages (e.g. powder for hot cappuccino beverages).

No. 1333/2008. These levels are listed in [Table 3.5](#). EFSA calculated the dietary exposure by combining these levels with individual food consumption data from the Comprehensive Database that, at the time, contained data from 40 dietary surveys in 22 European countries, covering infants (< 1 year), toddlers (1–2 years), children (3–9 years), adolescents (10–17 years) and adults (≥ 18–64 years and > 65 years). All 22 countries are members of the EU.

Dietary exposure to glycolipids was calculated for all individuals in the different age groups, irrespective of whether individuals reported consumption of non-alcoholic beverages, by multiplying the use level per food category by the mean summed consumption amounts of all foods within that category for each individual averaging over the number of survey days. Daily dietary exposure for each individual was subsequently determined by adding exposure estimates derived for each relevant food category for that individual and dividing it by the individual's body weight. This resulted in a distribution of individual mean dietary exposure estimates per day. The mean and P95 exposure estimates were calculated per survey and for each population group ([Table 3.6](#)). The food categories that contributed most to the mean dietary exposure to glycolipids were FC 14.1.4 Flavoured drinks and FC 14.5.2 Other.

EFSA considered the estimates of exposure for both use levels to be overestimates of the actual exposure to glycolipids in the general population if the food additive was used in the non-alcoholic beverages included in the assessment, because of the assumption that all beverages in a food category would contain glycolipids as antimicrobial preservative at the maximum or typical use level.

Table 3.6

Dietary exposure to glycolipids in the European population from use as a food additive^{a,b,c}

Exposure level	Dietary exposure (mg/kg bw per day) ^d					
	Infants (12 weeks– 11 months)	Toddlers (1–2 years)	Children (3–9 years)	Adolescents (10–17 years)	Adults (18–64 years)	Adults (≥ 65 years)
Maximum use level exposure assessment scenario						
Mean	0.01–0.26	0.02–1.07	0.07–1.00	0.06–0.57	0.05–0.38	0.03–0.19
High ^e	0.00 ^f –0.88	0.10–3.10	0.33–2.16	0.31–1.33	0.22–1.32	0.12–0.61
Typical use level exposure assessment scenario						
Mean	< 0.01–0.12	< 0.01–0.23	0.02–0.22	0.01–0.13	0.01–0.10	0.01–0.09
High ^e	0.00 ^f –0.38	0.02–0.64	0.08–0.47	0.06–0.29	0.05–0.27	0.05–0.19

Source: EFSA (233).

EFSA: European Food Safety Authority.

^a For the food categories and use levels used in this assessment, see Table 3.5.^b Dietary exposure was calculated according to a general population approach, meaning that all individuals in the different age groups were considered, irrespective of whether they reported consumption of non-alcoholic beverages that could contain glycolipids on at least one of the recording days in the dietary survey period.^c The European countries were Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Portugal, Romania, Slovenia, Spain and Sweden, all of which are members of the European Union.^d Range represents the lowest and highest estimates for each age group in all the dietary surveys included in the assessment.^e High exposure: 95th percentile.^f High exposure is zero when there are less than 5% consumers.

3.3 Overview of estimated dietary exposure

Table 3.7 summarizes the estimates of dietary exposure to glycolipids from their use as antimicrobial preservatives in non-alcoholic beverages as considered at the present meeting. The dietary exposure estimates summarized in Table 3.7 include those for the general and consumer-only populations. The Committee noted that the estimates for the United Kingdom represent a potential exposure to glycolipids through the consumption of non-alcoholic beverages, because they include the exposure via fruit and vegetable juices and nectars in which glycolipids are not permitted to be used in the United Kingdom.

To protect individuals who eat foods that may contain a food additive, consumer-only exposure could be estimated by including only individuals who report consumption of these foods. Such estimates were available for Australia, the United Kingdom and the USA (see Table 3.7). The estimates for Europe were for the general population, considering all individuals, irrespective of whether they reported consumption of non-alcoholic beverages that could contain glycolipids on at least 1 of the recording days in the dietary survey period. Despite this, the highest high estimate for Europe was similar to that for the USA and the highest high estimate of exposure for adults was calculated for Europe (see Table 3.7).

Table 3.7

Overview of estimated dietary exposure to glycolipids from their use as a food additive in non-alcoholic beverages^a

Country	Age group (years)	Population ^b	Source of concentrations ^c	Dietary exposure (mg/kg bw per day)	
				Mean	High ^{d,e}
Australia	All (≥ 2)	Consumer-only	MPLs	0.30	0.67 ^f
		Consumer-only	Usual use levels	0.20	0.45
United Kingdom ^g	Children (2–18)	Consumer-only	Maximum use levels	0.4–1.1	1.0–3.0
	Adults (≥ 19)	Consumer-only	Maximum use levels	0.2	0.7
	All (≥ 2)	Consumer-only	Maximum use levels	0.4	1.2
USA	Children (2–18)	Consumer-only	Maximum use levels	0.4–1.2	1.0–3.3
	Adults (≥ 19)	Consumer-only	Maximum use levels	0.3	0.9
	All (≥ 2)	Consumer-only	Maximum use levels	0.4	1.2
Europe ^h	Children (1–17)	General	Maximum use levels	0.01–1.07	0–3.1
		General	Typical use levels	< 0.01–0.23	0–0.64
	Adults (≥ 18)	General	Maximum use levels	0.03–0.38	0.12–1.3
		General	Typical use levels	0.01–0.10	0.05–0.27

Source: EFSA (233), FSANZ (232), sponsor (United Kingdom and the USA)

bw: body weight; FSANZ: Food Standards Australia New Zealand; MPL: maximum permitted level; USA: United States of America.

^a Exposure was calculated by multiplying the consumption amounts of non-alcoholic beverages, averaged across the survey days, by MPLs or use levels. Table 3.1 lists the beverages included and levels used for Australia, Table 3.3 those for the United Kingdom and the USA, and Table 3.5 those for Europe.

^b Dietary exposure for Australia, the United Kingdom and the USA are consumer-only estimates, including only those individuals who reported consumption of non-alcoholic beverages that could contain glycolipids on at least one of the recording days in the dietary survey period (2 days for Australia, 4 days for the United Kingdom and 2 days for the USA). The estimates for Europe were for the general population, including all individuals irrespective of whether they reported consumption of non-alcoholic beverages that could contain glycolipids on at least one of the recording days in the dietary survey period.

^c Usual and typical use levels both refer to the common use level at which the additive is expected to be used in foods within a food category.

^d High exposure: 95th percentile.

^e High exposure is zero when there are less than 5% consumers.

^f High exposure: 90th percentile.

^g Estimates for the United Kingdom are potential estimates, because they include the exposure to glycolipids via fruit and vegetable juices and nectars in which the food additive is not permitted for use in the United Kingdom.

^h The European countries were Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Netherlands (Kingdom of the), Portugal, Romania, Slovenia, Spain and Sweden, all of which are members of the European Union.

Additionally, the Committee noted that the exposure to additives that can be used in non-alcoholic beverages should preferably be estimated using a brand-loyal scenario, to address the fact that consumers are likely to be loyal to a certain brand of these drinks. However, an assessment based on only maximum use levels can be seen as a conservative brand-loyal assessment assuming that individuals are loyal to all non-alcoholic beverages they consume. The assessment also included countries with high consumption levels of these beverages, such as the United Kingdom and the USA.

The Committee noted differences between the maximum use levels provided by the sponsor (see [Table 3.3](#)) and those considered in the estimates for Australia (see [Table 3.1](#)) and for Europe (see [Table 3.5](#)). Most notably, no exposure to glycolipids through the consumption of fruit juices and nectars was considered for Europe, whereas the sponsor provided maximum use levels of 100 mg/kg for fruit juices and 80 mg/kg for fruit nectars. For Australia, these levels were 20 mg/kg in clear fruit and vegetable juices and nectars and 100 mg/kg in turbid fruit and vegetable juices. The Committee also noted that glycolipids may be added at up to 100 mg/kg in fruit and vegetable juices in Canada and the USA.

Based on the overview of dietary exposure estimates of glycolipids, the high (P95) dietary exposure could be up to 3.3 mg/kg bw per day for children aged 2–5 years in the USA and 1.3 mg/kg bw per day in adults (≥ 18 years) in Europe (see [Tables 3.4](#) and [3.7](#)). The Committee considered all estimates of dietary exposure to glycolipids to be overestimates of the actual mean or high dietary exposure in the general and consumer-only populations, as (i) it is highly unlikely that glycolipids are added to all beverages when there exist other food additives with the same function; and (ii) if used, glycolipids may not be added at the maximum use level.

4. Comments

4.1 Biochemical aspects

The main components of IMD AM-1 are three closely structurally related glycolipid congeners (Fig. 1.1). A series of pharmacokinetic and mass balance studies in rats with labelled and unlabelled glycolipids demonstrated that these intact lipid congeners are not absorbed. Instead, a proportion of the glycolipids is hydrolysed by intestinal microflora to their components, namely glucose, xylose, acetate, isovalerate and LCFA (211–214). These normal components of the diet are readily absorbed, and any remaining unchanged glycolipids are eliminated in faeces (74–83%).

The oral bioavailability was 0.01% of two of the IDM AM-1 congeners (IMD-028835 and IMD-026045), and 0.001% for IMD-027455.

4.2 Toxicological studies

The sponsor submitted reports of several toxicological studies on glycolipids. The test item used in the submitted toxicological studies was representative of the article of commerce, with the exception of one of the three pharmacokinetic studies.

There were no clinical signs of toxicity of glycolipids following acute exposure to 2000 mg/kg bw (216) or 10 000 mg/kg bw (217).

There were no toxicologically relevant effects in male and female rats that received up to 1.5% of the test item in drinking water (corresponding to doses of up to 1201 and 1423 mg/kg bw per day for males and females, respectively) for 90 days. The minor reductions in body weight and body weight gain were most pronounced at the start of the study and were not considered adverse (221,222). The Committee identified a NOAEL of 1201 mg/kg bw per day.

Male and female dogs were orally exposed to glycolipids at doses of 150, 500 or 1000 mg/kg bw per day for 90 days. The only test-item-related effect observed was a reduction in feed intake with a corresponding reduction in body weight gain (approximately 30% lower than controls) among female dogs exposed to glycolipids at 1000 mg/kg bw per day (219,220). Given the absence of any systemic exposure to glycolipids, the Committee considered that this observation was unlikely to be adverse and concluded that the NOAEL for this study was 1000 mg/kg bw per day, the highest dose tested.

The Committee evaluated reports of two reproductive and developmental toxicity studies on glycolipids in rats. The reproductive toxicity was investigated in a two-generation study where glycolipids were administered daily to rats at doses of up to 1000 mg/kg bw per day for the F0 and F1 generations (227,228). There were no test-item-related effects on body weight, body weight gain, feed consumption or feed efficiency in the F0 generation in any of the groups. A lower mean body weight gain was observed in males in the 150, 500 and 1000 mg/kg bw per day groups during the first week of the F1 generation. No differences in body weight gain in these groups were observed throughout the remainder of the study. There were no test-item-related effects on feed consumption or feed efficiency in F1 males and females.

A dose-related increase in the incidence of rales was observed for F0 and F1 males and females, but was not considered toxicologically relevant. There were no effects on F0 and F1 male and female mating and fertility parameters, on organ weight, or on macroscopic or microscopic findings. There were no effects on litter size, sex ratio, postnatal survival and pup weight.

In the prenatal developmental study, the test item was administered daily to female rats at doses of up to 1000 mg/kg bw per day during GD 6–19 (228,230). There were no significant reductions in body weight, feed consumption, reproductive parameters or fetal weight. A dose-related increase in the incidence of rales was noted following treatment, but did not persist. At caesarean section, no differences between the control and the treated groups were observed in the fetuses. The Committee identified a NOAEL of 1000 mg/kg bw per day, the highest dose tested, in both reproductive and developmental toxicity studies.

The test item was negative in a bacterial reverse mutation test (223,224), an in vitro micronucleus assay in human lymphocytes (224,225), a gene mutation study in L5178Y mouse lymphoma cells (224,226) and a mammalian in vivo

micronucleus assay (217). Consequently, the Committee had no concerns about the potential for genotoxicity of glycolipids.

4.3 Observations in humans

No data on the effects of oral exposure to glycolipids in humans were available.

4.4 Assessment of dietary exposure

Glycolipids are currently not included in the GSFA. For the current meeting, the sponsor requested the use of glycolipids as an antimicrobial preservative in non-alcoholic beverages at maximum use levels ranging from 25 to 100 mg/kg. The Committee evaluated estimates of dietary exposure to glycolipids from these beverages based on the use levels as submitted by the sponsor for the United Kingdom and the USA. In addition, dietary exposure estimates were evaluated for Australia (232) and Europe (233).

High (P95) dietary exposure estimates of glycolipids could be up to 3.3 mg/kg bw per day for children aged 2–5 years in the USA and 1.3 mg/kg bw per day for adults in Europe. The Committee considered that these high estimates of dietary exposure were the most suitable estimates currently available for evaluating dietary exposure to glycolipids, as they include all beverages to which glycolipids may be added at the requested maximum use levels. However, the Committee acknowledged that it is highly unlikely that glycolipids will be added to all beverages and, even if added, may not be at the maximum use level. These estimates are therefore overestimates of the actual exposure to glycolipids through the consumption of non-alcoholic beverages.

5. Evaluation

The Committee evaluated toxicological studies submitted by the sponsor to assess the safety of glycolipids. The Committee noted that the test materials used in the toxicological studies were representative of the article of commerce.

The studies on glycolipids evaluated by the Committee included acute toxicity, short-term toxicity, genotoxicity, and reproductive and developmental toxicity. Adverse effects were not noted at any dose tested.

A proportion of the glycolipids is hydrolysed by intestinal microflora to their components, namely glucose, xylose, acetate, isovalerate and LCFA, and the remaining unchanged glycolipids are eliminated in the faeces. The Committee further noted that the hydrolysis products are normal components of the human diet. The Committee therefore established an ADI “not specified”¹³ for glycolipids.

¹³ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

The Committee noted the high dietary exposure estimates of 3.3 mg/kg bw per day for children aged 2–5 years in the USA and 1.3 mg/kg bw per day for adults in Europe. The Committee concluded that these dietary exposure estimates did not raise any safety concerns.

A new specifications monograph and a chemical and technical assessment were prepared.

Rosemary extract (addendum)

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1. Explanation

Rosemary extract (INS No. 392) is an antioxidant food additive obtained from ground dried leaves of *Rosmarinus officinalis*. The antioxidant properties of rosemary extract are primarily attributed to its phenolic diterpene content, namely carnosic acid and carnosol. Rosemary also contains several volatile components that contribute to its characteristic flavour. The rosemary extract for

use as an antioxidant has a minimum ratio of total content of carnosic acid and carnosol to total volatile components of 15:1.

The Committee previously evaluated rosemary extract at its Eighty-second (108) and Eighty-seventh meetings (234). At the Eighty-second meeting, the Committee established a temporary ADI of 0–0.3 mg/kg bw for rosemary extract, expressed as carnosic acid plus carnosol. This ADI was based on a NOAEL of 64 mg/kg bw per day, the highest dose tested in a 90-day toxicity study in rats. An uncertainty factor of 200 was used, which included an uncertainty factor of 100 for inter- and intraspecies differences and an additional uncertainty factor of 2 to account for the temporary designation of the ADI, pending the submission of studies to elucidate the potential developmental and reproductive toxicity of rosemary extract. An additional uncertainty factor to account for the lack of a chronic toxicity study was not considered necessary, based on the absence of adverse effects in the short-term toxicity studies at doses up to and including the highest dose tested.

At the Eighty-seventh meeting, the Committee evaluated a submitted study on the reproductive and developmental toxicity of an acetone-based rosemary extract and several relevant studies published after the Eighty-second meeting of the Committee. The Committee concluded that the new studies provided evidence for the absence of reproductive toxicity, but not for the absence of developmental toxicity. The Committee retained the temporary ADI of 0–0.3 mg/kg bw, pending the submission of studies on the developmental toxicity of rosemary extract and studies to elucidate whether the effects noted on rodent pup thyroid hormone levels can be replicated.

Estimated mean and high-percentile dietary exposures to carnosic acid plus carnosol from use of rosemary extract as a food additive for all countries assessed based on typical use levels did not exceed the temporary ADI of 0–0.3 mg/kg bw, whereas the estimated dietary exposures for children from naturally occurring sources combined with food additive use were up to 0.42 mg/kg bw per day, which exceeds the ADI. The Committee did not consider the exceedance of the ADI to be a safety concern since the temporary ADI was based on the highest dose tested in a short-term toxicity study in rats. Further, in the newly submitted reproductive and developmental toxicity screening study, no effects on reproductive toxicity or on parental animals were observed at 316 mg/kg bw per day, the highest dose tested.

Rosemary extract was placed on the agenda of the present meeting at the request of the CCFA at its Fifty-fourth session (44) for a safety assessment to evaluate the developmental toxicity and the effects on rodent pup thyroid hormone levels.

The Committee evaluated unpublished studies with rosemary extract submitted by the sponsor, including a prenatal developmental toxicity study, a study on thyroid hormone levels and liver enzyme induction in immature rats,

and an in vitro study on the effects on liver enzymes. In addition, the Committee evaluated studies from the published literature submitted by the sponsor.

The Committee also conducted a literature search using the search terms “rosemary extract” OR “*Rosmarinus officinalis*” (title, abstract, keywords) AND “toxicity” (all fields) in Pubmed, Embase and Scopus for the period 2019–2025. The relevant studies are also summarized below. Two publications (235,236) describe studies previously evaluated by the Committee and were not reviewed again. An additional literature search was conducted to identify any further studies on dietary exposure to rosemary extract published after 2019 that were not previously considered by the Committee. One additional study was identified for inclusion in the dietary exposure assessment (237).

1.1 Chemical and technical considerations

No new information was submitted.

2. Biological data

2.1 Biochemical aspects

2.1.1 Absorption, distribution and elimination

(a) In vitro

(i) Carnosic acid

The absorption of carnosic acid was studied using the human colorectal adenocarcinoma cell line Caco-2 (238). Carnosic acid was added at concentrations of 50, 75 or 100 μM to the apical (AP) compartment of a permeable membrane support that was seeded with Caco-2 cells. The concentrations were based on the results of a 2,5-diphenyl-2H-tetrazolium bromide assay that showed that carnosic acid did not affect the cell viability of Caco-2 cells at 100 μM ($> 90.01\%$ viability compared with the untreated cells). Samples of the solution at the basolateral (BL) compartment were taken 0, 0.5, 1, 1.5, 2, 2.5 and 3 hours after exposure. Carnosic acid concentration in the samples was measured using HPLC. Bioavailability was measured by assessing the apparent permeability coefficient (P_{app}). The permeability in both directions ranged between 0.12×10^{-6} and 2.53×10^{-6} cm/s, which indicated that carnosic acid is a moderately permeable compound. The efflux ratio ($P_{\text{app(BL-AP)}}/P_{\text{app(AP-BL)}}$) was generally less than 1 at carnosic acid concentrations of 50 and 75 μM (except at 3 hours with 50 μM), indicating that lower doses of carnosic acid are mainly passively absorbed. At 100 μM , the efflux ratios increased to 1.52–1.70 after 2 hours of incubation, suggesting that carnosic acid could undergo active efflux at higher concentrations. Addition of the P-glycoprotein (P-gp) inhibitors verapamil and PEG 400 resulted in a 1.26- to 4.98-fold increase in P_{app} from AP to BL side showing that P-gp transporters can play a role in the efflux of carnosic acid. In the presence of ethylenediaminetetraacetic acid, which

increases paracellular permeability, a 1.5-fold increase in P_{app} from AP to BL side was observed, indicating that carnosic acid can be transported via a paracellular route. The results indicate that absorption and transportation of carnosic acid in Caco-2 cells mainly involve passive diffusion, but P-gp transporters may also be involved.

(b) In vivo

(i) Human: rosemary extract

One healthy male volunteer (aged 27 years) received a single oral dose of 1000 mg rosemary extract (containing at least 10% carnosic acid) to test an analytical method developed to measure carnosic acid in blood plasma (239). Venous blood samples were collected before dosing and at several timepoints after dosing (15, 30 and 45 minutes; 1, 2, 3, 4, 5 and 6 hours; and once per day for 10 days). Blood samples were centrifuged to obtain blood plasma. Carnosic acid was extracted from the blood plasma samples and quantified using HPLC. The result showed that carnosic acid was absorbed quickly after administration of rosemary extract. T_{max} of 1 hour and C_{max} of 2.27 ng/mL were calculated by the authors. However, based on the graph included in the publication, T_{max} seems to be around 2 hours and C_{max} around 2.5 ng/mL.

(ii) Mice: carnosic acid

The absorption and distribution of carnosic acid was studied in male ICR mice (age 5–6 weeks) (238). A total of 16 mice were divided into two groups: a treatment group receiving 50 mg/kg bw per day carnosic acid and a control group receiving only saline by oral gavage for 7 days. The animals received feed and water ad libitum. The animals were killed 4 hours after the last dose and blood, major organs (heart, liver, lung, stomach, spleen, kidney, small intestine, caecum and colon) and contents of the stomach, small intestine, caecum and colon were collected. Carnosic acid concentrations in serum, tissue and contents of digestive tract were measured by HPLC. Body weight and liver and spleen weight were not statistically significantly different between the treatment and the control groups. No differences in behaviour or appearance were observed. The recovery rate of carnosic acid in the collected samples ranged between 79.1 and 113%. Tissue concentrations of carnosic acid were highest in the caecum (2965 nmol/g) followed by the colon (179 nmol/g), stomach (36 nmol/g), spleen (13.4 nmol/g) and liver (10.6 nmol/g). Concentrations in the kidney, heart, lung and small intestine were lower (between 0.66 and 1.87 nmol/g). The concentration in serum was 15.2 nmol/mL. The concentrations in the content of the digestive tract varied from 74.54 nmol/g in the colonic content to 13.93, 1.08 and 0.15 nmol/g for the small intestine, stomach and caecum, respectively.

(iii) Mice: carnosol

The kinetics of carnosol were studied in male C57BL/6 mice (age 8 weeks) (240). A total of 36 mice received a single dose of carnosol at 100 mg/kg bw via oral gavage. The animals received standard chow and water ad libitum. Blood was collected from the retro-orbital vein at 15 and 30 minutes and 1, 2, 4, 6, 12 and 24 hours post-dosing. Blood samples were centrifuged to obtain blood plasma, to which an internal standard solution (0.1% formic acid in acetonitrile spiked with butyl paraben) was added to deproteinize the plasma sample. This mixture was centrifuged and the supernatant was analysed using liquid chromatography tandem mass spectrometry (LC-MS/MS). Carnosic acid (LOQ, 15 ng/mL; LOD not provided) was not detected at any timepoint, indicating that carnosol was not metabolized to detectable levels of carnosic acid. Carnosol was absorbed quickly and the maximum plasma concentration ($C_{\max}$) of 1.156 μM was reached 30 minutes post-dosing. The half-life ($t_{1/2}$) was 45.2 hours and the area under the curve from 0 to 24 hours was 9.7 μM hours/mL. The Committee noted a slight discrepancy between the values for $C_{\max}$ (1.156 μM versus 1.196 μM) and $t_{1/2}$ (45.2 versus 43.1) in the text and the table in the publication.

(iv) Mice: rosemary extract

In a kinetic study with rosemary extract (40% [w/w] carnosic acid and 5% [w/w] carnosol), male C57BL/6 mice (age 5–6 weeks) received a single dose of 100 mg/kg bw rosemary extract in cottonseed oil via oral gavage (241). A total of 32 mice were divided into two separate groups, but the difference in treatment between the two groups was not specified. The animals received standard chow and water ad libitum. Blood was collected from the retro-orbital vein at 15 and 30 minutes and 1, 2, 4, 6, 12 and 24 hours post-dosing. Blood samples were centrifuged to obtain blood plasma to which an internal standard solution (0.1% formic acid and 0.5 mg/mL butyl paraben in acetonitrile) was added to deproteinize the plasma sample. Carnosic acid and carnosol were analysed using LC-MS/MS. Both carnosic acid and carnosol were absorbed quickly after administration of rosemary extract, and the maximum plasma concentrations of 54 and 5 μM , respectively, were reached within 15 minutes post-dosing. The Committee noted that the maximum concentrations were observed at the first timepoint measured, which makes the estimation of the maximum concentrations uncertain. The half-life ($t_{1/2}$) was 3.5 and 7.5 hours and the areas under the curve were 120 and 14 μM hours/mL for carnosic acid and carnosol, respectively.

2.1.2 Biotransformation

No data were available.

2.1.3 Effects on enzymes and other biochemical parameters

(a) In vitro

(i) Rosemary extract

The effect of rosemary acetone extract (INS No. 392; 53.7% carnosic acid and 4.1% carnosol) on CYP and uridine diphosphate glucuronosyltransferase (UGT) 1A gene expression was investigated using 3D spheroid cultures of primary human, mouse and rat hepatocytes (242). The cells were exposed to 0, 0.01, 0.1 or 1 µg/mL of rosemary extract in a serum-free culture medium at a final DMSO concentration of 0.1% for 72 hours. As positive controls, rifampicin (10 µM), 6-(4-chlorophenyl)imidazo[2,1-b][1,3]thiazole-5-carbaldehyde-O-(3,4-dichlorobenzyl)oxime (CITCO; 1 µM) and β-naphthoflavone (BNF; 25 µM) were used. Rifampicin is a pregnane X receptor (PXR) ligand, which induces CYP2B6, CYP2C19, CYP3A4, UGT1A1, UGT1A4, cap3a11 (mouse), cyp2c39 (mouse), cyp3a2 (rat), cyp3a23/3a1 (rat) and cyp2e1 (rat). CITCO is a constitutive androstane receptor ligand, which also binds to PXR and induces CYP2B6. BNF is an aryl hydrocarbon receptor ligand, which induces CYP1A1.

Cytotoxicity was assessed by measuring ATP levels in spheroids exposed for 72 hours. Up to eight spheroids were measured per concentration per cell type for each compound. Cell viability was 95% or higher for all rosemary extract concentrations and for rifampicin, whereas BNF and CITCO led to statistically significant reductions in viability (up to 62% viability for BNF and up to 72% viability for CITCO). After 72 hours, spheroids were harvested for gene expression analysis in 100 µL RNA lysis buffer and stored until RNA isolation. Three replicates were used per treatment and cell type. Reverse transcription was performed using 1 µL RNA. Transcription levels were quantified for the human genes *CYP1A1*, *CYP2A6*, *CYP2B6*, *CYP2C19*, *CYP2E1*, *CYP3A4* and *CYP4A11* and *UGT1A1*, *UGT1A3*, *UGT1A4*, *UGT1A6* and *UGT1A9*. Corresponding homologous genes for mice and rats were also quantified, that is, *Cyp1a1*, *Cyp2a5*, *Cyp2b10*, *Cyp2c39*, *Cyp2e1*, *Cyp3a11*, *Cyp4a10*, *Ugt1a1*, *Ugt1a2*, *Ugt1a5*, *Ugt1a6a* and *Ugt1a9* for mice and *Cyp1a1*, *Cyp2a1*, *Cyp2a2*, *Cyp2b1*, *Cyp2c11*, *Cyp2e1*, *Cyp3a2*, *Cyp3a23/3a1*, *Cyp4a1*, *Ugt1a1*, *Ugt1a2*, *Ugt1a5*, *Ugt1a6* and *Ugt1a9* for rats. *ACTB*, *GADPH*, *HPRT1* and *YWHAZ* and their mouse and rat homologous genes were used as reference genes to normalize the results. A statistical comparison was performed of the normalized values for the treatment groups and control groups using heteroscedastic unpaired two-tailed *t*-tests. In this study, induction of gene expression was considered biologically relevant if the *P*-value was below 0.05 and the fold-change compared with the negative control was larger than 1.5.

In the primary human hepatocytes, CYP1A1 expression was statistically significantly increased by 3.1-fold at 1 µg/mL rosemary extract compared with the negative control. CYP3A4 expression was dose-dependently increased by 1.2,

1.5 or 2.3-fold at 0.01, 0.1 or 1 µg/mL rosemary extract, respectively, reaching statistical significance at the two highest concentrations. A statistically significant 1.9-fold increase in CYP2C19 expression was observed at 1 µg/mL rosemary extract. According to the study authors, the small induction by rosemary extract is unlikely to be biologically relevant. For CYP2B6 expression a small, statistically significant, 1.3-fold increase was observed at 1 µg/mL rosemary extract. The small increase by rosemary extract (< 1.5-fold) was not considered biologically relevant by the study authors. No statistically significant differences were observed for CYP2E1, CYP4A11 or CYP2A6. Rosemary extract at 1 µg/mL statistically significantly induced UGT1A1 expression by 1.5-fold; rifampicin, BNF and CITCO statistically significantly induced UGT1A1 expression by 1.9-, 1.7- and 1.7-fold, respectively. UGT1A4 expression was statistically significantly increased by 1.3- and 1.9-fold at 0.1 and 1 µg/mL rosemary extract compared with the negative control. No statistically significant differences were observed with rosemary extract for induction of UGT1A3, UGT1A6 and UGT1A9 expression.

In rat hepatocytes, a statistically significant 1.8-fold increase in induced gene expression was observed for Cyp3a23/3a1, the rat homolog of CYP3A4, at 1 µg/mL rosemary extract. At 0.01 µg/mL rosemary extract, Cyp3a23/3a1 expression was statistically significantly reduced by a factor of 0.3. Rosemary extract did not induce any of the Ugt1a genes analysed.

In mouse hepatocytes, the expression of Cyp3a11, the mouse homolog of CYP3A4, was statistically significantly induced by 2.4-fold at 1 µg/mL rosemary extract. The expression of Cyp1a1 was decreased at 0.01 µg/mL rosemary extract; however, this was considered coincidental as this effect was not observed at higher concentrations. Expression of other genes was maximally induced 1.4-fold by rosemary extract and these differences were not statistically significant.

In summary, exposure to rosemary extract resulted in the induced expression of CYP3A4 in human primary hepatocytes and the induced expression of the homologous genes Cyp3a11 and Cyp3a23/3a1 in primary hepatocytes from mice or rats, respectively, although at the lowest dose tested a statistically significantly reduced expression was observed in primary rat hepatocytes. In human primary hepatocytes, rosemary extract also induced expression of CYP1A1, CYP2C19 (although the induction was minor), UGT1A1 and UGT1A4, but no other enzymes were induced in the primary hepatocytes from mice or rats.

In a screening study on possible herb–drug interactions, Husain et al. (243) evaluated an ethanolic extract of rosemary plant material (not further specified) in several in vitro assays. In a luciferase reporter gene assay to measure PXR activation, HepG2 cells were exposed to 0, 6.66, 20 or 60 µg/mL rosemary extract. A statistically significantly higher (up to two- to threefold) increased transcriptional activity of human PXR was observed. In the aryl hydrocarbon receptor reporter gene assay, human AhR reporter cells were exposed to

0, 3.33, 10 or 30 µg/mL rosemary extract. A statistically significantly higher (up to around fivefold) AhR activation was observed compared with the negative control. In an assay using baculosomes expressing the individual human CYP1A2 and CYP3A4 enzymes, rosemary extract inhibited CYP1A2 and CYP3A4 catalytic activity with IC₅₀ values (concentration at which 50% inhibition is observed) of 10–12.5 µg/mL and 15 and 20 µg/mL, respectively. In the rhodamine-123 uptake assay to measure P-gp inhibition, P-gp-overexpressing multidrug resistance protein 1 Madin-Darby canine kidney cells were exposed to 0, 12.5, 25 or 50 µg/mL rosemary extract. The intracellular uptake of rhodamine-123 was statistically significantly increased at the mid and high doses; however, there was no dose–response relationship.

Sánchez-Martínez et al. (244) tested a supercritical-fluid extract of rosemary leaves, containing mainly carnosic acid (39.40%) and carnosol (18.71%) in an in vitro anticholinesterase assay, and determined an IC₅₀ value of 107.85 µg/mL. Carnosol and carnosic acid as well as many other components of the extract were shown to have potential to cross the blood–brain barrier based on the results from a parallel artificial membrane permeability assay mimicking the blood–brain barrier.

(ii) Carnosol

The inhibitory potential of carnosol on several human CYP enzymes was studied using baculosomes expressing individual human CYP isoforms (240). The baculosomes were incubated with carnosol in concentrations ranging from 10 pM to 100 µM for 10 minutes, and then specific substrates were added. Enzyme activity was measured via fluorescent readings after 10 minutes for CYP1A2 and CYP3A4, 30 minutes for CYP2C19, and 60 minutes for CYP2C9 and CYP2D6. According to the results section of the publication, IC₅₀ values were 83.6, 9.3, 8.8, 8.8 and 4.4 µM for CYP1A2, CYP3A4, CYP2C9, CYP2D6 and CYP2C19, respectively. The Committee noted that different numbers are provided in supplementary table 1 (“Inhibitory concentration (IC₅₀) of carnosol in P450 baculosomes”), namely 116.30, 5.38, 10.01, 16.51 and 9.23 µM, respectively. The inhibitory potential of carnosol on protein expression of transporters and metabolic enzymes was studied in HepG2 cells and in human colon cancer (HT-29) cells. The cells were treated with carnosol at 10 µM for 72 hours with fresh carnosol spiked media every 24 hours. Cell lysates were collected and protein expression was measured using a western blot. Protein expression of the transporters MRP1, MRP2 and multidrug resistance-associated protein 1, and of the enzymes sulfotransferase (SULT)1E1, SULT1A1 and CYP1A2, was measured in HepG2 cells, and protein expression of the transporter MRP2 and the enzymes SULT1A1, CYP3A4 and UGT2B4 was measured in HT-29 cells. Protein expression was normalized to β-actin and compared with the DMSO control. Induced expression (> 50%) was observed in HepG2 cells for MRP1 and

MRP2 and in HT-29 cells for UGT2B4 and CYP3A4, but none of these increases was statistically significant.

(b) In vivo

(i) Rats

The effects of rosemary extract (INS No. 392; 53.7% carnosic acid and 4.1% carnosol) on *Cyp* and *UGT1A* gene expression and CYP and UGT1A protein expression were investigated in immature rats (245). The measurements were performed in liver samples from an investigative study on the effects on thyroid hormones by Anonymous (246), described in [Section 2.2.5](#) below. In this study, groups of 20 female pregnant Han Wistar rats received rosemary extract at doses of 0, 1800, 3000 or 4200 mg/kg in the diet (equal to 0, 143, 230 or 355 mg/kg bw per day of rosemary extract and 0, 83, 133 or 205 mg/kg bw per day of carnosol and carnosic acid) from GD 6 to GD 19, and at doses of 0, 800, 1300 or 1900 mg/kg (equal to 0, 125, 217 or 304 mg/kg bw per day of rosemary extract and 0, 72, 125 or 176 mg/kg bw per day of carnosol and carnosic acid) from GD 20 throughout lactation to account for increased feed intake. The F1 offspring were indirectly exposed in utero and during lactation, and may have consumed a small amount of formulated diet containing rosemary extract from approximately PND 15–16 until weaning. The study was certified for compliance with GLP and Quality Assurance (QA). Livers were collected from pups on PND 13 or after reaching sexual maturity (PND 43 for females and PND 50 for males) and were immediately snap frozen. For gene expression analysis, four individual liver samples per timepoint, treatment group and sex were used. Total RNA was isolated and reverse transcription was performed in 1 μ L RNA, quantifying transcription levels of the genes *Cyp1a1*, *Cyp2a1*, *Cyp2a2*, *Cyp2b1*, *Cyp2c11*, *Cyp2e1*, *Cyp3a2*, *Cyp3a23/3a1*, *Cyp4a1*, *Ugt1a1*, *Ugt1a2*, *Ugt1a5*, *Ugt1a6* and *Ugt1a9*. To normalize the data, *Actb*, *Gadph*, *Hprt1* and *Ywhaz* were used as reference genes. For protein expression analysis, microsomes were prepared from four individual liver samples per treatment group for timepoint PND 43 (females) and PND 50 (males). For timepoint PND 13, microsomes were prepared from four replicate samples obtained by pooling five to six livers because of the small size of the individual livers at that stage. LC-MS/MS was performed to quantify the expression of the proteins *Cyp1a1*, *Cyp2a1*, *Cyp2a2*, *Cyp2b1*, *Cyp2c11*, *Cyp2e1*, *Cyp3a23/3a1*, *Cyp4a10*, *Ugt1a1*, *Ugt1a2*, *Ugt1a5*, *Ugt1a6* and *Ugt1a9*. For each protein, two internal peptide standards were included to check the validity of the analysis. For *Ugt1a2* and *Ugt1a9*, both peptide standards were below the LOD in all samples; these results therefore could not be quantified. Normalized values were used for statistical comparison of the treatment groups and the control groups using heteroscedastic unpaired two-tailed *t*-tests. Induction was considered potentially biologically relevant when there was a statistically significant ($P < 0.05$) and at least 1.5-fold difference.

The only statistically significant increase (> 1.5-fold) in CYP gene expression was a 1.6-fold increase in *Cyp2b1* in males of the low-dose group at PND 50, but this was considered an incidental finding as it was not observed in the mid- or high-dose groups. The Committee noted that there were several statistically significant decreases in gene expression of more than 1.5-fold (i.e. corresponding to a factor of ≤ 0.67 of the control value), but these were not mentioned or discussed by the study author. *Cyp2a1* was reduced in males in the high-dose group at PND 13. *Cyp2b1* was reduced in high-dose females at PND 13, in all male treatment groups at PND 13 and in mid-dose females at PND 43. *Cyp2c11* was decreased in mid- and high-dose females at PND 43. *Cyp3a2* was decreased in high-dose females and males at PND 13 and in all female treatment groups at PND 43. *Cyp3a23/3a1* was reduced in low-dose males at PND 13 and in mid-dose females at PND 43.

For Ugt1a gene expression, the only statistically significant increased (> 1.5-fold) expression was observed for Ugt1a6 (2.1-fold) in females at PND 13, but this was also only observed in the low-dose group and therefore considered an incidental finding. For Ugt1a gene expression, there were also several statistically significant decreases in gene expression of more than 1.5-fold that were not mentioned or discussed by the study author. This was the case for Ugt1a2 in low-dose females at PND 13, for Ugt1a5 in all male treatment groups and in high-dose females at PND 13, and for Ugt1a9 in all female treatment groups at PND 43.

The only statistically significant increases in protein expression (> 1.5-fold) were observed in low-dose females at PND 13 for Cyp2e1 (1.6-fold), Cyp4a10 (1.5-fold), Ugt1a5 (1.6-fold) and Ugt1a6 (1.6-fold). Since these increases were only observed at the low dose, these were not considered to be biologically relevant. Statistically significant decreases in protein expression (> 1.5-fold) were observed for Ugt1a1 in high-dose males at PND 13 and for Ugt1a5 in low- and high-dose males at PND 50. According to the study author, a biologically relevant, although statistically non-significant, increase in CYP2A2 protein expression was observed for females at PND 13. The fold increases ($\pm$ standard error of the mean [SEM]) were 1.70 ± 0.67 , 1.59 ± 0.90 and 5.02 ± 3.46 for the low-, mid- and high-dose groups, respectively. This was not observed in males at PND 13. In females at PND 43, CYP2A2 protein expression was 2.17 ± 1.19 , 2.99 ± 0.87 and 1.07 ± 0.38 in low-, mid- and high-dose groups, whereas CYP2A2 protein expression was decreased (not statistically significantly) in males at PND 50. Based on the extremely large variation in CYP2A2 protein expression in high-dose females at PND 13 (SEMs ranging from just over 1.0- to 8.5-fold increase, compared with control) and the absence of a dose-response relationship in females at PND 43 and in males, the Committee is not convinced of the biological relevance of the increase in CYP2A2 protein expression. Some decreases in gene expression and protein expression were observed. These were mostly incidental, and their biological relevance is unclear.

2.2 Toxicological studies

2.2.1 Acute toxicity

Naqvi et al. (247) performed an acute toxicity study with a hydro-ethanolic (30:70) rosemary extract (not further specified). According to the study authors, this study was conducted in accordance with OECD Test Guideline No. 420 (248), but the study description is not very detailed. Groups of five BALB/c mice (sex not reported) received a single oral dose of rosemary extract at 0 (vehicle), 1000, 3000 or 5000 mg/kg bw. The animals were observed for signs of toxicity, including blindness, piloerection and exploratory activity for 6 hours, and mortality checks were conducted from 48 hours until 14 days after the experiment. No signs of restlessness, seizures, piloerection and blindness were observed at 6 and 48 hours and no mortality was observed up to 14 days after the experiment (247). This indicates that the acute toxicity of this extract is low ($LD_{50} > 5000$ mg/kg bw).

2.2.2 Short-term studies of toxicity

(a) Rats

Salokhe et al. (249) studied an ethanolic extract of powdered rosemary leaves (carnosic acid and carnosol content unknown) in a 28-day oral repeat-dose toxicity study. According to the study authors, this study was conducted in accordance with OECD Test Guideline No. 407 (250), with minor modifications. However, the Committee noted substantial deviations in the study design (e.g. only a limited number of organs were examined). The study was not conducted in accordance with GLP. Groups of five male and five female Wistar rats received rosemary extract at doses of 300, 500 or 1000 mg/kg bw per day via oral gavage; the control group received water (vehicle). The animals were aged approximately 6 months at the start of dosing, and were housed under standard laboratory conditions with feed and water provided ad libitum. Animals were observed for clinical signs throughout the study. Feed and water consumption were recorded daily and body weight was recorded weekly. At the end of the dosing period, blood samples were collected via retro-orbital puncture for haematology (RBC count, [differential] WBC count, haemoglobin, haematocrit, platelet count, mean platelet volume, mean corpuscular volume, mean corpuscular haemoglobin, mean corpuscular haemoglobin concentration and red cell distribution width) and clinical biochemistry (glucose, blood urea nitrogen, creatinine, AST, ALT, total cholesterol, triglycerides, ALP, total protein, albumin, direct and total bilirubin, chloride, potassium and sodium) analysis. The Committee noted that no results were provided for mean platelet volume, red cell distribution width, glucose, total cholesterol, triglycerides, direct bilirubin, chloride, potassium and sodium. Selected organs (heart, lungs, kidneys and liver) were weighed, and the liver and kidneys were examined microscopically. There were no signs of toxicity and no effects were observed on feed and water intake. The Committee noted

that body weight was statistically significantly increased on days 14 and 28 in all female treatment groups (182–206 g versus 163 g in controls on day 14, and 191–201 g versus 165 g in controls on day 28) and on day 28 in high-dose males (201 g versus 165 g); however, there was already a large deviation in the average body weight on day 0 between the control (160 g) and the treatment (180–203 g) groups.

According to the study authors, no statistically significant changes were observed in any of the haematological parameters. However, the Committee noted that a dose-dependent decrease in RBC count of up to 25% was observed in males, but not in females. In addition, mean corpuscular volume was decreased by up to 35% in all treatment groups in both sexes, but without a dose–response relationship. A statistically significant and dose-dependent increase in total protein was observed in both males of up to 15% and females of up to 26% in the mid- and high-dose groups. Albumin levels were statistically significantly decreased in the low- and mid-dose groups, but this was not observed in the high-dose group. According to the study authors, the increase in total protein could be ascribed to the protein content of the plant extract, but the Committee considered that this is not substantiated by information on the protein content of the plant extract. Urea nitrogen was statistically significantly decreased in males by up to 33% and females by up to 22% in all dose groups, but without a clear dose–response relationship. If blood total protein is elevated, it is likely that blood urea nitrogen would also be elevated, which is not the case. In the absence of related effects, the effects on total protein and urea nitrogen were not considered adverse by the Committee. An incidental statistically significant increase in ALP was observed in low-dose females only. Relative heart, lung, liver and kidney weights were not affected by the treatment (absolute organ weights were not reported). No histopathological changes were reported in the kidney and liver. The study authors considered that the NOAEL of this study was 1000 mg/kg bw per day, the highest dose tested.

A repeat-dose oral toxicity study (90 days, with a 4-week recovery period) with an ethanolic rosemary extract standardized to 50% ursolic acid (ursolic acid 50–60%, betulinic acid 30–40% and oleanolic acid 5–10%) was available (251), as well as a 90-day study with ursolic acid in Han Wistar rats (252). Since the composition of the test compounds in these studies differs substantially from the composition of the food additive, these studies were not summarized.

Taha and Saleh (253) evaluated the anti-hyperlipidemic properties of an unspecified aqueous extract of powdered rosemary leaves in male rats in a study that was poorly reported and not conducted in accordance with GLP. Because of a lack of clarity about the methodology of the study and the fact that the study was not designed to investigate potential adverse effects of rosemary extract, the Committee did not consider this study of relevance for the safety assessment of rosemary extract.

(b) Piglets

Yang et al. (254) evaluated the effect of a rosemary extract on growth performance, nutrient digestibility, antioxidant capacity, intestinal morphology and microbiota of weaned piglets (age 23 ± 1 days). The main component of the rosemary extract (or of the total phenols in the rosemary extract; this is not clear from the study description) was carnosic acid ($\geq 60\%$). The study was not conducted in compliance with regulatory test guidelines or GLP. Groups of 48 piglets (eight pens each with six piglets, three male and three female) received a basal diet supplemented with rosemary extract at doses of 0, 100, 200 or 400 mg/kg feed per day for 21 days (equal to approximately 4, 9 or 18 mg/kg bw per day). The animals had access to feed and water ad libitum. Individual body weights were measured at the start and end of the experiment and feed consumption was recorded per pen. The average daily gain, average daily feed intake and feed-to-gain ratio were calculated. Incidences of diarrhoea were recorded. From day 14 to 21, the digestibility of dry matter, crude protein, ether extract and gross energy were measured through addition of the indigestible marker chromium oxide in the diets. On days 20 and 21, faecal samples were collected from one male and one female per pen. At the end of the study, blood samples were collected from the cervical vein from eight animals per treatment group. From the same animals, the livers were collected to study antioxidant ability (MDA, SOD, glutathione peroxidase, catalase and total antioxidant capacity) and the intestinal tracts were collected for morphological examination and gut microbial composition analysis using real-time polymerase chain reaction (PCR).

No differences in feed-to-gain ratio were observed and only the mid-dose group showed statistically significantly higher final body weight, average daily gain and average daily feed intake compared with the control group. The proportion of days in which piglets showed clinical signs of diarrhoea was statistically significantly decreased in the mid- and high-dose groups. The digestibility of crude protein and gross energy was increased in all treatment groups, but no effect was observed on the digestibility of dry matter and ether extract. In the treatment groups, statistically significant changes in the intestinal morphology were observed, including increased villus height, decreased crypt depth and increased villus height/crypt depth ratio in the jejunum and ileum. No effects were observed in the duodenum. Total antioxidant capacity, glutathione peroxidase serum and SOD activity were statistically significantly increased in serum and liver, whereas catalase activity was only statistically significantly increased in serum. MDA was statistically significantly decreased in serum and liver. Microbial composition was statistically significantly different in the treatment groups compared with the control, as the amounts of *Bifidobacterium* and *Bacteroidetes* increased and the amount of *Escherichia coli* decreased. No changes were observed in the amounts of *Lactobacillus* and *Firmicutes*.

The Committee noted that this study focused on some aspects related to growth performance and gut improvement and was not designed to investigate the potential adverse effects of the extract; the Committee therefore did not consider this study to be of relevance for the safety assessment of rosemary extract.

2.2.3 Long-term studies of toxicity and carcinogenicity

No new data were identified.

2.2.4 Genotoxicity

A limited bacterial reverse mutation assay was performed in *Salmonella typhimurium* strains TA98 and T100 with one concentration (5%, not further specified) of a rosemary extract prepared by maceration. The main components of the extract were gallic acid (26.4%), rosmarinic acid (18.0%) and luteolin-3-O-acetyl-O-glucuronide (14.2%); carnosol was detected but was below the LOQ, carnosic acid was not detected and 12-O-methyl carnosic acid was detected in low levels (0.5%). The results were negative according to the study authors, but these were not reported (255).

2.2.5 Reproductive and developmental toxicity

(a) Reproductive toxicity

No new data were identified.

(b) Developmental toxicity

(i) Rats

In a prenatal developmental toxicity study, groups of 22 pregnant Han Wistar rats received rosemary extract (INS No. 392; acetone extract containing 53.7% carnosic acid and 4.1% carnosol) at doses of 0, 1800, 3000 or 4200 mg/kg in the diet (equal to 132, 217 or 304 mg/kg bw per day of rosemary extract or 76, 125 or 176 mg/kg bw per day of carnosol and carnosic acid), from GD 6 to 21 (256). The study was conducted according to OECD Test Guideline No. 414 (191), and was certified for compliance with GLP and QA. Clinical observations, body weight and feed consumption were recorded. On GD 21, blood samples were taken from the dams for analysis of the thyroid hormones T₃, T₄ and TSH, and animals were killed for reproductive assessment and fetal examination.

There were no test-substance-related deaths or clinical signs in the study. A small reduction in maternal body weight gain was observed during GD 6–7 in the mid- and high-dose groups of 2 and 4 g, respectively. This incidental decrease in body weight was accompanied by a small but statistically significant decrease in feed consumption during GD 6 and 7 in the high-dose group. There was no effect on maternal body weight gain and feed consumption (GD 6–21).

The incidental change in body weight and feed consumption was therefore not considered adverse. No effects were observed on gravid uterine weight. Maternal serum concentrations of T₃ and T₄ on GD 21 showed no statistically significant differences between the control and treatment groups, but serum concentrations of TSH were statistically significantly increased in the mid- and high-dose groups compared with the control group. The average TSH serum concentrations were 568, 752, 824 and 873 pg/mL for the control, low-, mid- and high-dose groups, respectively. All TSH serum concentrations were within the historical control range (291–2986 pg/mL, 2.5–97.5th percentile; median 891 pg/mL) with the exception of four animals in the control group, one animal in the low-dose group and one animal in the high-dose group. The study authors therefore reasoned that the statistically significant differences observed were likely the result of the controls having low values rather than the treatment. In addition, no treatment-related effects were observed on thyroid gland weight or during macroscopic and microscopic examinations of the thyroid glands. The only macroscopic findings in the dams were one incidence of thyroids with abnormal colour, one incidence of abnormal caecum content and one incidence of an enlarged placenta in the high-dose group. These incidental findings were not considered treatment related.

No treatment-related effects were seen on the number of pregnancies, mean number of implantation sites, resorptions, live young, pre- or post-implantation losses, sex ratio, average placental, litter and fetal weights, or anogenital distance of the fetuses. Several skeletal and visceral abnormalities were observed in the fetuses. Complete *situs inversus* was seen in one female fetus in each test group and thoracic *situs inversus* was seen in one male fetus in the high-dose group. The frequency of complete *situs inversus* was comparable with that for the historical control data (one fetus in one litter), whereas thoracic *situs inversus* was not reported in the historical control data. However, since this was only observed in one fetus, it was considered incidental. The incidence of supernumerary 14th ribs was increased in the mid- and high-dose groups (23 fetuses in 12 litters and 24 fetuses in 10 litters for the mid- and high-dose groups, respectively, compared with 12 fetuses in seven litters in the control group). The incidences were within the historical control range (4–29 fetuses in 3–15 litters) and were considered unrelated to the treatment. Short supernumerary cervical ribs were seen in six fetuses from five litters in the high-dose group compared with three fetuses from three litters in the control group. The incidence in the high-dose group only slightly exceeded the historical control range (0–5 fetuses, 0–4 litters) and this observation was not considered to be treatment related. The Committee considered these findings to be incidental and not related to the test item. There were no adverse test-substance-related effects on any maternal or fetal parameters. The NOAEL for maternal and developmental toxicity was the

highest dose tested, that is, 304 mg/kg bw per day expressed as rosemary extract, and 176 mg/kg bw per day expressed as the sum of carnosol and carnosic acid.

To study the potential effect of rosemary extract (INS No. 392; acetone extract containing 53.7% carnosic acid and 4.1% carnosol) on thyroid hormone levels in immature rats, groups of 20 female pregnant Han Wistar rats received rosemary extract at doses of 0, 1800, 3000 or 4200 mg/kg in the diet (equal to 0, 143, 230 or 355 mg/kg bw per day of rosemary extract or 0, 83, 133 or 205 mg/kg bw per day of carnosol and carnosic acid) during GD 6–19, and at doses of 0, 800, 1300 or 1900 mg/kg in the diet (equal to 0, 125, 217 or 304 mg/kg bw per day of rosemary extract or 0, 72, 125 or 176 mg/kg bw per day of carnosol and carnosic acid) from GD 20 throughout lactation to account for increased feed intake (246). The F1 offspring did not directly receive rosemary extract, but they were indirectly exposed in utero and during lactation and may have consumed a small amount of formulated diet containing rosemary extract from approximately PND 15–16 until weaning. This study was designed to address the recommendation of the Committee at its Eighty-seventh meeting to determine whether the effects noted on pup thyroid hormone levels can be replicated (234), and therefore did not follow a specific OECD test guideline. The study was certified for compliance with GLP and QA.

All F0 females were checked twice daily for mortality and clinical signs of toxicity. Detailed physical examinations were conducted on GD 3, 5, 18 and 20 and lactation day (LD) 1, 7, 14 and 21. The body weight of the dams was recorded on GD 3, 6, 10, 14, 17 and 20, LD 1, 4, 7, 11, 14, 18 and 21, and at necropsy. Feed consumption was recorded between the body weight timepoints (i.e. GD 3–6, 6–10 etc.). Gestation length was recorded and parturition observations were conducted three times daily from GD 20 onwards to monitor the progress and completion of parturition, number of live and dead offspring, and any difficulties. Dams were killed and full macroscopic necropsy was performed at LD 21, on the day their last offspring died or when they reached GD 25 with no viable litter. The number of implantation sites was recorded.

For the F1 animals, clinical observations were conducted 24 hours after birth and daily thereafter. Body weight was recorded on PND 1, 4, 7, 13, 17 and 21, twice weekly from PND 21 to termination (as well as on the day that sexual maturation was attained), and at necropsy. Feed consumption was recorded twice weekly from PND 21 to termination. Mortality was checked daily to monitor changes in litter size. The sex ratio of each litter was recorded on PND 1, 4, 13 and 21. From PND 25 (females) and PND 38 (males) onwards, daily examinations for sexual maturation (balano-preputial separation for males; vaginal opening for females) were conducted. On PND 4, 13 and 21 and after sexual maturation, a subset of F1 was killed. On PND 4, litters were reduced to eight pups (four males and four females when possible) by random culling. On PND 13, up to two male and female animals per litter were killed. On PND 21, litters were culled

to select in total 20 males and 20 females per group that were kept until sexual maturation. Detailed physical examinations were conducted once per week for these F1 selected animals. The selected animals were killed on PND 43 (female) and 50 (male), or on the day of sexual maturation if sexual maturation had not taken place by PND 43 or 50.

Blood samples were collected for analysis of T_3 , T_4 and TSH. On PND 4, samples from culled male and female pups were pooled and analysed for T_3 and T_4 only, because of small sample size. On PND 13, individual samples were collected from all culled pups. At sexual maturation, individual samples were collected from pups surviving to scheduled termination. Externally abnormal F1 offspring culled on PND 4, all F1 offspring culled on PND 13, unselected F1 offspring culled on PND 21 and selected F1 animals were subject to a detailed necropsy. In addition, histology and pathology of the liver and thyroid were conducted for the control and high-dose groups. Remaining liver tissue from all dose groups was frozen and dispatched for further separate analyses to quantify *Cyp* and *Ugt1a* gene expression and *CYP* and *UGT1A* protein expression (245).

In the F0 generation, no treatment-related deaths or clinical signs occurred. Four dams were killed before study termination for welfare reasons (one dam) and total litter loss (three dams: one control dam, one mid-dose dam and one high-dose dam). Body weight gain was statistically significantly higher in the mid- and high-dose groups during GD 10–14 and in all treatment groups during GD 14–17, but overall body weight gain was not statistically significantly different from the control group. Feed consumption was slightly, but statistically significantly, higher in the high-dose group compared with the control group during GD 10–17; however, the average feed consumption during GD 6–20 was not different between the groups. During lactation, overall body weight gain was statistically significantly higher in the high-dose group compared with the control group (38 g versus 27 g). This difference was most likely the result of a greater reduction in group mean body weight during LD 14–21 in controls of 13 g compared with the treatment groups (reductions of 9, 7 and 4 g in the low-, mid- and high-dose groups, respectively). Body weight gain was slightly increased in the low- and mid-dose groups, but the difference was not statistically significant.

There were very few macroscopic findings in the dams: one incidence of abnormally coloured adrenals and enlarged thyroid in one high-dose animal, and one incidence of threadlike uterus in the control group. These incidental findings were not considered to be related to the test item.

There was no effect on gestation length, gestation index, mean number of implantation sites, live young, litter size, offspring survival or sex ratio. Three dams experienced total litter loss, which occurred once in the control, mid-dose and high-dose groups and was not considered to be related to the test item.

No treatment-related clinical signs were observed in pups. In male offspring of the high-dose group, a small but statistically significant lower (by

6%) body weight and lower (by 6%) overall body weight gain was observed at the end of the study, resulting from a lower body weight gain during PND 7–13. A statistically significant lower body weight gain was also observed in female offspring during PND 1–4 (all treatment groups) and PND 7–13 (mid- and high-dose groups); however, body weight was not statistically significantly different between the treatment and control groups at the end of the study. This small change in body weight in male pups was not considered adverse. There was no effect on sexual maturation.

Absolute liver weights were statistically significantly lower on PND 13 in all treatment groups in both males by up to 16% and females by up to 14%, and relative liver weights were statistically significantly lower in all treatment groups for females by up to 11% and in high-dose males by 11%. At sexual maturation (PND 43 for female offspring and PND 50 for male offspring) there were no statistically significant differences. The study author indicates that it is likely that these differences were caused by the differences in body weight, rather than a direct effect related to the test item. The transient decreases in liver weight were not considered adverse. No treatment-related macroscopic findings were observed on PND 4, 13 or 21, or at sexual maturation. Histopathological findings in the liver were limited to four incidental findings, each occurring only once in two males and two females from the high-dose group. The Committee considered these findings unrelated to the test item. No histopathological findings were observed in the thyroids.

The results from the thyroid hormone analysis are given in [Table 2.1](#). TSH was not measured at PND 4. At PND 13, serum TSH levels were statistically significantly lower in male offspring in the mid-dose group only. No differences were observed in TSH levels for female offspring at PND 13 or for male and female offspring at sexual maturation.

At PND 4 there were no statistically significant differences in serum T_3 and T_4 levels between the treatment groups and the control group. At PND 13, the T_3 mean level in high-dose females was statistically significantly lower by 16% compared with the controls. For male PND 13 offspring, a smaller decrease of 12% was observed in the T_3 level at the high dose, which was not statistically significant. T_4 levels were reduced in PND 13 males by 12, 7 and 23% in the low, mid- and high-dose groups, respectively, compared with controls (high-dose was statistically significant). In female PND 13 offspring, statistically significant reductions in T_4 were seen in all treatment groups (of 12, 13 and 24% in low-, mid- and high-dose groups, respectively) compared with the controls. At sexual maturation (PND 43 for female offspring and PND 50 for male offspring), no statistically significant differences were observed for T_3 , T_4 and TSH for either male or female offspring. There were no differences in absolute and relative thyroid or parathyroid weights in male and female offspring at termination.

Table 2.1

Thyroid hormone (T₃, T₄, TSH) levels (in ng/mL mean $\pm$ SD) in serum of male and female rats in the control and different treatment groups at PND 4, 13 and sexual maturation (PND 43 for females and PND 50 for males)

Hormone	Timepoint, sex	Control group (ng/mL)	Treatment groups (ng/mL)		
			Low dose	Mid dose	High dose
T ₃	PND 4, M and F	0.16 $\pm$ 0.03	0.20 $\pm$ 0.06	0.20 $\pm$ 0.02	0.19 $\pm$ 0.04
	PND 13, M	0.54 $\pm$ 0.09	0.52 $\pm$ 0.10	0.51 $\pm$ 0.07	0.48 $\pm$ 0.07
	PND 13, F	0.56 $\pm$ 0.09	0.51 $\pm$ 0.10	0.52 $\pm$ 0.06	0.47 $\pm$ 0.07**
	PND 50, M	0.71 $\pm$ 0.08	0.70 $\pm$ 0.10	0.68 $\pm$ 0.10	0.67 $\pm$ 0.09
	PND 43, F	0.90 $\pm$ 0.13	0.90 $\pm$ 0.19	0.88 $\pm$ 0.14	0.96 $\pm$ 0.21
T ₄	PND 4, M and F	13.6 $\pm$ 2.48	16.4 $\pm$ 4.41	15.2 $\pm$ 2.87	16.2 $\pm$ 2.35
	PND 13, M	42.3 $\pm$ 10.3	37.4 $\pm$ 6.25	39.5 $\pm$ 4.55	32.7 $\pm$ 5.93***
	PND 13, F	45.2 $\pm$ 6.84	39.8 $\pm$ 8.01*	39.1 $\pm$ 4.67*	34.2 $\pm$ 5.85***
	PND 50, M	61.0 $\pm$ 7.30	65.7 $\pm$ 10.1	58.5 $\pm$ 7.03	62.4 $\pm$ 7.47
	PND 43, F	40.9 $\pm$ 10.5	38.6 $\pm$ 9.63	41.6 $\pm$ 11.8	41.2 $\pm$ 10.4
TSH	PND 13, M	0.31 $\pm$ 0.09	0.27 $\pm$ 0.14	0.19 $\pm$ 0.11*	0.28 $\pm$ 0.30**a
	PND 13, F	0.49 $\pm$ 0.30	0.51 $\pm$ 0.34	0.48 $\pm$ 0.37	0.32 $\pm$ 0.23
	PND 50, M	0.67 $\pm$ 0.38	0.53 $\pm$ 0.24	0.53 $\pm$ 0.29	0.59 $\pm$ 0.29
	PND 43, F	0.28 $\pm$ 0.25	0.24 $\pm$ 0.21	0.20 $\pm$ 0.21	0.20 $\pm$ 0.13

F: female; M: male; PND: postnatal day; SD: standard deviation; T₃: triiodothyroxine; T₄: thyroxine; TSH: thyroid-stimulating hormone; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

^a According to the study report, this value was statistically significant ($P < 0.05$). The Committee noted that the standard deviations of values for the control and the high-dose group are fully overlapping.

In summary, the results of the thyroid hormone analysis showed statistically significantly lower hormone levels for TSH, T₃ and T₄ in some of the treatment groups at PND 13, but not at PND 4 or at sexual maturation. Since the effect of rosemary extract on thyroid hormones was transient and not accompanied by thyroid weight changes, histopathological findings in the thyroid or effects on sexual maturation, the study author did not consider the changes to be adverse.

The Committee noted that for T₄, this new study replicated the findings of the earlier study (236,257) evaluated previously by the Committee (258). Reductions in thyroid hormone levels, particularly during prenatal and/or postnatal development, are cause for concern. If these reductions occur at a specific stage of brain development, they may have an adverse effect. In the absence of studies on neurodevelopmental toxicity, the Committee noted that this transient reduction in T₄ levels could possibly lead to adverse neurodevelopmental effects.

Basheer et al. (259) evaluated the effect of a volatile oil extract of rosemary (main active components 1,8-cineole, α -pinene, camphor and β -pinene; carnosol and carnosic acid were not present in the extract) on some physiological parameters in lactating rats and their pups. The study was not conducted in compliance with any regulatory test guideline or GLP, and the reporting was limited. Groups of six pregnant albino rats (age 10–12 weeks) received volatile oil of rosemary at doses of 0 (vehicle control), 250, 500 or 1000 mg/kg bw per day via oral gavage for 21 days during the lactation period. The body weights of the dams and pups were recorded on LD 0, 7, 14 and 21. For the dams, follicle-stimulating hormone (FSH), luteinizing hormone (LH), oestrogen, prolactin and progesterone were analysed (no information on sampling was available). For the pups, mortality and timing of developmental milestones (appearance of auricle aperture, hair, teeth, eye-opening, testicle descent, vaginal opening, preputial separation) and ano-genital distance were recorded. It was not reported how many pups were included in the study.

The body weight of the dams was statistically significantly lower by up to 32% at all timepoints in the mid- and high-dose groups and at LD 14 and 21 in the low-dose group, compared with the control group. Sex hormone concentrations were statistically significantly lower in all dose groups for oestrogen (by up to 78%, not dose dependent), LH (by up to 75%, not dose dependent), FSH (by up to 82%, dose dependent) and prolactin (by up to 44%, not dose dependent), whereas the concentrations of progesterone were similar in all treated groups.

The body weight of the pups was statistically significantly lower at all timepoints in all dose groups (by up to 46%) and total mortality rate for pups at weaning was dose-dependently and statistically significantly increased (from 0.5 in the control group to 4.67 in the high-dose group). The day of auricle aperture was similar for the control and treatment groups; however, the day of appearance of hair, teeth and eye-opening and the day of testicle descent, preputial separation and vaginal opening were statistically significantly delayed in the mid- and high-dose groups. The ano-genital distance in male pups dose-dependently and statistically significantly decreased (by up to 33%) in all dose groups.

Adverse test-substance-related effects were found in all dose groups. The lowest-observed-adverse-effect level (LOAEL) was therefore 250 mg/kg bw per day, the lowest dose tested. The Committee noted that the composition of the volatile oil extract is substantially different from the additive, which is a solvent-based rosemary extract with carnosol and carnosic acid as main active compounds. Because the study also has several limitations, it was not considered relevant for the safety assessment of the use of rosemary extract as a food additive.

2.2.6 Special studies

(a) Immune response

Alzahri et al. (260) studied the effects of an aqueous rosemary extract (not further specified) on the thymus gland and on the levels of interleukin (IL) -10, IgM and IgG in serum of chick embryos. Groups of 20 fertilized eggs received 0.1 mL of low- (10%), medium- (50%) and high-concentration (100%) extract/egg by injection into the air sac on days 0 and 8 of incubation. On days 14 and 20 of incubation, serum and thymus glands were collected. The thymus was studied histologically and enzyme-linked immunoassays were performed to quantify IL-10, IgM and IgG in blood serum. In the thymus of chick embryos treated with high-concentration extract, a low density of epithelial cells and lymphocytes was observed on day 14, and a low cell density was observed in the cortex and the medulla on day 20. IL-10 levels at day 14 were dose-dependently decreased in all treatment groups compared with the controls, whereas on day 20 these levels were increased (but without a dose-response relationship). IgM levels were increased with the low-concentration extract only on day 14 and decreased in all treatment groups at day 20 (without a dose-response relationship). IgG levels were decreased in all treatment groups at day 14 and day 20 (without a dose-response relationship) (260).

(b) Thyroid function

Kasim et al. (261) evaluated the effect of an aqueous extract of leaves of rosemary (not further specified) on the thyroid gland in groups of seven hyperthyroid male Albino rats. Three groups were rendered hyperthyroid by daily subcutaneous administration of 0.5 mg/kg levothyroxine for 12 days. The negative control group did not receive levothyroxine. After 12 days, the negative control group and the positive (hyperthyroid) control group were given distilled water (vehicle) and the (hyperthyroid) treatment groups were given 10 mg/kg bw per day of propylthiouracil (PTU) or 10 mg/kg bw per day of rosemary extract for 12 days by oral gavage. At the end of the experiment blood samples were collected to analyse T_3 , T_4 and TSH, and the thyroid glands were examined. TSH concentrations were not statistically significantly different between the groups. T_3 and T_4 concentrations were statistically significantly higher in the positive (hyperthyroid) control group compared with the other groups. The T_3 concentration in the rosemary extract treatment group was not statistically significantly different from the negative control group, but statistically significantly higher than the PTU treatment group. The T_4 concentration in the rosemary extract treatment group was statistically significantly lower than in the negative control group, but statistically significantly higher than in the PTU treatment group. Adverse effects were seen in the thyroid gland in the positive control group, but these were diminished

in the PTU and rosemary extract treatment groups. No other parameters were investigated in this study. The Committee noted that only one dose of rosemary extract was studied and that the effects on thyroid hormone levels in healthy rats were not determined.

The same group evaluated the effect of an aqueous rosemary extract of dried leaves (not further specified) on sex hormones in hyperthyroid male rats (262) using the same study design. In this study, blood samples were collected to analyse LH, testosterone and FSH. Testosterone concentration was highest in the positive (hyperthyroid) control group, followed by the (hyperthyroid) rosemary treatment group. The testosterone concentration was much lower in the negative control and the (hyperthyroid) PTU treatment groups. All differences were statistically significant. FSH concentrations were not statistically significantly different between the positive (hyperthyroid) control and rosemary treatment group. The concentrations in the negative control group and PTU group were statistically significantly lower. The same pattern was seen for LH. The Committee noted that only one dose of rosemary extract was studied and that the effects of rosemary extract on sex hormone levels in healthy rats were not determined.

2.3 Observations in humans

Gómez de Cedrón et al. (263) evaluated the tolerability and safety as well as immunological effects of a food supplement containing rosemary extract (supercritical fluid extract containing 11.25 mg diterpene phenols, not further specified) and shark liver oil enriched in alkylglycols (150 mg; to enhance the bioavailability of the diterpenes) in a 6-week double-blind randomized parallel pilot study. A total of 60 healthy volunteers were randomly allocated to the control group (18 women and 12 men) or treatment group (18 women and 12 men), and received capsules with rosemary extract and shark liver oil or control capsules once daily for 6 weeks.

To assess the safety and tolerability, all volunteers completed questionnaires about capsule tolerance. At baseline and at the middle and end of the study, vital constants (systolic and diastolic blood pressure and heart rate) of the participants were monitored, bioelectric impedance analysis was conducted and blood samples were drawn. Morning urine was also collected at baseline and at the end of the study. A haematological (leucocytes, lymphocytes, neutrophils, monocytes, basophils, haemoglobin, erythrocytes, platelets) and biochemical (triglycerides, total cholesterol, low- and high-density lipoprotein, urea, creatinine, AST, ALT, ALP, γ -glutamyl transferase, bilirubin, urinary thromboxane B₂, total isoprostanes and C-reactive protein) analysis was performed. As well as safety, potential positive effects on immunological parameters and expression of genes related to inflammation and oxidative stress (among others) were studied.

No severe adverse effects were reported during the study. No effects were observed related to liver function (AST, ALT, ALP, γ -glutamyl transferase, bilirubin) and no differences were observed for vital constants, creatinine and oxidation markers in blood or urine. The Committee noted that because of the composition of the food supplement, the results of this study are of limited relevance for the evaluation of rosemary extract as a food additive.

3. Dietary exposure

3.1 Introduction

Rosemary extract as a food additive is not yet included in the GSFA, which defines the conditions of use and maximum permitted levels of food additives. However, it is permitted for use in many countries including Australia, Brazil, China, Japan, Mexico and New Zealand, and in Europe. The dietary exposure to rosemary extract (expressed as carnosic acid and carnosol) has been previously evaluated by the Committee at its Eighty-second and Eighty-seventh meetings (108,234). In order to complete the evaluation, the Committee conducted a literature search in PubMed using the search terms “dietary exposure of rosemary extract” or “estimates of dietary intake of rosemary extract” for articles published since 2018; however, no additional publications were identified. One published report was found from an internet search with more recent estimates of dietary exposure for Australia and New Zealand (237).

3.2 Previous JECFA assessments of dietary exposure

The Committee first evaluated chronic dietary exposure to rosemary extract (expressed as carnosic acid plus carnosol) at its Eighty-second meeting based on maximum permitted and proposed use levels, and noted that the dietary exposure estimates for high consumers of 0.09–0.81 mg/kg bw per day may exceed the upper bound of the temporary ADI by up to 2.7-fold (96). The Committee concluded that the dietary exposures were conservative and that this exceedance did not necessarily represent a safety concern, and requested data on the typical use levels in food in order to refine the dietary exposure estimates.

At the Eighty-seventh meeting, information on maximum proposed and typical use levels for the GSFA food categories were provided by the sponsor, and the Committee re-evaluated dietary exposure estimates of rosemary extract (expressed as carnosic acid plus carnosol) based on typical use levels submitted from Australia, New Zealand, the USA and Europe (258). The Committee noted that estimated mean and high-percentile dietary exposures to rosemary extract (expressed as carnosic acid plus carnosol) based on typical use levels did not exceed the upper end of the temporary ADI of 0–0.3 mg/kg bw per day. For children,

mean estimates of dietary exposure ranged between < 0.01 and $0.14 \text{ mg/kg bw per day}$ and high-percentile exposures ranged between < 0.01 and $0.30 \text{ mg/kg bw per day}$. For adults, mean estimates of dietary exposure ranged between < 0.01 and $0.05 \text{ mg/kg bw per day}$ and high-percentile exposures ranged between 0.01 and $0.12 \text{ mg/kg bw per day}$. The Committee also noted that dietary exposures for children from naturally occurring sources only for Europe (rosemary and other herbs) and Australia and New Zealand ranged between 0.0 and $0.34 \text{ mg/kg bw per day}$ for mean exposures and between 0.0 and $1.66 \text{ mg/kg bw per day}$ for high-percentile exposures. Estimated dietary exposures for adults ranged between 0.0 and $0.18 \text{ mg/kg bw per day}$ for mean exposures and between 0.0 and $0.52 \text{ mg/kg bw per day}$ for high-percentile exposures. The contribution from naturally occurring sources was $< 1\text{--}4\%$ for Australia and New Zealand, based on the distribution of dietary exposures for individuals, and $65\text{--}93\%$ for Europe, based on summing mean dietary exposures from added and natural sources. When dietary exposures from naturally occurring sources are combined with dietary exposures from added sources at typical use levels, EFSA estimated mean dietary exposures were up to $0.42 \text{ mg/kg bw per day}$ for children and up to $0.16 \text{ mg/kg bw per day}$ for adults.

3.3 Current assessment of dietary exposure

For the current assessment the Committee evaluated more recent estimates of dietary exposure for Australia and New Zealand (237) based on typical use levels. These estimates were based on an application requesting an extension of use for rosemary extract as a food additive in Australia and New Zealand to additional food classes (including colours and flavours, edible oils, flour products, noodles and pasta, ground poultry, fermented comminuted meat products and sausages; [Table 3.1](#)).

Because it was not calculated at the Eighty-seventh meeting, the Committee at the current meeting decided to estimate the high exposure from mean dietary exposure from EFSA data that were reported at the Eighty-seventh meeting. All dietary exposure reported from the Eighty-seventh meeting and from this current meeting are shown in [Table 3.2](#). The mean and 90th percentile estimates of dietary exposure to rosemary extract (as carnosic acid plus carnosol) as a food additive were up to 0.16 and $0.30 \text{ mg/kg bw per day}$ in children, and up to 0.082 and $0.15 \text{ mg/kg bw per day}$ in the general population. The Committee noted that these exposures were similar to those estimated by the Committee at its Eighty-seventh meeting. The Committee noted that when dietary exposures from naturally occurring sources are combined with dietary exposures from added sources at typical use levels, the mean and 90th percentile estimates of dietary exposure were up to 0.42 and $0.84 \text{ mg/kg bw per day}$ in children and up to 0.16 and $0.32 \text{ mg/kg bw per day}$ in the general population.

Table 3.1

Use level proposed for extension of use to additional food classes for rosemary extract as a food additive in Australia and New Zealand

Food category	Name	MPL (mg/kg)	Typical use level (mg/kg)	Proposed MPL for GSFA from sponsor (mg/kg) ^a
2.1	Only fish oil and algal oil	50	40	50
2.1 ^b	Fats and oils for the professional manufacture of heat-treated foods	50	40	50
2.2.2	Other fat and oil emulsions including spreads-only spreadable fats with fat content less than 80%	75	60	100
4.2.5.4	Nuts butters and nut spreads	50	25	200
5.4	Icings and frostings	20	10	—
6.3	Processed cereal and meal products-only grain bars, breakfast bars, breakfast cereals	50	25 (20) ^c	—
6.4	Flour products only for flour-based snacks e.g. pretzels, fritters and crackers; not noodles and pasta	10	5	—
6.4 ^b	Flour products including noodles and pasta	10	8	—
7.0 ^b	Breadcrumbs, tortillas (wheat or corn)	40	20	—
7.2	Fine bakery wares (e.g. biscuits, cakes, pastries)	40	20	200
8.1 ^b	Ground poultry	40	32	—
8.2	Processed meat, poultry and game products in whole cuts or pieces, fat content < 10%	15	7.5	—
8.2	Processed meat, poultry and game products in whole cuts or pieces, fat content > 10%	37.5	30 (18.75) ^c	—
8.2.3	Dried meat	150	120 (75) ^c	—
8.3.1 ^b	Non-heat-treated meat products-only dried sausages	40	30	—
8.3.1 ^b	Non-heat-treated meat products-only meat with a fat content < 10%, excluding dried sausages	100	80	100
8.3.1	Non-heat-treated meat products-only meat with a fat content > 10%, excluding dried sausages	37.5	18.8	150
8.3.1	Non-heat-treated meat products-only dehydrated meat	150	75	150
8.3.2	Heat-treated meat products-only meat with a fat content > 10%, excluding dried sausages	37.5	18.8	150
8.3.2	Heat-treated meat products-only dried meat	100	50	100
8.3.2	Heat-treated meat products-only dehydrated meat	150	75	150
12.0	Salts and condiments (not for sauces)	40	20	—
12.6	Sauces	50	25	100
15.1	Potato-flour-, cereal-flour- or starch-based snacks	20	10	50
15.2	Processed nuts	50	25	200

GSFA: General Standard for Food Additives; MPL: maximum permitted level.

^a From the report of the Eighty-seventh meeting (234).

^b Extension of use to the additional food category was proposed for Australia and New Zealand (237).

^c Use levels in parentheses were used at the Eighty-seventh meeting (234).

Table 3.2

Estimated dietary exposure to rosemary extract (expressed as carnosic acid plus carnosol) as a food additive for consumers only and from naturally occurring sources plus added sources from the Eighty-seventh meeting of the Committee, including new estimates from Australia and New Zealand

Population	Country or region	Age group	Dietary exposure (mg/kg bw per day)					
			Typical use level scenario		Dietary estimates from naturally occurring only		Naturally occurring plus typical use level scenario	
			Mean	High	Mean	High	Mean	High
Children	New Zealand	5–14 years	0.091	0.18	0.023 ^a	–	0.091 ^a	0.18 ^a
	New Zealand ^b	5–14 years	0.16	0.29	–	–	–	–
	USA	0–19 years	0.05–0.14	0.15–0.30	–	–	–	–
	Europe ^c	4 months–17 years	< 0.01–0.09	< 0.01–0.20	0.0–0.34	0.0–1.66	0.0–0.42	0–0.84 ^d
Adults	New Zealand	≥ 15 years	0.041	0.083	0.18 ^e	0.30 ^e	0.043 ^e	0.085 ^e
	New Zealand ^b	5–14 years	0.077	0.14	–	–	–	–
	USA	≥ 20 years	0.05	0.12	–	–	–	–
	Europe ^c	≥ 18 years	< 0.01–0.03	0.01–0.10	0.0–0.13	0.0–0.52	0.0–0.16	0–0.32 ^d
General population	Australia	≥ 2 years	0.043	0.093	0.048 ^f	0.081 ^f	0.045 ^f	0.096 ^f
	Australia ^b	≥ 2 years	0.082	0.15	–	–	–	–
	USA	All	0.06	0.14	–	–	–	–
Other	Europe	Supplement users only, ≥ 3 years	0.01–0.09	0.02–0.13	–	–	–	–

bw: body weight; USA: United States of America.

^a Proportion of consumers: naturally occurring only 0.1% ($n = 5$), natural plus added (both scenarios) 99% ($n = 3258$).

^b Based on typical use and requested extension of use.

^c The additive source is from the non-brand-loyal scenario.

^d High exposure is mean $\times 2$, approximating the 90th percentile (258).

^e Proportion of consumers: naturally occurring only 1% ($n = 52$), natural plus added (both scenarios) 99% ($n = 4620$).

^f Proportion of consumers: naturally occurring only 5.8% ($n = 477$), natural plus added (both scenarios) 99.6% ($n = 7701$).

4. Comments

4.1 Biochemical aspects

Several new studies were available. An in vitro study showed that the absorption and transportation of carnosic acid across Caco-2 cell monolayers mainly involves passive diffusion (238). In mice, carnosic acid was found in several tissues after 7 days of oral gavage exposure, with the highest tissue concentration in the caecum followed by the colon, stomach, spleen and liver (238). Carnosol was rapidly absorbed ($C_{\max}$ 30 min) and had a half-life of approximately 45 hours after a single oral administration in mice (240). This half-life for carnosol was longer

than that reported after oral gavage administration of rosemary extract in mice ($t_{1/2}$ 7.5 hours) (241) or in a study in rats with rosemary extract ($t_{1/2}$ 12–14 hours) evaluated previously by the Committee (258). Carnosic acid was not detected at any timepoint, indicating that carnosol was not metabolized to detectable concentrations of carnosic acid. Following oral administration of a rosemary extract (containing 40% w/w carnosic acid and 5% w/w carnosol) to mice, carnosol and carnosic acid were both rapidly absorbed ($C_{\max}$ < 15 minutes), but carnosic acid was eliminated more quickly than carnosol ($t_{1/2}$ 3.5 and 7.5 hours, respectively) (241).

Exposure to rosemary acetone extract (INS No. 392; 53.7% carnosic acid and 4.1% carnosol) induced the expression of CYP3A4 in 3D spheroid cultures of human primary hepatocytes and induced the expression of the homologous genes Cyp3a11 and Cyp3a23/3a1 in 3D spheroid cultures of primary hepatocytes from mice and rats, respectively. In human primary hepatocytes, rosemary extract also induced expression of CYP1A1, CYP2C19 (although the induction was minor), UGT1A1 and UGT1A4 (242). This indicates that there could be species differences in enzyme induction following exposure to rosemary extract. Previously, hepatic enzyme induction was reported in primary cultures of human hepatocytes following exposure to carnosic acid, as evidenced by upregulation of CYP2B6 and CYP3A4 mRNA levels. The induced expression of CYP3A4 is also in line with the results of a previous in vitro luciferase reporter gene assay that showed that carnosol is an activator for hPXR, mPXR and rPXR, whereas carnosic acid is a potent agonist of both hPXR and mPXR, but not rPXR (258).

In baculosomes expressing individual human CYP isoforms, carnosol inhibited CYP1A2, CYP3A4, CYP2C9, CYP2D6 and CYP2C19 (other CYPs were not investigated) (240). In another study using baculosomes, an ethanol extract of rosemary (carnosol and carnosic acid content not specified) inhibited CYP1A2 and CYP3A4 (243). No statistically significant changes were observed in protein expression of several transporters and metabolic enzymes in HepG2 and HT-29 cells exposed to carnosol (240).

The effects of rosemary extract (INS No. 392; 53.7% carnosic acid and 4.1% carnosol) on Cyp and UGT1A gene expression and CYP and UGT1A protein expression were investigated in rat pups (245). Measurements were performed in liver samples of pups from dams exposed to rosemary extract from GD 6 throughout lactation in a study on the effects of rosemary extract on thyroid hormone levels (245), further discussed below. Livers were collected from pups on PND 13 or after reaching sexual maturity. Any statistically significant increases (> 1.5-fold) in gene and protein expression were observed in the low-dose group only and were therefore not considered to be biologically relevant.

The Committee previously reviewed a study in adult female rats in which minimal increases (about 1.5-fold) were observed in total hepatic microsomal CYP content and in levels of hepatic CYP2A, CYP2C11, CYP2E1 and CYP4A

activity but not in CYP1A, CYP2B or CYP3A. The rats were treated with a supercritical CO₂ extract of rosemary (33% w/w carnosol plus carnosic acid content), at a dose equal to 195 mg/kg bw per day for 13 weeks (96). This enzyme induction was observed to be reversible following a 4-week treatment-free period. Elevated liver enzyme activity (glutathione-S-transferase and quinone reductase) was observed in mice and rats fed commercial extracts of rosemary in the diet at concentrations equivalent to up to 900 and 500 mg/kg bw per day for mice and rats, respectively, for 2–4 weeks, but not for carnosol (96).

4.2 Toxicological studies

Several additional toxicological studies were available. A hydro-ethanolic rosemary extract (carnosol and carnosic acid not specified) was of low acute oral toxicity in mice (LD₅₀ > 5000 mg/kg bw) (247).

No adverse effects were observed in a 28-day oral repeat-dose toxicity study with some deviations from the OECD Test Guideline No. 407 (250) in rats, conducted with an ethanolic extract of powdered rosemary leaves (carnosic acid and carnosol content unknown) (249). The NOAEL of this study was 1000 mg/kg bw per day, the highest dose tested.

In an oral prenatal developmental toxicity study in pregnant rats compliant with OECD Test Guideline No. 414 (191), rosemary extract (INS No. 392; acetone extract containing 53.7% carnosic acid and 4.1% carnosol) did not cause adverse effects on any maternal or fetal parameters. The NOAEL for maternal and developmental toxicity was the highest dose tested, that is, 304 mg/kg bw per day expressed as rosemary extract, and 176 mg/kg bw per day expressed as the sum of carnosol and carnosic acid.

In a study designed to address the recommendations of the Committee at its Eighty-seventh meeting (234), the potential effects of rosemary extract (INS No. 392; acetone extract containing 53.7% carnosic acid and 4.1% carnosol) on thyroid hormone levels in immature rats were examined (246). Rosemary extract was administered to female rats at dietary concentrations of 0, 1800, 3000 or 4200 mg/kg (equal to 0, 83, 133 or 205 mg/kg bw per day of carnosol and carnosic acid) from GD 6 to GD 19, and at doses of 0, 800, 1300 or 1900 mg/kg (equal to 0, 72, 125 or 176 mg/kg bw per day of carnosol and carnosic acid) from GD 20 throughout lactation to account for increased feed intake. Blood samples were collected for analysis of T₃, T₄ and TSH from subsets of pups on PND 4 (T₃ and T₄ only, because of small sample size), PND 13 and at termination (sexual maturation, PND 43 for males and PND 50 for females).

In the F1 generation, TSH levels were only statistically significantly decreased in males of the mid-dose group at PND 13. No differences were observed in TSH levels for female offspring at PND 13 or for male and female offspring at termination (TSH was not measured at PND 4). The Committee

noted that there was a large variation in the TSH measurements. At PND 4 and at termination, there were no statistically significant differences in serum T_3 and T_4 levels between the treated and control groups.

At PND 13, the mean T_3 level in females in the high-dose group was statistically significantly lower by 16% compared with the controls. For male offspring, a smaller decrease of 6% was observed in the T_3 level at the high dose, which was not statistically significant. T_4 levels were reduced in F1 males (by 12, 7 and 23% in the low-, mid- and high-dose groups, respectively; the high-dose reduction was statistically significant) compared with controls. In female offspring, statistically significant reductions in T_4 levels were observed in all treated groups (of 12, 13 and 24% in low-, mid- and high-dose groups, respectively) compared with controls. At termination, no statistically significant differences were observed for T_3 , T_4 and TSH levels for either male or female offspring. Since the effect of rosemary extract on thyroid hormones was transient and not accompanied by thyroid weight changes, histopathological findings in the thyroid or effects on sexual maturation, the study author did not consider the changes to be adverse.

The Committee noted that for T_4 , this new study replicated the findings of the earlier study (236,257) evaluated previously by the Committee (258), in which similar dietary concentrations of rosemary extract were given (2100, 3600 or 5000 mg/kg feed). In this earlier study, dose-related and statistically significant reductions in T_4 levels were seen in the F1 offspring on PND 13. In male F1 offspring, the mean T_4 reductions were 5, 17 and 27% in the low-, mid- and high-dose groups, respectively, compared with controls. In female F1 offspring, T_4 reductions of 6, 15 and 25% were observed in the low-, mid- and high-dose groups, respectively, compared with controls. All these reductions were statistically significant, except for in low-dose males. Similarly, in both studies, TSH and T_3 were not affected in a dose-related manner. The consistent results from these studies (236,246,257) show that the decrease in serum T_4 levels in pups at PND 13 is likely to be treatment related.

The classical thyroid inhibitor, PTU and other chemicals known to affect the thyroid, such as polybrominated diphenyl ethers, perchlorate, perfluorinated substances and triclosan, all reduce T_4 levels in the dams as well as the pups when treatment is administered during gestation and lactation. Although TSH levels are often unaffected, the reductions in T_4 are sustained throughout the treatment period (264) and are not transient as in the rosemary extract studies. In the study on rosemary extract evaluated previously by the Committee, paternal T_4 levels were measured at termination and were unaffected by treatment (236,257). In the later study (246), only parental females were treated and no measurements were made of maternal thyroid hormone levels. In the new prenatal developmental toxicity study (256), only females were treated and T_3 and T_4 levels measured at termination were unaffected.

Reductions in thyroid hormone levels, particularly during prenatal and/or postnatal development, are cause for concern. If these reductions occur at a specific stage of brain development, they may have an adverse effect. For example, in a publication by O'Shaughnessy et al. (265) using PTU as a thyroid inhibitor, a 40% decrease in pup serum T_4 on PND 14 was estimated to equate to a 70% decrease in pup brain T_4 . However, the magnitude, duration and timing of T_4 changes required to produce neurodevelopmental effects are not yet clear (266). There are several mechanisms by which T_4 may be reduced. In the absence of a clear effect on TSH levels and on thyroid weights and histopathology, decreases in T_4 could be caused by induction of liver enzymes, resulting in greater clearance of T_4 , rather than via a direct effect on the hypothalamic–pituitary–thyroid axis. In rats, the metabolism of thyroid hormones is mainly by phase II liver enzymes, UGTs (specifically UGT1A1 and UGT1A6) and SULTs. In the *in vitro* study by Anonymous (242) using primary rat hepatocyte cultures, rosemary extract induced protein expression of UGT1A1 but not UGT1A6, and did not induce gene expression for any of the UGT1A enzymes. *In vivo*, there were no dose-related effects on UGT1A protein expression or gene expression in PND 13 rat pups (245). The Committee also noted that no increases in liver weight or histopathological changes in the liver were observed in pups in that study, and that no clear effects were observed in CYP activity and protein expression in pups (245). Other possible mechanisms were not investigated. The mechanism and toxicological relevance of the transient reductions in T_4 levels on PND 13 therefore remains unclear.

In the absence of studies on neurodevelopmental toxicity, the Committee noted that this transient reduction in T_4 levels could possibly lead to adverse neurodevelopmental effects. The Committee performed dose–response modelling on the pup T_4 levels at PND 13 (see section on “Dose–response analysis” below).

Additional studies in pigs, rats and hyperthyroid rats were identified from the literature search (254,259,261,262). These were considered less relevant for the safety evaluation of rosemary extract as a food additive because of the test materials used and/or the study design.

4.3 Observations in humans

No new data relevant to the evaluation of rosemary extract as a food additive were available.

4.4 Assessment of dietary exposure

Dietary exposure to rosemary extract has been previously evaluated. The Committee at its Eighty-second meeting evaluated dietary exposure to rosemary extract (expressed as carnosic acid plus carnosol) based on maximum permitted and proposed levels (108). The Committee at its Eighty-seventh meeting assessed

dietary exposure to rosemary extract from naturally occurring sources, combined with dietary exposures from added sources at typical use levels (234).

The current Committee noted that additional exposure data, including a newly calculated dietary exposure to rosemary extract as a food additive from Australia and New Zealand based on typical use and proposed use levels (237), were available. The Committee also updated the previous exposure data from its Eighty-seventh meeting to include an estimation for high dietary exposure for Europe.

Overall, considering the exposure to the additive itself combined with the dietary exposure from naturally occurring sources, the high estimates were up to 0.84 mg/kg bw per day for children and up to 0.32 mg/kg bw per day for adults.

4.5 Dose–response analysis

The Committee performed benchmark dose modelling using the individual data on pup T_4 serum levels at PND 13 from the Anonymous study (246). Ideally, a numerical benchmark response (BMR) is chosen that would reflect the onset of an adverse effect relevant to humans. EFSA (267) has used a BMR of 20% for decreases in maternal T_4 levels based on changes resulting in neurodevelopmental effects in humans and in rat offspring, citing the following studies. A 25% decrease in maternal T_4 serum levels during the second trimester in humans resulted in neurodevelopmental and cognitive deficits in children (268). In rats, neurological effects were observed in dams and fetuses after a 20% decrease in maternal T_4 levels (269). In other studies in rats, a 10–17% decrease in maternal T_4 levels during pregnancy and lactation was associated with neurodevelopmental toxicity in rat offspring (270,271). The decreases in T_4 levels in pups were larger than the decreases in maternal T_4 levels in these studies.

The Committee is not aware of quantitative data relating decreases in pup T_4 levels, in the absence of reductions in maternal T_4 levels, to neurodevelopmental effects. In the absence of such data, the BMR of 20% used for decreases in maternal T_4 levels was also used for pups.

The analysis was performed using the benchmark dose (BMD) software PROAST, version 71.1 (RIVM, Bilthoven, the Kingdom of the Netherlands) and according to chapter 5 of EHC 240 (114). In the analysis, four models are included that meet the properties needed to analyse continuous data, including the estimation of the four model parameters described in EHC 240 (272). In the analysis, the sex of the pups was included as a covariate. The analysis fitted the Hill, exponential, inverse exponential and lognormal models. Model averaging was performed using 1000 bootstrap runs, and the relative weights for the Hill, exponential, inverse exponential and lognormal models were 0.2788, 0.2816, 0.2045 and 0.2352, respectively. Further details on the BMD analysis and the individual response data are included in [Annex 6](#).

Table 4.1

Model-averaged BMDLs and BMDUs for reductions in serum T₄ levels in pups on PND 13 following exposure of dams from GD 6 throughout lactation to rosemary extract, expressed in mg/kg bw per day carnosol and carnosic acid

Sex	BMDL ₂₀	BMDU ₂₀
Male	97.2	213
Female	126	247

BMDL₂₀/BMDU₂₀: lower/upper 95% confidence limit on the benchmark dose for a 20% response.

Table 4.1 shows the BMD values. The BMD confidence intervals (i.e. the interval between the lower 95% confidence limit [BMDL] and the upper 95% confidence limit [BMDU]) are approximately a factor 2 for both sexes, indicating that the data are sufficiently informative to derive a reliable BMDL₂₀ (BMDL for a 20% response).

The Committee selected the BMDL₂₀ of 97 mg/kg bw per day in male pups for consideration in the Evaluation.

5. Evaluation

No maternal or embryo-fetal toxicity was observed in a new prenatal developmental toxicity study in rats. A separate new study on thyroid hormone levels in rat pups from dams treated with rosemary extract in the diet during pregnancy and lactation confirmed the results of a previous study showing that rosemary extract exposure resulted in decreased serum T₄ levels on PND 13 in male and female pups. A BMDL₂₀ of 97 mg/kg bw per day expressed as carnosic acid plus carnosol was calculated for this study. This BMDL₂₀ is higher than the NOAEL of 64 mg/kg bw per day, expressed as carnosic acid plus carnosol, on which the current temporary ADI for rosemary extract is based. This NOAEL of 64 mg/kg bw per day was the highest dose tested in a 90-day toxicity study in rats. The Committee considered that this NOAEL of 64 mg/kg bw per day is still the most appropriate point of departure, since higher doses of rosemary extract have only been tested in studies of reproductive and developmental toxicity of shorter duration, and have not been tested in studies of chronic toxicity. In the derivation of the temporary ADI of 0–0.3 mg/kg bw per day, an additional uncertainty factor of 2 was incorporated into the overall uncertainty factor of 200 to account for the absence of studies to elucidate the potential developmental and reproductive toxicity of rosemary extract. This factor is no longer necessary. The temporary ADI of 0–0.3 mg/kg bw per day (expressed as carnosic acid and carnosol) was therefore withdrawn and an ADI of 0–0.6 mg/kg bw per day (expressed as carnosic acid and carnosol) was established.

Estimated high dietary exposures to rosemary extract (expressed as carnosic acid plus carnosol) based on naturally occurring sources combined with dietary exposures from added sources at typical use levels were up to 0.84 mg/kg bw per day for children and up to 0.32 mg/kg bw per day for adults. For children, these exceeded the ADI of 0–0.6 mg/kg bw. The ADI is based on the highest dose tested in a 90-day toxicity study in rats. The only adverse effect observed in reproductive and developmental toxicity studies in rats using higher doses was a reduction in T₄ hormone levels in rat pups after exposure during pregnancy and lactation. The MOE between the BMDL₂₀ for reduction in T₄ levels in pups and the estimated high dietary exposure for children is greater than 100, and is considered sufficient. The Committee therefore did not consider the slight exceedance of the ADI, which was only by children, as a safety concern. The Committee concluded that the use of rosemary extract as a food additive does not raise any safety concerns.

Thaumatococcus

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1. Explanation

At its Fifty-fourth session, the CCFA requested that the Committee conduct a safety assessment of thaumatin II and, if necessary, revise the specifications of thaumatin to include thaumatin II produced recombinantly in green plants such as spinach, lettuce, red beet and *Nicotiana benthamiana* (44).

Thaumatococcus II is one of the two major components of thaumatin (INS No. 957). Thaumatococcus is a mixture of intensely sweet proteins extracted with water from the arils of the fruit of the West African perennial plant *Thaumatococcus daniellii* (Benth). Thaumatococcus functions as a high-intensity sweetener and flavour enhancer.

Based on the information provided by the sponsor, the Committee evaluated thaumatococcus II produced by recombinant expression of the thaumatococcus II gene in transgenic lines of *Nicotiana tabacum*.

Thaumatococcus has previously been considered by the Committee at its Twenty-seventh and Twenty-ninth meetings (273,274). At its Twenty-ninth meeting, the Committee established an ADI “not specified”¹⁴ based on metabolism of normal food components and a lack of adverse effects in toxicological and human studies.

No new toxicological studies on thaumatococcus or recombinant thaumatococcus II were submitted by the sponsor. The only information submitted specific to thaumatococcus II was a bioinformatic analysis for allergens. Additionally, information was submitted pertaining to the residuals present, which included identifying the residual host plant proteins in the final product and their bioinformatic analyses for comparison to allergens.

A literature search was conducted spanning the period 1985–February 2025 in PubMed and Google, using the search terms “thaumatococcus”, “toxicity”, “acute toxicity”, “reproductive toxicity”, “developmental toxicity”, “genotoxicity”, “chronic toxicity”, “carcinogenicity” and “allergenicity”. The search yielded 872 references. One new study was identified that was relevant to the assessment and that had not been previously evaluated by the Committee.

A literature search was also undertaken to identify publications that included concentration or use level data for thaumatococcus or estimates of dietary exposure to thaumatococcus (see Section 3.1 for details). A small number of publications was identified and reviewed by the Committee.

¹⁴ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

1.1 Genetic information

N. tabacum cultivar Burley B5 plants that are devoid of endogenous alkaloids, nicotine and anabasine are used as the host to produce thaumatin II in the seeds. A modified thaumatin II gene is inserted into the host DNA via *Agrobacterium tumefaciens*-mediated transformation. This is accomplished with a plasmid carrying cassettes for the expression of DL-phosphinothricin resistance as a selection marker, and the expression of thaumatin II protein in the seeds. The presence of the thaumatin II transgene, genomic insertion sites and DNA copy number in the transformants was confirmed by PCR and DNA sequencing in the T0 and T1 generation candidates. The insertions were confirmed to be in different loci, and no tandem or inverted repeats were detected. Harvested T2–T3 generation plants were analysed to verify that there was no segregation of the transgene and then used for seed production by self-pollination. T4 generation seeds form the basis for plant propagation and thaumatin II production. The *N. tabacum* seeds are periodically assessed for germination rate, absence of pathogens, retention of genomic insertions and thaumatin II protein expression.

1.2 Characterization of thaumatin

Thaumatococcus (INS No. 957) is a 2:1 mixture of thaumatin I and thaumatin II. The chemistry and applications of thaumatin have been extensively reviewed (275–282). The proteins share similar properties, amino acid composition, sweetness and molecular weight (of about 22 kDa), only differing by five amino acids. Each consists of a single polypeptide chain of 207 amino acids with eight intramolecular disulfide bonds (280). X-ray crystallography studies of thaumatin I and II have elucidated their backbone structure (276,283,284). Circular dichroism analysis of thaumatin I (285) indicated a few α -helices, but numerous β -pleated sheets and bends. Disruption as a result of heat denaturation or cleavage of disulfide bonds leads to a loss of sweetness, suggesting that the tertiary structure of both the proteins is essential for taste modulation (277,286,287).

1.3 Characterization of thaumatin II

The amino acid sequence of seed-expressed thaumatin II is identical to that from *T. daniellii*.

The thaumatin II article of commerce is an odourless, white to off-white powder that is highly water soluble (> 20% w/v) and 3000–10 000 times sweeter than sucrose (288). The amino acid sequence and molecular mass were verified to match that of a thaumatin II working standard by four different analytical methods, namely: (i) capillary gel electrophoresis (CGE); (ii) matrix-assisted laser desorption ionization (MALDI) time-of-flight mass spectrometry (MS); (iii) in-source decay (ISD) MS; and (iv) tryptic MALDI MS peptide mapping.

The average molecular masses determined by MALDI MS matched the values in reference databases. Results from ISD-MS analysis of thaumatin II confirmed the mass of the protein to be 22 287 Da, and the lack of truncations or terminal modifications. MALDI MS of the tryptic digest of purified thaumatin II confirmed that accessory sequences and the signal peptide were cleaved to yield the expected mature protein. In totality, the average molecular mass of $22\,287 \pm 21.5$ Da from these methods matches the values in the literature (289,290) for thaumatin II from *T. danielli* (22 293 Da).

The purity of thaumatin II was confirmed by CGE and HPLC to be 95.1–96.3% and 95.4–97.2%, respectively.

1.4 Chemical and technical considerations

The tobacco plants expressing thaumatin II in the seeds are grown using standard agricultural methods (291), then harvested and cleaned according to current Good Agricultural and Collection Practices (292). The seeds are crushed and cold-pressed to remove the oil. The defatted seed cake is treated with an acid solution ($\text{pH} < 4.5$) to remove host proteins and other seed components. The thaumatin II in the resultant supernatant is further purified by salt precipitation and multiple filtration steps. The article of commerce is stabilized and standardized with food-grade ingredients; it is either spray-dried or lyophilized as a dry, white to off-white powder.

The total protein content for the article of commerce is determined by colourimetry, using the Bradford assay and a bovine serum albumin standard. Thaumatin II purity is determined by HPLC and ultraviolet–visible spectroscopy. The percent total thaumatin II protein in the bulk product (w/w) is calculated based on the data obtained for total protein and purity of thaumatin II. The tobacco-seed-produced thaumatin II does not contain any residual DNA or any tobacco alkaloids in the final product.

The specifications for thaumatin II include levels for protein (not less than 75% calculated based on the data obtained for total protein [$\geq 80\%$], purity of thaumatin II [$\geq 95\%$], pH [3.0–6.0], moisture [$< 5\%$ w/w] and ash [$< 10\%$ w/w]), limits for heavy metals (≤ 0.1 mg/kg each for Cd and Hg, 0.2 mg/kg for Pb and 1 mg/kg for As) and microorganisms (*Escherichia coli*, *Pseudomonas aeruginosa*, *Salmonella* and coagulase-positive *Staphylococcus aureus* absent in 25 g final product). Thaumatin II is stable at ambient temperatures for at least 6 months.

2. Biological data

The predominant proteins in food additive INS No. 957 are thaumatin I and thaumatin II. The proteins share similar properties, amino acid composition, sweetness and molecular weight (of about 22 kDa), only differing by five amino

acids. The amino acid sequence of thaumatin II produced by recombinant methods is identical to that of thaumatin II extracted from the arils of the fruit of *Thaumatococcus daniellii* (Benth). HPLC demonstrated that thaumatin II comprises more than 95% of the total protein.

On that basis, the toxicological database for thaumatin was considered relevant to the evaluation of the safety of thaumatin II produced recombinantly in *N. tabacum*. With the exception of one 13-week study in rats, all of the available studies had been evaluated by the previous Committee.

At its Twenty-ninth meeting (293), the Committee evaluated studies on thaumatin that included metabolism, acute toxicity, short-term toxicity, teratogenicity, mutagenicity, allergenicity and studies in humans. The Committee also noted the long history of human use of the fruit from *T. daniellii* as a sweetener and the absence of toxic effects following its ingestion. Additional details of these studies were provided in the published review of Higginbotham et al. (294). The thaumatin test materials used had a protein content of 78–85%.

2.1 Biochemical aspects

2.1.1 Absorption, distribution and elimination

No new studies were available.

2.1.2 Metabolism

No relevant new studies were available. The previous Committee (293) evaluated the metabolism of thaumatin in vitro and in vivo.

(a) In vitro studies

In an in vitro study using a rapid multienzyme system with purified mammalian digestive enzymes (trypsin, chymotrypsin, peptidase), thaumatin was digested more rapidly than egg albumin (293,295).

EFSA (296) reviewed an unpublished report investigating the in vitro digestibility of thaumatin compared with whey protein. Sodium dodecyl-sulfate polyacrylamide gel electrophoresis (SDS-PAGE) analysis showed that thaumatin and whey protein were readily hydrolysed by pepsin and pancreatin. This study was not available to the Committee.

(b) In vivo studies

To assess the absorption of thaumatin proteins, a nitrogen balance study was performed in rats (294). Ten male Charles River rats (body weight 140–160 g) were fed a diet containing 10% protein from egg albumin for 10 days, followed by a diet containing 5% egg albumin and 5% thaumatin for another 10 days. The presence of egg albumin was needed to compensate for the lack of histidine in

thaumatin, an essential amino acid in the rat. A second group of 10 male rats was given the same diets in reverse order. Using this cross-over design, each animal served as its own control. During the last 5 days of each 10-day diet exposure, food consumption was measured, and food and excreta were analysed for nitrogen using the Kjeldahl method. The total nitrogen intake (N) and the nitrogen eliminated in the faeces (F) were calculated for each animal. Protein digestibility (D) was calculated using the formula $D = (N-F)/F$ and expressed as a percentage. Nitrogen digestibility was approximately 90% for both diets, indicating that the digestibility of thaumatin is comparable to that of egg albumin and that the majority of these proteins are metabolized and absorbed.

2.2 Toxicological studies

2.2.1 Acute toxicity

The acute oral toxicity of thaumatin in mice and rats was low (293). The results are summarized in [Table 2.1](#).

Table 2.1

Acute toxicity of thaumatin

Species	Sex	Test substance	Route	LD ₅₀ (mg/kg bw)	Reference
Mouse	Not specified	Thaumatococcus (78–85% protein)	Gavage	> 20 000	Ben-Dyke (297)
Rat	Not specified	Thaumatococcus (78–85% protein)	Gavage	> 20 000	Ben-Dyke and Joseph (298)

LD₅₀: median lethal dose.

2.2.2 Short-term studies of toxicity

At its Twenty-ninth meeting the Committee evaluated two 13-week studies in rats, a 4-week follow-up study in rats investigating thyroid function and a 13-week study in dogs (293,294). These studies were conducted before the development of OECD guidelines for toxicity testing and GLP.

An additional 13-week study in rats conducted according to guidelines and GLP, not previously considered by the Committee, was identified in a literature search (299).

(a) Rats

In a preliminary 13-week study previously evaluated by the Committee (293), male and female CD rats (10 per sex per group) were fed thaumatin at levels of 0, 1.0, 4.0 or 8.0% w/w in the diet (equivalent to 0, 1000, 4000 or 8000 mg/kg bw per day). The control groups were given the basal diet supplemented with 8.0% casein w/w to compensate for the high protein intake in thaumatin-treated animals. No deaths occurred during the study. Compared with the casein-fed

controls, food consumption was 5–11% lower in mid- and high-dose males and females. No changes in body weight were observed in females. Mean body weight gains of mid- and high-dose males were 6 and 9% lower than in controls. A statistically significant decrease in haemoglobin was observed in high-dose females; however, the previous Committee noted that the values were within the range normally found for animals of comparable age and weight. A statistically significant increase in both the absolute (14%) and relative (17%) liver weights was observed in high-dose females. No treatment-related adverse macroscopic or microscopic effects were observed in any of the treated groups. No differences were observed in tests for hepatic or renal function. Antibodies to thaumatococcus were not detected in the serum of treated animals in week 13.

In a second 13-week study (294) previously evaluated by the Committee, thaumatococcus was fed to male and female CD rats (20 per sex per group) at dietary levels of 0, 0.3, 1.0 or 3.0% w/w (equal to 0, 243, 861 or 2417 mg/kg bw per day for males and 0, 293, 1000 or 2822 mg/kg bw per day for females). Animals were assessed daily for mortality and clinical signs. Body weights and food consumption were recorded weekly. Water intake was measured on three consecutive days, before treatment and again during weeks 6 and 12. Ophthalmoscopy was conducted before treatment on all animals and during week 12 for control and high-dose groups. Blood samples were obtained during weeks 4 and 12 (10 per sex per group) for assessment of haematological and blood chemistry parameters. Urine collected during the final week of treatment was analysed for the presence of blood, glucose, protein, ketones and bilirubin. At the end of the study animals were killed and subjected to gross necropsy. Selected tissues were collected, weighed and examined microscopically. Statistical analyses were performed using the Student's *t*-test.

No mortality or clinical signs were observed during the treatment period. Food consumption in high-dose females was 7% lower than in controls. No treatment-related changes in body weight were observed.

Haematocrit (the proportion of RBCs in the blood) showed significant increases of 8 and 13% at week 12 in mid- and high-dose males, respectively. Significant reductions of 10 and 12% were seen in low- and mid-dose females, respectively. A dose-related reduction of up to 60% in triglyceride levels was observed in females during weeks 4 and 12 of treatment. Treated females showed an 8% increase in kidney weight, which was a 13% increase relative to body weight, at all doses. Thyroid weights were significantly higher in all treated male groups but within the historical control range. Thyroid weights were significantly lower in all treated female groups, both in absolute and relative weight. [Table 2.2](#) summarizes the changes observed.

Table 2.2

Summary of changes observed in 13-week dietary rat study

Dose (mg/kg bw)	Haematocrit (%)		Triglycerides (mg/100 mL)		Kidney weight (g/100 g bw)		Thyroid weight (g/100 g bw)	
	Males	Females	Males	Females	Males	Females	Males	Females
243	–	↓	–	↓	–	↑	↑	↓
861	↑	↓	–	↓	–	↑	↑	↓
2417	↑	–	–	↓	–	↑	↑	↓

No differences were seen upon ophthalmoscopic examination, macroscopic examination of all animals at necropsy, or microscopic examination of tissue sections from the control and high-dose groups. Microscopic examination of the thyroid from all animals in all groups did not detect any adverse effects with treatment.

To investigate possible hypo- or hyperthyroid effects, an additional rat study was conducted. Male and female CD rats (10 per sex per group) were fed 3% thaumatin or 3% egg albumin in the diet for 4 weeks. At the end of the treatment period, no statistically significant differences in thyroid hormone levels were observed between the two groups, based on the levels of T_3 and T_4 measured.

At its Twenty-ninth meeting the Committee noted that, although variations in thyroid weights were demonstrated in a 13-week rat study (increases in males and decreases in females), no treatment-related differences were observed upon histological analysis or in thyroid hormone levels (T_3 and T_4). The NOAEL was 3% w/w in the feed, the highest dietary concentration tested, equivalent to 2417 mg/kg bw per day.

A short-term study in Crj:CD (SD) rats (299) that investigated the safety of irradiating thaumatin for sterilization was not previously evaluated by the Committee.

Thaumatococcus (molecular weight approximately 22 kDa; purity not specified; sterilized by electron beam irradiation) was administered to male and female rats (10 per sex per group) in the diet at 0, 0.3, 1.0 or 3.0% w/w for 13 weeks (equal to 0, 260, 788 or 2502 mg/kg bw per day for males and 0, 299, 1042 or 2889 mg/kg bw per day for females). An additional group of rats received 3.0% non-irradiated thaumatin in the diet. The study was conducted in compliance with the GLP standards (300) and guidelines (301) of the Japanese Ministry of Health and Welfare. No mortality or treatment-related clinical signs were observed. No differences in food or water consumption or in body weight were observed between control and treated groups. No treatment-related adverse effects were observed for organ weights, urinalysis, ophthalmoscopic examination, haematology or blood biochemistry, or upon macroscopic or microscopic

examination at necropsy. The NOAEL was 3.0% w/w in the feed, the highest dietary concentration tested, equivalent to 2502 mg/kg bw per day.

(b) Dogs

In a study previously evaluated by the Committee, thaumatococcus was fed to male and female beagle dogs (four per sex per group) at dietary levels of 0, 0.3, 1.0 or 3.0% w/w (equal to 0, 140, 450 or 1400 mg/kg bw per day) for 13 weeks (294). Animals were assessed daily for mortality and clinical signs. Body weights and food consumption were recorded weekly. Ophthalmoscopy was conducted before treatment on all animals and during week 12 for control and high-dose groups. Blood samples were obtained during weeks 4 and 12 and assessed for haematological and blood biochemistry parameters. Urine collected during the final week of treatment was analysed for the presence of blood, glucose, protein, ketones and bilirubin. Animals were killed at the end of the study, and gross necropsy was conducted. Organs were weighed and tissue samples underwent histological examination for all animals.

No deaths occurred during the study period and no adverse clinical signs were observed. Food and water consumption were not affected by treatment. Slight decreases (values not specified) in haemoglobin concentrations, erythrocyte counts and haematocrit were observed in high-dose males during weeks 4 and 12 of treatment. However, these values were stated to be within the range of historical controls. High-dose males showed an increase in absolute but not relative liver weight. No differences were seen in other blood parameters, urinalysis, ophthalmoscopic examination, macroscopic examination of all animals at necropsy, or microscopic examination of tissue sections from the control and high-dose groups. The previous Committee established a NOAEL of 3.0% w/w in the diet, equivalent to 1400 mg/kg bw per day.

2.2.3 Long-term studies of toxicity and carcinogenicity

No studies were available.

2.2.4 Genotoxicity

At its Twenty-ninth meeting (293), the Committee reviewed two genotoxicity studies that were conducted before the development of OECD or GLP guidelines for genotoxicity testing. [Table 2.3](#) summarizes the results.

Table 2.3

Genotoxicity of thaumatin in vitro and in vivo

End-point	Test system	Route of administration	Concentration or dose	Result	Reference
In vitro reverse mutation	<i>Salmonella typhimurium</i> TA98, TA100, TA1535 and TA1537, and <i>Escherichia coli</i> WP2	Not applicable	0.05–50 mg/plate $\pm$ S9	Negative ^a	Higginbotham et al. (294)
In vivo dominant-lethal assay	Male CD-1 mouse	Gavage	200 or 2000 mg/kg bw per day (5 days)	Negative ^b	Higginbotham et al. (294)

^a Positive controls (methyl nitrosoguanidine, 2-nitrofluorene, 9-amino-acridine, benzo[a]pyrene and 2-aminoanthracene) confirmed the validity of the assay.

^b Following treatment, each male was paired with three untreated females for up to 1 week. New females were introduced each week following treatment, for seven consecutive weeks. Females were killed 4, 11 or 18 days post-mating. Compared with the negative control, no adverse effects were seen in the thaumatin group with regards to mating, fertility, development and survival of zygotes, or pre- and post-implantation losses. The positive control (trimethyl phosphate) had an adverse effect on all parameters.

2.2.5 Reproductive and developmental toxicity

In a study previously reviewed by the Committee, pregnant CD rats (20 per group) received thaumatin by gavage during GD 6–15 at doses of 0, 200, 600 or 2000 mg/kg bw per day (294). The control group received the vehicle (distilled water). All rats were killed on GD 21. No treatment-related effects were observed on the number of corpora lutea, the number and condition of the implantations, pre- and post-implantation losses, or litter size. Thaumatin treatment did not affect fetal weight or increase the frequency of visceral or skeletal abnormalities.

2.2.6 Special studies**(a) Allergenicity**

Bioinformatic analysis of the amino acid sequence for thaumatin II (302) using the AllergenOnline database (31) did not show sequence homology with any known allergens. The search incorporated a FASTA full-length scan, an 80-mer sliding window sequence search with a 35% identity cut-off limit (303) and an 8-mer exact sequence match search.

At its Twenty-ninth meeting (293), the Committee evaluated studies on potential allergenicity that measured the release of histamine, directly or indirectly.

(i) In vitro

In a study previously evaluated by the Committee, peritoneal mast cells obtained from control rats were incubated at 37 °C for 5 minutes with a range of concentrations of thaumatin or the positive control synthetic polypeptide, synacthen (294). The amount of histamine released into the incubation medium was measured fluorometrically. The positive control protein liberated 50% of the

available histamine from the purified rat mast cells at a concentration of 2 μ M. A concentration of 1 mM of thaumatin was required to elicit the same response.

(ii) Rats

In a study previously evaluated by the Committee, rats (five per group) received subcutaneous injections at four separate sites of either 10 μ g thaumatin or 10 μ g egg albumin, together with Freund's adjuvant (294). Animals were killed after 11 or 12 days. The passive subcutaneous anaphylaxis test was conducted, in which serum (0.1 mL) of these sensitized animals was administered intradermally to control rats who then received an intravenous injection of thaumatin (2 mg) and Evans blue dye (2%) 48 hours later. Thaumatin produced a weaker response than egg albumin, based on induction of a blueing response from release of histamine.

(iii) Guinea pigs

In a study previously evaluated by the Committee, guinea pigs (five per group) received intramuscular injections of either 50 mg thaumatin or 50 mg egg albumin, together with an adjuvant (1.3% alhydrogel) (294). Animals were killed between 12 and 30 days later. Sections of ileum from each sensitized animal were transferred to a Schultze-Dale bath (containing oxygenated saline solution) and challenged with either thaumatin or egg albumin at a range of doses. In this assay, tissue contraction is elicited by the reaction of antibody with antigen and involves the release of histamine (304). The minimum dose of thaumatin capable of inducing a response was 250 ng, which was comparable to the results with egg albumin. When guinea pigs were sensitized with the protein and Freund's complete adjuvant, thaumatin was slightly more sensitizing than egg albumin.

2.3 Observations in humans

At its Twenty-ninth meeting the Committee evaluated several studies conducted in humans. The Committee also noted the long history of consumption of thaumatin in the absence of evidence of toxic effects on humans.

2.3.1 Clinical studies

In a study previously evaluated by the Committee, the effects of thaumatin on haematological and blood biochemistry parameters were investigated in a clinical study (293). Volunteers (18 men, 12 women) were randomly assigned to two groups. Each participant ingested one capsule each morning, containing either 280 mg thaumatin or 210 mg egg albumin, for 13 consecutive weeks. Blood was collected and analysed prior to the start of the trial and after 4, 8 and 12 weeks. No treatment-related changes were observed in either of the groups.

A literature search identified a clinical study not previously evaluated by the Committee, which investigated the glycaemic effect of thaumatin (305). Volunteers (10 healthy, eight diabetic) ingested 9.62 mg thaumatin as a liquid solution each day for 4 weeks. After 1 and 4 weeks, blood glucose levels were measured prior to dosing and up to 2 hours post-dosing. No effect on glucose levels was measured in the healthy subjects; a decrease was observed in the diabetic subjects. No adverse reactions were reported.

2.3.2 Allergenicity

In a study previously evaluated by the Committee, gelatine capsules containing thaumatin or lactose were given to volunteers for a period of 14 days at a dose of 100 mg/day, using a double-blind cross-over design (294). The four women and six men were randomly assigned to one of the two groups (five per group). One group received thaumatin during the first 14-day period, followed by lactose for a second 14-day period; the other group received lactose first and then thaumatin. Before the study, the subjects were prick-tested for common allergens and thaumatin. No sensitization with skin prick testing was observed either as a result of the prick test exposure with thaumatin before the study began, or following the oral exposure to thaumatin. Additionally, blood was taken from the volunteers at the start and end of the 28-day study period. No antibodies to thaumatin were detected in the sera by use of the passive subcutaneous anaphylaxis test in a baboon or rhesus monkey (i.e. no blueing response).

In a study previously evaluated by the Committee, thaumatin was assessed for oral sensitivity and irritation using chewing gum containing thaumatin at 150 ppm (293). Volunteers (25; sex not specified) each chewed five gum sticks (5.3 g each) per day, for 15 minutes per stick, over a period of 28 days. Daily exposure to thaumatin was calculated to be 3.98 mg per day. The control group (25; sex not specified) received chewing gum that did not contain thaumatin. No visible signs of irritant or allergic responses were detected on the oral mucosa of either group. No weal or flare reactions were observed in any volunteers with prick-testing, either before or after the treatment period.

In a study previously evaluated by the Committee, prick-testing was conducted in laboratory personnel who had inhaled thaumatin intermittently for periods of up to 7 years (294). About one-half (67/140) responded to common inhalant allergens. A positive response to thaumatin was observed in 13 subjects, all except one of whom were atopic or allergic.

2.4 Thaumatin II impurities

Thaumatin II is expressed in the seeds of *N. tabacum* that are naturally devoid of alkaloid toxicants (alkaloids nicotine, nornicotine and anabasin), ensuring there are no residual host alkaloids in the product. The *N. tabacum*-expressed

thaumatin II product contains more than 95% thaumatin II of total protein. The residual protein (< 5%) in the final product comprises host proteins derived from tobacco seed.

The residual tobacco seed proteins present in the thaumatin II product were identified by the sponsor; most are common to other plant species consumed in the human diet. The amino acid sequences of the residual tobacco seed proteins were compared with known allergens using the AllergenOnline (31) and Allermatch (306) databases. No proteins of allergenic concern were identified. One protein had a moderate similarity to an allergic protein from tomato seed. No residual DNA was detected in the final product (307).

Other host impurities present may include fatty acids, carbohydrates, minerals and ash, all of which are found in many foods and are not of toxicological concern.

2.4.1 Tobacco seed safety

The nutritive properties and chemical composition of tobacco seeds have been examined, and its use as a feed source for animals proposed (308). Protein-rich tobacco seed cake, the fraction remaining after grinding and oil extraction, has been licensed as a feed supplement in the EU since 2010 (309).

No adverse effects were observed in rats, pigs or lambs consuming tobacco seeds as feed (310–313). Rossi et al. (311) investigated the Burley variety tobacco seed cake as a protein source for piglets during the weaning phase. When healthy piglets were fed 4% tobacco seed cake for 43 days, there were no significant differences in growth, average daily feed intake, feed efficiency or typical blood parameters used to measure general health status compared with controls.

3. Dietary exposure

3.1 Introduction

Thaumatococcus was assessed by the Committee at its Twenty-ninth meeting in 1985 (274,293); however, that Committee did not undertake an evaluation of dietary exposure. There was only one reference to dietary exposure from a study submitted to WHO, which noted an anticipated maximum daily intake of 1–2 mg/person per day (293). That Committee also commented that thaumatococcus makes an insignificant contribution to normal protein intakes.

For the current meeting the Committee evaluated a number of submissions that included information related to the use of and dietary exposure to thaumatococcus. Submissions were received from the Calorie Control Council, ICA, ICGA and the International Council of Beverages Associations (ICBA).

A literature search was also undertaken to identify publications that included estimates of dietary exposure to thaumatococcus. EBSCO Discovery Service

was used, and databases including Medline, Food Science Source, Food Science and Technology Abstracts, and Science Direct were searched as well as a number of science, toxicological, food, nutrition and public health journals. Search terms included “thaumatin” and “dietary intake”, “intake”, “dietary exposure”, “exposure”, “consumption” or “risk assessment”. A literature search was also conducted to identify publications containing use level or concentration data. Using the same databases noted above, search terms included “thaumatin” and “concentration”, “content”, “occurrence”, “use level” or “quantification”. These search terms were also used in a general internet search to capture other articles or reports not included in the scientific literature.

The literature searches resulted in a small number of additional publications on use levels or concentrations of thaumatin, or dietary exposure estimates, not already submitted to the Committee. The reference lists of publications reviewed were also screened to identify additional relevant articles. Review papers assessing intense sweeteners were reviewed and did not include any additional information in relation to dietary exposure to thaumatin (314,315). Original publications that were not in English were not included in those assessed by the Committee.

Some of the estimates of dietary exposure to thaumatin and use level information did not specify the form of thaumatin to which they related; they described the food additive simply as “thaumatin”. The producer of the thaumatin II under evaluation notes that the new production process yields thaumatin products of equal sweetness potency to products produced by old production methods, and can therefore be used at the same rates in foods (307). For the purpose of this assessment, dietary exposure estimates and use level or concentration data reviewed for thaumatin have therefore been assumed by the Committee to be equally relevant to thaumatin II, and reflective of estimates of dietary exposure to and use levels of thaumatin II.

3.2 Use of thaumatin

Thaumatin is permitted for use globally as a food additive including as a sweetener and flavour enhancer. It can be used singly or in combination with sugar or other caloric or non- or low-caloric sweetening agents (307). For example, thaumatin was noted to occur with steviol glycosides in a table-top sweetener product (316).

Within the GSFA (317), thaumatin (INS No. 957) is permitted for use as a flavour enhancer and sweetener at GMP in a number of specified food categories including: 14.1.5 Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa; 4.2.2.7 Fermented vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweed products, excluding fermented soybean products of food categories 06.8.6, 06.8.7, 12.9.1, 12.9.2.1 and 12.9.2.3; 11.4 Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings); and 12.1.2 Salt substitutes.

Thaumatococcus is a food additive that is also included in table 3 of the GSFA (317), meaning it can be used in accordance with GMP in over 60 other food categories including cereal-based products, water-based flavoured drinks, confectionery (includes chewing gum), fruit and vegetable products, processed meats, preserved fish and products, condiments and sauces, dairy products, table-top sweeteners, dietary supplements and prepared foods.

Permissions for use of thaumatococcus also exist in specific countries including Australia and New Zealand; Brazil; Canada; China; Hong Kong, Special Administrative Region (China); Israel; Japan; Mexico; the Republic of Korea; Singapore; South Africa; Switzerland; the USA; and the EU (318).

In Australia and New Zealand, thaumatococcus is permitted for use as a sweetener at GMP in over 50 food classes including dairy products, edible oils and oil products, processed fruit and vegetables and products, chocolate and cocoa products, confectionery, cereal products, processed meats and seafood, table-top sweeteners, special-purpose foods, beverages and mixed foods (14,15). A study assessing food labels and the use of intense sweeteners in Australia and New Zealand (319) indicated that there was a low use of thaumatococcus in foods: four out of 205 (2.0%) products reviewed in Australia and two out of 190 (1.1%) products reviewed in New Zealand were found to contain thaumatococcus.

In the Republic of Korea, one study analysed foods from a local market to determine the level of thaumatococcus in foods for which thaumatococcus was included on the ingredient list (320). In 14 samples of Soju (an alcoholic beverage), thaumatococcus was either not detected or detected at a concentration of up to 0.14 mg/kg, with a mean of 0.11 mg/kg for the three positive samples. In two samples of fruit and vegetable beverages, thaumatococcus levels were measured as 2.81 or 3.95 mg/kg (average 3.38 mg/kg). In two samples of confectionery, thaumatococcus was either not detected or measured at 28.54 mg/kg.

In the USA, thaumatococcus II has been granted GRAS approval as a sweetener (288) and flavour modifier (288). Intended GMP use levels in a range of foods include wine, beer and other fermented or distilled beverages (5 mg/kg); jams, jellies and marmalades (5 mg/kg); potato-based and similar snacks (10 mg/kg); breakfast cereals (10 mg/kg); chewing gum (10 mg/kg); ice cream and other edible ices (50 mg/kg); cocoa and chocolate products (50 mg/kg); breath mints and similar (50 mg/kg); confectionery (50 mg/kg); decorations, coating and fillings (50 mg/kg); fine bakery items (30 mg/kg); food supplements (capsules, tablets, syrups and similar) (400 mg/kg); and table-top sweeteners (288). Use of thaumatococcus II as a flavour modifier has been noted to be between 1 and 150 mg/kg, with most foods having an average use level of 2–7 mg/kg (321).

In the EU, thaumatococcus is permitted for use as a sweetener and flavour enhancer. Permissions are for 15 food categories with maximum levels ranging from 0.5 to 400 mg/kg and at GMP (*quantum satis*) in three categories of table-top

sweeteners (318). Only one use level of thaumatin was submitted to EFSA (318) for table-top sweetener tablets: 100 mg/kg. A review of sweeteners published in 2018 (314) noted no use in Belgium based on a food label survey. Thaumatin was not identified in commercial brands marketed in Portugal (322). Thaumatin was not present in any sugar-free products consumed in Italy (106 products tested, including beverages, table-top sweeteners, confectionery, chewing gum, yoghurts, jam and mousse) according to a survey used for a dietary exposure assessment for Italian teenagers (323). A later food label survey also in Italy only found use of thaumatin in one table-top sweetener out of 326 products (316).

The Mintel database was evaluated for the period 2013–2017 to identify products launched in Europe that contain thaumatin (324). The results showed that 3.7% of artificial sweeteners, 1.2% of other natural sweeteners, 0.9% of gum, 0.4% of meal replacements and other drinks, 0.2% of carbonated beverages, 0.1% of energy drinks, 0.1% of confectionery and less than 0.1% of snack, cereal and energy bars contained thaumatin. EFSA (318) also reviewed Mintel data from the period 2016–2021 and reported 34 thaumatin-containing foods out of over 35 000, with the highest proportion of a subcategory of foods containing thaumatin being 3.2% (table-top sweeteners). Other categories had proportions of products containing thaumatin between 0.01% (for desserts) and 0.1% (supplements, confectionery, gum and energy drinks).

Some use level data were provided in submissions to the Committee. The ICGA submitted reported or intended use levels of thaumatin II in chewing gum at a minimum of 10 mg/kg, typical use levels of 100 mg/kg and maximum use levels of 1000 mg/kg. The ICA submitted information noting thaumatin II is used as a sweetener in confectionery (GSFA category 5.2, including hard candy, soft candy and nougats) and decorations (GSFA category 5.4) at concentrations of 100–1000 mg/kg. The ICBA reported no use of thaumatin II in GSFA categories 14.1.4 Water-based beverages, 14.1.5 hot beverages or 6.8.1 soy-based beverages (as at November 2024).

In a review paper, Tennant (315) noted thaumatin use from the International Sweeteners Association of 0.01 mg per table-top sweetener tablet.

3.3 Estimates of dietary exposure to thaumatin

3.3.1 Scope of the dietary exposure assessment

As the Committee had not previously undertaken a detailed assessment of dietary exposure to thaumatin, the scope of this assessment included all studies or assessments available. To ensure that the Committee considered the whole body of evidence in its assessment, there were no exclusions based on methodology or data used, or date of the study. The methodology, data used, availability of multiple estimates of dietary exposure in the same country or region, and robustness of the assessments reviewed determined which studies would be used for the Evaluation.

The assessment by the current Committee included an evaluation of information submitted to the Committee, information available in the scientific literature and other published reports. The current Committee reviewed estimates of dietary exposure derived using screening methods (e.g. the poundage data method) or based on individual dietary records. The Committee did not calculate estimates of dietary exposure using national food consumption datasets from the FAO/WHO Cifocoss database (18). This was because there are extensive permissions for use of thaumatin in a large number of food categories within the GSFA, all of which are at GMP (for which a numerical concentration would be required to calculate dietary exposure), and use level data were only available for a small number of food categories. It would therefore not be possible to calculate a reliable estimate of dietary exposure. In addition, the Committee did not calculate international estimates of dietary exposure using the GEMS/Food cluster diets, as there are many permissions for use of thaumatin in processed foods and the consumption data in the cluster diets are primarily for raw commodities and some semi-processed commodities.

3.3.2 Estimates of dietary exposure

The estimates of dietary exposure to thaumatin reviewed used a range of different dietary exposure assessment methodologies, included different combinations of foods (from many foods to single foods only) and were for a range of different population groups. The studies reviewed included assessments published from 2015 to more recent assessments in 2024.

A study in Brazil assessed the proportion of Brazilian pregnant women aged 18 years and older consuming low-calorie sweeteners (325). It was estimated that 0% were identified as consuming thaumatin under Scenario 1 (thaumatin was specifically named in the ingredient list on a product label); 0.7% were predicted to have consumed a food product containing thaumatin under Scenario 2 (included in the foods in Scenario 1, in addition to all fruit-flavoured drink mixes and all foods named e.g. “diet”, “lite”, “no added sugar”, etc.); and 27.1% of respondents were consumers of thaumatin under Scenario 3 (all foods included in scenarios 1 and 2 and all food groups that could possibly contain the sweetener). The collection of label and industry data on use of intense sweeteners in a range of products was undertaken during August–October 2020. A label database from 2017 was also used for Scenario 3 and, where use of the sweetener was uncertain, it was assumed to contain it. Food consumption data were collected from two 24-hour recalls.

A summary of the dietary exposure assessments for thaumatin reviewed by the current Committee is provided in [Table 3.1](#).

Table 3.1

Reported dietary exposures to thaumatin for consumer-only population (unless otherwise specified)

Country or region; source	Population group (sample size, if reported)	Key aspects of the dietary exposure assessment	Estimated dietary exposure (mg/kg bw per day)	
			Mean	High
Italy; Le Donne et al. (316)	≥ 3 years (n = 3270)	Food consumption data from the Italian national survey (INRAN-SCAI 2005–2006); 3-day diary. A tier 2 assessment was undertaken (i.e. based on	Step 1 ≥ 3 years: 0.01	Step 1 ≥ 3 years: 0.06; ^a 0.13 ^b
	Children: 3–9.9 years (n = 193)	MPLs) with two scenarios. Step 1: all food categories and EU MPLs (excluding table-top sweeteners because of GMP permission); or Step 2: only food categories containing thaumatin according to the label survey with EU MPLs (excluding table-top sweeteners because of GMP permission). Label survey conducted in Rome, May–August 2014.	Children: 0.04	Children: 0.17; ^a 0.35 ^b
	Adolescents: 10–17.9 years (n = 247)	Exposures averaged per individual over all days of the nutrition survey.	Adolescents: 0.03	Adolescents: 0.10; ^a 0.20 ^b
	Adults: 18–64.9 years (n = 2312)		Adults: 0.01	Adults: 0.05; ^a 0.10 ^b
	Older adults: > 65 years (n = 518)		Older adults: 0.01	Older adults: 0.03; ^a 0.08 ^b
Republic of Korea; Kim et al. (326)	Population	Poundage data (annual production plus imports minus exports divided by number in population and 365 days per year). Per capita estimates.	0.000085	–
USA; FDA (321)	Population	Use as a flavour modifier at up to 7 mg/kg thaumatin all foods, with average per capita food consumption values published in the USDA ERS Food Availability Data System of 0.56 kg/person per day. Chewing gum only at 150 mg/kg thaumatin and 5 g/person per day consumption. Per person estimates. Application rates of thaumatin used to predict dietary exposures expressed as thaumatin II.	All foods: 0.07 ^c Chewing gum: 0.013 ^c Total: 0.08 ^c	–
USA; FDA (307,327)	≥ 2 years	Use as a sweetener at 10 mg/kg in nine food categories (water-based beverages, carbonated beverages, fruit juices and smoothies, fruit juice drinks, vegetable drinks, flavoured milks, fermented dairy, ready-to-drink teas and coffee, imitation dairy beverages), with consumption per food category from NHANES 2013–2016; 2-day average exposures; conducted using Exponent Inc. FARE software. GRN 910 Supplement II. Proposed application rate of thaumatin II used to predict dietary exposures expressed as thaumatin II.	0.082	0.172 ^d
EU; EFSA (328)	Toddlers, children, adolescents, adults, older adults (≥ 65 years) ^e	Comprehensive food consumption database; 2 days of 24-hour recall data or more. Concentrations from EU permissions (current MPLs), proposed extensions of use and submitted use levels. MPL scenario (all authorized uses at that time); other scenario was current MPLs and at the proposed changes of uses, and use levels provided by the applicant.	MPL scenario Toddlers: 0.02–0.12 Children: 0.04–0.13 Adolescents: 0.02–0.05 Adults: 0.01–0.03 Older adults: 0.003–0.02	MPL scenario^a Toddlers: 0.17–0.44 Children: 0.13–0.42 Adolescents: 0.07–0.22 Adults: 0.04–1.03 Older adults: 0.04–0.63

Table 3.1. (continued)

Country or region; source	Population group (sample size, if reported)	Key aspects of the dietary exposure assessment	Estimated dietary exposure (mg/kg bw per day)	
			Mean	High
EU; EFSA (328) (Continued)		EFSA FAIM model used for dietary exposure calculations. Exposures averaged over consumption days.	MPL+extension+use scenario Toddlers: 0.09–0.27 Children: 0.13–0.34 Adolescents: 0.07–0.16 Adults: 0.03–0.12 Older adults: 0.03–0.10	MPL+extension+use scenario^a Toddlers: 0.25–0.51 Children: 0.21–0.64 Adolescents: 0.13–0.32 Adults: 0.09–1.10 Older adults: 0.09–0.71
EU; Tennant and Vlachou (324)	Infants, toddlers, children, adolescents, adults, older adults (> 65 years) ^c	EFSA FAIM 2.0 used for dietary exposure calculations. Food consumption data from the Comprehensive Food Consumption Database; 2 days of 24-hour recall data or more. Concentrations from EU permissions and use levels from FoodDrinkEurope across 2018. Maximum levels scenario used: MPL values only. Niche scenarios: categories with < 1% products noted in Mintel database with concentration set to zero, typical use for tablet table-top sweeteners.	MPL scenario Infants: 0.002–0.019 Toddlers: 0.018–0.099 Children: 0.02–0.098 Adolescents: 0.012–0.04 Adults: 0.005–0.024 Older adults: 0.002–0.026 Niche scenarios Infants: < 0.001 Toddlers: < 0.001 Children: < 0.001 Adolescents: < 0.001 Adults: < 0.001 Older adults: < 0.001	MPL scenario^a Infants: 0.006–0.089 Toddlers: 0.10–0.277 Children: 0.057–0.277 Adolescents: 0.04–0.119 Adults: 0.024–0.075 Older adults: 0.01–0.058 Niche scenarios^a Infants: < 0.001 Toddlers: < 0.001 Children: < 0.001 Adolescents: < 0.001 Adults: < 0.001–0.003 Older adults: < 0.001–0.004
EU (318)	Infants, toddlers, children, adolescents, adults, older adults (≥ 65 years) ^c	Comprehensive food consumption database; 2 days of 24-hour recall data or more. Concentrations from EU permissions (current MPLs) and submitted use levels for table-top sweeteners. MPL scenario included 12 categories out of the 15 with authorized uses (all MPLs except tabletop sweeteners at use level of 100 mg/kg). Refined brand-loyal scenario included only table-top sweeteners (use level 100 mg/kg). Exposures averaged over consumption days.	MPL scenario Infants: < 0.01–0.01 Toddlers: 0.01–0.04 Children: < 0.01–0.07 Adolescents: < 0.01–0.08 Adults: < 0.01–0.05 Older adults: < 0.01–0.03 Refined scenario Infants: no exposure Toddlers: no exposure Children: < 0.01 Adolescents: < 0.01–0.01 Adults: < 0.01 Older adults: < 0.01	MPL scenario^a Infants: 0.03 Toddlers: 0.01–0.09 Children: 0.01–0.29 Adolescents: 0.01–0.33 Adults: < 0.01–0.27 Older adults: < 0.01–0.11 Refined scenario^a Infants: no exposure Toddlers: no exposure Children: < 0.01 Adolescents: < 0.01 Adults: < 0.01–0.02 Older adults: < 0.01–0.01

EFSA: European Food Safety Authority; EU: European Union; FAIM: Food Additives Intake Model; FARE: Foods Analysis and Residues Evaluation Program; FDA: United States Food and Drug Administration; GMP: Good Manufacturing Practice; GRAS: Generally Regarded as Safe; GRN: GRAS Notice; MPL: maximum permitted level; NHANES: National Health and Nutrition Examination Survey; USA: United States of America; USDA ERS: United States Department of Agriculture Economic Research Service.

^a 95th percentile.

^b 99th percentile.

^c Converted by the Committee to a body weight basis using a body weight of 60 kg.

^d 90th percentile.

^e Infants: 3–11 months; toddlers: 12–35 months; children: 3–9 years; adolescents: 10–17 years; adults: 18–64 years.

One estimate of dietary exposure based on a screening methodology using poundage data for the Republic of Korea was reviewed. The dietary exposure to thaumatin was estimated to be 0.000 085 mg/kg bw per day (326). For the population of the USA, use of thaumatin as a flavour modifier based on per capita food consumption data was estimated to result in a dietary exposure of 0.08 mg/kg bw per person per day (321).

A number of estimates of dietary exposure to thaumatin were calculated using individual dietary records. These were from studies that included 24-hour recalls or food records (or food diaries), and had 2 days or more of food consumption data. Concentration information used were either MPLs or proposed or actual use levels. Data indicating whether there was use in a food category were also used to refine estimates of dietary exposure. Estimates reviewed were from the USA and Europe. Mean and high dietary exposures were reported, as were exposures that included estimates for children (age < 17 years), adults (≥ 18 years) or the general population as a whole. For all the estimates of dietary exposure to thaumatin based on individual dietary records, mean estimates ranged up to 0.34 mg/kg bw per day for children, 0.12 mg/kg bw per day for adults and 0.082 mg/kg bw per day for the population as a whole. Estimates of high dietary exposure ranged up to 0.64 mg/kg bw per day for children, 1.1 mg/kg bw per day for adults and 0.172 mg/kg bw per day for the population as a whole.

3.3.3 Conclusions from the dietary exposure assessment

Certain criteria were applied to determine the estimates of dietary exposure that would be most appropriate for the Evaluation. This included estimates based on individual dietary records. For the high estimates of dietary exposure, all percentile estimates were included (e.g. P90 and P95). In addition, where there were more than one set of estimates for a particular country or region, the most recent set was used (if it was determined to be robust) in order to ensure that dietary exposure estimates were based on the latest datasets. This was the case for the estimates including European countries.

For the Evaluation, dietary exposures based on a range of foods containing thaumatin at MPLs (except for a use level for table-top sweeteners) for mean exposures could be up to 0.07 and 0.05 mg/kg bw per day for children and adults, respectively, and for high exposures up to 0.33 and 0.27 mg/kg bw per day for children and adults, respectively. Dietary exposures based on limited known use in table-top sweeteners could be up to 0.01 mg/kg bw per day for mean exposures and 0.02 mg/kg bw per day for high exposures. If thaumatin is used in combination with other sweeteners, use levels and therefore estimates of dietary exposure could be lower than those reported.

A comparison of estimates of dietary exposure reviewed by the current Committee was made with the anticipated daily intake noted by the Committee at its Twenty-ninth meeting of 1–2 mg/person per day (equivalent to 0.017–0.03 mg/kg bw per day based on a mean body weight of 60 kg). The estimates of dietary exposure reviewed by the current Committee were higher (by around 10-fold) than the initial estimate when using MPLs; however, dietary exposure estimates based on the refined or typical use scenarios were equal to or lower than the initial estimates.

There was a limited number of dietary exposure estimates for thaumatococcus, meaning that there were several WHO regions for which estimates were not available. These included the WHO African Region, Eastern Mediterranean Region and Western Pacific Region.

3.3.4 Major food contributors to dietary exposure

Information was provided on food contributors to thaumatococcus for assessments from the EU only. The main contributing foods vary depending on the concentration data used (MPL versus use levels; use levels as a sweetener versus as a flavour enhancer), the range of foods included in the assessment and the population group assessed.

For the regulatory maximum level scenario in the EU (318), where 12 food groups were included in the assessment, the main contributors were reported as food supplements (all age groups except infants and toddlers), flavoured drinks (most age groups from children and older), confectionery (toddlers, children and adults) and table-top sweeteners (adults and older adults). A use level of 400 mg/kg was used for food supplements and 100 mg/kg for table-top sweeteners; all other foods had a concentration assigned of 50 mg/kg or less. For the refined scenario, the only contribution was from table-top sweeteners (with no contribution for infants or toddlers).

For the MPL scenario in the EU (328), edible ices (14–68%) and cocoa and chocolate products (9–60%) were main contributors across all population groups; flavoured fermented milks were also the main contributor for toddlers (11–59%) and food supplements for older adults (23–63%). For the MPL, extension of use and use level scenario, fine bakery wares (5–66%) were a main contributor across all population groups assessed, with alcoholic beverages contributing for adults and older adults (5–41%) and processed fruit and vegetables (7–46%) for toddlers.

4. Comments

4.1 Biochemical aspects

In studies previously reviewed by the Committee, thaumatin was digested more rapidly than egg albumin in vitro by proteolytic enzymes (trypsin, chymotrypsin, peptidase) (293). A nitrogen balance study conducted in rats showed that the digestibility of thaumatin was similar to that of egg albumin; the majority of thaumatin proteins (90%) are metabolized and absorbed similarly to other dietary proteins (294).

4.2 Toxicological studies

The toxicological database for thaumatin was considered relevant to the evaluation of the safety of thaumatin II produced recombinantly in *N. tabacum*. With the exception of one 13-week study in rats, the studies were evaluated by the previous Committee. Additional details of these studies were provided in the published review of Higginbotham et al. (294). The thaumatin test materials used had a protein content of 78–85%.

The oral LD₅₀ of thaumatin was more than 20 000 mg/kg bw in mice and rats (293, 294).

In a 13-week rat study the NOAEL was 3.0% w/w in the diet, the highest concentration tested (equal to 2417 mg/kg bw per day) (293,294). In a 13-week dietary study in rats not previously evaluated by the Committee (299), no adverse effects were observed at the highest concentration of 3.0% w/w in the diet (equal to 2502 mg/kg bw per day). In a 13-week dog study the NOAEL was 3% in the diet, the highest concentration tested (equal to 1400 mg/kg bw per day).

No long-term or carcinogenicity studies were available.

Thaumatin was not mutagenic in a bacterial reverse mutation test at concentrations of up to 50 mg/plate or in a dominant-lethal assay in mice at doses of up to 2000 mg/kg bw per day (293,294). Although these studies evaluated the potential for mutagenicity and did not address other genotoxicity end-points (e.g. clastogenicity), the Committee noted that thaumatin is readily digestible to normal food components for which genotoxic properties would not be anticipated.

No new studies were available for reproductive and developmental toxicity. No effects were observed on testis, uterus or ovaries in short-term repeat-dose toxicity studies (293,294,299). The NOAEL in a developmental toxicity study in rats evaluated by the previous Committee was 2000 mg/kg bw per day, the highest dose tested.

Bioinformatics analysis conducted by the sponsor on the amino acid sequence for thaumatin II did not identify any matches with known allergens. The Committee noted that thaumatin is quickly degraded by digestive enzymes.

The Committee noted that most of the residual tobacco seed proteins present are common to other plant species consumed in the human diet. Additionally, no adverse effects were observed in rats, pigs or lambs consuming tobacco seeds as feed (310–313).

4.3 Observations in humans

In a clinical study evaluated by the previous Committee, no treatment-related changes were observed on haematological and blood biochemistry parameters in volunteers that consumed 280 mg thaumatin daily for 13 consecutive weeks (293,294). The previous Committee noted that no reactions to thaumatin were detected with skin prick testing after 14 days of oral administration of 100 mg/day of thaumatin (293,294).

4.4 Assessment of dietary exposure

A dietary exposure assessment for thaumatin had not previously been undertaken by the Committee. The Committee evaluated information on use levels and dietary exposure that was submitted for consideration and obtained through a literature search. For the purposes of this assessment, where the food additive reviewed was referred to as thaumatin, it was assumed that the information was equally relevant to thaumatococcus II.

Thaumatococcus is permitted for use as a flavour enhancer and sweetener in the GSFA and in many countries. It is commonly permitted for use under conditions of GMP in a range of food categories including beverages, table-top sweeteners, confectionery, chocolate and cocoa products, food supplements, fruit and vegetable products, dairy products, condiments and sauces. Evaluation of the use of thaumatococcus via food product databases or label surveys from Australia, Brazil, New Zealand, the Republic of Korea and Europe shows that a small proportion of food products that are on the world-wide market contain the sweetener. It is most commonly found in table-top sweeteners, with other uses in beverages and confectionery and in lower proportions of snack bars, desserts, meal replacements and supplements.

Estimates of dietary exposure to thaumatococcus from the Republic of Korea based on poundage data, and from the USA and Europe based on individual dietary records, were reviewed by the Committee at the current meeting. For the Evaluation, dietary exposures based on a range of foods containing thaumatococcus at MPLs (except for a use level for table-top sweeteners) could be for mean exposures up to 0.07 mg/kg bw per day for children and 0.05 mg/kg bw per day for adults, and for high exposures up to 0.33 mg/kg bw per day for children and 0.27 mg/kg bw per day for adults.

Major contributors to dietary exposure when a range of foods were included in the assessment included food supplements for children (excluding infants and toddlers) and adults, flavoured drinks and confectionery for children, and table-top sweeteners for adults.

5. Evaluation

Thaumatococcus consists primarily of thaumatin I and thaumatin II proteins (at a ratio of 2:1) together with minor amounts of plant constituents. These proteins have similar properties, amino acid composition, sweetness and molecular weight (of about 22 kDa), and only differ by five amino acids. The amino acid sequence of thaumatin II produced by recombinant methods in *Nicotiana tabacum* is identical to that of thaumatin II extracted from the arils of the fruit of *Thaumatococcus daniellii*. On this basis, the toxicological database for thaumatin was considered relevant to the evaluation of the safety of thaumatin II produced recombinantly in *N. tabacum*.

At its Twenty-ninth meeting, the previous Committee established an ADI “not specified”¹⁵ for thaumatin (INS No. 957) obtained from *T. daniellii*.

One new 13-week toxicity study in rats was available for thaumatin that supports the previous evaluation. The NOAEL was 3.0% w/w in the diet, equal to 2502 mg/kg bw per day, the highest dose tested.

Based on the amino acid sequence similarity of thaumatin I, thaumatin II and recombinant thaumatin II, the Committee revised the ADI “not specified” for thaumatin (INS No. 957) to a group ADI “not specified” including thaumatin (INS No. 957) and thaumatin II produced by recombinant methods in the seeds of the tobacco plant *N. tabacum*.

The high dietary exposure to thaumatin was estimated to be up to 0.33 mg/kg bw per day for children. The Committee concluded that the dietary exposure to thaumatin does not present a safety concern.

A new specifications monograph for thaumatin II and a chemical and technical assessment were prepared.

5.1 Recommendations

The Committee recommended that the specifications for thaumatin (INS No. 957) be updated.

¹⁵ The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

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Annex 1

Reports and other documents resulting from previous meetings of the Joint FAO/WHO Expert Committee on Food Additives

Table A1.1
Publications resulting from previous JECFA meetings

JECFA meeting no. (year)	Topic	Publications resulting from meeting (year of publication)
1 (1956)	Food additives	TRS 129 (1957), NMRS 15 (1957)
2 (1957)	Food additives	TRS 144 (1958), NMRS 17 (1958)
3 (1958)	Food additives	Specifications for identity and purity of food additives, Vol. I (1962)
4 (1959)	Food additives	Specifications for identity and purity of food additives, Vol. II (1963)
5 (1960)	Food additives	TRS 220 (1961), NMRS 29 (1961)
6 (1961)	Antimicrobials and antioxidants	TRS 228 (1962), NMRS 31 (1962)
7 (1963)	Food additives	TRS 281 (1964), NMRS 35 (1964)
8 (1964)	Food additives	TRS 309 (1965), NMRS 38 (1965), NMRS 38B (1966)
9 (1965)	Food additives	TRS 339 (1966), NMRS 40 (1966), NMRS 40A, B, C (1966)
10 (1966)	Food additives	TRS 373 (1967), NMRS 43 (1967)
11 (1967)	Food additives	TRS 383 (1968), NMRS 44 (1968), NMRS 44A (1968), NMRS 44B (1969)
12 (1968)	Food additives	TRS 430 (1969), NMRS 45 (1969), NMRS 45A (1969)
13 (1969)	Food additives	TRS 445 (1970), NMRS 46 (1970), NMRS 46A, 46B (1970)
14 (1970)	Food additives	TRS 462 (1971), NMRS 48 (1971), NMRS 48A, B, C (1971), FAS 1, 2, 3 (1972)
15 (1971)	Food additives	TRS 488 (1972), NMRS 50 (1972), NMRS 50A, B, C (1972)
16 (1972)	Food additives and contaminants	TRS 505 (1972), NMRS 51 (1972), NMRS 51A (1972), FAS 4 (1972)
17 (1973)	Food additives	TRS 539 (1974), NMRS 53 (1974), NMRS 53A (1974), FAS 5 (1974)
18 (1973)	Food additives	TRS 557 (1974), NMRS 54 (1974), NMRS 54A, B (1975), FAS 6, 7 (1975)
19 (1974)	Food additives	TRS 576 (1975), NMRS 55 (1975), NMRS 55A (1975), NMRS 55B (1976), FAS 8 (1975), FAS 9 (1976)
20 (1975)	Food additives	TRS 599 (1976), FNM 1 (1976), FNM 1B (1977), FAS 10 (1976), FAS 11 (1977)
21 (1977)	Food additives	TRS 617 (1977), NMRS 57 (1977), FAS 12 (1977)
22 (1978)	Food additives	TRS 631 (1978), FAS 13 (1978), FNP 4 (1978), FNR 7 (1978),
23 (1979)	Food additives	TRS 648 (1980), FAS 14 (1980), FNP 12 (1979)
24 (1980)	Food additives	TRS 653 (1981), FAS 15 (1980), FNP 17 (1980)
25 (1981)	Food additives	TRS 669 (1981), FAS 16 (1981), FNP 19 (1981)
26 (1982)	Food additives and contaminants	TRS 683 (1982), FAS 17 (1982), FNP 25 (1982)
27 (1983)	Food additives and contaminants	TRS 696 (1984), FAS 18 (1983), FNP 28 (1983), FNP 5 (Revision 1) (1983)
28 (1984)	Food additives and contaminants	TRS 710 (1984), FAS 19 (1984), FNP 31/1 (1984), FNP 31/2 (1984)

JECFA meeting no. (year)	Topic	Publications resulting from meeting (year of publication)
29 (1985)	Food additives and contaminants	TRS 733 (1986), FAS 20 (1987), FNP 34 (1986)
30 (1986)	Food additives and contaminants	TRS 751 (1987), FAS 21 (1987), FNP 37 (1986)
—	—	Principles for the safety assessment of food additives and contaminants in food. WHO Environmental Health Criteria, No. 70, 1987 (https://apps.who.int/iris/handle/10665/37578)
31 (1987)	Food additives and contaminants	TRS 759 (1987), FAS 22 (1988), FNP 38 (1988)
32 (1987)	Veterinary drug residues in food	TRS 763 (1988), FAS 23 (1988), FNP 41 (1988)
33 (1988)	Food additives and contaminants	TRS 776 (1989), FAS 24 (1989)
34 (1989)	Veterinary drug residues in food	TRS 788 (1989), FAS 25 (1990), FNP 41/2 (1990)
35 (1989)	Food additives and contaminants	TRS 789 (1990), FAS 26 (1990), FNP 49 (1990)
36 (1990)	Veterinary drug residues in food	TRS 799 (1990), FAS 27 (1991), FNP 41/3 (1991)
37 (1990)	Food additives and contaminants	TRS 806 (1991), FAS 28 (1991)
—	—	Compendium of food additive specifications, from 1992 (https://www.fao.org/food/food-safety-quality/scientific-advice/jecfa/jecfa-additives/en/)
38 (1991)	Veterinary drug residues in food	TRS 815 (1991), FAS 29 (1991), FNP 41/1 (1991), FNP 5 (Revision 2) (1991)
39 (1992)	Food additives and naturally occurring toxicants	TRS 828 (1992), FAS 30 (1993)
40 (1992)	Veterinary drug residues in food	TRS 832 (1993), FAS 31 (1993), FNP 41/5
41 (1993)	Food additives and contaminants	TRS 837 (1993), FAS 32 (1993)
42 (1994)	Veterinary drug residues in food	TRS 851 (1995), FAS 33 (1994), FNP 41/6 (1994)
43 (1994)	Veterinary drug residues in food	TRS 855 (1995), FAS 34 (1995), FNP 41/7 (1995)
44 (1995)	Food additives and contaminants	TRS 859 (1995), FAS 35 (1996)
45 (1995)	Veterinary drug residues in food	TRS 864 (1996), FAS 36 (1996), FNP 41/8 (1996)
46 (1996)	Food additives and contaminants	TRS 868 (1997), FAS 37 (1997)
47 (1996)	Veterinary drug residues in food	TRS 876 (1998), FAS 38 (1998), FNP 41/9 (1997)
48 (1997)	Veterinary drug residues in food	TRS 879 (1998), FAS 39 (1997), FNP 41/10 (1998)
49 (1997)	Food additives and contaminants	TRS 901 (1999), FAS 40 (1998)
50 (1998)	Veterinary drug residues in food	TRS 888 (1999), FAS 41 (1998), FNP 41/11 (1999)
51 (1998)	Food additives	TRS 891 (2000), FAS 42 (1999)
52 (1999)	Veterinary drug residues in food	TRS 893 (2000), FAS 43 (2000), FNP 41/12 (2000)
53 (1999)	Food additives and contaminants	TRS 896 (2000), FAS 44 (2000)
54 (2000)	Veterinary drug residues in food	TRS 900 (2001), FAS 45 (2000), FNP 41/13 (2000)
55 (2000)	Food additives and contaminants	TRS 901 (2001), FAS 46 (2001)
56 (2001)	Mycotoxins in food	TRS 906 (2002), FAS 47 (2001), FNP 74 (2001)
57 (2001)	Food additives and contaminants	TRS 909 (2002), FAS 48 (2002)
58 (2002)	Veterinary drug residues in food	TRS 911 (2002), FAS 49 (2002), FNP 41/14 (2002)
59 (2002)	Food additives	TRS 913 (2002), FAS 50 (2003)

JECFA meeting no. (year)	Topic	Publications resulting from meeting (year of publication)
60 (2002)	Veterinary drug residues in food	TRS 918 (2003), FAS 51 (2003), FNP 41/15 (2003)
61 (2003)	Food additives and contaminants	TRS 922 (2004), FAS 52 (2004)
62 (2004)	Veterinary drug residues in food	TRS 925 (2004), FAS 53 (2005), FNP 41/16 (2004)
63 (2004)	Food additives	TRS 928 (2005), FAS 54 (2005)
64 (2005)	Food additives	TRS 930 (2005), FAS 55 (2006), FNP 82 (2006)
65 (2006)	Food additives	TRS 934 (2006), FAS 56 (2006), Mono 1(1–4) (2006)
66 (2006)	Veterinary drug residues in food	TRS 939 (2006), FAS 57 (2006), Mono 2 (2006)
67 (2007)	Food additives and contaminants	TRS 940 (2007), FAS 58 (2007), Mono 3 (2007)
68 (2007)	Food additives and contaminants	TRS 947 (2008), FAS 59 (2008), Mono 4 (2007)
69 (2008)	Food additives	TRS 952 (2009), FAS 60 (2009), Mono 5 (2009)
70 (2008)	Veterinary drug residues in food	TRS 954 (2009), FAS 61 (2009), Mono 6 (2009)
71 (2009)	Food additives	TRS 956 (2010), FAS 62 (2010), Mono 7 (2009)
72 (2010)	Contaminants in food	TRS 959 (2011), FAS 63 (2011), Mono 8 (2011), Mono 9 (2010)
73 (2010)	Food additives and contaminants	TRS 960 (2011), FAS 64 (2011), Mono 10 (2010)
74 (2011)	Veterinary drug residues in food	TRS 966 (2012), FAS 65 (2011), Mono 11 (2011)
75 (2011)	Veterinary drug residues in food	TRS 969 (2012), FAS 66 (2012), Mono 12 (2012)
76 (2012)	Food additives	TRS 974 (2012), FAS 67 (2012), Mono 13 (2012)
77 (2013)	Food additives and contaminants	TRS 983 (2014), FAS 68 (2013), Mono 14 (2013)
78 (2013)	Veterinary drug residues in food	TRS 988 (2014), FAS 69 (2014), Mono 15 (2014)
79 (2014)	Food additives	TRS 990 (2015), FAS 70 (2015), Mono 16 (2014)
80 (2015)	Food additives and contaminants	TRS 995 (2016), FAS 71 (2015), FAS 71/1 (2016), Mono 17 (2015)
81 (2015)	Veterinary drug residues in food	TRS 997 (2016), FAS 72 (2016), Mono 18 (2016)
82 (2016)	Food additives	TRS 1000 (2016), FAS 73 (2017), Mono 19 (2016)
83 (2016)	Contaminants in food	TRS 1002 (2017), FAS 74 (2018)
84 (2017)	Food additives	TRS 1007 (2017), FAS 75 (2019), Mono 20 (2017)
85 (2017)	Veterinary drug residues in food	TRS 1008 (2018), FAS 76 (2019), Mono 21 (2018)
86 (2018)	Food additives	TRS 1014 (2019), FAS 77 (2019), Mono 22 (2018)
87 (2019)	Food additives	TRS 1020 (2019), 258 (2020), Mono 23 (2019)
88 (2019)	Veterinary drug residues in food	TRS 1023 (2020), FAS 79 (2020), Mono 24 (2020)
89 (2020)	Food additives	TRS 1027 (2021), FAS 80 (2021), Mono 25 (2021)
90 (2020)	Contaminants in food	TRS 1032 (2022), FAS 81 (2022)
91 (2021)	Food additives and contaminants	TRS 1036 (2022), FAS 82 (2023), Mono 26 (2021)
92 (2021)	Food additives	TRS 1037 (2022), FAS 83 (2023), Mono 27 (2022)
93 (2022)	Contaminants in food	TRS 1040 (2023), FAS 84, Mono 28 (2023)
94 (2022)	Veterinary drug residues in food	TRS 1041 (2022), FAS 85 (2023), Mono 29 (2023)
95 (2022)	Food additives	TRS 1042 (2023), FAS 86 (2023), Mono 30 (2023)

JECFA meeting no. (year)	Topic	Publications resulting from meeting (year of publication)
96 (2023)	Food additives	TRS 1050 (2023), FAS 87 (2024), Mono 31 (2023)
97 (2023)	Food additives	TRS 1051 (2024), FAS 88 (2024), Mono 32 (2024)
98 (2024)	Veterinary drug residues in food	TRS 1055 (2024), FAS 89 (2025), Mono 33 (2024)
99 (2024)	Food additives	TRS 1056 (2024), FAS 90 (2025), Mono 34 (2025)

FAO: Food and Agriculture Organization of the United Nations; FAS: WHO Food Additives Series; FNP: FAO Food and Nutrition Paper; JECFA: Joint FAO/WHO Expert Committee on Food Additives; Mono: FAO JECFA Monograph; NMRS: FAO Nutrition Meetings Report Series; TRS: WHO Technical Report Series; WHO: World Health Organization.

Notes: (i) The JECFA report that is accepted on the last day of the meeting is published as part of the WHO Technical Report Series. (ii) The JECFA monograph that includes the background information on which the report is based is published as part of the WHO Food Additives Series. (iii) Revised and new specifications are published as part of the FAO Food and Nutrition Papers and FAO JECFA Monographs.

Annex 2

List of acronyms used in the monographs

3D	three-dimensional
ADI	acceptable daily intake
ALP	alkaline phosphatase
ALT	alanine aminotransferase
AMP	adenosine-5'-monophosphate
ANIT	a-naphthalene isothiocyanate
AP	apical (compartment)
APTT	activated partial thromboplastin time
ASD	autism spectrum disorder
AST	aspartate aminotransferase
BL	basolateral (compartment)
BMDL ₍₂₀₎	lower 95% confidence limit on the benchmark dose (for a 20% response)
BMDU ₍₂₀₎	upper 95% confidence limit on the benchmark dose (for a 20% response)
BMR	benchmark response
BNF	β-naphthoflavone
bw	body weight
CAS	Chemical Abstracts Service
CCFA	Codex Committee on Food Additives
CCl ₄	carbon tetrachloride
CGE	capillary gel electrophoresis
CIFOCosS	FAO/WHO chronic individual food consumption database – summary statistics
CITCO	6-(4-chlorophenyl)imidazo[2,1-b][1,3]thiazole-5-carbaldehyde-O-(3,4-dichlorobenzyl)oxime
C _{max}	maximum concentration
CO ₂	carbon dioxide
CYP	cytochrome P450
DMSO	dimethyl sulfoxide

DSS	dioctyl sodium sulfosuccinate
EC	Enzyme Commission
EER	estimated energy requirement
EFSA	European Food Safety Authority
EHC 240	Environmental Health Criteria 240
ELISA	enzyme-linked immunosorbent assay
EMA	European Medicines Agency
EOGRT	extended one-generation reproductive toxicity
EU	European Union
F344	Fischer 344
FAO	Food and Agriculture Organization of the United Nations
FC	food category
FDA	United States Food and Drug Administration
FSANZ	Food Standards Australia New Zealand
FSH	follicle-stimulating hormone
FSMP	formula for special medical purposes
GD	gestational day
GEMS	Global Environment Monitoring System
GIT	gastrointestinal tract
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practice
GRAS	Generally Regarded as Safe
GSFA	General Standard for Food Additives
HDCA	hyodeoxycholic acid
HepG2	human liver cancer cell line
HPLC	high-performance liquid chromatography
HT-29	human colon cancer cell line
IC ₅₀	concentration at which 50% inhibition is observed
ICA	International Confectionery Association
ICBA	International Council of Beverages Associations

ICGA	International Chewing Gum Association
ICGMA	International Council of Grocery Manufacturers Associations
IFAC	International Food Additives Council
Ig	immunoglobulin
IL	interleukin
INS	International Numbering System for Food Additives
ISD	in-source decay
JECFA	Joint FAO/WHO Expert Committee on Food Additives
LC	liquid chromatography
LCFA	long-chain fatty acid
IcFOS	long-chain fructo-oligosaccharides
LD	lactation day
LD ₅₀	median lethal dose
LH	luteinizing hormone
LOAEL	lowest-observed-adverse-effect level
LOD	limit of detection
LOQ	limit of quantification
MALDI	matrix-assisted laser desorption ionization
MDA	malondialdehyde
MN-RET	micronucleated reticulocyte
MOE	margin of exposure
MPL	maximum permitted level
mRNA	messenger ribonucleic acid
MRP	multidrug resistance protein
MS	mass spectrometry
NHANES	National Health and Nutrition Examination Survey [USA]
NHNS	National Health and Nutrition Survey [Japan]
NOAEL	no-observed-adverse-effect level
OECD	Organisation for Economic Co-operation and Development
P95	95th percentile
PCR	polymerase chain reaction

PEG	polyethylene glycol
PND	postnatal day
PNP	<i>p</i> -nitrophenol
PNPG	<i>p</i> -nitrophenyl- α -D-glucopyranoside
PTU	propylthiouracil
PXR	pregnane X receptor
QA	Quality Assurance
QWBA	quantitative whole-body autoradiography
RBC	red blood cell
RNA	ribonucleic acid
RTD	ready to drink
SCE	sister chromatid exchange
scGOS	short-chain galacto-oligosaccharides
SD	standard deviation
SEM	standard error of the mean
SGF	simulated gastric fluid
SIF	simulated intestinal fluid
SOD	superoxide dismutase
$t_{1/2}$	half-life
T ₃	triiodothyronine
T ₄	thyroxine
TBIL	total bilirubin
TCA	taurocholic acid
TCDCA	taurochenodeoxycholic acid
THDCA	taurohyodeoxycholic acid
$T_{\max}$	time to reach maximum concentration
TMDI	theoretical maximum daily intake
TOS	total organic solids
TSH	thyroid-stimulating hormone
UGT	uridine diphosphate glucuronosyltransferase

USA	United States of America
VOC	volatile organic chemical
WBC	white blood cell
WHO	World Health Organization

Annex 3

Participants of the one-hundredth meeting of the Joint FAO/WHO Expert Committee on Food Additives

Rome, 10–19 June 2025

Members

Dr S. Barlow, Brighton, East Sussex, United Kingdom (*invited by WHO*)

Dr D. Benford, Cheddington, Buckinghamshire, United Kingdom (*Vice-Chairperson*)
(*invited by WHO*)

Dr R. Cantrill, Bedford, Nova Scotia, Canada (Chairperson) (*invited by FAO*)

Dr E. Dessipri, General Chemical State Laboratory, Athens, Greece (*invited by FAO*)

Dr M. DiNovi, Baltimore (MD), USA (*invited by FAO*)

Dr N. Fletcher, Food Standards Australia New Zealand, Majura Park, Australian Capital Territory, Australia (*invited by WHO*)

Dr D.E. Folmer, Office of Post-Market Assessment, Office of Food Chemical Safety, Dietary Supplements and Innovation, Human Foods Program, United States Food and Drug Administration, College Park (MD), USA (*Joint Rapporteur*) (*invited by FAO*)

Ms T. Hambridge, Food Standards Australia New Zealand, Majura Park, Australian Capital Territory, Australia (*invited by FAO*)

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Ms K. Laurvick, United States Pharmacopeia, Rockville (MD), USA (*invited by FAO*)

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Dr U. Mueller, Perth, Western Australia, Australia (Joint Rapporteur) (*invited by WHO*)

Dr J. Schlatter, Zurich, Switzerland (*invited by WHO*)

Dr J.R. Srinivasan, Division of Cosmetics, Office of Cosmetics and Colors, United States Food and Drug Administration, College Park (MD), USA (*invited by FAO*)

Additional experts invited by FAO and WHO

Dr F. Aguilar M., Chessy, France (*invited by WHO*)

Dr P.E. Boon, Department for Chemical Food Safety, Centre for Prevention, Lifestyle and Health, National Institute for Public Health and the Environment, Bilthoven, the Kingdom of the Netherlands (*invited by FAO*)

Ms M.-A. Hammer, Ottawa, Ontario, Canada (*invited by WHO*)

Dr S.V. Kabadi, Office of Pre-Market Additive Safety, Office of Food Chemical Safety, Dietary Supplements and Innovation, Human Foods Program, United States Food and Drug Administration, College Park (MD), USA (*invited by FAO*)

Dr F.-Q. Li, China National Center for Food Safety Risk Assessment, Beijing, China (*invited by FAO*)

Dr A.-K. Lundebye, Institute of Marine Research, Bergen, Norway (*invited by WHO*)

Dr N. Sugimoto, Division of Food Additives, National Institute of Health Sciences, Kanagawa, Japan (*invited by FAO*)

Dr S. West-Barnette, Office of Pre-Market Additive Safety, Office of Food Chemical Safety, Dietary Supplements and Innovation, Human Foods Program, United States Food and Drug Administration, College Park (MD), USA (*invited by FAO*)

Dr H.-J. Yoon, Department of Food Safety and Regulatory Science, Chung-Ang University, Republic of Korea (*invited by WHO*)

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Ms N.Y. Ho, Department of Nutrition and Food Safety, World Health Organization, Geneva, Switzerland (*WHO Consultant*)

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Mr K. Petersen, Department of Nutrition and Food Safety, World Health Organization, Geneva, Switzerland (*WHO Joint Secretary*)

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Ms A. Vlachou, Agrifood Systems and Food Safety Division, Food and Agriculture Organization of the United Nations, Rome, Italy (*FAO Joint Secretary*)

Ms L.-P. Zhang, Food and Agriculture Organization of the United Nations, Rome, Italy (*Codex Secretariat*)

Annex 4

Toxicological information and information on specifications

Table A4.1
Food additives and processing aids evaluated toxicologically, assessed for dietary exposure and specifications revised

Food additive	Specifications	ADIs and other conclusions on toxicology and dietary exposure
Adipates	No ^a	The Committee noted that the estimates of dietary exposure calculated at the current meeting are overestimated for a number of reasons. The Committee concluded that most of these conservative dietary exposure estimates greatly exceed the current group ADI of 0–5 mg/kg bw. Further, the Committee noted that there is the potential for exceedances of the current group ADI as a result of normal daily consumption amounts of an individual food. The Committee noted that hazard assessment procedures have evolved and additional relevant data may have become available since the numerical ADI was established at its Ninth meeting in 1965, indicating that a re-evaluation of the toxicology of adipic acid is needed.
Amyloglucosidase (JECFA95-4) from <i>Rasamsonia emersonii</i> expressed in <i>Aspergillus niger</i>	R	The Committee concluded that dietary exposure to this enzyme is not anticipated to pose a risk for allergenicity. Based on the absence of any treatment-related adverse effects in a 13-week oral toxicity study in rats, the previous Committee identified a NOAEL of 1500 mg TOS/kg bw per day, the highest dose tested. The previous Committee concluded that the dietary exposure estimate of 9 mg TOS/kg bw per day was appropriate for use in the evaluation. Comparison of the NOAEL of 1500 mg/kg bw per day with the dietary exposure estimate of 9 mg TOS/kg bw per day yields an MOE of more than 160. Based on this MOE and a lack of concern for allergenicity and genotoxicity, the Committee removed the temporary designation of the previous ADI and established an ADI “not specified”^{nb} for amyloglucosidase (JECFA95-4) from <i>R. emersonii</i> expressed in <i>A. niger</i> when used at the levels specified and in accordance with current GMP.
Ascorbyl palmitate	R	The Committee noted new studies that showed that ascorbyl palmitate was rapidly and extensively hydrolysed pre-systemically to ascorbic acid and palmitic acid. Hydrolysis of ascorbyl palmitate is mediated by carboxylesterases, which are also expressed in neonates and infants, albeit at lower levels than in adults. On that basis, the Committee considered that systemic exposure to ascorbyl palmitate would be low for all population subgroups. The Committee withdrew the previously established ADI of 0–1.25 mg/kg bw for ascorbyl palmitate or ascorbyl stearate, or the sum of both. The Committee noted that the previous ADI for ascorbyl palmitate or ascorbyl stearate, or the sum of both, was based on toxicological data for ascorbyl palmitate, and therefore established a group ADI “not specified”^{nb} for ascorbyl palmitate or ascorbyl stearate, or the sum of both.

Food additive	Specifications	ADIs and other conclusions on toxicology and dietary exposure
Carob bean gum	No ^c	The Committee previously assigned an ADI "not specified"^b to carob bean gum. However, an ADI does not apply to infants younger than 12 weeks since they might be at risk at lower levels of exposure compared with older age groups. To complete the evaluation, the Committee therefore requested studies in neonatal animals adequate for the evaluation of the safety of carob bean gum in infant formula. Newly submitted unpublished data obtained from a neonatal pig study evaluated at the present meeting allowed the Committee to identify a NOAEL of 2400 mg/kg bw per day, the highest dose tested. Based on the estimated high dietary exposures for infants, the Committee noted that MOEs calculated for the use of carob bean gum at 1000 mg/L and 6000 mg/L in infant formula were 9 and 1.5, respectively. The MOE for the high exposure calculated for infants from the use of carob bean gum at 10 000 mg/L in infant formula was less than 1. The Committee concluded that the MOEs based on carob bean gum concentrations of 1000 and 6000 mg/L in infant formula indicate a low risk; however, the MOE based on a concentration of 10 000 mg/L indicates a potential risk. Based on the available data and considering the short exposure period involved, the Committee concluded that there is no concern for the disruption of the intestinal microbiota by carob bean gum used as a thickener in infant formula.
Diocetyl sodium sulfosuccinate	No	At its Forty-fourth meeting in 1995, the Committee established an ADI of 0–0.1 mg/kg bw for DSS. The high (P95) dietary exposure estimates of DSS of up to 0.30 mg/kg bw per day for children aged 1–17 years could exceed this ADI. The estimates for adults were all below the ADI. Considering that (i) the dietary exposure estimates are conservative because of the assumptions that all soft drinks that could contain DSS do contain this food additive (although other food additives that perform the same function in foods are available) and, when DSS is used, it will always be present at the maximum and/or average use levels; and (ii) the ranges of exposure estimates based on only maximum use levels were comparable to those based on a combination of maximum and average use levels, the Committee concluded that the use of DSS as a food additive in soft drinks and fruit-flavoured water-based desserts does not pose a safety concern.
Gardenia (Genipin) Blue	N	The Committee identified a NOAEL of 50 mg/kg bw per day for geniposide based on hepatotoxic effects observed at the next higher dose in a 26-week oral toxicity study in rats. The Committee also identified a NOAEL of 50 mg/kg bw per day for genipin based on hepatotoxic effects observed at the highest dose in a 21-day oral toxicity study in mice. No longer-term oral toxicity studies on genipin were available. The Committee selected the NOAEL of 700 mg/kg bw per day for the Gardenia Blue polymer from the carcinogenicity study as a point of departure for the safety evaluation of Gardenia (Genipin) Blue. The Committee established an ADI of 0–7 mg/kg bw per day for the Gardenia (Genipin) Blue, expressed on the basis of the Gardenia Blue polymer, by applying an uncertainty factor of 100 to the NOAEL to account for inter- and intraspecies variabilities. Comparison of the NOAEL of 50 mg/kg bw per day with the theoretical high dietary exposure to geniposide of 1 µg/kg bw per day yields an MOE of 50 000. Based on this MOE and no concern for genotoxicity, the Committee concluded that dietary exposure to geniposide as an impurity in Gardenia (Genipin) Blue does not pose a safety concern.

Food additive	Specifications	ADIs and other conclusions on toxicology and dietary exposure
Glycolipids	N	A proportion of the glycolipids is hydrolysed by intestinal microflora to their components, namely glucose, xylose, acetate, isovalerate and long-chain fatty acids, and the remaining unchanged glycolipids are eliminated in the faeces. The Committee further noted that the hydrolysis products are normal components of the human diet. The Committee therefore established an ADI “not specified”^a for glycolipids. The Committee noted the high dietary exposure estimates of 3.3 mg/kg bw per day for children aged 2–5 years in the USA and 1.3 mg/kg bw per day for adults in Europe. The Committee concluded that these dietary exposure estimates did not raise any safety concerns.
Rosemary extract	No	A BMDL₂₀ of 97 mg/kg bw per day expressed as carnosic acid plus carnosol was calculated for a new study on thyroid hormone levels in rat pups from dams treated with rosemary extract in the diet during pregnancy and lactation. This BMDL₂₀ is higher than the NOAEL of 64 mg/kg bw per day, expressed as carnosic acid plus carnosol, on which the current temporary ADI for rosemary extract is based. In the derivation of the temporary ADI of 0–0.3 mg/kg bw per day, an additional uncertainty factor of 2 was incorporated into the overall uncertainty factor of 200 to account for the absence of studies to elucidate the potential developmental and reproductive toxicity of rosemary extract. This factor is no longer necessary. The temporary ADI of 0–0.3 mg/kg bw per day (expressed as carnosic acid and carnosol) was therefore withdrawn and an ADI of 0–0.6 mg/kg bw per day (expressed as carnosic acid and carnosol) was established. Estimated high dietary exposures to rosemary extract (expressed as carnosic acid plus carnosol) based on naturally occurring sources combined with dietary exposures from added sources at typical use levels were up to 0.84 mg/kg bw per day for children and up to 0.32 mg/kg bw per day for adults. The MOE between the BMDL₂₀ for reduction in T₄ levels in pups and the estimated high dietary exposure for children is greater than 100, and is considered sufficient. The Committee therefore did not consider the slight exceedance of the ADI, which was only by children, as a safety concern.
Thaumatococcus	N	Based on the amino acid sequence similarity of thaumatococcus I, thaumatococcus II and recombinant thaumatococcus II, the Committee revised the ADI “not specified”^a for thaumatococcus (INS No. 957) to a group ADI “not specified” including thaumatococcus (INS No. 957) and thaumatococcus II produced by recombinant methods in the seeds of the tobacco plant <i>Nicotiana tabacum</i>. The high dietary exposure to thaumatococcus was estimated to be up to 0.33 mg/kg bw per day for children. The Committee concluded that the dietary exposure to thaumatococcus does not present a safety concern.

ADI: acceptable daily intake; BMDL₂₀: lower 95% confidence limit on the benchmark dose for a 20% response; DSS: dioctyl sodium sulfosuccinate; GMP: Good Manufacturing Practices; INS: International Numbering System for Food Additives; MOE: margin of exposure; N: new specifications; NOAEL: no-observed-adverse-effect level; R: revised specifications; TOS: total organic solids; USA: United States of America.

^a Information is required to complete the specifications.

^b The reader is referred to the Technical Report of the Eighty-seventh JECFA meeting for clarification of the term ADI “not specified”.

^c No new toxicological information was available for this evaluation that indicated a need for revision of the ADI “not specified” established previously by the Committee. An errata item was prepared for the existing specifications monographs.

Table A4.2
Food additives and processing aids considered for specifications only

Food additive	Specifications	Details
α -Amylase (JECFA95-1) from <i>Geobacillus stearothermophilus</i> expressed in <i>Bacillus licheniformis</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.
α -Amylase (JECFA95-2) from <i>Geobacillus stearothermophilus</i> expressed in <i>Bacillus licheniformis</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.
α -Amylase (JECFA95-3) from <i>Rhizomucor pusillus</i> expressed in <i>Aspergillus niger</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.
Asparaginase (JECFA95-5) from <i>Pyrococcus furiosus</i> expressed in <i>Bacillus subtilis</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.
β -Amylase (JECFA95-6) from <i>Bacillus flexus</i> expressed in <i>Bacillus licheniformis</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.
Gellan gum	N, R	The Committee generated new specifications for low-acyl clarified gellan gum (INS No. 418(ii)) and revised the existing specifications monograph for gellan gum (INS No. 418(i)).
Xylanase (JECFA95-9) from <i>Bacillus licheniformis</i> expressed in <i>Bacillus licheniformis</i>	R	The specifications monograph was revised and the tentative status was removed. As a result, the temporary designation of ADI “not specified” established at the Ninety-fifth meeting was removed.

N: new specifications; R: revised specifications.

Annex 5

Dioctyl sodium sulfosuccinate: supplementary material

Table A5.1
Soft drinks considered by the Committee for estimating the dietary exposure to dioctyl sodium sulfosuccinate in Europe using the DietEx Exposure tool

Food category	Code ^a	Description
Carbonated soft drinks (fruit flavoured)	A03EB	Fruit soft drinks, currant red
	A03EC	Fruit soft drinks, currant white
	A03ED	Fruit soft drinks, gooseberry
	A03EE	Fruit soft drinks, grape red
	A03EF	Fruit soft drinks, grape white
	A03EG	Fruit soft drinks, grapefruit
	A03EH	Fruit soft drinks, mango
	A03EJ	Fruit soft drinks, mulberry
	A03EK	Fruit soft drinks, nectarine
	A03EL	Fruit soft drinks, orange
	A03EM	Fruit soft drinks, papaya
	A03EN	Fruit soft drinks, peach
	A03EP	Fruit soft drinks, pear
	A03EQ	Fruit soft drinks, pineapple
	A03ER	Fruit soft drinks, plum
	A03ES	Fruit soft drinks, pomegranate
	A03ET	Fruit soft drinks, raspberry
	A03EV	Fruit soft drinks, mixed fruit
	A03FV	Diet soft drinks with fruit juice
	A03EY	Soft drink with bitter principle
	A03EZ	Soft drinks, apple flavour
	A03FA	Soft drink, apricot flavour
	A03FB	Soft drinks, banana flavour
	A03FC	Soft drink, cherry flavour
	A03FE	Soft drinks, grapefruit flavour

Food category	Code ^a	Description
Carbonated soft drinks (not fruit flavoured)	A03FF	Soft drink, lemon flavour
	A03FG	Soft drinks, lime flavour
	A03FH	Soft drink, mixed flavours
	A03FJ	Soft drinks, orange flavour
	A03FK	Soft drink, pear flavours
	A03FL	Soft drinks, pineapple flavour
	A03FM	Soft drink, raspberry flavours
	A03FN	Soft drinks, mango flavour
	A03FX	Diet soft drinks with flavours
	A03FP	Cola beverages, decaffeinated
	A03FR	Cola beverages, caffeinic
	A03FS	Cola mix, flavoured cola
Non-carbonated soft drinks (fruit flavoured)	A03FD	Soft drinks, flavoured with herbs
	A03GB	Isotonic and sport drinks
Non-carbonated soft drinks (not fruit flavoured)	A16YN	Fermented functional drinks (including fermented infusions)
	A03GA	Energy drinks

^a Food code according the FoodEx2 classification.

Table A5.2

Estimates of dietary exposure to diethyl sodium sulfosuccinate among children aged 1–2 years according to the maximum exposure scenario calculated by the Committee using the DietEx Exposure tool

Country	Dietary survey	Total no. individuals	Dietary exposure (mg/kg bw per day)	
			Mean	High ^{a,b}
Belgium	Regional Flanders	36	0.06	0.19 ^c
Bulgaria	NUTRICHILD	428	< 0.01	0.06
Cyprus	National Dietary Survey of the Children of Cyprus	275	< 0.01	0
Denmark	Danish National Dietary Survey among Infants and Young Children 2006–2007	917	0.01	0.05
Estonia	National Dietary Survey among children up to 10 years old and breastfeeding mothers in Estonia	268	0.02	0.08
Finland	Diabetes Prediction and Prevention Nutrition Study (DIPP) 2001–2009	500	< 0.01	0
France	French National Dietary Survey (INCA3, 2014–2015)	139	0.02	0.10
Germany	Consumption Survey of Food Intake among Infants and Young Children	348	0.08	0.30
Hungary	Hungarian National Food Consumption Survey	535	< 0.01	0.05
Italy	Italian National Dietary Survey in children aged 3 months to 9 years	322	< 0.01	0
Latvia	Latvian National Dietary Survey	242	< 0.01	0.04
Netherlands (Kingdom of the)	Dutch National Food Consumption Survey 2012–2016 (DNFCS)	440	0.06	0.25
Portugal	National Food, Nutrition and Physical Activity Survey of the Portuguese general population	571	< 0.01	0.05
Slovenia	Slovenian National Food Consumption Survey	343	< 0.01	0.05
Spain	Spanish National Dietary Survey in children and adolescents	327	< 0.01	0

bw: body weight

^a High exposure: 95th percentile.

^b High exposure equal to 0 mg/kg bw per day occurs when consumption of carbonated and/or non-carbonated soft drinks is reported by less than 5% of the participants of a dietary survey.

^c For a number of observations lower than 60, the 95th percentile may not be statistically robust.

Annex 6

Rosemary extract: supplementary material

Table A4.1 lists the data used in the dose–response modelling (1). The data are deemed suitable for modelling. A sufficient number of dose groups (four) was available. The presence of a trend, dose–response relationship and measured responses in the range of the required BMR of 20% are evaluated based on the results of the analysis.

Table A4.1
Individual serum T₄ levels in male and female pups on PND 13 following exposure of dams to rosemary extract from GD 6 throughout lactation; dose expressed as the sum of carnosol and carnosic acid (1)

Dose (mg/kg bw per day)	T ₄ (pg/mL)		Dose (mg/kg bw per day)	T ₄ (pg/mL)	
	Male	Female		Male	Female
0	40 600	37 000	125	44 000	46 500
0	40 800	36 400	125	42 700	40 300
0	30 300	45 500	125	36 800	47 200
0	40 400	43 800	125	46 300	34 600
0	35 100	48 800	125	31 700	39 400
0	40 200	45 300	125	35 700	46 300
0	37 500	38 600	125	39 000	40 500
0	43 200	47 500	125	33 100	43 400
0	33 600	45 500	125	38 300	39 300
0	57 600	57 500	125	44 400	33 900
0	38 900	47 300	125	40 500	36 300
0	69 000	57 100	125	37 500	35 800
0	43 300	37 700	125	32 100	37 400
72	45 800	42 900	125	34 000	46 700
72	28 900	43 200	176	27 700	34 700
72	45 000	41 500	176	36 300	36 800
72	33 400	40 400	176	32 400	40 300
72	32 800	35 800	176	35 200	22 500
72	35 400	33 500	176	22 900	26 900
72	39 100	29 800	176	25 600	37 300
72	31 000	42 400	176	22 700	41 300

Dose (mg/kg bw per day)	T ₄ (pg/mL)		Dose (mg/kg bw per day)	T ₄ (pg/mL)	
	Male	Female		Male	Female
72	37 300	32 500	176	36 600	32 200
72	35 000	37 000	176	35 900	25 700
72	34 400	36 300	176	38 600	34 600
72	45 300	28 400	176	40 600	37 400
72	45 500	34 600	176	34 800	33 200
72	28 700	52 700	176	29 800	28 700
72	44 000	49 300	176	38 200	44 300
125	40 100	40 500	176	34 700	41 100
125	40 200	37 500	176	36 800	33 900
125	38 000	32 100	176	40 300	34 200
125	35 700	34 000	176	22 500	29 800

bw: body weight; GND: gestational day; PND: postnatal day; T₄: thyroxine.

According to EHC 240 (2), the null and full models were fitted to the data to determine the presence of a trend and the goodness of fit. For each of the four continuous models (Hill, exponential, inverse exponential and lognormal model), their three- and four-parameter variants were fitted, represented as m3 and m5, respectively, in Table A4.2. In addition, each model was fitted with and without sex as covariate on parameter *a* (background) or *b* (sensitivity). No constraints were applied to steepness parameter *d* (3). In general, continuous responses are considered to be lognormally distributed (4). Regarding the current T₄ data, this assumption was confirmed by quantile-quantile plots of the residuals after analysis (Fig. A4.1). Quantile-quantile plots of the regression residuals are positioned on the unity line, indicating that the data are lognormally distributed.

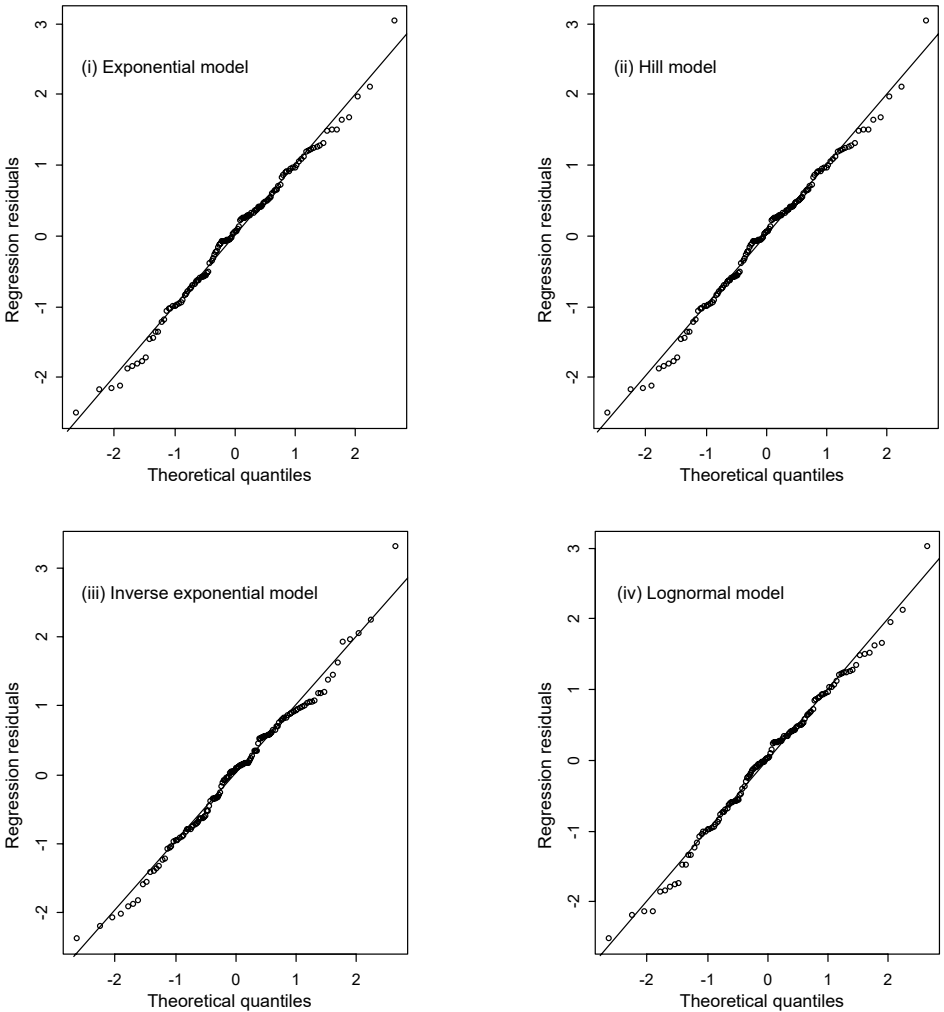
Table A4.2
The fitted models included in the BMD analysis of T₄ serum concentration in pups on PND 13

Model	Log-likelihood	No. parameters	Akaike information criterion
Full model	47.64	9	−77.28
Full model- <i>v</i>	47.75	10	−75.50
Null model	29.25	2	−54.50
Null model- <i>a</i>	29.91	3	−53.82
Exponential m3	42.71	4	−77.42
Exponential m3-<i>a</i>	43.9	5	−77.80
Exponential m3- <i>b</i>	43.18	5	−76.36
Exponential m3- <i>ab</i>	43.98	6	−75.96
Exponential m5	42.71	5	−75.42
Exponential m5- <i>a</i>	43.89	6	−75.78
Exponential m5- <i>b</i>	43.17	6	−74.34
Exponential m5- <i>ab</i>	43.98	7	−73.96
Hill m3	42.71	4	−77.42
Hill m3-<i>a</i>	43.89	5	−77.78
Hill m3- <i>b</i>	43.18	5	−76.36
Hill m3- <i>ab</i>	43.98	6	−75.96
Hill m5	42.7	5	−75.40
Hill m5- <i>a</i>	43.88	6	−75.76
Hill m5- <i>b</i>	43.17	6	−74.34
Hill m5- <i>ab</i>	43.97	7	−73.94
Inverse exponential m3	42.3	4	−76.60
Inverse exponential m3-<i>a</i>	43.58	5	−76.74
Inverse exponential m3- <i>b</i>	42.62	5	−75.24
Inverse exponential m3- <i>ab</i>	43.38	6	−74.76
Inverse exponential m5	42.31	5	−74.62
Inverse exponential m5- <i>a</i>	43.35	6	−74.70
Inverse exponential m5- <i>b</i>	42.61	6	−73.22
Inverse exponential m5- <i>ab</i>	43.36	7	−72.72
Lognormal m3	42.53	4	−77.06
Lognormal m3-<i>a</i>	43.72	5	−77.44
Lognormal m3- <i>b</i>	43.02	5	−76.04
Lognormal m3- <i>ab</i>	43.78	6	−75.56
Lognormal m5	42.48	5	−74.96
Lognormal m5- <i>a</i>	43.67	6	−75.34

Lognormal m5- <i>b</i>	42.98	6	-73.96
Lognormal m5- <i>ab</i>	43.78	7	-73.56

a, *b* and *v* indicate the model parameter (background, sensitivity and residual variance, respectively) on which the covariate sex was applied; BMD: benchmark dose; m3, m5: three- and four-parameter variants of the models; PND: postnatal day; T₄: thyroxine.

Fig. A4.1
Quantile-quantile plots of the regression residuals of the (i) exponential, (ii) Hill, (iii) inverse exponential and (iv) lognormal models.

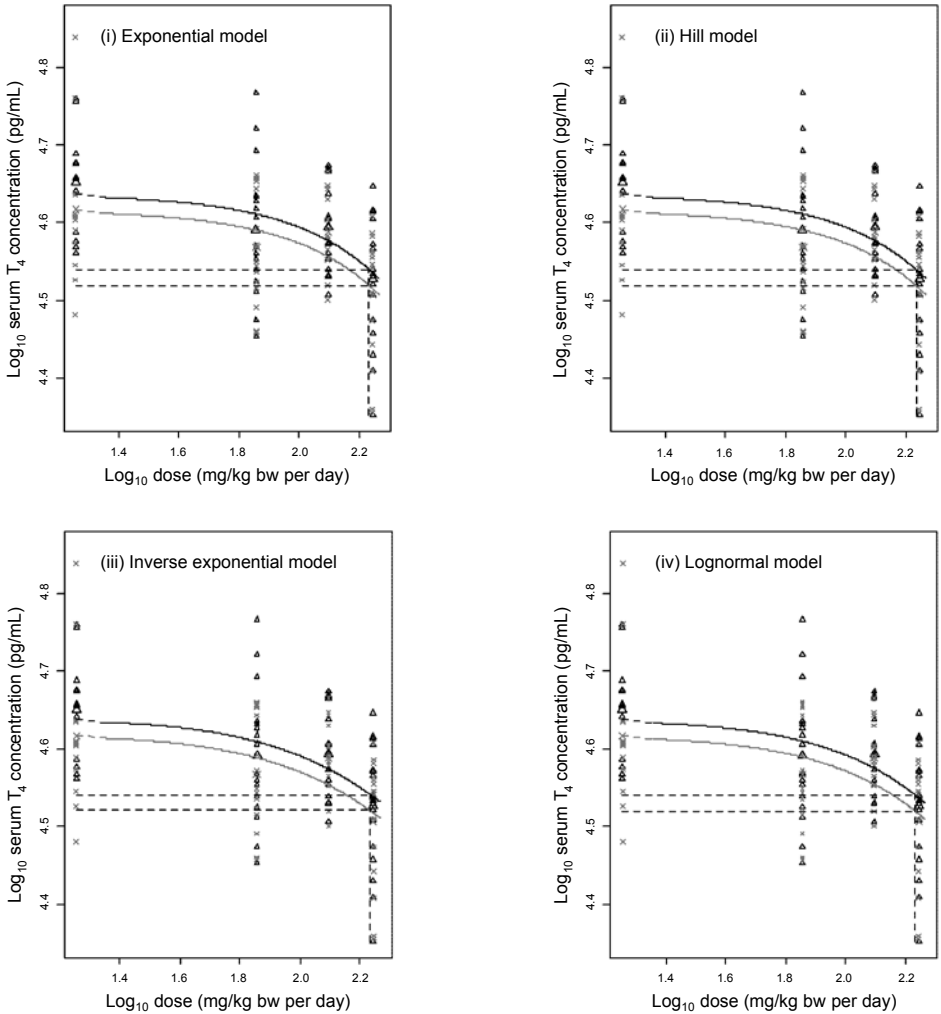


Suitability and choices of models are based on the Akaike information criterion (AIC), where a lower (more negative) value indicates a better description of the data. Comparison of the AICs of the full model and the full model with a covariate on the residual variance ν indicates that there is no difference in residual variance between males and females. A covariate is therefore not applied in further analysis.

For each of the four models the three-parameter model (m3) with a covariate on background provides the lowest AIC ([Table A4.2](#), in bold). This indicates that sexes differ in background T_4 levels but are equally sensitive. It is noted that different applications of covariates within each three-parameter model result in minor (< 2) differences in AICs, indicating that the differences in background are minor, and that differences in sensitivity cannot be excluded. The higher AICs of the four-parameter models (m5) indicate that the data do not provide sufficient information to estimate a maximum (levelling-off) response at high doses. The lower AIC (by more than two points difference) in m3 for each of the four models compared with the null model indicates the presence of a significant trend. The similar AIC (less than two points difference) in m3 for each of the four models compared with the full model indicates a sufficient goodness of fit.

The four selected models are plotted in [Fig. A4.2](#) (see [Table A4.3](#) for parameter estimates and resulting $BMDL_{20}$ and $BMDU_{20}$ values). All models converged, indicating that the data are sufficiently informative to estimate the required parameter values. It can also be observed that sufficient suitable data are available in the range of the required BMR of 20%.

Fig. A4.2
Dose–response of serum T₄ concentrations versus dose on double log scales for the (i) exponential, (ii) Hill, (iii) inverse exponential and (iv) lognormal models, resulting in the best (lowest) AIC (see Table A4.2). Triangles and upper curves correspond to female pups and crosses and lower curves to males. Small symbols are individual measurements (see Table A4.1) and larger symbols are the medians per dose group. See Table A4.3 for parameter estimates and resulting BMDL₂₀ and BMDU₂₀ values.



AIC: Akaike information criterion; BMDL₂₀/BMDU₂₀: lower/upper 95% confidence limit on the benchmark dose for a 20% response.

Table A4.3
Parameter estimates and outcomes using PROAST version 71.1 for Fig. A4.2.

Model	(i) Exponential	(ii) Hill	(iii) Inverse exponential	(iv) Lognormal
$f(x)$	$a \left(c^{1-\exp(-(x/b)^d)} \right)$	$a \left(c^{x^d/(b^d+x^d)} \right)$	$a \left(c^{\exp(-(x/b)^{-d})} \right)$	$a \left(c^{\Phi(\ln(b)+d \ln(x))} \right)$
Covariate	Sex	Sex	Sex	Sex
Parameter estimates				
a (female)	43 320	43 320	43 510	43 430
a (male)	41 310	41 310	41 490	41 410
b	5 164	5 131	173 700	0.006 796
c	Fixed at 1×10^{-18}	Fixed at 1×10^{-18}	Fixed at 1×10^{-18}	Fixed at 1×10^{-18}
d	1.534	1.536	0.2389	0.4745
Residual variance (var)	0.028 51	0.028 51	0.028 66	0.028 6
Log-likelihood	43.9	43.89	43.58	43.72
AIC	-77.8	-77.78	-77.16	-77.44
BMR (%)	20	20	20	20
BMD (mg/kg bw per day)	172	172	172	172
BMDL (mg/kg bw per day)	129	129	126	127
BMDU (mg/kg bw per day)	227	228	244	236

AIC: Akaike information criterion; BMD: benchmark dose; BMDL/BMDU: lower/upper 95% confidence limit on the benchmark dose; BMR: benchmark response.

In model averaging, in each of the 1000 runs newly bootstrapped data are modelled using all model and covariate combinations listed in [Table A4.2](#), as depicted in [Fig. A4.3](#). In several runs this resulted in different sensitivities between male and female pups, and therefore different model-averaged BMDL and BMDU for males and females ([Table A4.4](#)) compared with the results of the individual models ([Table A4.5](#)). It also resulted in a lower model-averaged BMDL for male pups ([Table A4.4](#)) compared with the BMDLs of the individual models ([Table A4.5](#)).

Fig. A4.3
Bootstrap curves based on model averaging using 1000 bootstrap runs for the benchmark dose (BMD) analysis of T_4 serum levels in pups on PND 13. Triangles and upper (black) curves correspond to median T_4 levels in female pups, and crosses and lower (grey) curves to levels in males. Open circles of corresponding shades at the bottom of the plot indicate the derived BMD for each bootstrap run.

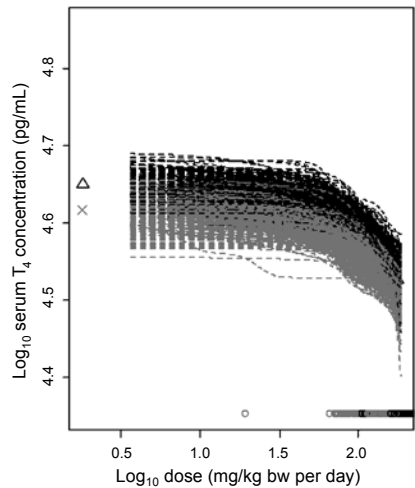


Table A4.4
Model-averaged BMDLs and BMDUs for reductions in serum T_4 concentrations in pups on PND 13 following exposure of dams from GD 6 throughout lactation to rosemary extract, expressed in mg/kg bw per day carnosol and carnolic acid

Sex	BMDL ₂₀ (mg/kg bw per day)	BMDU ₂₀ (mg/kg bw per day)
Male	97.2	213
Female	126	247

BMDL₂₀/BMDU₂₀: lower/upper 95% confidence limit on the benchmark dose for a 20% response; GD: gestational day; PND: postnatal day; T_4 : thyroxine.

Table A4.5
The relative model weight for the BMD analysis of T_4 serum levels in pups on PND 13

Model	Weight	BMDL ₂₀ (mg/kg bw per day)	BMDU ₂₀ (mg/kg bw per day)
Exponential	0.2816	129	227
Hill	0.2788	129	228
Inverse exponential	0.2045	126	244
Lognormal	0.2352	127	236

BMD: benchmark dose; BMDL₂₀/BMDU₂₀: lower/upper 95% confidence limit on the benchmark dose for a 20% response; PND: postnatal day; T_4 : thyroxine.

References

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An abstract graphic consisting of numerous thin, white, wavy lines that flow from the top left towards the bottom right, creating a sense of movement and depth against a solid yellow background.

This volume contains monographs prepared at the one-hundredth meeting of the Joint FAO/WHO Expert Committee on Food Additives (JECFA), which met at FAO headquarters in Rome, Italy, on 10–19 June 2025.

The toxicological and dietary exposure monographs in this volume summarize the safety and dietary exposure data on seven specific food additives: amyloglucosidase (JECFA95-4) from *Rasamsonia emersonii* expressed in *Aspergillus niger* (the safety evaluation of which could not be completed at the Ninety-fifth meeting), ascorbyl palmitate, carob bean gum, Gardenia (Genipin) Blue, glycolipids, rosemary extract and thaumatin II. The dietary exposure data are summarized for an additional two food additives: adipates and dioctyl sodium sulfosuccinate.

This volume and others in the WHO Food Additives series contain information that is useful to those who produce and use food additives and veterinary drugs, those involved in the control of contaminants in food, government and food regulatory officers, industrial testing laboratories, toxicological laboratories and universities.